



On-Chain Operator Program

From zero to your own on-chain bank · Master File

Draft · 2026-09-24

Educational content only. Not financial advice. No results are guaranteed.

On-Chain Operator Program — Master File



ON-CHAIN OPERATOR
PROGRAM

Contents

1. Build status
2. Offer, decisions & curriculum
3. Whop setup
4. Complete mastery map (every topic, where it's taught, at what level)
5. Whop store listing (logo, images, copy)
6. Operations kit (capstones, application, call script, emails, live tier, terms)
7. Worksheets & templates
8. Funnel changes
9. Video scripts (VSLs, welcome, module intros) & course video plan
10. Course content (92 lessons + 15 Mastery Starters)
11. Skills archive
12. Build prompt (hand this to Grok or Claude Code)

1. Build status

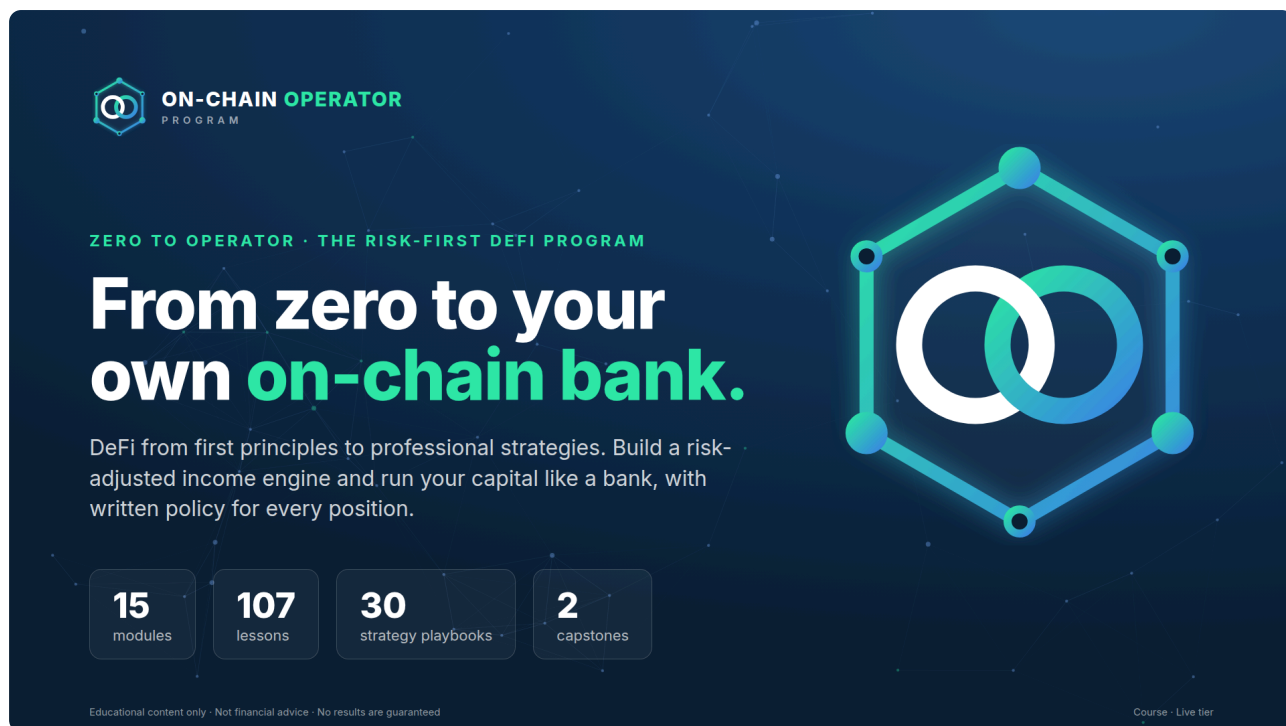
Everything in one file: `ON-CHAIN-OPERATOR-PROGRAM.md`, also as `On-Chain-Operator-Program.docx` (Word) and `On-Chain-Operator-Program.pdf`. Rebuild: `python3 programs/defi-program/build_master.py`, then `cd programs/defi-program/export && npm install && npm run build`. **To finish the build with AI:** paste `BUILD-PROMPT.md` into Whop's AI (Grok 4.6) and attach the files it lists.

File	Phase	Status
<code>01-offer-and-curriculum.md</code>	1–2 Offer + curriculum	Decisions locked; curriculum expanded to 6 stages · 15 modules · 107 lessons
<code>02-sample-lesson-amm-math.md</code>	Lesson 2.2	Written
<code>03-defi-strategy-mastery.md</code>	Lesson 8.3 (strategy library)	Written
<code>04-funnel-changes.md</code>	4 Funnel wiring	Draft, not applied to Zapier
<code>05-whop-setup.md</code>	3 Whop setup	Ready
<code>06-whop-store-listing.md</code>	Store product listing: copy + image upload map	Ready to paste (3 placeholders)
<code>assets/</code>	55 images: logo system (8), store icon + banner + 7 gallery (9), 15 module banners, 17 diagrams, 6 charts	Rendered by <code>export/build_images.js</code>
<code>Day-1-Setup-Kit.pdf</code>	Printable beginner setup checklist (generated from Module 0)	Built by <code>npm run build</code>
<code>07-program-operations.md</code>	Capstones & rubrics, certification, application form & scoring, call script, member emails, live tier, community, terms, 14-day refund policy, support, launch plan	Ready
<code>10-video-production-plan.md</code>	Every course video: 121 lesson/starter videos (≤ 25 min each) with script files, structure and render commands	Scripts ready
<code>video-scripts/lessons/</code>	121 lesson video scripts (JSON scenes, rendered by <code>export/build_video.js</code>)	Generated by <code>export/build_lesson_scripts.py</code>
<code>09-mastery-map.md</code>	Complete topic map: every DeFi mastery topic, where it's taught, at what level; section plan	Ready
<code>sections/</code>	The course split into 6 stage files (generated by <code>build_master.py</code>) for review and hand-off in parts	Generated

File	Phase	Status
08-worksheets.md	17 worksheet templates used by lessons and capstones	Ready
video/	18 videos: main VSL, 30s vertical VSL, welcome, 15 module intros + SCRIPTS.md	Rendered by export/build_video.js
lessons/module-00-crypto-from-zero.md	Module 0 (8) + Day-1 Setup Kit	Written
lessons/module-01-foundations-safety.md	Module 1 (6)	Written
lessons/module-02-trading-on-chain.md	Module 2 (4 + sample 2.2)	Written
lessons/module-03... module-08	Modules 3–8 (lesson 8.3 = 03-...)	Written
lessons/module-09-defi-vs-grid-bots.md	Module 9 (3)	Written
lessons/module-10-advanced-yield-engineering.md	Module 10 (6)	Written
lessons/module-11-hedging-risk-engineering.md	Module 11 (5)	Written
lessons/module-12-own-bank.md	Module 12 (6)	Written
lessons/module-13-income-engine.md	Module 13 (5)	Written
lessons/module-14-automation-mastery.md	Module 14 (4)	Written

All 107 lessons (15 expert) and 15 Mastery Starters are written (Modules 0–14, plus lesson 8.3 in 03-defi-strategy-mastery.md). Full topic coverage: 09-mastery-map.md. Stage-by-stage section files: sections/. Course videos: every lesson and Mastery Starter has a ready-to-render script (video-scripts/lessons/, plan in 10-video-production-plan.md).

2. Offer, decisions & curriculum



The graphic is a dark blue rectangular banner. In the top left corner is the 'ON-CHAIN OPERATOR PROGRAM' logo, featuring a stylized 'O' with a green and blue geometric design. To the right of the logo, the text 'ON-CHAIN OPERATOR PROGRAM' is written in white and green. Below this, the text 'ZERO TO OPERATOR · THE RISK-FIRST DEFI PROGRAM' is in green. The main headline 'From zero to your own on-chain bank.' is in large white and green font. Below the headline, a paragraph describes the program: 'DeFi from first principles to professional strategies. Build a risk-adjusted income engine and run your capital like a bank, with written policy for every position.' To the right of the text is a large, glowing green and blue geometric logo consisting of a hexagon with internal circles. At the bottom, four white boxes contain the following statistics: '15 modules', '107 lessons', '30 strategy playbooks', and '2 capstones'. At the very bottom, small text reads 'Educational content only · Not financial advice · No results are guaranteed' on the left and 'Course · Live tier' on the right.

ON-CHAIN OPERATOR PROGRAM

ZERO TO OPERATOR · THE RISK-FIRST DEFI PROGRAM

From zero to your own on-chain bank.

DeFi from first principles to professional strategies. Build a risk-adjusted income engine and run your capital like a bank, with written policy for every position.

15 modules

107 lessons

30 strategy playbooks

2 capstones

Educational content only · Not financial advice · No results are guaranteed

Course · Live tier

Status: **draft, nothing created on Whop yet.** Items marked **DECIDE** need your call before phase 3 (Whop setup).

Decisions so far (2026-09-24)

Item	Decision
Name	On-Chain Operator Program
Format	Two tiers: self-paced course + a higher tier with live sessions
Sales path	Application → call → checkout
Audience	Everyone at once: GBB customers, Elite Intel members and cold ad traffic from launch
Price	Course \$15,000 one-time · Live tier on application
Refund policy	14-day conditional refund (≤ 10% of lessons, no capstone, no live session used); full text in 07-program-operations.md §9
Starter tier / GBB pricing conflict	Open
Structure	15 modules, 107 lessons (incl. 15 expert) + a Mastery Starter per module, 6 stages: zero to operator (expanded 2026-09-24)

What \$15,000 changes (recommendations, your call):

- It becomes the top of the ladder (above Grid Bot Elite at \$2,497), so it needs a **high-ticket sales path**: cold traffic → application → call → checkout, not straight-to-checkout ads. `gbb-new-Lead`

scoring (capital 20k+ = 4) becomes the qualifier.

- Buyers at this price expect access to you: the live tier (calls, reviews, capstone feedback) carries most of the value. The course alone is hard to justify at \$15k.
- High-ticket crypto education attracts consumer-protection scrutiny. Keep every page and call script free of income/return claims, and put terms, refund policy and "educational, not financial advice" at checkout.

Built from: Notion → ATLAS → *DeFi & On-Chain (Complete Module)* (42 chapters, 16 strategy frameworks, 20 on-chain metrics) and the existing Grid Bot Builder offer stack.

Phase 1 — Offer

Positioning

Grid Bot Builder automates trading **on exchanges**. The DeFi program teaches people to operate **on-chain** safely: research a protocol, size a position, know the exit. Your module already has a strong line to lead with:

"If you can't explain the unwind, you don't understand the position yet."

Promise (skills, not returns): *Research any DeFi protocol with a 6-step risk loop, understand exactly where yield comes from, and never sign a transaction you can't unwind.*

Working name — DECIDE

Superseded: see "Decisions so far" at the top. Kept for the record.

Option	Why
DeFi Operator Blueprint (recommended)	Matches "Grid Bot Building Blueprint" on Whop; "operator" = process, not hype
Risk-First DeFi	States the differentiator
On-Chain Operator Program	Neutral, broader

Who it's for — DECIDE

Superseded: see "Decisions so far" at the top. Kept for the record.

Audience	Fit	How they arrive
GBB customers (recommended primary)	Already trust you, already think in systems/risk	Cross-sell email ~14 days after GBB onboarding
Elite Intel Community members	Warm, recurring, already reading intel briefs	In-community offer + LAUNCH-style code
Cold ad traffic	Largest pool, needs most education	Separate ad campaign → lead skill with defi-lead tag

Recommendation: launch to GBB customers + Elite Intel members first (warm, lower CAC, real feedback), open to cold traffic once conversion is proven.

Format — DECIDE

Superseded: see "Decisions so far" at the top. Kept for the record.

- **Core (recommended):** self-paced Whop course (15 modules below) + worksheets (the two Notion trackers as templates) + quizzes + capstone.
- **Optional higher tier:** Core + live monthly DeFi Q&A / portfolio review. Your Calendly already has `vip-defi-consult` and `defi` links — are those for this program? (The GBB skills are told never to use them, so they're free for DeFi.)

Pricing — DECIDE (proposal only)

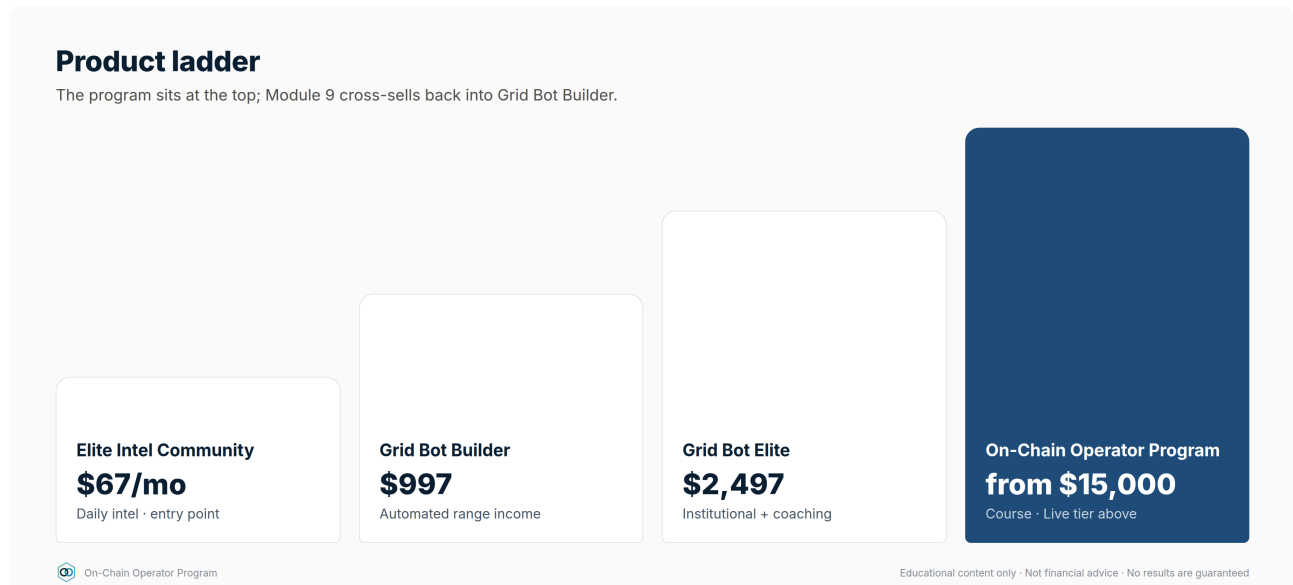
Superseded: see "Decisions so far" at the top. Kept for the record. Mirror GBB's one-time + 3-pay pattern, priced **below** GBB so it works as a cross-sell rather than competing with it:

Plan	Proposal	Reasoning
One-time	\$497	Half of GBB; education product, not software
3-pay	3 × \$197 (\$591)	~19% payment-plan premium, in line with GBB's 20% (3 × \$399 vs \$997)
GBB customer price	\$397 or bundle	Rewards the cross-sell
Higher tier (if live calls)	\$997	Only if you'll run the calls

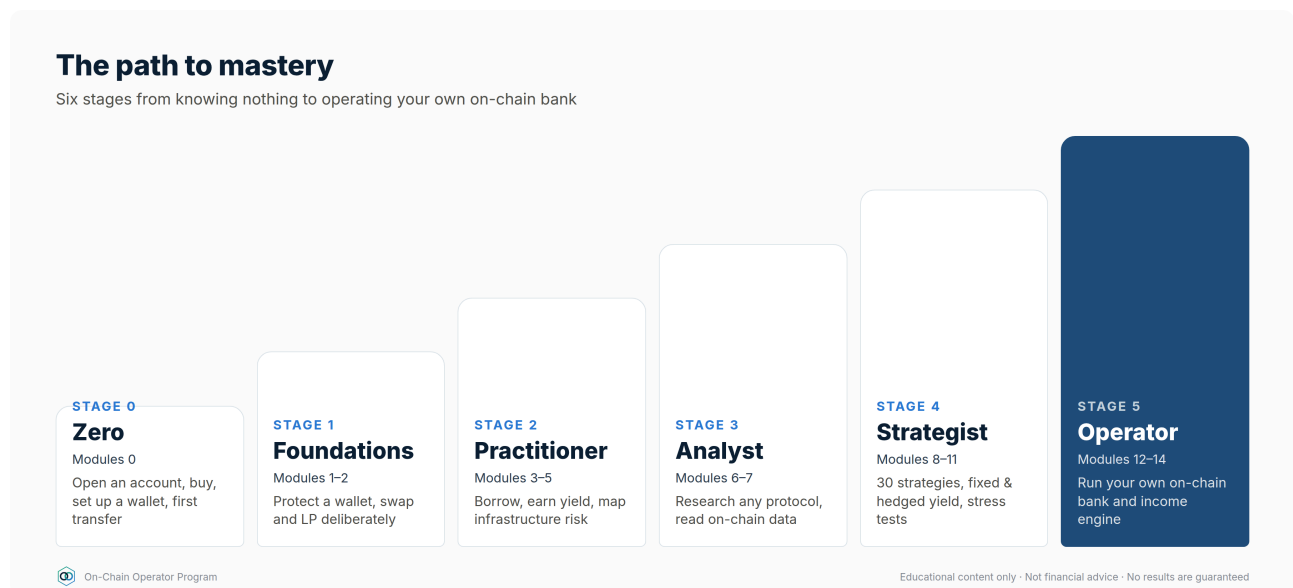
Refund policy — DECIDE: GBB has 30-day money-back on the software. For a course, suggest **14 days, if under 30% of lessons completed**.

Still open from before: gbb ad performance mentions a "Grid Bot Starter" (A\$47–97) and GBB at A\$497. Other skills say \$997, no Starter. Which is current? It affects where DeFi sits in the ladder.

Ladder after launch



Phase 2 — Curriculum: zero to mastery



Expanded 2026-09-24: the program now takes someone who knows nothing all the way to operating as their own bank. That's **15 modules and 107 lessons in 6 stages**, with expert lessons (marked *expert*) in every stage from Foundations up, and **every module opens with a Mastery Starter** (lesson N.0): a plain-English primer, the words you need, a first safe step and a beginner → practitioner → master ladder, so anyone can enter any module from zero. Modules 1–9 keep their numbers. Module 0 and Modules 10–14 are new. All 42 ATLAS chapters are still used exactly once; the new modules are original content.

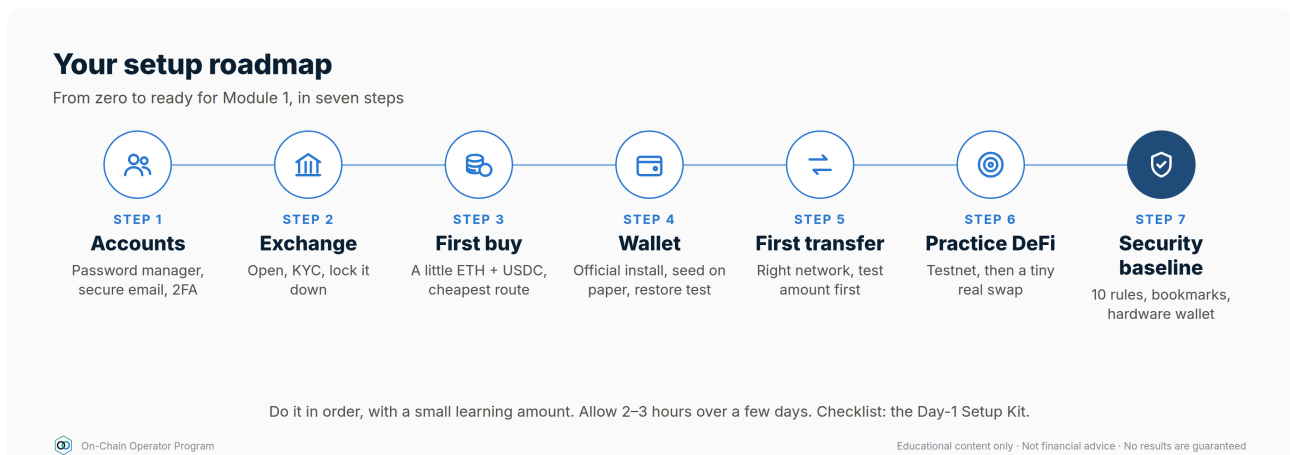
Every lesson follows one template: **Objective** → **Explanation** → **Worked example** → **Checklist** → **3 quiz questions**, with at least one diagram or chart. A full sample lesson is in `02-sample-lesson-amm-math.md`.

Stage	Modules	You can...
0 • Zero	0	Open and secure an account, buy, set up and back up a wallet, make a first transfer and a first practice DeFi transaction
1 • Foundations	1–2	Protect a wallet, read a transaction, swap and LP deliberately
2 • Practitioner	3–5	Borrow, earn yield and map infrastructure risk
3 • Analyst	6–7	Research any protocol and read on-chain data
4 • Strategist	8–11	Run the 30-strategy library, engineer fixed and hedged yield, stress-test a portfolio
5 • Operator	12–14	Run your own on-chain bank: balance sheet, credit line, income engine, automation

Stage 0 • Zero

Module 0 — Crypto From Zero (8 lessons + Mastery Starter, new)

Outcome: go from never having owned crypto to a secured exchange account, a backed-up wallet, a first transfer and a first practice DeFi transaction. Ends with the **Day-1 Setup Kit** checklist.



#	Lesson	Source
0.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
0.1	Money, ledgers and why blockchains exist	new
0.2	Opening and securing an exchange account (KYC, 2FA, allowlists)	new
0.3	Buying your first crypto without overpaying	new
0.4	Exchange account vs your own wallet: who holds the keys?	new
0.5	Setting up your wallet and backing it up (restore test)	new
0.6	Networks, gas and your first transfer	new
0.7	Your first DeFi steps (testnet practice first)	new
0.8	Your security baseline, and the language of DeFi	new

Stage 1 · Foundations

Module 1 — Foundations & Safety (9 lessons + Mastery Starter)

Outcome: set up and use a wallet safely; know what can go irreversibly wrong.

#	Lesson	ATLAS ch.
1.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
1.1	What DeFi is — and the risk-first mindset	1
1.2	How a transaction actually happens (gas, nonce, finality)	2
1.3	Wallets, keys, hardware & multisig	3
1.4	Tokens, approvals & allowances	4
1.5	Stablecoins and how they break	5
1.6	Scam defence: phishing, drainers, address poisoning	36
1.7	Reading signatures and simulating transactions	new
1.8	Privacy and physical security	new
1.9	Smart accounts and account abstraction (<i>expert</i>)	new

Assets: before-signing checklist (ch. 42) introduced here and reused in every module.

Module 2 — Trading On-Chain (8 lessons + Mastery Starter)

Outcome: execute a swap deliberately, understanding price impact and MEV.

#	Lesson	ATLAS ch.
2.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
2.1	DEXs, aggregators & routing	6
2.2	AMM mathematics ($x \cdot y = k$) — <i>sample lesson</i>	7
2.3	Providing liquidity	8
2.4	Impermanent loss & true LP P&L	9
2.5	MEV and how to protect your trades	35
2.6	Advanced execution: limit, TWAP and intent-based orders	new
2.7	Advanced AMM design and LVR (<i>expert</i>)	new
2.8	The MEV supply chain (<i>expert</i>)	new

Assets: AMM flow diagram, 100 ETH / 300k USDC worked example, LP-vs-hold benchmark.

Stage 2 · Practitioner

Module 3 — Lending & Leverage (7 lessons + Mastery Starter)

Outcome: borrow against collateral with a buffer and a written defence plan.

#	Lesson	ATLAS ch.
3.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
3.1	How lending markets work (utilisation, rate curves)	10
3.2	LTV, liquidation threshold & health factor	11
3.3	Liquidations and cascades	12
3.4	Borrowing strategies & looping	13
3.5	Perpetual futures and margin on-chain	new
3.6	Lending design deep dive: e-mode, caps, auctions, soft liquidation, bad debt (<i>expert</i>)	new
3.7	CDP stablecoins: minting your own dollars (<i>expert</i>)	new

Assets: liquidation feedback-loop diagram, Aave health-factor primary source.

Module 4 — Yield (8 lessons + Mastery Starter)

Outcome: split any APY into organic vs subsidised, and name the risk being paid for.

#	Lesson	ATLAS ch.
4.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
4.1	Yield farming: base yield vs emissions	14
4.2	Native staking	15
4.3	Liquid staking (LSTs)	16
4.4	Restaking & shared security	17
4.5	Vaults & yield optimisers	18
4.6	Airdrops & points — opportunity cost	37
4.7	Stablecoin savings rates and yield-bearing stablecoins	new
4.8	Tokenized treasuries and real-world assets (RWAs)	new

Module 5 — Infrastructure Risk (8 lessons + Mastery Starter)

Outcome: map every bridge, oracle, L2 and contract a position depends on.

#	Lesson	ATLAS ch.
5.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
5.1	Bridges and trust assumptions	19
5.2	Layer 2s, sequencers & withdrawal paths	20

#	Lesson	ATLAS ch.
5.3	Oracles, TWAPs & manipulation	21
5.4	Smart contracts: state, proxies, admin keys	22
5.5	Smart-contract risk & what audits don't prove	23
5.6	Beyond Ethereum: Solana, Bitcoin and other ecosystems	new
5.7	Operating across chains: gas, routes and chain abstraction	new
5.8	Reading smart-contract code: enough to verify claims (<i>expert</i>)	new

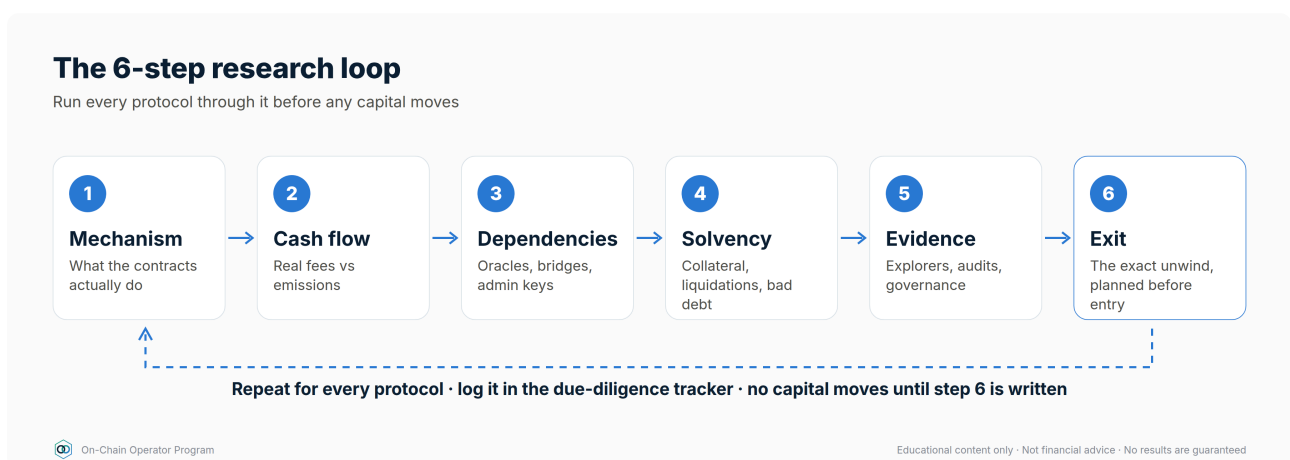
Stage 3 · Analyst

Module 6 — Protocol Research (7 lessons + Mastery Starter)

Outcome: complete a full due-diligence file on a real protocol.

#	Lesson	ATLAS ch.
6.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
6.1	Protocol due diligence	24
6.2	Tokenomics: supply, unlocks, FDV, value capture	25
6.3	Governance & DAOs	26
6.4	The on-chain research workflow	41
6.5	Case studies: how DeFi failures happened	new
6.6	Vote-escrow tokenomics, bribes and governance markets (<i>expert</i>)	new
6.7	Valuing DeFi protocols: fees, revenue, earnings, multiples (<i>expert</i>)	new

Assets: **DeFi Protocol Due Diligence Tracker** (Notion) + `defi-due-diligence` 6-step loop.



Module 7 — On-Chain Analytics (9 lessons + Mastery Starter)

Outcome: read on-chain data without over-interpreting it.

#	Lesson	ATLAS ch.
7.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
7.1	On-chain data foundations (address ≠ person)	27
7.2	Block explorer mastery	28
7.3	Exchange flows	29
7.4	Whale & entity analysis	30
7.5	Holder & supply metrics (MVRV, SOPR, HODL waves)	31
7.6	Network activity	32
7.7	DEX & liquidity analytics	33
7.8	Derivatives on-chain (perps, funding, OI)	34
7.9	Querying chain data yourself (SQL, indexers) (<i>expert</i>)	new

Assets: 20-metric directory as a downloadable reference card.

Stage 4 · Strategist

Module 8 — The DeFi Operating System (7 lessons + Mastery Starter)

Outcome: a written portfolio plan with risk buckets, limits and an emergency plan.

#	Lesson	ATLAS ch.
8.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
8.1	DeFi portfolio construction	38
8.2	The DeFi risk framework	39
8.3	Strategy library (30 strategies in 7 levels, Core → Expert), full content in 03-defi-strategy-mastery.md	40
8.4	The operating playbook: deploy, monitor, respond, review	42
8.5	Measuring performance honestly	new
8.6	Psychology and discipline	new
8.7	Quantitative risk: volatility, VaR, drawdown, correlation, sizing (<i>expert</i>)	new

Module 9 — DeFi vs Grid Bots (3 lessons + Mastery Starter, new content)

Outcome: choose the right tool for the market — and the natural bridge into GBB.

#	Lesson	ATLAS ch.
9.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
9.1	LP position vs grid bot: same idea (sell high/buy low inside a range), different risks	new

#	Lesson	ATLAS ch.
9.2	When a CEX grid beats on-chain LP, and when it doesn't (fees, custody, IL vs inventory risk, regime)	new
9.3	Building a combined system: grid bots for range markets, DeFi for yield/collateral	new

Cross-sell: GBB checkout link for non-customers; Elite Intel invite for everyone.

Module 10 — Advanced Yield Engineering (9 lessons + Mastery Starter, new)

Outcome: build fixed, hedged and structured yield, and know exactly what each one is short.

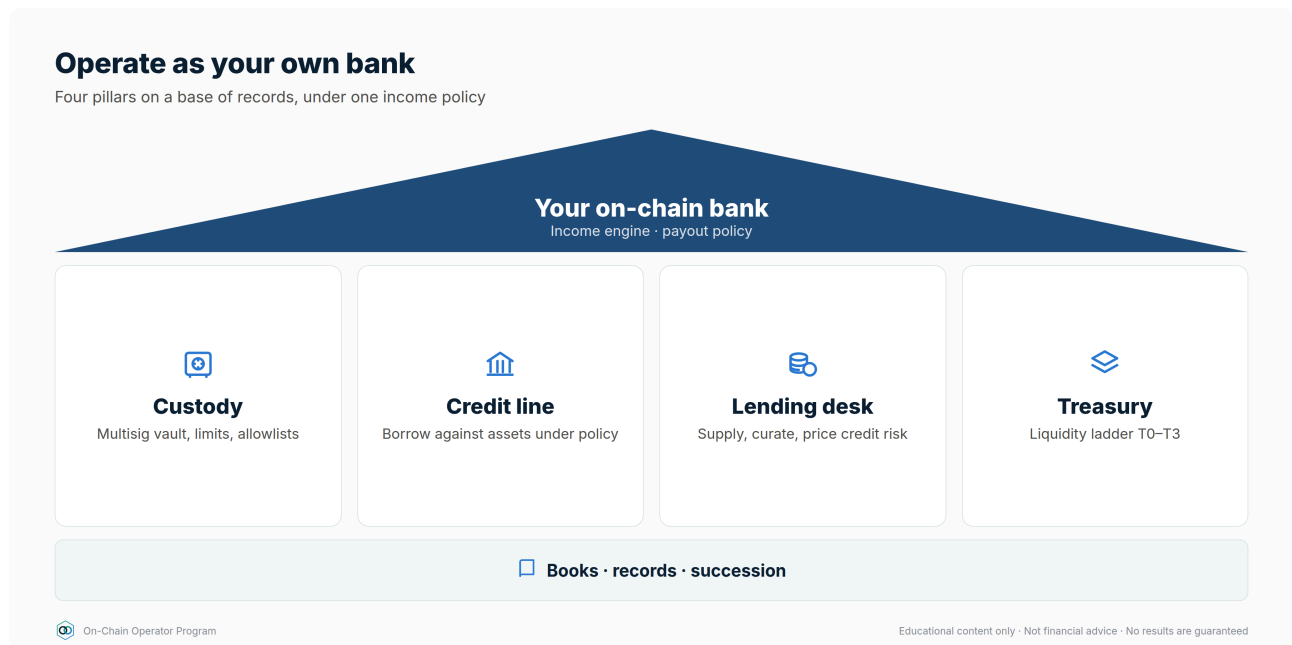
#	Lesson	Source
10.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
10.1	Fixed-rate yield: principal and yield tokens (PT/YT)	new
10.2	Cash-and-carry basis trades	new
10.3	Delta-neutral funding carry, done properly	new
10.4	Options income: covered calls, cash-secured puts, options vaults	new
10.5	Active concentrated-liquidity management	new
10.6	Restaking and points: pricing speculative yield	new
10.7	Being the house: perp-exchange liquidity vaults (<i>expert</i>)	new
10.8	DeFi rates: term structure, fixed vs floating (<i>expert</i>)	new
10.9	Peg and redemption arbitrage (<i>expert</i>)	new

Module 11 — Hedging & Risk Engineering (5 lessons + Mastery Starter, new)

Outcome: hedge the risks you don't want, stress-test the portfolio, and have an incident plan ready.

#	Lesson	Source
11.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
11.1	Hedging price exposure with perps and options	new
11.2	Depeg, protocol and smart-contract cover	new
11.3	Liquidation protection: buffers, alerts and automated deleveraging	new
11.4	Stress-testing a portfolio: -50% days, depegs, rate spikes	new
11.5	Incident response: what to do in the first 60 minutes	new

Stage 5 · Operator — operate as your own bank



Module 12 — Operate as Your Own Bank (8 lessons + Mastery Starter, new)

Outcome: run your crypto like a bank runs its book: a balance sheet, custody policy, a credit line, a liquidity ladder and records.

#	Lesson	Source
12.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
12.1	Your balance sheet: assets, liabilities, equity	new
12.2	Custody architecture: vault, multisig and spending policies	new
12.3	The credit line: borrowing against your assets like a bank client	new
12.4	Treasury and the liquidity ladder	new
12.5	Being the lender: supplying, curating and pricing credit risk	new
12.6	Books, records and succession: if you're gone, can your family recover it?	new
12.7	Tax, regulation and compliance awareness	new
12.8	Institutional-grade custody: MPC, custodians, policy engines (<i>expert</i>)	new

Module 13 — The Income Engine (5 lessons + Mastery Starter, new)

Outcome: design an income portfolio, measure expected income after expected losses, and set a payout you can sustain.

#	Lesson	Source
13.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
13.1	Income sources ranked by durability	new

#	Lesson	Source
13.2	Risk-adjusted yield: subtracting expected losses	new
13.3	Building the income portfolio	new
13.4	The payout policy: how much you can take out	new
13.5	Scaling, compounding and the annual review	new

Module 14 — Automation & Mastery (6 lessons + Mastery Starter, new)

Outcome: monitor and automate safely, run multisig operations, and complete the operator capstone.

#	Lesson	Source
14.0	Mastery Starter: plain-English primer, key words, first safe step, mastery ladder	new
14.1	Monitoring: dashboards, alerts and on-chain watchers	new
14.2	Automation: keepers, bots and agents without handing over the keys	new
14.3	Operating procedures: multisig signing, change control, reviews	new
14.4	Operator capstone and certification	new
14.5	Building your own tools: reading contracts, data and simple scripts	new
14.6	Searchers, keepers and arbitrage bots (<i>expert</i>)	new

Capstones

- **Analyst capstone (after Module 7):** a complete due-diligence file on one real protocol.
- **Operator capstone (Module 14):** a full personal "bank": balance sheet, custody policy, credit-line policy, income portfolio with risk-adjusted expected income, payout policy, stress test and incident plan. Reviewed on the Live tier.

Totals

15 modules · 107 lessons (42 ATLAS + 65 new, 15 of them expert) + 15 Mastery Starters · 323 quiz questions · 30 strategy playbooks in 7 levels · 2 capstones · strategy calculator with 20 commands. Full topic coverage: [09-mastery-map.md](#).

Decided

All launch decisions are made: pricing (Course \$15,000; Live on application), the 14-day conditional refund policy, lifetime access, and the 6-stage structure.

3. Whop setup

Decisions locked (2026-09-24)

Item	Decision
Name	On-Chain Operator Program
Tier 1 — Course (self-paced, 15 modules, 107 lessons + Mastery Starters, worksheets, 2 capstones)	\$15,000 one-time (no payment plan)
Tier 2 — Live (course + live sessions)	Price on application (set on the sales call)
Audience	Everyone at launch (GBB customers, Elite Intel members, cold ads)
Sales path	Application → call → checkout (no straight-to-checkout ads)
Call link	<code>vip-defi-consult</code> Calendly (verify the exact URL is live)

Refund policy: 14-day conditional refund (`07-program-operations.md` \$9). Live tier: price on application.

Step 1 — You, in the Whop dashboard

1. Create a product named **On-Chain Operator Program**. Upload the icon, banner and gallery images and paste the copy from `06-whop-store-listing.md`.
2. Add a course experience and create the 9 module sections (titles in `01-offer-and-curriculum.md`).
3. Add terms of sale: the 14-day conditional refund policy and "Educational content only. Not financial advice. No results are guaranteed."
4. Send me the product ID (starts with `prod_`).

Step 2 — Me, via Zapier (after you confirm)

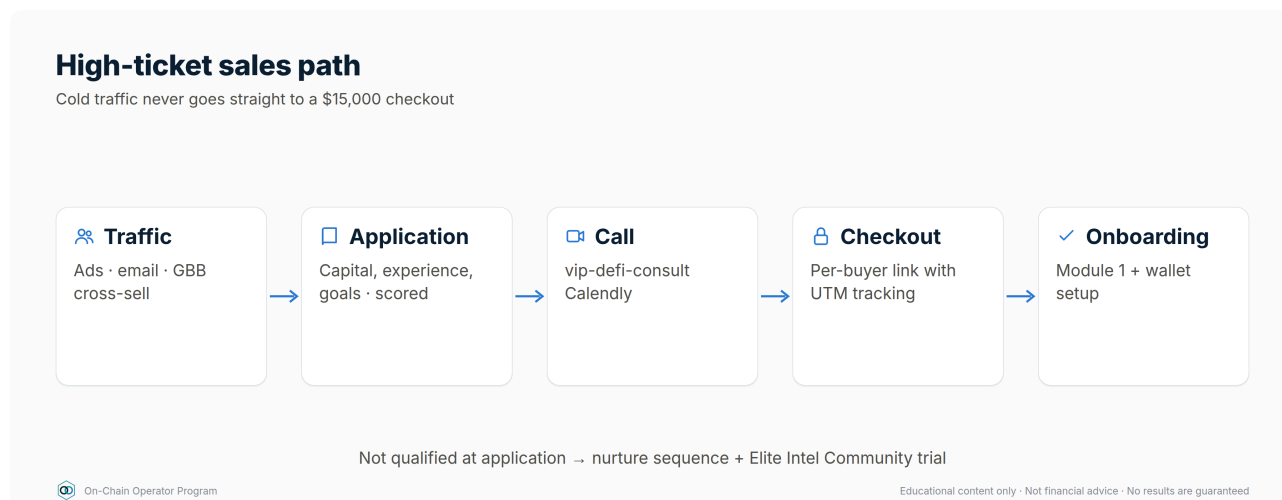
Action	Zapier tool	Values
Course plan	<code>whop_create_plan</code>	product = your <code>prod_</code> ID, one-time, \$15,000, hidden until launch
Live-tier plan	<code>whop_create_plan</code>	once the price is set
Checkout links per campaign	<code>whop_create_checkout_session</code>	metadata <code>utm_campaign</code> , <code>utm_adset</code> , <code>utm_creative</code> so ad reports attribute sales
Applications	<code>whop_create_lead</code>	one lead per application, product-specific

Nothing gets created until you've seen the exact values and said yes.

Step 3 — Upload content

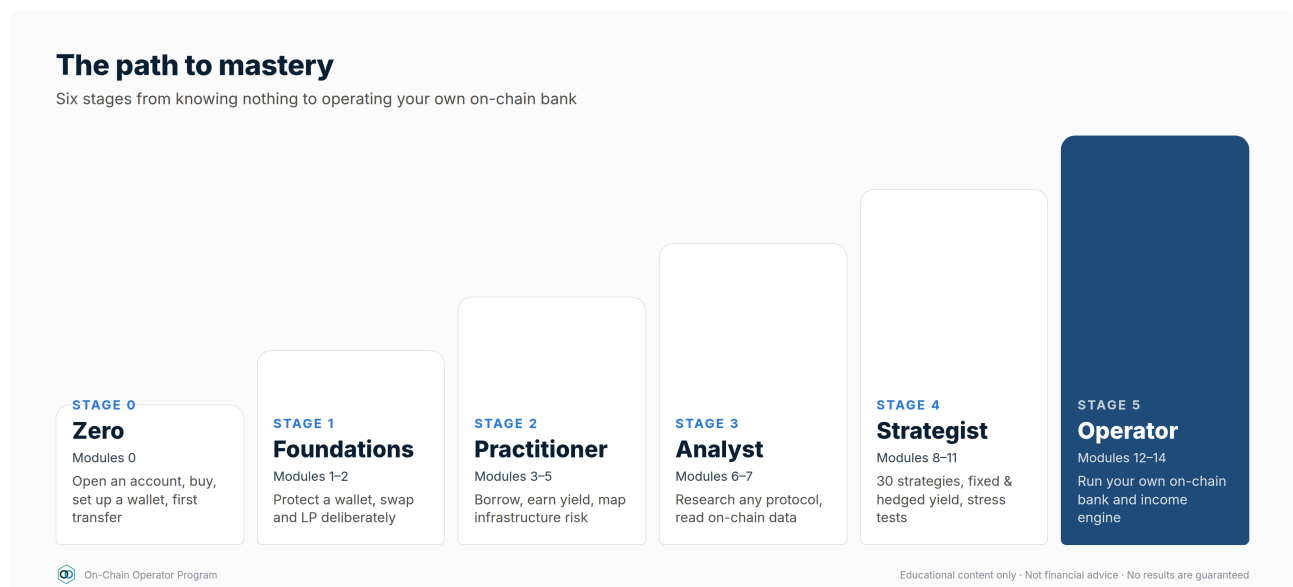
All 107 lessons and 15 Mastery Starters are written and ready to paste in (lessons/ , 02-... , 03-...), with module banners, diagrams, the Day-1 Setup Kit PDF, the welcome video and 15 module intro videos (video/). Course layout and drip plan: 07-program-operations.md section 1.

Step 4 — High-ticket sales path



Suggested qualification (adjust as you like): lead score ≥ 6 (capital 20k+ / major exchange / some experience) **and** completed application.

4. Complete mastery map



This is the full syllabus by **topic**: every subject a complete DeFi mastery course needs, where it's taught, and at what level. Use it to check coverage, find a topic quickly, or plan a learner's path.

Levels: **B** = Beginner (Mastery Starter or foundation lesson) · **P** = Practitioner (can do it safely) · **M** = Master (can design, size, hedge and teach it).

How mastery works in every module

Every module follows the same path, so a complete beginner can start anywhere:

1. **Mastery Starter (N.0)**: 60-second plain-English version, key words, prerequisites, a first safe step, and the module's own beginner → practitioner → master ladder.
2. **Lessons (N.1...)**: Objective → Explanation → Worked example → Checklist → 3-question quiz, each with at least one image.
3. **Module practical**: do it for real, with small amounts.
4. **Mastered when...**: each starter states the test (e.g. "you can open, monitor and unwind a borrow without approaching liquidation").
5. **Capstones**: analyst (after Module 7), operator (Module 14), graded against rubrics (07-program-operations.md).

Topic map by domain

1. Money, blockchains and the basics

Topic	Where	Level
What money, ledgers and blockchains are	0.0, 0.1	B
BTC, ETH, stablecoins, smart contracts, DeFi	0.1, 1.1	B

Topic	Where	Level
The DeFi stack and risk-first mindset	1.1	B
Transactions, gas, nonces, finality	1.2	B
Glossary of core terms	0.8, every N.0	B

2. Accounts, custody and security

Topic	Where	Level
Exchange accounts, KYC, 2FA, allowlists	0.2	B
Custodial vs self-custody	0.4	B
Wallet setup, seed backup, restore test	0.5	B
Keys, hardware wallets, multisig, smart wallets	1.3	B–P
Approvals, allowances, Permit signatures	1.4	B–P
Scam defence: phishing, drainers, address poisoning	1.6, 0.8	B
Reading signatures, simulation, blind signing, account delegation	1.7	P
Privacy and physical security	1.8	P
Smart accounts, account abstraction, passkeys, social recovery	1.9	M
Bank-grade custody architecture, limits, allowlists, timelocks	12.2	M
Multisig signing procedures and change control	14.3	M
Institutional custody: MPC, qualified custodians, policy engines	12.8	M
Incident response and fresh-wallet procedure	11.5	M

3. Trading and execution

Topic	Where	Level
Buying first crypto; market vs limit orders; fees	0.3	B
DEXs, aggregators, routing	2.1	B
AMM mathematics ($x \cdot y = k$), price impact	2.2	P
MEV and trade protection	2.5	P
Limit, TWAP and intent-based orders	2.6	P–M
MEV supply chain: searchers, builders, relays, private order flow, rebates	2.8	M

4. Liquidity providing

Topic	Where	Level
Providing liquidity and fee APR	2.3	B–P
Impermanent loss and true LP P&L	2.4	P
AMM designs (StableSwap, weighted, hooks) and LVR	2.7	M

Topic	Where	Level
Concentrated liquidity and range management	Strategy #5, 10.5	M
LP vs grid bot	9.1–9.3	P–M

5. Lending, borrowing and leverage

Topic	Where	Level
Money markets, utilisation, rate curves	3.1	B–P
LTV, liquidation threshold, health factor	3.2	P
Liquidations and cascades	3.3	P
Borrowing strategies and looping	3.4	P–M
Lending internals: e-mode, isolation, caps, auctions, soft liquidation, bad debt	3.6	M
CDP stablecoins: minting against collateral, PSMs	3.7, strategy #26	M
Credit lines and credit policy	12.3	M
Being the lender: curated vaults, credit risk	12.5	M

6. Derivatives

Topic	Where	Level
Perpetual futures, margin, liquidation price	3.5	P
Funding, open interest, positioning data	7.8	P
Basis (cash-and-carry)	10.2	M
Delta-neutral funding carry	10.3	M
Options: covered calls, cash-secured puts, vaults	10.4	M
Hedging with perps and options	11.1	M
Perp liquidity vaults (being the house)	10.7, strategy #28	M

7. Yield

Topic	Where	Level
Where returns come from (5 sources)	8.3 Part 1, 4.0	B
Farming: base yield vs emissions	4.1	B–P
Staking and liquid staking	4.2, 4.3	B–P
Restaking and points	4.4, 10.6	P–M
Vaults and optimisers	4.5	P
Airdrops as expected-value bets	4.6	P
Fixed-rate yield (PT/YT)	10.1	M
Vote-escrow, gauges and bribe income	6.6, strategy #27	M

Topic	Where	Level
Rates: term structure, fixed vs floating	10.8, strategy #30	M
Peg and redemption arbitrage	10.9, strategy #29	M

8. Stablecoins, cash and real-world assets

Topic	Where	Level
Stablecoin designs and depegs	1.5	B
Savings rates and yield-bearing stablecoins	4.7	P
Tokenized treasuries and RWAs	4.8	P
DeFi cash management and the liquidity ladder	Strategy #14, 12.4	M

9. Infrastructure and multi-chain

Topic	Where	Level
Networks and first transfers	0.6	B
Bridges and trust models	5.1	P
Layer 2s, sequencers, withdrawals	5.2	P
Oracles and manipulation	5.3	P
Smart contracts, proxies, admin keys, timelocks	5.4	P
Contract risk and reading audits	5.5	P–M
Reading contract code to verify claims	5.8	M
Solana, Bitcoin in DeFi, other ecosystems	5.6	P
Cross-chain gas, routes, chain abstraction	5.7	P–M

10. Research and analytics

Topic	Where	Level
Protocol due diligence (6-step loop)	6.1	P
Tokenomics, FDV, unlocks, value capture	6.2	P
Governance and DAOs	6.3	P
Research workflow and theses	6.4	P–M
Failure case studies and patterns (7 cases, incl. Black Thursday 2020)	6.5	P–M
Protocol valuation: fees, revenue, earnings, multiples	6.7	M
On-chain data, explorers, flows, whales	7.1–7.4	P
Holder/supply metrics, network activity, DEX analytics	7.5–7.7	P–M
Querying chain data yourself (SQL, indexers)	7.9	M

11. Strategy, portfolio and performance

Topic	Where	Level
Portfolio buckets and caps	8.1	P
Risk register	8.2	P
30-strategy library (7 levels)	8.3	P–M
Operating playbook: deploy, monitor, respond, review	8.4	P
Performance measurement: TWR, benchmarks, attribution	8.5	M
Psychology and discipline	8.6	P–M
Quantitative risk: volatility, VaR, drawdown, correlation, sizing	8.7	M
Stress testing	11.4	M
Protection: cover, liquidation automation	11.2, 11.3	M

12. Operating as your own bank

Topic	Where	Level
Balance sheet, equity, LTV, runway	12.1	M
Custody, credit, ladder and lending policies	12.2–12.5	M
Books, records and succession	12.6	M
Tax, regulation and compliance awareness	12.7	P–M

13. Income

Topic	Where	Level
Income sources by durability	13.1	P
Risk-adjusted yield	13.2	P–M
Building the income portfolio	13.3	M
Payout policy	13.4	M
Scaling, compounding, annual review	13.5	M

14. Automation and tools

Topic	Where	Level
Monitoring and alerts	14.1	P–M
Automation with scoped permissions	14.2	M
Building read-only tools, APIs, RPC	14.5	M
Searchers, keepers and arbitrage bots	14.6	M
Operator capstone and certification	14.4	M

Coverage check

Every domain has a **Beginner** entry point (a Mastery Starter or foundation lesson), **Practitioner** lessons with safe hands-on practice, and **Master** lessons that design, size and hedge. The totals:

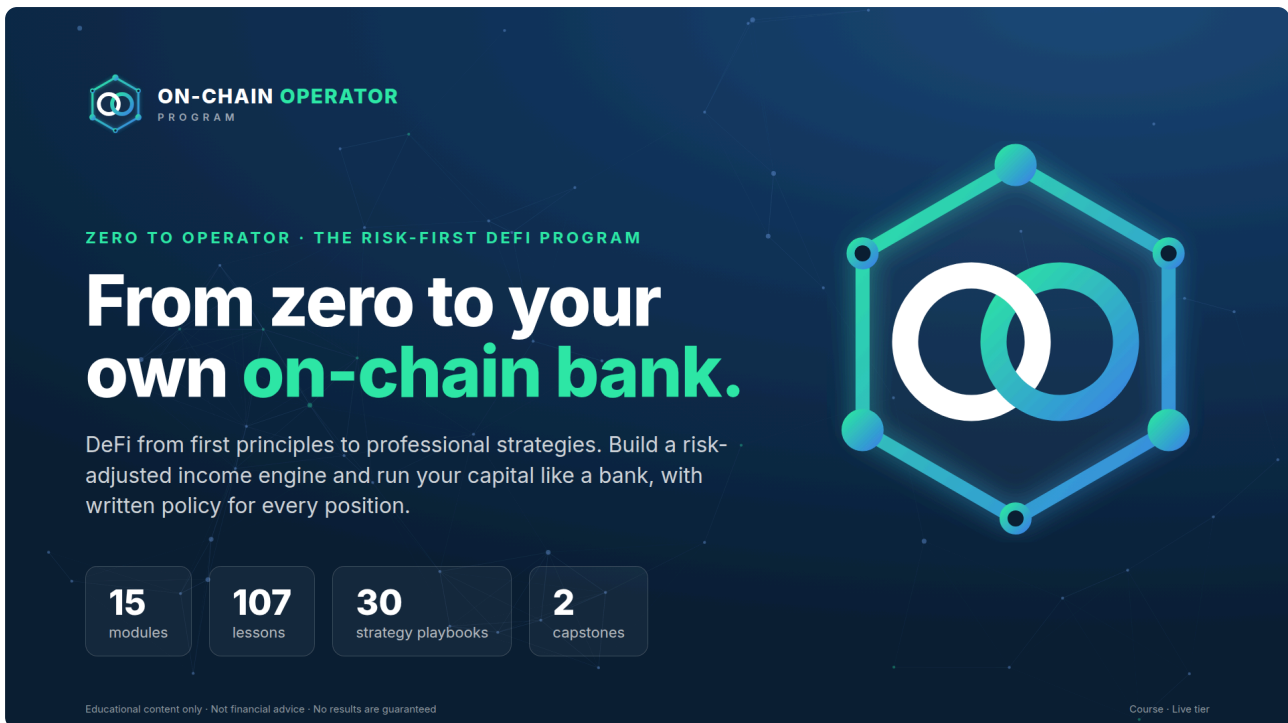
- **15 Mastery Starters** (one per module)
- **107 lessons** across 14 domains, including **15 expert lessons** on DeFi's machinery (AMM design and LVR, MEV supply chain, lending internals, CDPs, smart accounts, contract reading, ve/bribe markets, valuation, SQL analytics, quantitative risk, perp vaults, rates, arbitrage, institutional custody, searchers)
- **323 quiz questions, 30 strategy playbooks in 7 levels, 17 worksheets, 20 calculators**
- **2 capstones** and **3 certification levels**

Section plan (for building and reviewing in parts)

The program is split into six section files, one per stage, in `sections/`, regenerated by `build_master.py`. Each can be reviewed or handed to an AI on its own:

Section file	Stage	Modules	Lessons (+ starters)
<code>sections/section-0-zero.md</code>	0 · Zero	0	8 (+1)
<code>sections/section-1-foundations.md</code>	1 · Foundations	1–2	17 (+2)
<code>sections/section-2-practitioner.md</code>	2 · Practitioner	3–5	23 (+3)
<code>sections/section-3-analyst.md</code>	3 · Analyst	6–7	16 (+2)
<code>sections/section-4-strategist.md</code>	4 · Strategist	8–11	24 (+4)
<code>sections/section-5-operator.md</code>	5 · Operator	12–14	19 (+3)

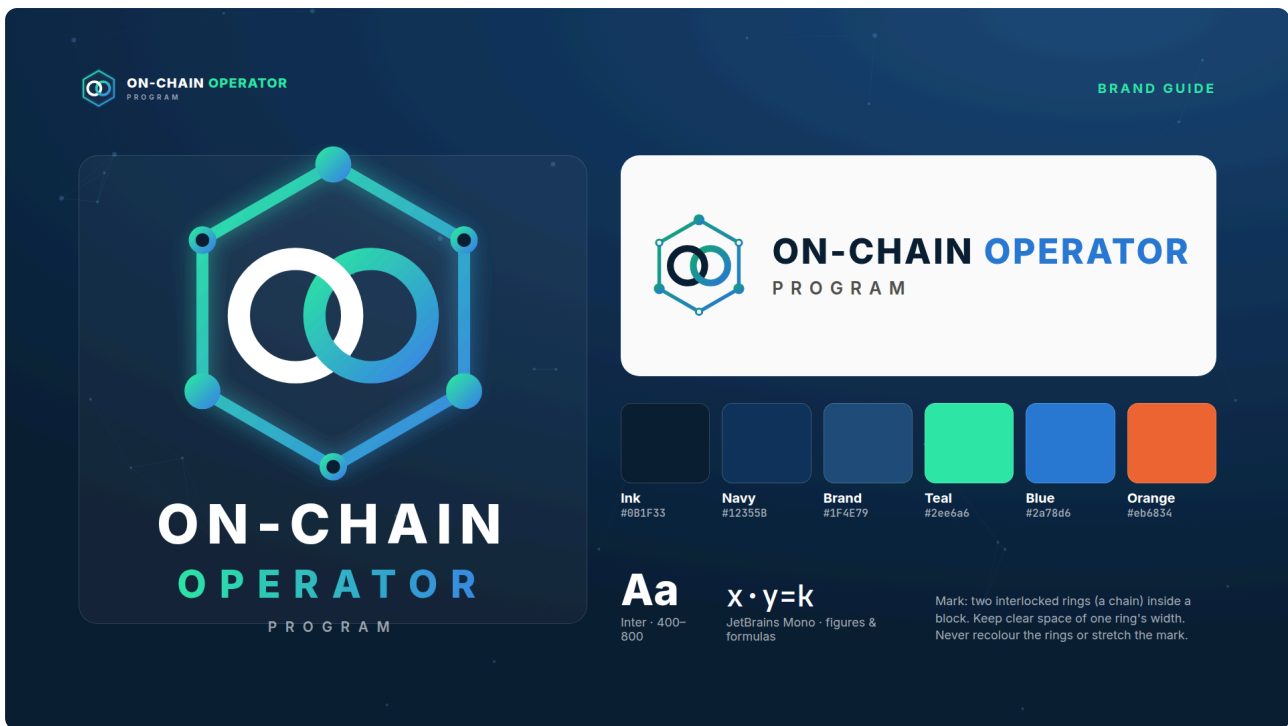
5. Whop store listing



Everything needed to add the program to your Whop store as a new product: logo, images, copy and upload order. **Nothing is published yet.** Create the product, paste this in, and publish when ready.

Check Whop's current image size limits in the dashboard when you upload. These files are produced at standard sizes (1024×1024 square, 1920×1080 16:9) and can be re-exported at any size with `npm run images`.

1. The logo



The mark is **two interlocked chain rings inside a hexagonal block**, with nodes on its corners: a chain (on-chain), a block, and a network. The rings link, like the two halves of the program: DeFi strategy and running your own bank.

File	Use
assets/brand/logo-icon-1024.png	Whop product icon , social avatars (stacked lockup on navy)
assets/brand/logo-mark-dark-1024.png	App icon / square badge without text
assets/brand/logo-mark-transparent-dark-1024.png	Mark over dark backgrounds (transparent PNG)
assets/brand/logo-mark-transparent-light-1024.png	Mark over light backgrounds (transparent PNG)
assets/brand/logo-horizontal-dark.png	Headers, email banners on dark
assets/brand/logo-horizontal-light.png	Documents, invoices, light pages
assets/brand/favicon-256.png	Favicon / tiny sizes (no corner nodes)
assets/brand/brand-guide.png	Colours, type and usage rules

Rules: keep clear space of one ring's width around the mark; never recolour the rings, stretch the mark or put the light version on a dark background.

2. Images: upload map

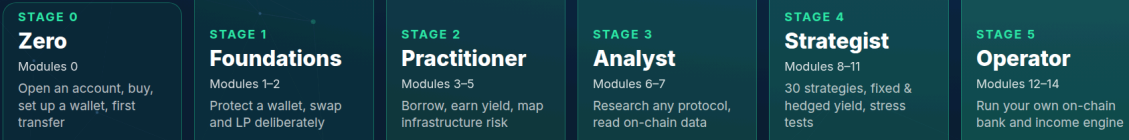
Whop slot	File	Size
Product icon / logo	assets/store/icon-1024.png (same as brand/logo-icon-1024.png)	1024×1024
Cover / hero banner	assets/store/banner-1920x1080.png	1920×1080
Gallery 1	assets/store/gallery-01-path-to-mastery.png	1920×1080
Gallery 2	assets/store/gallery-02-curriculum.png	1920×1080
Gallery 3	assets/store/gallery-03-own-bank.png	1920×1080
Gallery 4	assets/store/gallery-04-strategy-levels.png	1920×1080
Gallery 5	assets/store/gallery-05-research-loop.png	1920×1080
Gallery 6	assets/store/gallery-06-grid-vs-lp.png	1920×1080
Gallery 7	assets/store/gallery-07-whats-included.png	1920×1080
Listing video (VSL)	video/vsl-main.mp4 (≈90s, 1920×1080, captioned)	16:9
Ads / Reels / Shorts	video/vsl-short-vertical.mp4 (≈25s, 1080×1920)	9:16
Course module headers	assets/modules/module-00.png ... module-14.png	1600×500 @2x
In-lesson diagrams & charts	assets/diagrams/*.png, assets/charts/*.png	1800 wide @2x



ON-CHAIN OPERATOR

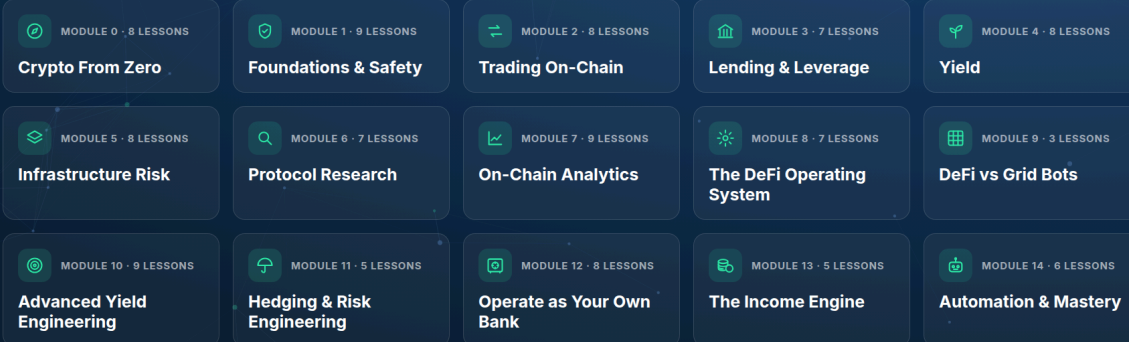
PROGRAM

From knowing nothing to operating your own bank



Educational content only - Not financial advice - No results are guaranteed

15 modules · 107 lessons · 2 capstones

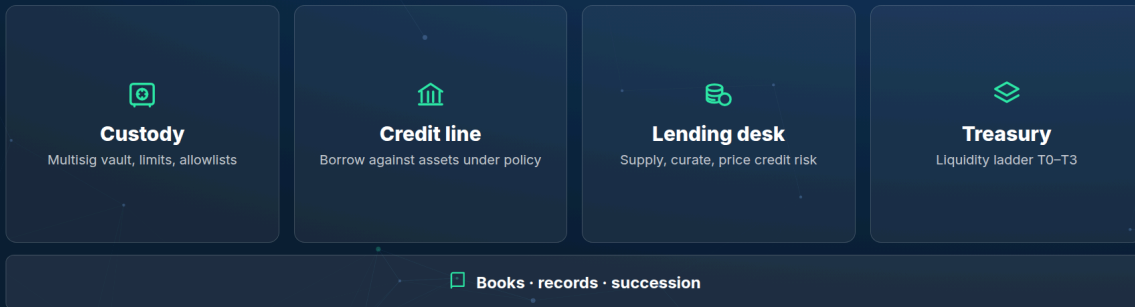


Educational content only - Not financial advice - No results are guaranteed

Operate as your own bank

Your on-chain bank

Income engine · payout policy



Educational content only · Not financial advice · No results are guaranteed

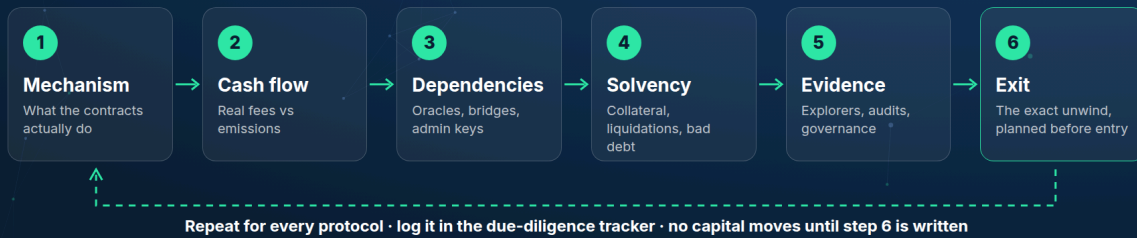
30 strategies, 7 levels, risk first

- 1 Core** Stable lending · Staking & LSTs · Collateral borrowing
- 2 Liquidity** 50/50 LP · Concentrated LP · Stable-stable LP
- 3 Yield** Vaults · Incentive farming · Airdrops & points
- 4 Advanced** Leveraged loops · LST loops · Funding carry · LP hedge
- 5 Treasury & research** Cash management · Accumulation & growth screens
- 6 Professional** PT/YT fixed rate · Basis · Options · Credit lines · Lending vaults · Hedged restaking
- 7 Expert** CDP minting · Bribe markets · Perp vaults · Peg arbitrage · Rates positioning

Graduate a level only when the one below is routine. Leverage and carry come last.

Educational content only · Not financial advice · No results are guaranteed

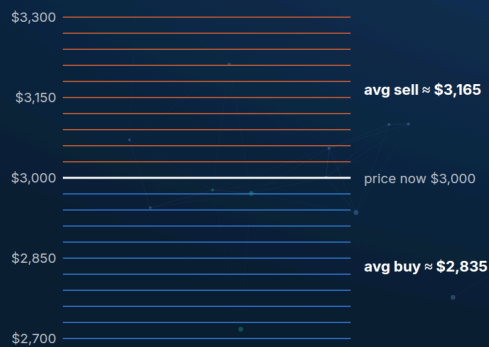
The 6-step research loop



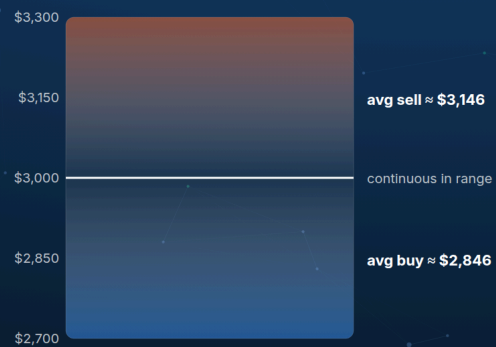
Educational content only · Not financial advice · No results are guaranteed

A DeFi LP is an on-chain grid bot

Grid bot · 20 levels, \$30 apart



Concentrated LP · same range



Blue = buys as price falls · Orange = sells as price rises · Range \$2,700–\$3,300, ETH at \$3,000. Illustrative.

Educational content only · Not financial advice · No results are guaranteed

Two ways to join

SELF-PACED

Course

- 15 modules, 107 lessons, zero to operator
- ⊙ 30 strategy playbooks
- 🔍 Due-diligence & bank-policy worksheets
- 📈 Strategy, income & bank calculators
- 🔒 Analyst + operator capstones

COURSE + COACHING

Live

- ✓ Everything in Course
- 🗣️ Live group sessions
- 👤 Capstone & portfolio reviews
- 📄 Own-bank policy review
- 💰 Income-engine & payout review

Educational content only · Not financial advice · No results are guaranteed

3. Listing copy

Product name: On-Chain Operator Program

Tagline (short): From zero to your own on-chain bank.

Short description (≈150 characters): DeFi from first principles to professional strategies. Build a risk-adjusted income engine and run your capital like a bank. 15 modules · 107 lessons.

Headline: From zero to your own on-chain bank.

Full description:

Most people enter DeFi through a number: an APY. Operators start with a different question: *what am I being paid to risk, and how do I get out?*

The On-Chain Operator Program takes you from never having owned crypto to running your capital the way a bank runs its book: a balance sheet, a custody policy, a credit line, a liquidity ladder, and an income engine with a payout you can sustain.

Six stages, 15 modules and 107 lessons, each module opening with a beginner Mastery Starter: wallets and safety, trading and liquidity, lending and yield, protocol research and on-chain analytics, 25 strategy playbooks up to fixed-rate, basis, options and hedged yield, and finally operating your own on-chain bank. Every strategy is taught with its maths, its kill rules and exactly how it loses money.

What you'll learn

- Stage 0: from knowing nothing to set up. Secure accounts, open an exchange, buy your first crypto without overpaying, set up and back up a wallet, make your first transfer on the right network, and practise DeFi on a free test network (with the Day-1 Setup Kit)

- Stage 1: protect a wallet, read any transaction, swap and provide liquidity deliberately
- Stage 2–3: lending, health factors and liquidations; yield, staking and restaking; a 6-step research loop for any protocol; on-chain analytics
- Stage 4: 30 strategy playbooks in 7 levels, including fixed-rate yield (PT/YT), cash-and-carry basis, delta-neutral funding carry, covered calls and cash-secured puts, hedging and stress testing
- Expert lessons in every stage on how DeFi's machinery really works: AMM design and loss-versus-rebalancing, the MEV supply chain, lending internals, minting stablecoins against collateral, smart accounts, reading contract code, vote-escrow and bribe markets, protocol valuation, querying chain data with SQL, quantitative risk, perp-exchange vaults, interest-rate curves, arbitrage, institutional custody and searcher bots
- Stage 5: operate as your own bank. Balance sheet, multisig custody, a credit line against your assets, a liquidity ladder, lending-desk decisions, books and succession, and an income engine that pays out less than it expects to earn

What's included

	Course	Live
15 modules, 107 lessons, zero to operator		
A Mastery Starter in every module: start any topic from zero	✓	✓
30 strategy playbooks with maths and kill rules	✓	✓
Printable Day-1 Setup Kit for complete beginners	✓	✓
Welcome video, 15 module intros and a narrated video for every lesson	✓	✓
Checklists and quizzes in every lesson	✓	✓
Due-diligence, pre-launch and bank-policy worksheets	✓	✓
Strategy, income and balance-sheet calculators	✓	✓
Analyst and operator capstones	✓	✓
Live group sessions		✓
Capstone and portfolio reviews		✓
Own-bank policy and income-engine reviews		✓
Price	\$15,000 one-time	On application

Who it's for

- Complete beginners who want to learn DeFi properly, from zero, with a process
- Traders and Grid Bot Builder customers ready to operate on-chain
- Holders of meaningful crypto who want income and liquidity from it without taking risks they don't understand

Who it's not for

- Anyone looking for guaranteed returns or signals to copy
- Anyone unwilling to self-custody or follow a written policy

How to join: Apply → short call → enrolment link. (Application link: <DEFI_APPLICATION_URL>)

FAQ

- *I've never owned crypto. Is this for me?* Yes. Module 0 assumes zero knowledge and walks you through every setup step in order: securing your email, opening and locking down an exchange account, your first purchase, setting up and backing up a wallet, your first transfer, and a first practice DeFi transaction on a free test network. It ends with a printable Day-1 Setup Kit checklist.
- *Do you manage my funds or need my keys?* Never. You keep custody. We will never ask for a seed phrase, private key or API key.
- *Will I make money / earn an income?* The program teaches you to build and run an income portfolio and to measure it honestly: expected income after expected losses, with a payout below that. It doesn't promise or project returns, and every strategy is taught alongside how it loses money.
- *What does "operate as your own bank" mean?* Running your crypto with the disciplines a bank uses: a balance sheet, custody controls, a credit policy, liquidity management and records. It's a method, not a licence or a financial service.
- *How does this relate to Grid Bot Builder?* Module 9 shows how on-chain liquidity and exchange grid bots do similar jobs, and how to run both in one portfolio.
- *Refunds?* 14-day conditional refund: full refund within 14 days of purchase if you've completed no more than 10% of lessons, haven't submitted a capstone and haven't used a live session or review. Duplicate purchases and unfixable access problems are always refunded. No refunds based on investment results. Full policy at checkout.

Footer disclaimer (put on the listing and at checkout): Educational content only. Not financial, tax or legal advice. Digital assets are volatile and you can lose some or all of your capital. No results or income are guaranteed.

4. Store setup checklist

- ☐ Product created in the Whop dashboard as **hidden**, named "On-Chain Operator Program"
- ☐ Logo icon, banner and 7 gallery images uploaded in the order above
- ☐ Copy pasted with the refund policy; `<DEFI_APPLICATION_URL>` filled
- ☐ Course experience added with 15 modules; module headers uploaded; lessons pasted with their diagrams
- ☐ Plans created via Zapier (course \$15,000 one-time; live tier once priced), approved first
- ☐ Checkout tested end to end on a hidden plan
- ☐ Listing reviewed for any income or return claims before going public

6. Operations kit

Everything around the lessons needed to sell, deliver and run the program.

Contents: 1. Whop course layout · 2. Capstones & rubrics · 3. Certification · 4. Application form & scoring · 5. Sales call script · 6. Member email sequence · 7. Live tier design · 8. Community rules · 9. Terms, disclaimers & refunds · 10. Support & FAQ · 11. Launch plan

1. Whop course layout

Whop section	Contents	Video
Start here	Welcome, how the program works, Day-1 Setup Kit (PDF)	video/welcome.mp4
Module 0–14	One chapter per module; one lesson per page, with images and its own narrated video (≤ 25 min); module practical at the end	video/module-NN-intro.mp4 at the top of each module; video/lesson-NN-M.mp4 on each lesson page (plan: 10-video-production-plan.md)
Tools	Strategy calculator (defi_calc.py) instructions, worksheet templates (08-worksheets.md)	n/a
Capstones	Briefs, rubrics, submission form	n/a
Live (Live tier only)	Session calendar, recordings, review booking	n/a

Drip (recommended): release a stage at a time as the previous stage's quiz is passed, not by date. Beginners shouldn't reach leverage before safety.

2. Capstones & rubrics

Analyst capstone (after Module 7)

Brief: a due-diligence file on one real protocol: the 6-step loop (Module 6), tokenomics and governance, and an on-chain section (Module 7), ending in a verdict with size, conditions and exit.

Criterion	Weight	Pass standard
Mechanism and cash flow correctly explained	20%	Fees vs emissions separated with sources
Dependency map complete	20%	Oracles, bridges, admin/timelock, stablecoins, other protocols
Evidence quality	20%	Primary sources + at least two independent data views
On-chain analysis without over-interpretation	15%	Caveats stated for every metric

Criterion	Weight	Pass standard
Verdict: size, conditions, exit	25%	Actionable, with kill rules and an unwind path

Pass: $\geq 70\%$. Resubmission allowed.

Operator capstone (Module 14.4)

Brief: the ten-section personal bank (see 14.4). No keys, seeds or account access anywhere.

Criterion	Weight	Pass standard
Balance sheet & policies internally consistent	20%	LTV, HF and runway match the credit and ladder policies
Custody & security	15%	Multisig or equivalent, limits, a tested recovery, succession
Income engine honesty	20%	Loss assumptions stated; payout $\leq 80\%$ of expected income
Stress test & fixes	20%	Five scenarios, each failure fixed
Operations	15%	Alerts mapped to actions; automation scoped; signing procedure
Clarity	10%	A stranger could run it from the document

Pass: $\geq 70\%$. Distinction: $\geq 90\%$ plus one quarter of books kept to standard.

3. Certification

- **On-Chain Analyst:** analyst capstone passed
- **On-Chain Operator:** operator capstone passed
- **On-Chain Operator, with distinction:** as above Certificates state that they reflect course completion only, not a licence or a professional qualification.

4. Application form & scoring

Form (Typeform or Whop form):

1. Name, email, country
2. Experience: never owned crypto / some / active trader / advanced
3. Capital you plan to operate: < \$10k / \$10–50k / \$50–250k / \$250k+
4. Goal: learn safely from zero / earn income / borrow against assets / run a structured portfolio / other
5. Do you already use Grid Bot Builder? (yes/no)
6. Why now? (short text)
7. Acknowledgement (required): *"I understand this is education, not financial advice, that no returns are promised, and I've read the 14-day conditional refund policy."*

Scoring (plugs into `gbb-new-lead` as `interest = defi`):

Answer	Points
Capital \$250k+ / \$50–250k / \$10–50k / < \$10k	4 / 3 / 1 / 0
Clear goal (not "get rich quick")	2
GBB customer	1
Thoughtful "why now"	1

≥ 6 → invite to call · 3–5 → call optional + nurture · < 3 → nurture + Elite Intel Community trial.

Disqualify anyone seeking guaranteed returns or asking the program to manage funds.

5. Sales call script (30 min)

- Open (2 min):** "This call is to see if the program fits you. If it doesn't, I'll tell you."
- Where are you now? (8 min):** experience, current setup, what's gone wrong before, goal.
- Where do you want to be? (5 min):** in a year, what does "working" look like? (Steer from return targets to capabilities: a safe setup, research ability, income policy, credit line.)
- The gap (5 min):** reflect back what's missing: process, safety, policy.
- The program (5 min):** stages, what they'll build, Course vs Live, capstones. Show the path image.
- Fit and objections (4 min):** see below.
- Close (1 min):** "Would you like the enrolment link?" If yes, send a per-buyer checkout link (with UTM metadata).

Objection answers (never make return promises):

- "Will I make my money back?" "I can't promise returns and nobody honestly can. What you'll leave with is a process to research, size and exit positions, and an income policy that doesn't depend on luck."
- "It's expensive." "It is. If the cost would strain you, don't do it now; start with the free Day-1 Setup Kit and Elite Intel."
- "I've lost money in crypto before." "Most losses come from setup mistakes, leverage and yields nobody explained. Those are the first things we fix."

Never: promise returns, pressure with fake deadlines, or ask for keys or account access.

6. Member email sequence (after purchase)

Day	Subject	Body (plain text, < 120 words)
0	Welcome, operator	Access link · watch the welcome video · start Module 0 · print the Day-1 Setup Kit · "we'll never ask for your seed phrase"
2	Your Day-1 Setup Kit	"Tick Part A today. Secure your email before anything else."
7	Did your restore test pass?	Nudge for Lesson 0.5; link to support
14	Stage 1 unlocked	What Modules 1–2 build; Live tier: first session date
30	Your first month	Progress check; invite to book a check-in (Live)

Day	Subject	Body (plain text, < 120 words)
60	Halfway to analyst	Analyst capstone brief
90	From analyst to operator	Stage 4–5 preview; operator capstone brief

7. Live tier design (price on application, set on the sales call)

Component	Proposed	Notes
Group sessions	2 × 60 min per month	One teaching deep-dive, one Q&A
Capstone reviews	Written review + 20-min call, each capstone	Uses the rubrics above
Portfolio/policy review	1 per quarter, 30 min	Reviews policies, never gives personalised investment advice
Protocol research clinic	Monthly, recorded	Walks through a due-diligence file live
Recordings	All sessions	Stored in the Live section

Boundaries: education and process review only; no personalised financial advice, no trade calls, no fund management.

8. Community rules

1. Education, not signals. No "buy X now" posts.
2. Never ask for or share seed phrases, keys or account access. Anyone who asks is removed.
3. No DMs offering "help", "recovery" or investments. Report them.
4. No referral links or shilling.
5. Show your working: numbers from the calculators, sources linked.
6. Be kind to beginners. Everyone started at Module 0.

9. Terms, disclaimers & refunds

Disclaimer (listing, checkout, every module): *The On-Chain Operator Program is educational content only. It is not financial, investment, tax or legal advice, and nothing in it is a recommendation to buy, sell or hold any asset. Digital assets are volatile and you can lose some or all of your capital. No results or income are promised or guaranteed. Examples are illustrative. "Operate as your own bank" describes a personal method, not a licence or financial service. We will never ask for your seed phrase, private keys or account access.*

Refund policy: 14-day conditional refund

1. **Eligibility.** You can request a full refund within **14 days of purchase** if **all** of these are true:
 - you've completed **no more than 10% of the lessons** (10 of 107; Mastery Starters don't count),
 - you haven't submitted a capstone, and
 - you haven't attended a live session or used a review (Live tier).

2. **Always refunded:** duplicate purchases, and access problems we can't fix within 7 days of you reporting them (whatever your progress).
3. **Not refunded:** requests after 14 days or beyond the progress limit, and any request based on investment or trading results. This is an education program, and no returns are promised.
4. **Live tier and upgrades:** the same 14-day conditions apply from the purchase or upgrade date; if you upgrade Course → Live, the price difference is refundable under the same conditions.
5. **How to request:** email support with your order ID and the reason. Refunds go back to the original payment method through Whop, normally within 5–10 business days. Access ends when the refund is issued.
6. **Chargebacks:** please contact support first. A chargeback ends access immediately.
7. **Your rights:** this policy doesn't limit any rights you have under consumer law where you live.

The policy is shown on the listing, at checkout and in the application acknowledgement.

Other terms to include: lifetime access to course content, no sharing of accounts, content copyright, conduct rules (section 8), Live-tier boundaries (section 7).

10. Support & FAQ

- Support channel: email (response within 2 business days) + community.
- **Top questions:** "I sent crypto on the wrong network" (check whether the address exists on that network; contact the receiving service; prevention in 0.6) · "I lost my seed phrase" (if the wallet is still accessible, move funds to a new wallet with a new seed now) · "A support person DM'd me" (it's a scam; block) · "Is X protocol safe?" (run the 6-step loop; staff don't give recommendations).

11. Launch plan

1. Create the product; upload logo, banner and gallery images; paste the listing and the refund policy.
2. Upload the course: 15 modules, lesson pages with images, intro videos, Day-1 Setup Kit.
3. Create the plans: course \$15,000 one-time; Live tier priced on application.
4. Publish the application form; connect scoring to `gbb-new-lead`.
5. Test end to end: application → call → per-buyer checkout → onboarding email → access.
6. Soft launch to the warm list (GBB customers, Elite Intel) and cold ads at the same time (per decision), with the VSLs.
7. Weekly: `gbb-ad-performance` and `gbb-pipeline-check` including DeFi.

7. Worksheets & templates

Copy these into Notion, a spreadsheet or a document. Numbers come from `defi_calc.py`. **Never** write seed phrases, private keys or passwords in any worksheet.

W1 · Records sheet (from Lesson 0.3, keep forever)

Date	Action (buy/sell/transfer/swap/fee/income)	Asset	Amount	Price (USD)	Fee	Network	Tx link	Notes
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W2 · Position journal (Lesson 8.4)

Position	Protocol / chain	Contracts	Amount	Entry date/price	Profit engine	Thesis	Kill rules	Approvals	Weekly result
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W3 · Due-diligence file (Module 6; mirrors the Notion tracker)

- Protocol · chain · TVL (source/date) · risk rating · verdict
- 1 Mechanism · 2 Cash flow (organic vs subsidised) · 3 Dependencies · 4 Solvency · 5 Evidence (links) · 6 Exit
- Tokenomics: market cap, FDV, 12-month unlocks, value capture
- Governance: voting concentration, timelock
- On-chain: holders, flows, activity, depth, derivatives
- Monitoring triggers

W4 · Risk register (Lesson 8.2)

Position	Risk surface	Description	Likelihood 1–5	Impact 1–5	Score	Response
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W5 · Portfolio plan (Lesson 8.1)

Bucket	Target %	Current %	Holdings	Caps (protocol / chain / issuer / bridge)
Rebalance rule: _____				

W6 · Balance sheet (Lesson 12.1)

Assets	Value	Liabilities	Value
Collateral		Loans	
Yield positions		Margin	
Reserve (T0+T1)			
Total assets		Total liabilities	

Assets	Value	Liabilities	Value
Equity = ____ · LTV = ____ · HF = ____ · Runway (months) = ____			

W7 · Custody policy (Lesson 12.2)

- Vault: type (multisig m-of-n) · key locations (general, not exact) · allowlisted destinations · timelock
- Operating: wallet type · daily limit · protocols allowed
- Hot: float size · refill schedule
- Recovery test log: date · path tested · result

W8 · Credit policy (Lesson 12.3)

Max LTV ____% · HF floor ____ · alert levels ____ / ____ · actions at each level · fixed or variable (why) · repayment source · collateral allowed

W9 · Liquidity ladder (Lesson 12.4)

Tier	Access	Holds	Target (months of obligations)	Current	Refill rule
T0	Seconds				
T1	Hours				
T2	Scheduled				
T3	Days–weeks				

W10 · Lending policy (Lesson 12.5)

Market/vault	Collateral	Oracle	Liquidation LTV	Curator	Loss prob. (assumed)	LGD	Cap
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W11 · Income portfolio & payout policy (Module 13)

`defi_calc.py income --pos "name:amount:yield:loss_prob:lgd" ... --payout 0.7` Blended risk-adjusted yield ____% · expected income ____/yr · payout ratio ____% · monthly payout ____ · review rule _____

W12 · Stress test (Lesson 11.4)

Scenario	Loss	Liquidations?	Liquidity OK?	Payout OK?	Fix
Crypto –50%					
Stablecoin –10%					
Borrow rate 20%					
Largest protocol hacked					

Scenario	Loss	Liquidations?	Liquidity OK?	Payout OK?	Fix
Main L2 halted 48h					

W13 · Incident playbook (Lesson 11.5)

Official channels per protocol (bookmarked) · containment steps per incident type · fresh-wallet procedure · who to contact · journal template

W14 · Alert sheet (Lesson 14.1)

Alert	Threshold	Tool	Action
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W15 · Automation permissions (Lesson 14.2)

Bot/automation	Permission	Contract	Cap	Expiry	Revoke method
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W16 · Annual review (Lesson 13.5)

Equity start/end · LTV · runway · income expected vs realised by source · losses & near-misses · payout paid vs policy · updated loss assumptions · custody & succession drill done?

W17 · Letter of instruction (Lesson 12.6), stored apart from any key

What exists (wallet types, multisig setup) · where documents are · who the co-signer/professional is · how the executor should proceed · **no seeds, no keys, no passwords**

8. Funnel changes (draft, not applied)

Proposed changes to the four live Zapier skills so they handle the DeFi program alongside Grid Bot Builder. **Nothing has been changed in Zapier.** Once you approve, the edits go into Zapier (`update_zapier_skill`) and are mirrored into `.claude/skills/gbb-*`.

Product name: **On-Chain Operator Program**, course \$15,000 one-time, sold through application → call → checkout (see `05-whop-setup.md`). This changes the emails below: hot/warm DeFi leads get the **application + call** link, not a direct checkout link. Live tier: price on application. Link placeholders (filled at setup): `<DEFI_CHECKOUT_URL>`, `<DEFI_PLAN_IDS>`, `<DEFI_ACCESS_URL>` (the program's Whop hub link), `<DEFI_APPLICATION_URL>` (Typeform application), `<DEFI_CALL_LINK>` (your `vip-defi-consult` Calendly link; verify the URL).

1. gbb new lead → handles DeFi interest

Add fixed values

- DeFi product: `On-Chain Operator Program`, \$15,000 one-time; checkout `<DEFI_CHECKOUT_URL>`.
- Mailchimp tags: `defi-lead`, `defi-nurture`.
- HubSpot deal name: `DEFI - <First Last>`.

Add an interest field to the form / intake: `interest = grid | defi | both`.

Routing change (after scoring, step 3):

interest	Deal(s) created	Tags	Email
grid	GBB - ... (unchanged)	gbb-lead (+ hot/nurture)	unchanged
defi	DEFI - ..., amount \$15,000	defi-lead (+ defi-nurture if cold)	DeFi email by tier (below)
both	GBB deal only; DeFi mentioned as next step	gbb-lead, defi-lead	GBB email + one line on DeFi

Scoring stays the same. Capital and experience predict fit for both products.

Draft emails (plain text, under 120 words, no profit promises)

DeFi — warm/hot

Hi {first}, Thanks for your interest in On-Chain Operator Program. It's a step-by-step program for operating in DeFi safely: how to research a protocol, where yield actually comes from, and how to plan the exit before you enter. 15 modules and 107 lessons, from zero to running your own on-chain bank. It's application-only. Apply here: `<DEFI_APPLICATION_URL>` Qualified applicants book a call to see if it's the right fit: `<DEFI_CALL_LINK>` Stewart *Educational content, not financial advice.*

DeFi — cold (nurture)

Hi {first}, One rule before you put anything into DeFi: if you can't explain how you'd get out, don't get in. Over the next few emails I'll share the checklist we use before signing any transaction. If you want daily market intel in the meantime, Elite Intel Community has a 3-day free trial: <https://elite-intel-community.whop.site/> Stewart

2. gbb customer onboarding → DeFi branch + cross-sell

When product = **On-Chain Operator Program**:

1. HubSpot: lifecycle customer ; deal DEFI - ... → closedwon, amount = actual plan price.
2. Mailchimp: remove defi-lead , add defi-customer .
3. Gmail onboarding (draft below).
4. Sheets → Payments (product = DeFi) and ExcludeList.
5. Elite Intel Community invite, same rules as GBB (separate email, paid trial, never granted free).
6. If *not* a GBB customer: tag gbb-crosssell-candidate (no email yet; see the sequence below).

DeFi onboarding email

Welcome to On-Chain Operator Program, {first}. Your access is live: <DEFI_ACCESS_URL> (log in with this email). Start with Module 1 — Foundations & Safety. Do the practical (3-wallet setup) before anything else. Two rules for the whole program:

1. Never share your seed phrase or private keys. We will never ask.
2. Run the before-signing checklist every time. Reply to this email if you get stuck. Stewart

When product = **Grid Bot Builder** (existing flow, plus one addition):

- Tag defi-crosssell-candidate . The cross-sell email goes **14 days after onboarding**, not in the welcome sequence, so they're set up and trading first.

GBB → DeFi cross-sell (day 14)

Hi {first}, By now your grid bots should be running. Here's where most of our builders go next: putting idle stablecoins and reserves to work on-chain, without taking on risks they don't understand. On-Chain Operator Program covers this, including a module on how a DeFi liquidity position is basically an on-chain grid bot, and when each one wins. It's application-only: <DEFI_APPLICATION_URL> Stewart

3. gbb ad performance → report DeFi

- Add **On-Chain Operator Program** to "products" with its plan IDs.
- Per campaign: DeFi sales (# and \$), CAC-to-DeFi.
- New cross-sell metric: **GBB → DeFi attach rate** = GBB buyers who later buy DeFi ÷ GBB buyers ≥ 14 days old.

- Budget rule unchanged (it uses total net Whop revenue).

4. gbb pipeline check → DeFi deals

- Include deals starting `DEFI -` with the same stages.
- Revenue by product adds DeFi.
- Stale DeFi leads → offer `defi-nurture` tagging (same confirm rule).

New Zapier skill (proposed): `defi-crosssell-sweep`

Weekly: find `defi-crosssell-candidate` contacts whose GBB onboarding is ≥ 14 days old and who haven't bought DeFi → show the list → on a yes, send the day-14 email and remove the tag. Same guardrails: no Slack/Discord/Skool, confirm before sending, no profit promises.

9. Video scripts

Generated by `export/build_video.js` from `export/videos.js` (edit the scripts there, then rebuild).
Narration: Kokoro TTS, voice `af_heart` (offline, Apache-2.0 model). To use your own voice, record the VO lines below and replace the audio track, or re-run with `VOICE=am_michael` etc.

Compliance: no income or return claims, no fake urgency, keys never requested, disclaimer on screen.

VSL: main (sales page & store listing)

File: `video/vsl-main.mp4` · 1920×1080 · 94.9s · Use: Whop store listing video, sales page hero, application page.

#	Time	Visual	Voice-over
1	0.0–9.4s	"12% APY" struck out → "What am I being paid to risk?"	Most people enter DeFi through a number. An APY. Operators start with a different question. What am I being paid to risk?
2	9.4–15.4s	And how do I get out?	And how do I get out? If you can't explain the exit, you don't understand the position yet.
3	15.4–25.5s	Without a process, DeFi is a maze: Wallets, seed phrases and networks · Approvals you can't read · Yields you can't explain · Mistakes that can't be undone	Without a process, DeFi is a maze. Wallets, networks, approvals, and yields nobody can explain. And on-chain, a mistake can't be undone.
4	25.5–34.2s	Logo reveal + "From zero to your own on-chain bank."	The On-Chain Operator Program takes you from zero, never having owned crypto, to running your capital like your own on-chain bank.
5	34.2–47.9s	The path: <code>assets/store/gallery-01-path-to-mastery.png</code>	6 stages. 15 modules. 107 lessons, each module opening with a beginner starter. You start by setting everything up safely, step by step, with a Day-1 Setup Kit.
6	47.9–58.7s	Strategy library: <code>assets/store/gallery-04-strategy-levels.png</code>	Then you learn to research any protocol, and work through 30 strategy playbooks. Each one with its maths, its exit rules, and exactly how it loses money.
7	58.7–67.4s	Stage five: <code>assets/store/gallery-03-own-bank.png</code>	Finally, you build your own bank. A balance sheet. Multisig custody. A credit line against your assets. A liquidity ladder.
8	67.4–76.4s	The income engine: <code>assets/charts/income-waterfall.png</code>	And an income engine that measures what you expect to earn after expected losses, and pays out less than that. Never from principal.

#	Time	Visual	Voice-over
9	76.4– 87.1s	What you won't get: Signals to copy · Guaranteed returns · Anyone asking for your keys	What you won't get: signals to copy, guaranteed returns, or anyone asking for your keys. Just a process you can explain, position by position.
10	87.1– 94.9s	Logo + "Apply to join" button · Application-only · Course and Live tiers	The On-Chain Operator Program is application-only. Apply today, and we'll see if it's the right fit for you.

VSL: 30-second vertical cut (ads, Reels, Shorts, TikTok)

File: `video/vsl-short-vertical.mp4` · 1080×1920 · 23.2s · Use: Paid social and organic short-form.
Upload as 9:16.

#	Time	Visual	Voice-over
1	0.0–5.9s	"12% APY" struck out → "Paid to risk what?"	12% APY. The real question is: what are you being paid to risk?
2	5.9– 12.4s	Logo reveal + "From zero to your own on-chain bank."	The On-Chain Operator Program takes you from zero to running your crypto like your own bank.
3	12.4– 19.5s	107 lessons · 30 strategy playbooks · 0 guaranteed returns	107 lessons. 30 strategy playbooks. And zero promises. Just a process.
4	19.5– 23.2s	Logo + "Apply to join" button · Application-only	Apply to join the On-Chain Operator Program.

Welcome video (inside the program, top of Module 0)

File: `video/welcome.mp4` · 1920×1080 · 44.8s · Use: First thing a new member sees after purchase.

#	Time	Visual	Voice-over
1	0.0– 5.3s	Logo reveal + "Welcome, operator."	Welcome to the On-Chain Operator Program. Here's how to get the most from it.
2	5.3– 15.0s	How it works: <code>assets/diagrams/path-to-mastery.png</code>	The program runs in six stages. Do them in order. Each stage unlocks the next, from your first wallet to running your own on-chain bank.
3	15.0– 25.9s	Every lesson, the same shape: Objective · Explanation · Worked example · Checklist · Three-question quiz	Every lesson has the same shape. An objective, an explanation, a worked example, a checklist, and a short quiz. Do the checklist. That's where the learning sticks.
4	25.9– 33.4s	Start here: <code>assets/diagrams/setup-roadmap.png</code>	Start with Module 0 and the Day-1 Setup Kit. Use a small amount you can afford to lose while you learn.
5	33.4– 41.3s	One rule, forever: / never share your seed phrase.	And one rule, forever. Never share your seed phrase. Nobody from this program, or anywhere else, will ever ask for it.

#	Time	Visual	Voice-over
6	41.3–44.8s	Logo + "Open Module 0" button · Educational content only · Not financial advice	Open Module 0, and let's begin.

Module 0 intro: Crypto From Zero

File: `video/module-00-intro.mp4` · 1920×1080 · 30.3s · Use: Top of Module 0 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.4s	Module 0: <code>assets/modules/module-00.png</code>	Module 0. Crypto From Zero.
2	3.4–19.3s	The goal	The goal: go from never having owned crypto to a secured exchange account, a backed-up wallet, a first transfer and a first practice DeFi transaction. Ends with the Day-1 Setup Kit checklist.
3	19.3–27.1s	Mastery Starter + 8 lessons: 0.0 Mastery Starter · 0.1 Money, ledgers and why blockchains exist · 0.2 Opening and securing an exchange account · 0.3 Buying your first crypto without overpaying · 0.4 Exchange account vs your own wallet: who holds the keys? · 0.5 Setting up your wallet and backing it up · 0.6 Networks, gas and your first transfer · 0.7 Your first DeFi steps · 0.8 Your security baseline, and the language of DeFi	8 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	27.1–30.3s	Logo + "Start lesson 0.1" button · Educational content only · Not financial advice	Start with lesson 0.1.

Module 1 intro: Foundations & Safety

File: `video/module-01-intro.mp4` · 1920×1080 · 20.3s · Use: Top of Module 1 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.7s	Module 1: <code>assets/modules/module-01.png</code>	Module 1. Foundations & Safety.
2	3.7–9.4s	The goal	The goal: set up and use a wallet safely; know what can go irreversibly wrong.

#	Time	Visual	Voice-over
3	9.4–17.4s	Mastery Starter + 9 lessons: 1.0 Mastery Starter · 1.1 What DeFi is — and the risk-first mindset · 1.2 How a transaction actually happens · 1.3 Wallets, keys, hardware & multisig · 1.4 Tokens, approvals & allowances · 1.5 Stablecoins and how they break · 1.6 Scam defence: phishing, drainers, address poisoning · 1.7 Reading signatures and simulating transactions · 1.8 Privacy and physical security · 1.9 Smart accounts and account abstraction	9 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	17.4–20.3s	Logo + "Start lesson 1.1" button · Educational content only · Not financial advice	Start with lesson 1.1.

Module 2 intro: Trading On-Chain

File: `video/module-02-intro.mp4` · 1920×1080 · 19.9s · Use: Top of Module 2 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.1s	Module 2: <code>assets/modules/module-02.png</code>	Module 2. Trading On-Chain.
2	3.1–9.3s	The goal	The goal: execute a swap deliberately, understanding price impact and MEV.
3	9.3–17.1s	Mastery Starter + 8 lessons: 2.0 Mastery Starter · 2.1 DEXs, aggregators & routing · 2.2 AMM mathematics · 2.3 Providing liquidity · 2.4 Impermanent loss & true LP P&L · 2.5 MEV and how to protect your trades · 2.6 Advanced execution: limit, TWAP and intent-based orders · 2.7 Advanced AMM design and LVR · 2.8 The MEV supply chain	8 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	17.1–19.9s	Logo + "Start lesson 2.1" button · Educational content only · Not financial advice	Start with lesson 2.1.

Module 3 intro: Lending & Leverage

File: `video/module-03-intro.mp4` · 1920×1080 · 19.6s · Use: Top of Module 3 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.5s	Module 3: <code>assets/modules/module-03.png</code>	Module 3. Lending & Leverage.
2	3.5–8.7s	The goal	The goal: borrow against collateral with a buffer and a written defence plan.

#	Time	Visual	Voice-over
3	8.7–16.7s	Mastery Starter + 7 lessons: 3.0 Mastery Starter · 3.1 How lending markets work · 3.2 LTV, liquidation threshold & health factor · 3.3 Liquidations and cascades · 3.4 Borrowing strategies & looping · 3.5 Perpetual futures and margin on-chain · 3.6 Lending design deep dive: e-mode, caps, auctions, soft liquidation, bad debt · 3.7 CDP stablecoins: minting your own dollars	7 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	16.7–19.6s	Logo + "Start lesson 3.1" button · Educational content only · Not financial advice	Start with lesson 3.1.

Module 4 intro: Yield

File: `video/module-04-intro.mp4` · 1920×1080 · 20.9s · Use: Top of Module 4 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–2.8s	Module 4: <code>assets/modules/module-04.png</code>	Module 4. Yield.
2	2.8–10.2s	The goal	The goal: split any APY into organic vs subsidised, and name the risk being paid for.
3	10.2–18.1s	Mastery Starter + 8 lessons: 4.0 Mastery Starter · 4.1 Yield farming: base yield vs emissions · 4.2 Native staking · 4.3 Liquid staking · 4.4 Restaking & shared security · 4.5 Vaults & yield optimisers · 4.6 Airdrops & points — opportunity cost · 4.7 Stablecoin savings rates and yield-bearing stablecoins · 4.8 Tokenized treasuries and real-world assets	8 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	18.1–20.9s	Logo + "Start lesson 4.1" button · Educational content only · Not financial advice	Start with lesson 4.1.

Module 5 intro: Infrastructure Risk

File: `video/module-05-intro.mp4` · 1920×1080 · 20.6s · Use: Top of Module 5 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.4s	Module 5: <code>assets/modules/module-05.png</code>	Module 5. Infrastructure Risk.
2	3.4–9.8s	The goal	The goal: map every bridge, oracle, L2 and contract a position depends on.

#	Time	Visual	Voice-over
3	9.8–17.7s	Mastery Starter + 8 lessons: 5.0 Mastery Starter · 5.1 Bridges and trust assumptions · 5.2 Layer 2s, sequencers & withdrawal paths · 5.3 Oracles, TWAPs & manipulation · 5.4 Smart contracts: state, proxies, admin keys · 5.5 Smart-contract risk & what audits don't prove · 5.6 Beyond Ethereum: Solana, Bitcoin and other ecosystems · 5.7 Operating across chains: gas, routes and chain abstraction · 5.8 Reading smart-contract code: enough to verify claims	8 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	17.7–20.6s	Logo + "Start lesson 5.1" button · Educational content only · Not financial advice	Start with lesson 5.1.

Module 6 intro: Protocol Research

File: `video/module-06-intro.mp4` · 1920×1080 · 19.2s · Use: Top of Module 6 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.4s	Module 6: <code>assets/modules/module-06.png</code>	Module 6. Protocol Research.
2	3.4–8.2s	The goal	The goal: complete a full due-diligence file on a real protocol.
3	8.2–16.2s	Mastery Starter + 7 lessons: 6.0 Mastery Starter · 6.1 Protocol due diligence · 6.2 Tokenomics: supply, unlocks, FDV, value capture · 6.3 Governance & DAOs · 6.4 The on-chain research workflow · 6.5 Case studies: how DeFi failures happened · 6.6 Vote-escrow tokenomics, bribes and governance markets · 6.7 Valuing DeFi protocols: fees, revenue, earnings, multiples	7 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	16.2–19.2s	Logo + "Start lesson 6.1" button · Educational content only · Not financial advice	Start with lesson 6.1.

Module 7 intro: On-Chain Analytics

File: `video/module-07-intro.mp4` · 1920×1080 · 18.7s · Use: Top of Module 7 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.4s	Module 7: <code>assets/modules/module-07.png</code>	Module 7. On-Chain Analytics.
2	3.4–7.7s	The goal	The goal: read on-chain data without over-interpreting it.
3	7.7–15.7s	Mastery Starter + 9 lessons: 7.0 Mastery Starter · 7.1 On-chain data foundations · 7.2 Block explorer mastery · 7.3 Exchange flows · 7.4 Whale & entity analysis · 7.5 Holder & supply metrics · 7.6 Network activity · 7.7 DEX & liquidity analytics · 7.8 Derivatives on-chain · 7.9 Querying chain data yourself	9 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.

#	Time	Visual	Voice-over
4	15.7–18.7s	Logo + "Start lesson 7.1" button · Educational content only · Not financial advice	Start with lesson 7.1.

Module 8 intro: The DeFi Operating System

File: `video/module-08-intro.mp4` · 1920×1080 · 21.2s · Use: Top of Module 8 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.9s	Module 8: <code>assets/modules/module-08.png</code>	Module 8. The DeFi Operating System.
2	3.9–10.3s	The goal	The goal: a written portfolio plan with risk buckets, limits and an emergency plan.
3	10.3–18.3s	Mastery Starter + 7 lessons: 8.0 Mastery Starter · 8.1 DeFi portfolio construction · 8.2 The DeFi risk framework · 8.3 Strategy library · 8.4 The operating playbook: deploy, monitor, respond, review · 8.5 Measuring performance honestly · 8.6 Psychology and discipline · 8.7 Quantitative risk: volatility, VaR, drawdown, correlation, sizing	7 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	18.3–21.2s	Logo + "Start lesson 8.1" button · Educational content only · Not financial advice	Start with lesson 8.1.

Module 9 intro: DeFi vs Grid Bots

File: `video/module-09-intro.mp4` · 1920×1080 · 20.5s · Use: Top of Module 9 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.7s	Module 9: <code>assets/modules/module-09.png</code>	Module 9. DeFi vs Grid Bots.
2	3.7–9.7s	The goal	The goal: choose the right tool for the market — and the natural bridge into GBB.
3	9.7–17.6s	Mastery Starter + 3 lessons: 9.0 Mastery Starter · 9.1 LP position vs grid bot: same idea (sell high/buy low inside a range), different risks · 9.2 When a CEX grid beats on-chain LP, and when it doesn't · 9.3 Building a combined system: grid bots for range markets, DeFi for yield/collateral	3 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	17.6–20.5s	Logo + "Start lesson 9.1" button · Educational content only · Not financial advice	Start with lesson 9.1.

Module 10 intro: Advanced Yield Engineering

File: `video/module-10-intro.mp4` · 1920×1080 · 21.1s · Use: Top of Module 10 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.7s	Module 10: <code>assets/modules/module-10.png</code>	Module 10. Advanced Yield Engineering.
2	3.7–10.2s	The goal	The goal: build fixed, hedged and structured yield, and know exactly what each one is short.
3	10.2–18.2s	Mastery Starter + 9 lessons: 10.0 Mastery Starter · 10.1 Fixed-rate yield: principal and yield tokens · 10.2 Cash-and-carry basis trades · 10.3 Delta-neutral funding carry, done properly · 10.4 Options income: covered calls, cash-secured puts, options vaults · 10.5 Active concentrated-liquidity management · 10.6 Restaking and points: pricing speculative yield · 10.7 Being the house: perp-exchange liquidity vaults · 10.8 DeFi rates: term structure, fixed vs floating · 10.9 Peg and redemption arbitrage	9 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	18.2–21.1s	Logo + "Start lesson 10.1" button · Educational content only · Not financial advice	Start with lesson 10.1.

Module 11 intro: Hedging & Risk Engineering

File: `video/module-11-intro.mp4` · 1920×1080 · 22.0s · Use: Top of Module 11 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–4.1s	Module 11: <code>assets/modules/module-11.png</code>	Module 11. Hedging & Risk Engineering.
2	4.1–10.8s	The goal	The goal: hedge the risks you don't want, stress-test the portfolio, and have an incident plan ready.
3	10.8–18.7s	Mastery Starter + 5 lessons: 11.0 Mastery Starter · 11.1 Hedging price exposure with perps and options · 11.2 Depeg, protocol and smart-contract cover · 11.3 Liquidation protection: buffers, alerts and automated deleveraging · 11.4 Stress-testing a portfolio: –50% days, depegs, rate spikes · 11.5 Incident response: what to do in the first 60 minutes	5 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	18.7–22.0s	Logo + "Start lesson 11.1" button · Educational content only · Not financial advice	Start with lesson 11.1.

Module 12 intro: Operate as Your Own Bank

File: `video/module-12-intro.mp4` · 1920×1080 · 23.8s · Use: Top of Module 12 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.7s	Module 12: <code>assets/modules/module-12.png</code>	Module 12. Operate as Your Own Bank.
2	3.7–13.1s	The goal	The goal: run your crypto like a bank runs its book: a balance sheet, custody policy, a credit line, a liquidity ladder and records.
3	13.1–20.9s	Mastery Starter + 8 lessons: 12.0 Mastery Starter · 12.1 Your balance sheet: assets, liabilities, equity · 12.2 Custody architecture: vault, multisig and spending policies · 12.3 The credit line: borrowing against your assets like a bank client · 12.4 Treasury and the liquidity ladder · 12.5 Being the lender: supplying, curating and pricing credit risk · 12.6 Books, records and succession: if you're gone, can your family recover it? · 12.7 Tax, regulation and compliance awareness · 12.8 Institutional-grade custody: MPC, custodians, policy engines	8 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	20.9–23.8s	Logo + "Start lesson 12.1" button · Educational content only · Not financial advice	Start with lesson 12.1.

Module 13 intro: The Income Engine

File: `video/module-13-intro.mp4` · 1920×1080 · 23.8s · Use: Top of Module 13 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.5s	Module 13: <code>assets/modules/module-13.png</code>	Module 13. The Income Engine.
2	3.5–12.8s	The goal	The goal: design an income portfolio, measure expected income after expected losses, and set a payout you can sustain.
3	12.8–20.7s	Mastery Starter + 5 lessons: 13.0 Mastery Starter · 13.1 Income sources ranked by durability · 13.2 Risk-adjusted yield: subtracting expected losses · 13.3 Building the income portfolio · 13.4 The payout policy: how much you can take out · 13.5 Scaling, compounding and the annual review	5 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	20.7–23.8s	Logo + "Start lesson 13.1" button · Educational content only · Not financial advice	Start with lesson 13.1.

Module 14 intro: Automation & Mastery

File: `video/module-14-intro.mp4` · 1920×1080 · 22.5s · Use: Top of Module 14 in the Whop course.

#	Time	Visual	Voice-over
1	0.0–3.9s	Module 14: <code>assets/modules/module-14.png</code>	Module 14. Automation & Mastery.
2	3.9–11.6s	The goal	The goal: monitor and automate safely, run multisig operations, and complete the operator capstone.
3	11.6–19.4s	Mastery Starter + 6 lessons: 14.0 Mastery Starter · 14.1 Monitoring: dashboards, alerts and on-chain watchers · 14.2 Automation: keepers, bots and agents without handing over the keys · 14.3 Operating procedures: multisig signing, change control, reviews · 14.4 Operator capstone and certification · 14.5 Building your own tools: reading contracts, data and simple scripts · 14.6 Searchers, keepers and arbitrage bots	6 lessons, plus a Mastery Starter if you're new to the topic. Each lesson ends with a checklist and a short quiz.
4	19.4–22.5s	Logo + "Start lesson 14.1" button · Educational content only · Not financial advice	Start with lesson 14.1.

9b. Course video production plan

Generated by `export/build_lesson_scripts.py` from the lesson files. Edit lessons, then re-run.

The rule

Every lesson and every Mastery Starter gets its own narrated video, each at most 25 minutes.
Typical length is 3–12 minutes; nothing may exceed 25 minutes (split into Part 1 / Part 2 if it would).

What exists

Video type	Count	Where	Status
VSL (main + vertical)	2	<code>video/vsl-*.mp4</code>	Rendered
Welcome	1	<code>video/welcome.mp4</code>	Rendered
Module intros	15	<code>video/module-NN-intro.mp4</code>	Rendered
Lesson videos (incl. Mastery Starters)	121	<code>video-scripts/lessons/*.json</code> → <code>video/lesson-*.mp4</code>	Scripts ready; 2 rendered as samples
Strategy library series (8.3)	7 (one per level)	to be scripted (see prompt)	To do

Estimated total runtime of lesson videos from the baseline scripts: **4.0 hours**.

Video structure (every lesson)

1. Title card: module banner + "Lesson N.M" + title
2. Objective (or, for Mastery Starters, the 60-second version)
3. Key ideas: one scene per group of points, narrated in full
4. Every diagram and chart from the lesson, shown full-screen and explained
5. Worked example, step by step, with the numbers on screen
6. Checklist read aloud
7. Quiz: question → pause → answer
8. Outro: "Do the checklist before moving on"

Brand: the same dark navy system, logo, captions burned in (most people watch muted), disclaimer footer. Narration: Kokoro TTS (`af_heart`) by default; swap for a human or other AI voice by replacing the audio track or changing `VOICE`.

How to render

```
cd programs/defi-program/export
npm install
python3 build_lesson_scripts.py      # regenerate scripts from lessons
node build_video.js lesson-01-3     # one lesson
node build_video.js lessons          # all lessons (several hours of rendering)
```

Every lesson video

Lesson	Title	Scenes	Est. length	Script	Rendered
0.0	Mastery Starter	8	1.5 min	video-scripts/lessons/lesson-00-0.json	
0.1	Money, ledgers and why blockchains exist	10	2.9 min	video-scripts/lessons/lesson-00-1.json	✓
0.2	Opening and securing an exchange account	11	3.0 min	video-scripts/lessons/lesson-00-2.json	
0.3	Buying your first crypto without overpaying	11	2.1 min	video-scripts/lessons/lesson-00-3.json	
0.4	Exchange account vs your own wallet: who holds the keys?	9	2.4 min	video-scripts/lessons/lesson-00-4.json	
0.5	Setting up your wallet and backing it up	11	2.9 min	video-scripts/lessons/lesson-00-5.json	
0.6	Networks, gas and your first transfer	11	3.1 min	video-scripts/lessons/lesson-00-6.json	
0.7	Your first DeFi steps (practice mode first)	11	2.8 min	video-scripts/lessons/lesson-00-7.json	
0.8	Your security baseline, and the language of DeFi	19	4.9 min	video-scripts/lessons/lesson-00-8.json	
1.0	Mastery Starter	8	1.4 min	video-scripts/lessons/lesson-01-0.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
1.1	What DeFi is, and the risk-first mindset	10	2.8 min	video-scripts/lessons/lesson-01-1.json	
1.2	How a transaction actually happens	9	2.4 min	video-scripts/lessons/lesson-01-2.json	
1.3	Wallets, keys, hardware & multisig	11	2.4 min	video-scripts/lessons/lesson-01-3.json	
1.4	Tokens, approvals & allowances	10	2.4 min	video-scripts/lessons/lesson-01-4.json	
1.5	Stablecoins and how they break	10	2.6 min	video-scripts/lessons/lesson-01-5.json	
1.6	Scam defence	9	2.4 min	video-scripts/lessons/lesson-01-6.json	
1.7	Reading signatures and simulating transactions	9	2.6 min	video-scripts/lessons/lesson-01-7.json	
1.8	Privacy and physical security	9	2.4 min	video-scripts/lessons/lesson-01-8.json	
1.9	Smart accounts and account abstraction	10	2.7 min	video-scripts/lessons/lesson-01-9.json	
2.0	Mastery Starter	8	1.4 min	video-scripts/lessons/lesson-02-0.json	
2.1	DEXs, aggregators & routing	11	2.2 min	video-scripts/lessons/lesson-02-1.json	
2.2	AMM Mathematics ($x \cdot y = k$)	4	0.3 min	video-scripts/lessons/lesson-02-2.json	
2.3	Providing liquidity	9	2.3 min	video-scripts/lessons/lesson-02-3.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
2.4	Impermanent loss & true LP P&L	11	2.4 min	video-scripts/lessons/lesson-02-4.json	
2.5	MEV and protecting your trades	10	2.1 min	video-scripts/lessons/lesson-02-5.json	
2.6	Advanced execution: limit, TWAP and intent-based orders	9	2.1 min	video-scripts/lessons/lesson-02-6.json	
2.7	Advanced AMM design and LVR	11	2.7 min	video-scripts/lessons/lesson-02-7.json	✓
2.8	The MEV supply chain	11	2.4 min	video-scripts/lessons/lesson-02-8.json	
3.0	Mastery Starter	8	1.3 min	video-scripts/lessons/lesson-03-0.json	
3.1	How lending markets work	10	2.0 min	video-scripts/lessons/lesson-03-1.json	
3.2	LTV, liquidation threshold & health factor	10	1.7 min	video-scripts/lessons/lesson-03-2.json	
3.3	Liquidations and cascades	9	1.9 min	video-scripts/lessons/lesson-03-3.json	
3.4	Borrowing strategies & looping	10	1.9 min	video-scripts/lessons/lesson-03-4.json	
3.5	Perpetual futures and margin on-chain	9	2.0 min	video-scripts/lessons/lesson-03-5.json	
3.6	Lending design deep dive: e-mode, caps, auctions, soft liquidation, bad debt	11	2.8 min	video-scripts/lessons/lesson-03-6.json	
3.7	CDP stablecoins: minting your own dollars	9	2.1 min	video-scripts/lessons/lesson-03-7.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
4.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-04-0.json	
4.1	Yield farming: base yield vs emissions	9	1.7 min	video-scripts/lessons/lesson-04-1.json	
4.2	Native staking	9	1.4 min	video-scripts/lessons/lesson-04-2.json	
4.3	Liquid staking (LSTs)	9	1.8 min	video-scripts/lessons/lesson-04-3.json	
4.4	Restaking & shared security	9	1.5 min	video-scripts/lessons/lesson-04-4.json	
4.5	Vaults & yield optimisers	9	1.6 min	video-scripts/lessons/lesson-04-5.json	
4.6	Airdrops & points: opportunity cost	9	1.4 min	video-scripts/lessons/lesson-04-6.json	
4.7	Stablecoin savings rates and yield-bearing stablecoins	9	2.2 min	video-scripts/lessons/lesson-04-7.json	
4.8	Tokenized treasuries and real-world assets (RWAs)	9	2.1 min	video-scripts/lessons/lesson-04-8.json	
5.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-05-0.json	
5.1	Bridges and trust assumptions	9	1.9 min	video-scripts/lessons/lesson-05-1.json	
5.2	Layer 2s, sequencers & withdrawal paths	9	1.9 min	video-scripts/lessons/lesson-05-2.json	
5.3	Oracles, TWAPs & manipulation	9	1.9 min	video-scripts/lessons/lesson-05-3.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
5.4	Smart contracts: state, proxies, admin keys	9	2.0 min	video-scripts/lessons/lesson-05-4.json	
5.5	Smart-contract risk & what audits don't prove	9	1.8 min	video-scripts/lessons/lesson-05-5.json	
5.6	Beyond Ethereum: Solana, Bitcoin and other ecosystems	9	2.3 min	video-scripts/lessons/lesson-05-6.json	
5.7	Operating across chains: gas, routes and chain abstraction	9	2.1 min	video-scripts/lessons/lesson-05-7.json	
5.8	Reading smart-contract code: enough to verify claims	10	2.1 min	video-scripts/lessons/lesson-05-8.json	
6.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-06-0.json	
6.1	Protocol due diligence	10	1.7 min	video-scripts/lessons/lesson-06-1.json	
6.2	Tokenomics: supply, unlocks, FDV, value capture	9	1.5 min	video-scripts/lessons/lesson-06-2.json	
6.3	Governance & DAOs	9	1.4 min	video-scripts/lessons/lesson-06-3.json	
6.4	The on-chain research workflow	9	1.4 min	video-scripts/lessons/lesson-06-4.json	
6.5	Case studies: how DeFi failures happened	11	2.4 min	video-scripts/lessons/lesson-06-5.json	
6.6	Vote-escrow tokenomics, bribes and governance markets	9	2.3 min	video-scripts/lessons/lesson-06-6.json	
6.7	Valuing DeFi protocols: fees, revenue, earnings and multiples	9	1.8 min	video-scripts/lessons/lesson-06-7.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
7.0	Mastery Starter	8	1.1 min	video-scripts/lessons/lesson-07-0.json	
7.1	On-chain data foundations	9	1.3 min	video-scripts/lessons/lesson-07-1.json	
7.2	Block explorer mastery	9	1.4 min	video-scripts/lessons/lesson-07-2.json	
7.3	Exchange flows	9	1.3 min	video-scripts/lessons/lesson-07-3.json	
7.4	Whale & entity analysis	9	1.2 min	video-scripts/lessons/lesson-07-4.json	
7.5	Holder & supply metrics	9	1.5 min	video-scripts/lessons/lesson-07-5.json	
7.6	Network activity	9	1.2 min	video-scripts/lessons/lesson-07-6.json	
7.7	DEX & liquidity analytics	9	1.4 min	video-scripts/lessons/lesson-07-7.json	
7.8	Derivatives on-chain	9	1.4 min	video-scripts/lessons/lesson-07-8.json	
7.9	Querying chain data yourself	9	1.7 min	video-scripts/lessons/lesson-07-9.json	
8.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-08-0.json	
8.1	DeFi portfolio construction	10	1.7 min	video-scripts/lessons/lesson-08-1.json	
8.2	The DeFi risk framework	9	1.6 min	video-scripts/lessons/lesson-08-2.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
8.4	The operating playbook: deploy, monitor, respond, review	9	1.6 min	video-scripts/lessons/lesson-08-4.json	
8.5	Measuring performance honestly	9	1.8 min	video-scripts/lessons/lesson-08-5.json	
8.6	Psychology and discipline	10	2.3 min	video-scripts/lessons/lesson-08-6.json	
8.7	Quantitative risk: volatility, VaR, drawdown, correlation and sizing	9	2.5 min	video-scripts/lessons/lesson-08-7.json	
9.0	Mastery Starter	8	1.3 min	video-scripts/lessons/lesson-09-0.json	
9.1	Same idea, different machine	12	3.2 min	video-scripts/lessons/lesson-09-1.json	
9.2	When each one wins	11	2.4 min	video-scripts/lessons/lesson-09-2.json	
9.3	Building a combined system	10	3.0 min	video-scripts/lessons/lesson-09-3.json	
10.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-10-0.json	
10.1	Fixed-rate yield: principal and yield tokens (PT/YT)	11	2.8 min	video-scripts/lessons/lesson-10-1.json	
10.2	Cash-and-carry basis trades	10	2.3 min	video-scripts/lessons/lesson-10-2.json	
10.3	Delta-neutral funding carry, done properly	11	2.1 min	video-scripts/lessons/lesson-10-3.json	
10.4	Options income: covered calls, cash-secured puts, options vaults	10	2.3 min	video-scripts/lessons/lesson-10-4.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
10.5	Active concentrated-liquidity management	11	2.2 min	video-scripts/lessons/lesson-10-5.json	
10.6	Restaking and points: pricing speculative yield	10	2.3 min	video-scripts/lessons/lesson-10-6.json	
10.7	Being the house: perp-exchange liquidity vaults	10	2.4 min	video-scripts/lessons/lesson-10-7.json	
10.8	DeFi rates: term structure, fixed vs floating	10	2.2 min	video-scripts/lessons/lesson-10-8.json	
10.9	Peg and redemption arbitrage	10	2.3 min	video-scripts/lessons/lesson-10-9.json	
11.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-11-0.json	
11.1	Hedging price exposure with perps and options	9	2.0 min	video-scripts/lessons/lesson-11-1.json	
11.2	Depeg, protocol and smart-contract cover	9	1.5 min	video-scripts/lessons/lesson-11-2.json	
11.3	Liquidation protection: buffers, alerts and automated deleveraging	9	1.6 min	video-scripts/lessons/lesson-11-3.json	
11.4	Stress-testing a portfolio	10	2.2 min	video-scripts/lessons/lesson-11-4.json	
11.5	Incident response: the first 60 minutes	9	2.3 min	video-scripts/lessons/lesson-11-5.json	
12.0	Mastery Starter	8	1.2 min	video-scripts/lessons/lesson-12-0.json	
12.1	Your balance sheet	9	1.9 min	video-scripts/lessons/lesson-12-1.json	

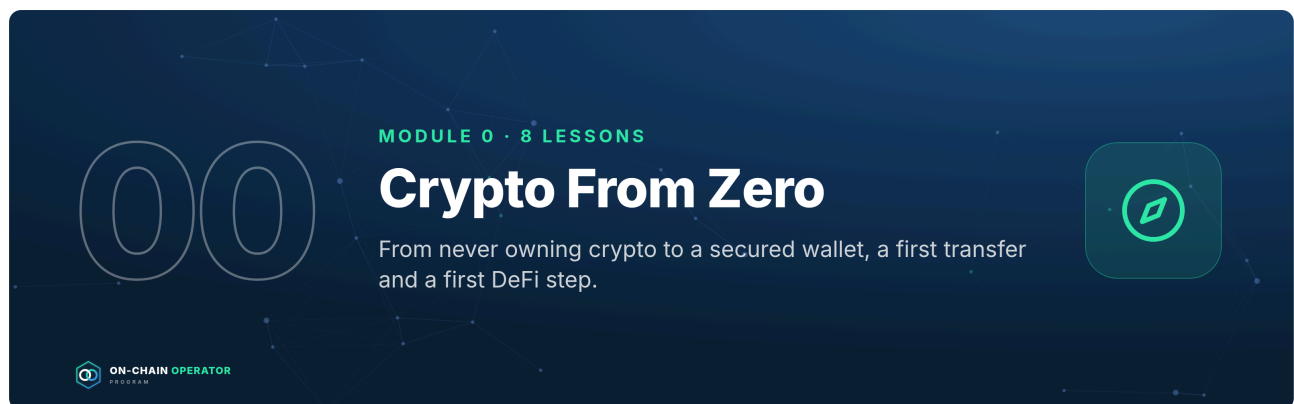
Lesson	Title	Scenes	Est. length	Script	Rendered
12.2	Custody architecture: vault, multisig and spending policies	11	2.2 min	video-scripts/lessons/lesson-12-2.json	
12.3	The credit line: borrowing like a bank client	10	2.6 min	video-scripts/lessons/lesson-12-3.json	
12.4	Treasury and the liquidity ladder	11	2.1 min	video-scripts/lessons/lesson-12-4.json	
12.5	Being the lender: supplying, curating and pricing credit risk	9	2.4 min	video-scripts/lessons/lesson-12-5.json	
12.6	Books, records and succession	10	2.4 min	video-scripts/lessons/lesson-12-6.json	
12.7	Tax, regulation and compliance awareness	9	2.6 min	video-scripts/lessons/lesson-12-7.json	
12.8	Institutional-grade custody: MPC, custodians and policy engines	10	2.2 min	video-scripts/lessons/lesson-12-8.json	
13.0	Mastery Starter	8	1.1 min	video-scripts/lessons/lesson-13-0.json	
13.1	Income sources ranked by durability	10	1.7 min	video-scripts/lessons/lesson-13-1.json	
13.2	Risk-adjusted yield: subtracting expected losses	10	1.6 min	video-scripts/lessons/lesson-13-2.json	
13.3	Building the income portfolio	11	1.9 min	video-scripts/lessons/lesson-13-3.json	
13.4	The payout policy: how much you can take out	9	1.8 min	video-scripts/lessons/lesson-13-4.json	
13.5	Scaling, compounding and the annual review	9	1.9 min	video-scripts/lessons/lesson-13-5.json	

Lesson	Title	Scenes	Est. length	Script	Rendered
14.0	Mastery Starter	8	1.0 min	video-scripts/lessons/lesson-14-0.json	
14.1	Monitoring: dashboards, alerts and on-chain watchers	9	1.7 min	video-scripts/lessons/lesson-14-1.json	
14.2	Automation: keepers, bots and agents without handing over the keys	9	1.9 min	video-scripts/lessons/lesson-14-2.json	
14.3	Operating procedures: multisig signing, change control, reviews	9	1.7 min	video-scripts/lessons/lesson-14-3.json	
14.4	Operator capstone and certification	10	1.7 min	video-scripts/lessons/lesson-14-4.json	
14.5	Building your own tools: reading contracts, data and simple scripts	9	2.1 min	video-scripts/lessons/lesson-14-5.json	
14.6	Searchers, keepers and arbitrage bots: how they work	9	2.3 min	video-scripts/lessons/lesson-14-6.json	

10. Course content

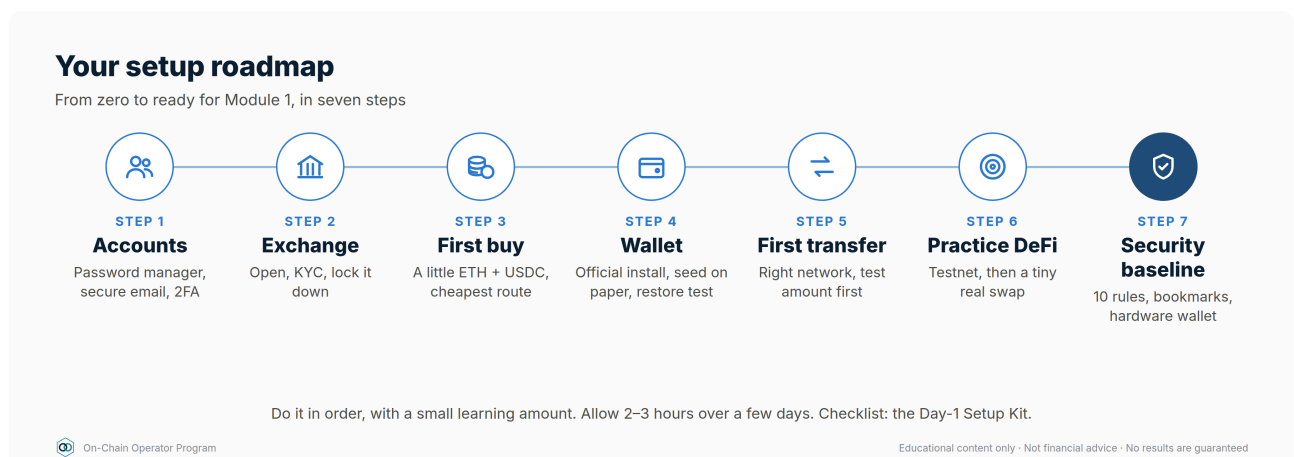
Finished lessons in curriculum order. Lesson 8.3 (strategy mastery) appears before Module 9.

Module 0 — Crypto From Zero



Outcome: go from never having owned crypto to a secured exchange account, a backed-up wallet, a first transfer you made yourself, and a first practice DeFi transaction. Stage 0 · Zero. No prior knowledge needed. Every new word is explained the first time it appears, and again in the glossary at the end. Educational content only. Not financial advice. Examples name well-known products only so you know what to look for. Check what's available and regulated where you live. Fees shown are illustrative; check your provider's fee page.

How to use this module: do it in order, with a small amount of money you can afford to lose while learning (for example \$50–\$200). Every lesson ends with a step you actually do. At the end you'll have finished the **Day-1 Setup Kit** (last section): the checklist that sets up everything the rest of the program needs.



Lesson 0.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Crypto is money recorded on a public ledger that nobody can quietly edit. You'll open an exchange account, buy a little crypto, move it into a wallet only you control, and make your first practice DeFi transaction, all with a small amount and a checklist.

Words you'll need

Term	Meaning
Blockchain	a shared ledger kept by thousands of computers
Wallet	an app or device holding the keys that control your crypto
Seed phrase	12–24 words that are your wallet. Never share
Exchange	a company that swaps your money for crypto
Gas	the network fee for a transaction

Before you start

- ☐ A phone, a computer, ID for KYC, a bank account
- ☐ \$50–\$200 you can afford to lose while learning
- ☐ 2–3 hours over a few days

Your first safe step

Read Lesson 0.1, then secure your email with a unique password and authenticator-app 2FA. No money needed yet.

The mastery ladder

Level	You can...
Beginner	Can explain blockchain, wallet and seed phrase in plain words
Practitioner	Has a secured exchange account and a restore-tested wallet
Master	Completes the whole Day-1 Setup Kit and can coach a friend through it

You've mastered this module when...

...every box of the Day-1 Setup Kit is ticked.

Lesson 0.1 — Money, ledgers and why blockchains exist

Objective

Explain in plain words what a blockchain is, what a cryptocurrency is, and what "DeFi" means.

Explanation

- **Money is a record of who owns what.** Your bank balance is a line in your bank's private ledger (record book). You trust the bank to keep it correct.
- **A blockchain** is a ledger that thousands of computers keep a copy of and agree on together. New entries are added in batches called **blocks**, each linked to the one before: a chain of blocks. Nobody can quietly change an old entry, because everyone else's copy would disagree.

- **A cryptocurrency** is a unit recorded on a blockchain. **Bitcoin (BTC)** was the first (2009). **Ether (ETH)** is the currency of the **Ethereum** blockchain, which also runs programs.
- **A smart contract** is a program that lives on a blockchain and follows fixed rules, for example "whoever deposits X gets Y back plus interest".
- **Stablecoins** are tokens designed to stay worth \$1 (e.g. USDC, USDT). You'll use them a lot, but they can fail (Lesson 1.5).
- **DeFi (decentralised finance)** is financial services (trading, lending, earning interest) run by smart contracts instead of banks.
- **The trade-off:** no bank in the middle means nobody can freeze your money, but also **nobody can reverse your mistakes**. That's why this program starts with safety.

Worked example

When Alice sends Bob 0.1 ETH, she doesn't send a file. She signs an instruction ("move 0.1 ETH from my address to Bob's"). The network checks she owns it, adds it to the next block, and every copy of the ledger updates. Bob now owns it. There is no "undo" button.

Checklist

- ☐ I can explain a blockchain as "a shared ledger nobody can quietly edit"
- ☐ I know the difference between BTC, ETH and a stablecoin
- ☐ I understand that crypto transactions can't be reversed

Quiz

Q: 1. Why can't someone quietly change an old blockchain entry?

Answer: Thousands of computers hold copies and would reject a version that doesn't match.

Q: 2. What is a stablecoin designed to do?

Answer: Stay worth about \$1 (or another currency). It can still fail.

Q: 3. What does DeFi replace, and what do you give up?

Answer: It replaces banks and brokers with smart contracts. You give up anyone who can reverse mistakes or help recover funds.

Lesson 0.2 — Opening and securing an exchange account

Objective

Open an account on a reputable exchange and lock it down before putting money in.

Explanation

An **exchange** is a company that lets you swap your normal money (dollars, pounds, euros) for crypto. It's your **on-ramp**. Choose one that is:

- **Licensed or registered where you live** (check your country's financial regulator)
- **Large and long-established**, with a good security record

- **Supports your bank's deposit method** and the coins/networks you'll use (ETH, USDC, and low-cost networks such as Arbitrum or Base)

Well-known examples include Coinbase, Kraken, Binance, Bybit and OKX, but availability differs by country. **Pick one that's legally available to you.**

You'll be asked for **KYC** ("know your customer"): ID and sometimes proof of address. This is normal and legally required for regulated exchanges.

Secure it before depositing:

1. **Unique, long password** stored in a **password manager** (e.g. Bitwarden, 1Password)
2. **Secure your email first.** Whoever controls your email can often reset your exchange account. Give it its own strong password and 2FA.
3. **Two-factor authentication (2FA)** with an **authenticator app** (e.g. Google Authenticator, Authy) or a **hardware security key**. **Avoid SMS codes:** phone numbers can be hijacked ("SIM swap").
4. **Save the 2FA backup codes** offline (on paper, stored safely)
5. Turn on the **anti-phishing code** if offered (a word that appears in every real email from the exchange)
6. Turn on the **withdrawal address allowlist** if offered (withdrawals can only go to addresses you pre-approved)

Worked example

Maria signs up, completes KYC in 10 minutes, then spends another 10 on security: password manager, authenticator-app 2FA on both her email and the exchange, backup codes on paper, and an anti-phishing code "blue-kettle". Later, an email arrives about a "suspicious login" with no "blue-kettle" in it. She knows it's fake and deletes it.

Checklist

- ☐ Exchange chosen: legally available and reputable where I live
- ☐ KYC complete
- ☐ Email secured with its own strong password + 2FA
- ☐ Exchange 2FA via authenticator app or security key (not SMS)
- ☐ Backup codes stored offline
- ☐ Anti-phishing code and withdrawal allowlist switched on (if offered)

Quiz

Q: 1. Why secure your email before the exchange?

Answer: Email is often how accounts are reset. Whoever controls it can take over the exchange account.

Q: 2. Why avoid SMS 2FA?

Answer: Attackers can hijack phone numbers (SIM swap) and receive your codes.

Q: 3. What does a withdrawal allowlist do?

Answer: Only lets withdrawals go to addresses you approved in advance.

Lesson 0.3 — Buying your first crypto without overpaying

Objective

Fund your account and make a first purchase at a sensible cost.

Explanation

- **Deposit** your money, usually by bank transfer (cheapest) or card (fastest, most expensive).
- **Instant buy / convert** buttons are simple but often include a **spread** (a price markup) plus a fee.
- The **trading screen** ("advanced" or "spot trading") uses an **order book**:
 - **Market order**: buy now at the best available price
 - **Limit order**: buy only at your chosen price or better. It often has a lower fee ("maker" fee).
- **What to buy first for this program**: a little **ETH** (you need it to pay network fees, called "gas") and some **USDC** (a dollar stablecoin for practice).

Worked example (illustrative fees)

Buying \$500 of ETH:

Route	Typical cost	You pay in fees
Card + instant buy	~3.99%	~\$19.95
Bank transfer + limit order	~0.4%	~\$2.00

Same ETH, ~\$18 difference. Your exchange's fee page has the real numbers.

Checklist

- ☐ Deposited by the cheapest method available to me
- ☐ Bought a small amount of ETH (for gas) and USDC (for practice)
- ☐ Tried a limit order on the trading screen
- ☐ Saved a record: date, amount, price, fees (you'll need records for tax)

Quiz

Q: 1. Why keep a little ETH even if you mostly want stablecoins?

Answer: You need ETH to pay network fees (gas) on Ethereum and most of its low-cost networks.

Q: 2. Market or limit order: which controls the price you pay?

Answer: A limit order.

Q: 3. Why keep records from day one?

Answer: Most countries tax crypto gains. Records of dates, amounts, prices and fees make that straightforward.

Lesson 0.4 — Exchange account vs your own wallet: who holds the keys?

Objective

Understand custody and decide what stays on the exchange and what moves to your own wallet.

Explanation

- **On an exchange (custodial):** the exchange holds your crypto and owes it to you. Easy, recoverable with ID, but if the exchange fails, freezes or is hacked, your money is at risk. *"Not your keys, not your coins."*
- **In your own wallet (self-custody):** you hold the **keys**, which are the secret codes that control your crypto. Nobody can freeze it; nobody can recover it for you.
- **A wallet doesn't "store" coins.** The coins are on the blockchain; the wallet holds the keys that let you move them. That's why a backup of the keys (your **seed phrase**) is the wallet.
- **Addresses:** your wallet gives you a public **address** (like `0x7a3F...9c21`) to receive funds. Safe to share, like an account number.
- **Seed phrase / recovery phrase:** 12 or 24 words that recreate your keys. **Never share it, never type it into a website, never photograph it.**

	Exchange	Your wallet
Who holds keys	The exchange	You
Recover if you forget password	Yes, with ID	Only with your seed phrase
Can be frozen	Yes	No
Can use DeFi	Limited	Yes
Your main risk	The company	Your own mistakes and scams

Checklist

- ☐ I can explain custodial vs self-custody
- ☐ I know my seed phrase **is** the wallet
- ☐ I know an address is safe to share and a seed phrase never is

Quiz

Q: 1. What does "not your keys, not your coins" mean?

Answer: If someone else holds the keys, you depend on them to give your crypto back.

Q: 2. Is your crypto stored inside your wallet app?

Answer: No. It's on the blockchain. The wallet holds the keys that control it.

Q: 3. Which can you share: your address or your seed phrase?

Answer: Your address. Never the seed phrase.

Lesson 0.5 — Setting up your wallet and backing it up

Objective

Install a wallet safely, back up the seed phrase correctly, and prove the backup works.

Explanation

Two kinds of wallet:

- **Software wallet** (browser extension or phone app, e.g. MetaMask, Rabby): free, convenient; keys live on your computer/phone.
- **Hardware wallet** (a small device, e.g. Ledger, Trezor): keys stay on the device; every transaction must be approved on its screen. **Recommended once you hold more than you'd be upset to lose.** Buy only from the manufacturer's official website, never second-hand.

Safe install:

1. Get the wallet from the **official website** (type the address yourself or use the link from the official site). Fake wallet apps and sponsored search ads are a common scam.
2. Create a **new** wallet. Never use a seed phrase someone else gave you or that came pre-printed in a box.
3. Write the seed phrase **on paper** (or stamp it into metal), in order, spelled exactly.
4. Store it somewhere private and safe from fire and water. For larger amounts, keep a second copy in a separate location.
5. **Never** store it in photos, email, notes apps, cloud drives or password managers.

Your seed phrase: do and don't

The 12–24 words are the wallet. Whoever has them has everything.

✓ Do

- ✓ Write it on paper or stamp it in metal, in order
- ✓ Store it privately, safe from fire and water
- ✓ Keep a second copy elsewhere for larger amounts
- ✓ Test a restore before depositing
- ✓ Use a hardware wallet once the amount matters

✗ Never

- ✗ Type it into any website or form
- ✗ Photograph it or keep it in notes, email or cloud
- ✗ Share it with anyone: no real support will ask
- ✗ Use a phrase that came pre-printed or from someone else
- ✗ Store it next to written instructions on how to use it

 On-Chain Operator Program

Educational content only · Not financial advice · No results are guaranteed

Prove the backup works (do this now, with nothing in the wallet): remove the wallet (or use a second device), choose "restore/import", enter your written seed phrase, and check the **same address** appears.

Worked example

Sam installs Rabby from its official site, writes 12 words on the card in his hardware wallet box (the card is blank, as it should be), restores on his laptop, sees the same address `0x4b...e1`, and only then sends money to it.

Checklist

- ☐ Wallet installed from the official website
- ☐ New seed phrase written on paper (or metal), never digital
- ☐ Restore test done: same address appears
- ☐ Hardware wallet planned for larger amounts (bought from the maker directly)

Quiz

Q: 1. A wallet box contains a card with 24 words already printed. What do you do?

Answer: Don't use it. It's a scam. Genuine devices generate your seed phrase themselves.

Q: 2. Why test the restore before depositing?

Answer: To prove your written backup is correct while nothing is at risk.

Q: 3. Is a cloud-drive photo of your seed phrase a safe backup?

Answer: No. Anyone who gets into that account can take everything.

Lesson 0.6 — Networks, gas and your first transfer

Objective

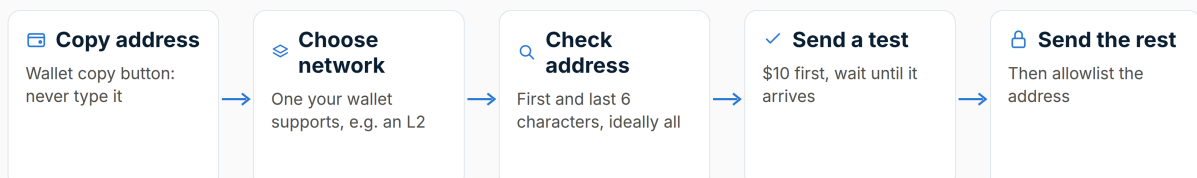
Move crypto from the exchange to your own wallet safely, on the right network, starting with a test amount.

Explanation

- **Networks (chains):** the same token (e.g. USDC) can exist on several blockchains: **Ethereum** (the main one, higher fees) and cheaper **Layer 2** networks such as **Arbitrum**, **Base** and **Optimism**.
- **The golden rule:** the **network you withdraw on must match a network your wallet (or the receiving service) supports**. Sending on the wrong network can mean lost funds.
- **Gas:** every transaction pays a small network fee in the network's gas token. On Ethereum and its Layer 2s that's **ETH**. Without a little ETH on the network you're using, you can't move anything.
- Your wallet address is usually the **same** on Ethereum and its Layer 2s, but your balances are separate per network. Switch networks in your wallet to see them.

Your first transfer: exchange → your wallet

The network must match on both sides. Send a small test first.



Keep a little ETH on that network for gas, or you can't move anything.

Step by step:

1. In your wallet, **copy your address** (use the copy button; never type it).
2. On the exchange, choose **Withdraw** → the coin → **pick the network** (for learning, a low-fee Layer 2 your wallet supports, e.g. Arbitrum or Base).
3. Paste the address. **Check the first 6 and last 6 characters** match your wallet. Better, check it all.
4. Send a **small test** first (e.g. \$10 of ETH).
5. Wait for it to arrive; see it in your wallet on that network.
6. Only then send the rest. Add the address to the exchange **allowlist**.

Worked example (illustrative)

Withdrawing \$100 of ETH to Arbitrum: exchange withdrawal fee ~\$0.10–\$1. Later, a swap on Arbitrum costs a few cents in gas. The same swap on Ethereum might cost a few dollars, more when the network is busy. That's why beginners learn on Layer 2s.

Checklist

- ☐ Network chosen on purpose, and supported by my wallet
- ☐ Address pasted, first and last 6 characters checked
- ☐ Test amount sent and received before the rest
- ☐ A little ETH held on that network for gas
- ☐ Address added to the exchange withdrawal allowlist

Quiz

Q: 1. What's the golden rule of withdrawals?

Answer: The network you send on must be one the receiving wallet or service supports.

Q: 2. You have USDC on Arbitrum but zero ETH there. Can you send it?

Answer: No. You need a little ETH on Arbitrum to pay gas.

Q: 3. Why send a test amount?

Answer: To confirm the address and network are right before risking the full amount.

Lesson 0.7 — Your first DeFi steps (practice mode first)

Objective

Connect your wallet to a DeFi app, understand what you're signing, and practise on a free test network before using real money.

Explanation

- **Connect wallet:** lets a site see your address. It can't move funds by itself.
- **Sign a message:** proves you own the address (e.g. to log in). Read it. Some "messages" are really permissions (Lesson 1.4).
- **Approve:** gives an app permission to move a specific token. Limit the amount.

- **Transaction:** actually does something (swap, deposit) and costs gas.
- **Testnets** are practice blockchains with free, worthless coins. Ethereum's main one is **Sepolia**. You get test ETH from a **faucet** (a free giveaway site). Use testnets to practise without risk.
- **Revoke** permissions you no longer need with a tool such as revoke.cash.

Practice run:

1. Add the Sepolia test network in your wallet (most wallets have a "show test networks" setting).
2. Get free test ETH from a Sepolia faucet.
3. Make a test swap on a testnet version of a well-known DEX (or send test ETH to yourself).
4. Look up the transaction on a block explorer (Lesson 1.2).
5. Then, on a Layer 2 with a **tiny** real amount: connect to a well-known app, read every prompt, do one small swap, then revoke the approval.

Worked example

Priya connects to a DEX on Arbitrum. Her wallet shows "Approve USDC: Unlimited". She edits it to \$20, swaps \$20 of USDC for ETH, pays ~\$0.05 in gas, checks the transaction on the explorer, then revokes the approval. Total cost: a few cents. Total lessons learned: five.

Checklist

- ☐ I know the difference between connect, sign, approve and transact
- ☐ Practised on the Sepolia testnet
- ☐ Made one small real swap on a Layer 2 with a limited approval
- ☐ Found my transaction on a block explorer
- ☐ Revoked the approval afterwards

Quiz

Q: 1. Can "connect wallet" move your funds?

Answer: No. It shares your address. Approvals and transactions are what move funds.

Q: 2. What is a testnet for?

Answer: Practising with worthless coins, so mistakes cost nothing.

Q: 3. What should you do with an approval after you're done?

Answer: Revoke it, or limit it to the amount needed in the first place.

Lesson 0.8 — Your security baseline, and the language of DeFi

Objective

Lock in the habits that prevent most losses, and learn the words used in the rest of the program.

Explanation — the 10 rules

1. **Never share your seed phrase.** No real person or company will ever ask for it.

2. **Bookmark** the sites you use; never reach them from ads, DMs or emails.
3. **Nobody legitimate DMs you first** offering help, investments or "recovery".
4. **Guaranteed returns are a scam.** Always.
5. **Read every wallet prompt** before approving: what, how much, to whom.
6. **Test first** with a small amount on anything new.
7. **Use a hardware wallet** once the amount matters to you.
8. **Keep devices updated**, and use a separate browser profile for crypto.
9. **Keep records** of every buy, sell, transfer and fee.
10. **If something feels urgent, stop.** Scammers create urgency; real opportunities wait.

Glossary (the words you'll meet next)

Word	Meaning
Address	Your public "account number" on a blockchain
Seed phrase	12–24 words that recreate your wallet's keys. Never share
Private key	The secret that signs transactions (your seed phrase generates it)
Custody	Who holds the keys: an exchange (custodial) or you (self-custody)
Gas	The network fee for a transaction, paid in the network's token
Network / chain	A blockchain, e.g. Ethereum, Arbitrum, Base
Layer 2 (L2)	A cheaper, faster network built on top of Ethereum
Token	A unit of value on a blockchain, e.g. USDC
Stablecoin	A token designed to hold a steady value, usually \$1
Smart contract	A program on a blockchain that follows fixed rules
DEX	Decentralised exchange: swap tokens from your wallet
Approval	Permission for an app to move a specific token
Block explorer	A website that shows every transaction (e.g. Etherscan)
Testnet / faucet	A practice blockchain / a site giving free test coins
APY	Annual percentage yield: yearly return including compounding
Liquidity	How easily something can be bought or sold without moving the price
KYC	ID checks required by regulated exchanges
2FA	Two-factor authentication: a second proof of identity at login

Checklist

- ☐ I've read the 10 rules and can repeat them
- ☐ Separate browser profile set up for crypto
- ☐ I know every word in the glossary

Quiz

Q: 1. Someone offers a "guaranteed 2% a day". What is it?

Answer: A scam. Guaranteed returns don't exist.

Q: 2. What's gas?

Answer: The network fee for a transaction, paid in the network's token.

Q: 3. A message says your wallet will be "suspended" unless you act in 1 hour. What do you do?

Answer: Stop. Urgency is a scam tactic. Wallets can't be suspended; check only through bookmarked official sites.

The Day-1 Setup Kit

Print this, and tick every box in order. When all boxes are ticked, you're ready for Module 1. Allow **2–3 hours** in total, spread over a few days (deposits and KYC can take time).

What you need: ID for KYC · a bank account · a smartphone for the authenticator app · a computer with an up-to-date browser · pen and paper (or a metal backup plate) · optional: a hardware wallet bought from the maker.

Part A — Accounts (≈ 45 min)

- ☐ Password manager installed; master password written down and stored safely
- ☐ Email secured: unique password + authenticator-app 2FA
- ☐ Authenticator app installed on your phone
- ☐ Exchange chosen (legally available to you, reputable)
- ☐ Exchange account opened and KYC completed
- ☐ Exchange 2FA on (authenticator or security key, not SMS); backup codes stored offline
- ☐ Anti-phishing code and withdrawal allowlist switched on

Part B — First purchase (≈ 20 min)

- ☐ Deposited a small learning amount by the cheapest method
- ☐ Bought a little ETH (gas) and some USDC (practice)
- ☐ Tried a limit order
- ☐ Started a simple records sheet: date, action, amount, price, fees

Part C — Your wallet (≈ 30 min)

- ☐ Wallet installed from the official website (separate browser profile)
- ☐ New seed phrase written on paper/metal, in order; never digital
- ☐ Restore test passed: same address appears
- ☐ Wallet address saved in the password manager's notes (address only, **never the seed**)

Part D — First transfer (≈ 20 min)

- ☐ Network chosen (low-fee L2 supported by your wallet)
- ☐ Test withdrawal (e.g. \$10 of ETH) sent and received
- ☐ Remaining learning funds sent; address added to the allowlist
- ☐ Transfer found on a block explorer

Part E — Practice DeFi (≈ 30 min)

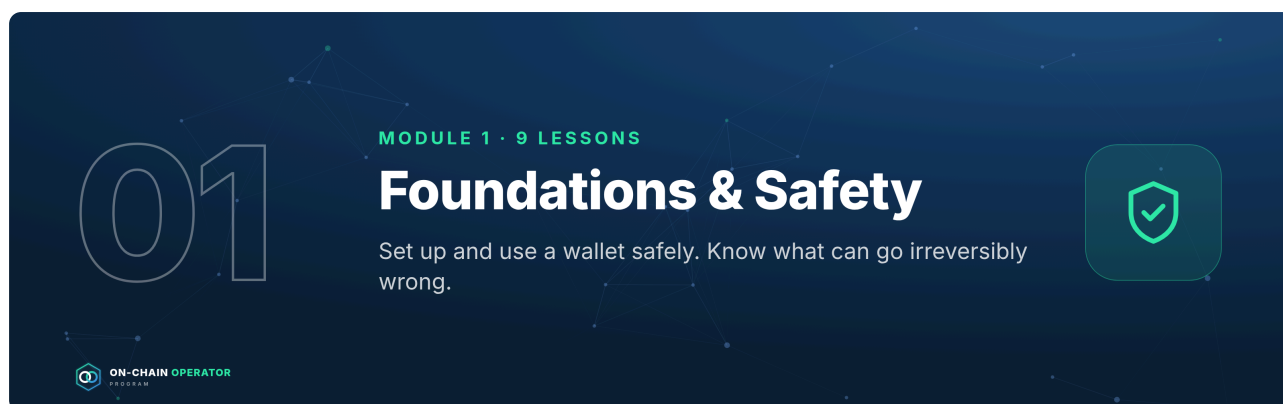
- ☐ Sepolia test network added; free test ETH from a faucet
- ☐ One practice transaction on the testnet
- ☐ One tiny real swap on an L2 with a limited approval
- ☐ Approval revoked afterwards

Part F — Security baseline (≈ 15 min)

- ☐ The 10 rules read and understood
- ☐ Sites bookmarked; no links from ads or DMs
- ☐ Hardware wallet ordered from the maker (when the amount will matter)
- ☐ Records sheet up to date

All ticked? You're ready for Module 1 — Foundations & Safety.

Module 1 — Foundations & Safety



Outcome: set up and use a wallet safely, and know what can go irreversibly wrong before any capital moves. Source: ATLAS "DeFi & On-Chain" ch. 1–5, 36. Educational content only. Not financial advice.

The **before-signing checklist** is introduced in this module and used in every module after it:

- ☐ Chain confirmed
- ☐ Contract address verified from an authoritative source
- ☐ Token and amount double-checked
- ☐ Spender / allowance reviewed
- ☐ Slippage set deliberately
- ☐ Gas estimate reviewed
- ☐ Intended outcome stated in one sentence

Lesson 1.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Most crypto losses aren't market losses; they're mistakes and scams: a seed phrase typed into a fake site, an approval signed without reading, the wrong network. This module builds the habits that prevent them.

Words you'll need

Term	Meaning
Private key	the secret that signs transactions
Approval	permission for an app to move a token
Multisig	a wallet that needs several keys to act
Phishing	a fake site or message built to steal
Simulation	a preview of what a transaction will change

Before you start

- ☐ Module 0 done (secured exchange, restore-tested wallet)
- ☐ A small amount in a wallet on a low-fee network

Your first safe step

Open your wallet's approvals (or a revoke tool), list every approval you've granted, and revoke any you don't use.

The mastery ladder

Level	You can...
Beginner	Uses a separate burner wallet and reads every prompt
Practitioner	Runs the before-signing checklist without notes; limited approvals; hardware wallet
Master	Multisig vault, simulation on every signature, private addresses, tested recovery

You've mastered this module when...

...you can take a new app from first visit to a limited, simulated, verified transaction, and revoke it after.

Lesson 1.1 — What DeFi is, and the risk-first mindset (*ch. 1*)

Objective

Describe the DeFi stack and explain why every yield is payment for a risk.

Explanation

DeFi replaces intermediaries (banks, brokers, exchanges) with **smart contracts**: public programs on a blockchain that hold assets and follow fixed rules. The stack, bottom to top:

1. **Blockchain**: records every balance and transaction (Ethereum, L2s, other chains).
2. **Smart contracts**: the protocol logic (a lending market, an exchange pool).
3. **Tokens**: the assets the contracts move.
4. **Wallets**: where *you* hold keys and sign actions.
5. **Interfaces**: websites that build transactions for you. They're a convenience, not the protocol itself.
6. **Data layers**: explorers and dashboards you use to verify what happened.

Three properties make DeFi powerful and dangerous:

- **Self-custody**: nobody can freeze your funds, and nobody can recover them either.
- **Composability**: protocols plug into each other. That makes products powerful, and a position can fail because of something it depends on.
- **Transparency**: you can verify everything on-chain. Transparent doesn't mean safe.

Risk-first rule: before comparing returns, list what could make you lose money. A higher yield means you're being paid to carry more risk, whether or not the website mentions it.

Worked example

A pool advertises **12% APY on a stablecoin**. Before depositing, ask:

- Where does the 12% come from? Borrowers paying interest, trading fees, or a reward token?
- Which stablecoin, and what backs it? (Lesson 1.5)
- Which contracts hold the money, and who can upgrade them?
- Is there a bridge or oracle involved?
- How do I withdraw, and could withdrawals be blocked?

If you can't answer these, you don't know what the 12% is paying you for.

Checklist

- ☐ I can name the six layers of the DeFi stack
- ☐ I can explain why self-custody cuts both ways
- ☐ Before any yield, I ask "what am I being paid to risk?"

Quiz

Q: 1. Why can composability increase risk?

Answer: A position inherits the failure risk of every protocol, asset, oracle and bridge it depends on.

Q: 2. Is a website the same thing as the protocol?

Answer: No. The interface only builds transactions. The contracts are the protocol, and interfaces can be faked or compromised.

Q: 3. What is the risk-first mindset in one sentence?

Answer: Identify what could make you lose money before you compare returns.

Lesson 1.2 — How a transaction actually happens (ch. 2)

Objective

Follow a transaction from signature to finality, and estimate its cost.

Explanation

Lifecycle: create → sign → broadcast → mempool (waiting) → included in a block → confirmed → final.

- **Gas** is the unit of computation. Cost = `gas used × gas price`. On Ethereum the price has a *base fee* (burned) plus a *priority tip* (to get included sooner). Gas price is quoted in **gwei** (1 gwei = 0.000000001 ETH).
- **Nonce**: each account's transactions are numbered in order. A stuck low-nonce transaction blocks all later ones until it confirms or is **replaced** (same nonce, higher fee).

- **Failed transactions still cost gas.** The network did the computation even though the result was reverted.
- **Finality:** the point after which a transaction can't realistically be reversed. It differs by chain, and L2s have their own rules (Module 5).
- **Explorers** (e.g. Etherscan) show status, fee, sender, contract called, token transfers and logs.

Worked example

A swap uses **150,000 gas** at **20 gwei**: $150,000 \times 20 = 3,000,000 \text{ gwei} = 0.003 \text{ ETH}$. At \$3,000/ETH that's **\$9**. The same swap during congestion at 80 gwei costs **\$36**. If you're compounding \$1.50 of rewards, the gas costs more than the rewards. That's why position size matters on-chain.

Checklist

- ☐ I check the fee estimate before signing
- ☐ I know how to find my transaction on an explorer
- ☐ I know a stuck transaction can be sped up or cancelled with the same nonce

Quiz

Q: 1. Your transaction reverted. Did you pay?

Answer: Yes, for the gas used up to the point of failure.

Q: 2. 200,000 gas at 10 gwei with ETH at \$2,500 costs?

Answer: $2,000,000 \text{ gwei} = 0.002 \text{ ETH} = \5 .

Q: 3. Why are your newer transactions all pending?

Answer: An earlier nonce is stuck. Replace or speed it up, and the rest follow.

Lesson 1.3 — Wallets, keys, hardware & multisig (ch. 3)

Objective

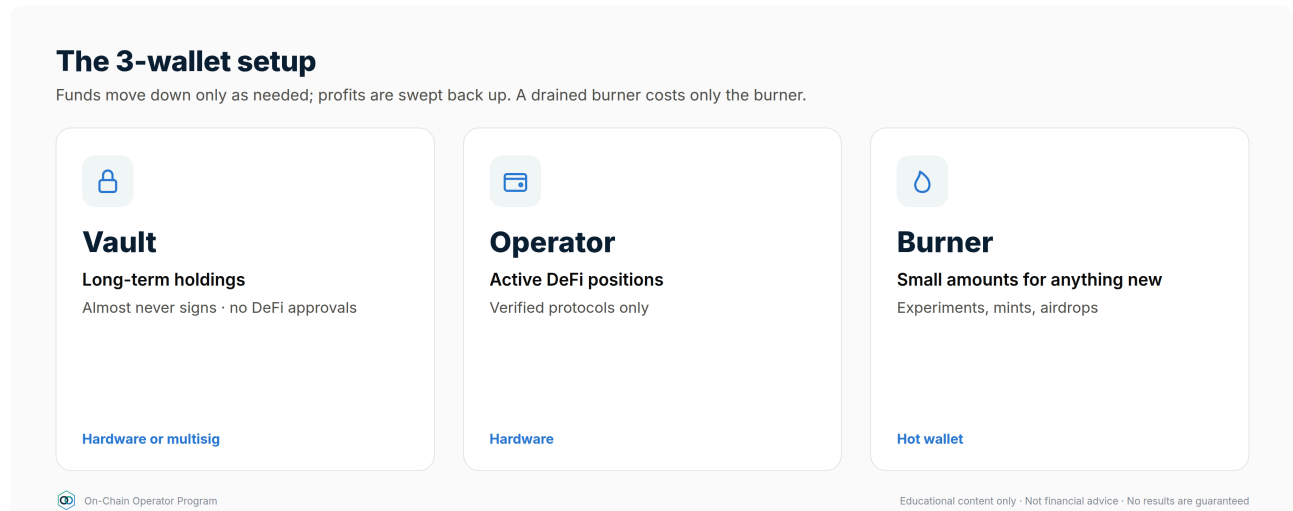
Set up a wallet structure where one mistake can't lose everything.

Explanation

- **Private key:** the authority to sign. Whoever has it controls the funds.
- **Seed phrase:** 12–24 words that recreate your keys. **Never type it into a website, never photograph it, never share it. No legitimate support team will ever ask for it.**
- **Hardware wallet:** keeps keys on a separate device. Signing requires physical confirmation.
- **Multisig:** needs M of N signers (e.g. 2 of 3). Good for large or shared funds.
- **Smart wallets:** contract accounts that can add spending limits, recovery and batching.

Compartmentalise: separate wallets by job, so a compromise only reaches one of them.

Worked example — the 3-wallet setup



Wallet	Holds	Signs	Device
Vault	Long-term holdings	Almost never. No DeFi approvals	Hardware (or multisig)
Operator	Active DeFi positions	Known, verified protocols only	Hardware
Burner	Small amounts for new apps, mints, airdrops	Anything experimental	Hot wallet

Move money *down* the chain (vault → operator → burner) only as needed, and sweep profits back *up*. If the burner gets drained, you lose only the burner's balance.

Checklist

- ☐ Seed phrase stored offline, in two separate physical places
- ☐ Vault wallet has never approved a DeFi contract
- ☐ Separate burner wallet for anything new
- ☐ I read every signing prompt on the hardware device screen, not just the website

Quiz

Q: 1. A "support agent" in DMs needs your seed phrase to fix a stuck transaction. What do you do?

Answer: Nothing. It's a scam, every time. Block and report.

Q: 2. Why keep the vault wallet free of approvals?

Answer: Approvals let contracts move your tokens. With none, a compromised protocol can't reach it.

Q: 3. What does 2-of-3 multisig protect against?

Answer: A single lost or compromised key. Two signers are still needed to move funds.

Lesson 1.4 — Tokens, approvals & allowances (ch. 4)

Objective

Read an approval request and know what you're allowing.

Explanation

- Most tokens on EVM chains are **ERC-20** (fungible). NFTs (ERC-721/1155) can also represent positions (e.g. concentrated LP).
- To let a protocol move your tokens you **approve** a **spender** for an **allowance**. The protocol then pulls tokens when you act.
- **Unlimited approvals** are convenient but let that contract move *all* of that token, now and in future. If the contract (or the approval) is exploited, the tokens can be taken.
- **Signature approvals (Permit / Permit2)** grant allowances with an off-chain signature, not a transaction. There's no gas, so they're easy to sign without noticing. Drainers abuse exactly this.
- **Wrapped / receipt tokens**: WETH, LP tokens and lending receipts represent a claim on something else.
- **Fake tokens**: anyone can create a token called "USDC". Only the **contract address** identifies it.

Worked example — reading an approval

Wallet shows: *"Allow 0x3fC9...a1B2 to spend your USDC. Amount: Unlimited."*

1. Is `0x3fC9...a1B2` the protocol's official router? Check the docs or explorer.
2. Do I need unlimited? Edit it to the amount of this deposit.
3. Is this USDC the real contract?
4. After finishing, revoke with a revoke tool (e.g. revoke.cash) if I won't be back.

Checklist

- ☐ Spender verified against official docs
- ☐ Allowance limited to what's needed
- ☐ Signature requests read as carefully as transactions
- ☐ Approvals reviewed and revoked monthly

Quiz

Q: 1. Why is a gasless "Permit" signature dangerous?

Answer: It can grant a token allowance without a transaction, so it's easy to sign without noticing. Drainers use it to take tokens.

Q: 2. How do you know a token is the real one?

Answer: By its contract address from an authoritative source, never by name or ticker.

Q: 3. When should you revoke an approval?

Answer: When you no longer use that protocol, or on a regular schedule.

Lesson 1.5 — Stablecoins and how they break (ch. 5)

Objective

Classify a stablecoin by design and name what could break its peg.

Explanation

Type	Backing	Main risks
Fiat-backed	Cash/treasuries held by an issuer	Issuer, bank and custodian risk; freezes; redemption limited to approved parties
Crypto-backed	Over-collateralised on-chain crypto	Collateral crash, liquidations, oracle failure
Synthetic / hedged	Derivative positions (e.g. spot + short perp)	Funding turns negative, exchange/venue risk, model risk
Algorithmic	Mostly its own mechanism / sister token	Reflexive death spiral when confidence breaks

Peg mechanics: if a stablecoin trades at \$0.98 and can be redeemed for \$1.00, arbitrageurs buy and redeem it, pushing the price back up. The peg is only as strong as **redemption access and reserve quality**.

History (research these as case studies):

- **May 2022:** TerraUSD (UST), an algorithmic design, lost its peg and collapsed toward zero.
- **March 2023:** USDC briefly traded well below \$1 after part of its reserves were exposed to the failed Silicon Valley Bank, then recovered once reserves were confirmed.

Worked example

A fiat-backed coin trades at **\$0.97**. Redemption is **\$1.00 minus a 0.1% fee**, but only for verified institutional minters.

- A minter makes ~2.9% per coin by buying and redeeming, so arbitrage *should* restore the peg.
- You can't redeem, so you rely on minters acting and on the reserves being real.
- If markets doubt the reserves, minters may not step in, and the discount can grow.

Checklist

- ☐ I know the type and backing of every stablecoin I hold
- ☐ I know who can redeem, and how
- ☐ I don't treat "different stablecoins" as diversified if they share issuers or collateral
- ☐ I have a depeg exit rule (e.g. sell or rotate below \$0.99 for longer than X hours)

Quiz

Q: 1. What keeps a fiat-backed stablecoin near \$1?

Answer: Redemption for \$1 of reserves, plus arbitrageurs who can redeem.

Q: 2. Why are algorithmic designs fragile?

Answer: They rely on confidence and their own token. When demand falls, the mechanism can amplify the fall.

Q: 3. Two stablecoins both backed by the same collateral. Diversified?

Answer: No. They share the same failure point.

Lesson 1.6 — Scam defence (ch. 36)

Objective

Recognise the common attacks before they cost you anything.

Explanation

Attack	How it works	Defence
Phishing site	Look-alike URL / sponsored search ad / fake app	Bookmarks only. Never click links from DMs or ads
Wallet drainer	You sign an approval, Permit or setApprovalForAll for an attacker	Read every prompt. Burner wallet for new sites
Address poisoning	Attacker sends \$0 transfers from an address that looks like one you use, so it appears in your history	Never copy addresses from history. Use a saved address book and check the <i>full</i> address
Fake token / airdrop	Unknown tokens appear in your wallet with a "claim" link	Ignore them. Don't interact or try to sell
Impersonation	"Support", "admin" or "project team" DMs you	Real teams never DM first and never ask for seeds
Too-good yields	New protocol, huge APY, anonymous team, no audit	Due diligence (Module 6). Burner wallet and small size, or skip

Worked example — address poisoning

You regularly send to `0x7a3F...9c21`. Today your history shows a transfer from `0x7a3F...9c21`, but the full address is `0x7a3F e0b4...d18 9c21`: same start and end, different middle. If you copy it from history you send your funds to the attacker. **Always paste from your address book and check the full address.**

Checklist

- ☐ Protocol sites bookmarked; never reached through search ads or DMs
- ☐ Address book in use; full address checked before sending
- ☐ Unknown tokens ignored

- ☐ DMs from "support" treated as scams by default
- ☐ New sites only with the burner wallet

Quiz

Q: 1. A new token worth "\$5,000" appears in your wallet with a claim site. What do you do?

Answer: Ignore it. It's bait, and interacting risks a drainer approval.

Q: 2. How does address poisoning trick people?

Answer: It puts a look-alike address in your history so you copy it by mistake.

Q: 3. What's the single best habit against drainers?

Answer: Reading every signature and approval prompt, and using a burner wallet for anything new.

Lesson 1.7 — Reading signatures and simulating transactions (new)

Objective

Know exactly what a signature or transaction will do before you approve it.

Explanation

- **Transaction simulation:** good wallets (and tools like transaction simulators) show the **expected balance changes** before you sign: "-100 USDC, +0.033 ETH". If the preview shows assets leaving that you didn't intend, stop.
- **Typed-data signatures (EIP-712):** structured messages (orders, Permits) that your wallet shows as fields. Read the **spender, token, amount** and **deadline**. A "Permit" or "Permit2" signature can hand over token access without any gas.
- **Blind signing:** when a hardware wallet can't decode a transaction, it shows only a hash. Keep blind signing **off** by default; turn it on only for a specific, verified action, then off again.
- **Account delegation (EIP-7702):** since Ethereum's 2025 Pectra upgrade, a normal wallet can sign an authorisation that delegates its behaviour to a smart contract. A malicious delegation can hand control of the whole account to an attacker. Only sign delegations from wallets and apps you trust, for a stated purpose.
- **Rule:** if you can't say in one sentence what a prompt does, don't sign it.

Worked example

A site asks you to "verify your wallet". The wallet shows a typed-data message: `Permit2 · token: USDC · spender: 0x9f...c3 · amount: 115792089... (unlimited) · deadline: 2030`. That isn't verification: it's unlimited USDC access for an unknown spender. Reject it, close the tab, and check your approvals.

Checklist

- ☐ My wallet shows simulated balance changes before signing
- ☐ I read spender, token, amount and deadline on every typed-data prompt
- ☐ Blind signing is off on my hardware wallet

- ☐ I never sign account-delegation requests from unknown sites

Quiz

Q: 1. What does transaction simulation show?

Answer: The expected balance changes before you sign.

Q: 2. Why is a Permit signature risky if you don't read it?

Answer: It can grant token access without a transaction or gas, so it's easy to approve by accident.

Q: 3. When should blind signing be on?

Answer: Only briefly, for a specific verified action, then switched off.

Lesson 1.8 — Privacy and physical security (new)

Objective

Keep your holdings private and your household safe as your on-chain wealth grows.

Explanation

- **Blockchains are public.** Anyone who links an address to you can see its balance and history, forever.
- **Link less:** don't post addresses, screenshots or wins publicly; use separate addresses for public activity (e.g. donations, NFTs) and for savings; be aware that exchange withdrawals link your identity (KYC) to the receiving address.
- **Name services** (e.g. an ENS name) are convenient but make an address easy to find and associate with you.
- **Physical risk ("wrench attacks"):** criminals target people known to hold crypto. Low profile is the first defence. Multisig with keys in separate locations means no single person can be forced to move everything.
- **Home and devices:** hardware wallets and backups out of sight; encrypted devices; don't discuss holdings with people who don't need to know.

Worked example

Alex posts a screenshot of a big win, with the address visible. Anyone can now see Alex's balance and follow every future move, and scammers and thieves now have a target. Better: no screenshots, a vault address that has never been shared, and a 2-of-3 multisig so no single key holder can be coerced into draining it.

Checklist

- ☐ Savings addresses have never been posted or linked publicly
- ☐ No holdings screenshots or wins on social media
- ☐ Vault keys held so no single person can move everything
- ☐ Backups stored discreetly, away from where people would look

Quiz

Q: 1. What does linking an address to your identity reveal?

Answer: Its full balance and history, now and in the future.

Q: 2. How does multisig help against physical threats?

Answer: No single person can be forced to move everything; other keys are elsewhere.

Q: 3. What links your identity to a wallet address?

Answer: For example, KYC exchange withdrawals, name services, or posting the address publicly.

Lesson 1.9 — Smart accounts and account abstraction (expert)

Objective

Understand how smart accounts change wallet security, and use their features without adding new risks.

Explanation

- A normal wallet (an **EOA**, externally owned account) is controlled by one private key. A **smart account** is a contract wallet whose rules you choose.
- **ERC-4337 (account abstraction)**: smart accounts without changing Ethereum itself. Transactions become "user operations" handled by **bundlers**; **paymasters** can pay gas for you (e.g. in USDC).
- **Features**: multisig and **social recovery** (trusted "guardians" can restore access), **spending limits**, **session keys** (Lesson 14.2), **batching** (approve + swap in one step), **passkeys** (log in with your device's biometrics instead of a seed phrase).
- **EIP-7702 (2025)**: lets an ordinary wallet temporarily or permanently delegate to smart-account code: powerful, and a phishing target (Lesson 1.7).
- **New risks**: the account's contract code, module/plugin permissions, guardian collusion, and differences between chains (the same address may not exist on every chain).

The 3-wallet setup

Funds move down only as needed; profits are swept back up. A drained burner costs only the burner.



Vault

Long-term holdings

Almost never signs · no DeFi approvals

Hardware or multisig



Operator

Active DeFi positions

Verified protocols only

Hardware



Burner

Small amounts for anything new

Experiments, mints, airdrops

Hot wallet

 On-Chain Operator Program

Educational content only · Not financial advice · No results are guaranteed

Worked example

Your operating wallet becomes a smart account with: a passkey on your phone plus a hardware key (2-of-2 for large moves); a \$2,000/day limit for small moves with just the passkey; 3 guardians (2-of-3) who can recover access after a 48-hour delay you can cancel. Lose your phone: recover via guardians. Phone stolen: the thief is capped at \$2,000/day and you have 48 hours to cancel a recovery you didn't start.

Checklist

- ☐ I know which of my wallets are EOAs and which are smart accounts
- ☐ Guardians chosen so no two could plausibly collude; recovery has a cancel delay
- ☐ Module/plugin permissions reviewed like approvals
- ☐ Smart account deployed (or deployable) on every chain I send to

Quiz

Q: 1. What does a paymaster do?

Answer: Pays gas on your behalf, e.g. letting you pay fees in USDC.

Q: 2. What is social recovery?

Answer: Trusted guardians can restore access to your account, usually after a delay.

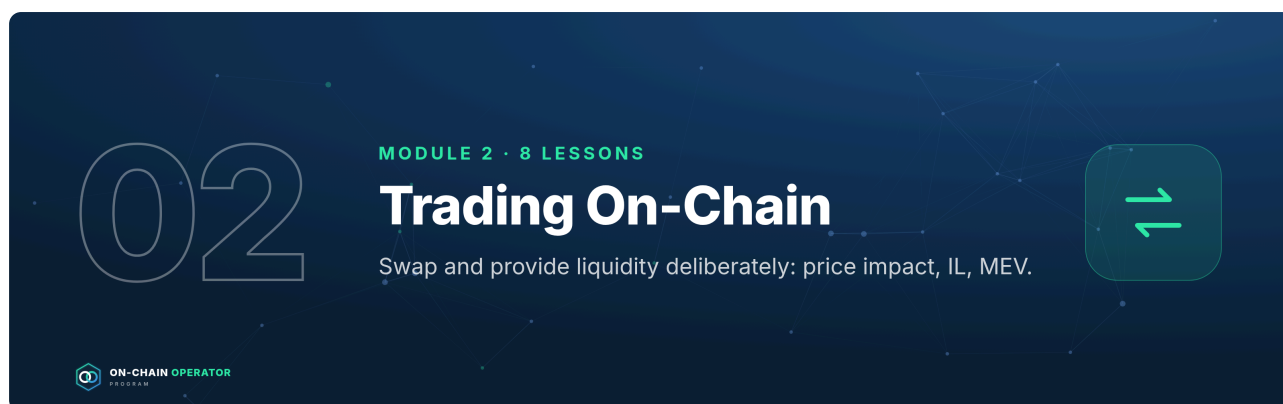
Q: 3. Why check the smart account exists on the destination chain?

Answer: Smart-account addresses may not be deployed on every chain; funds sent there may be hard to access.

Module 1 practical

1. Set up the 3-wallet structure (vault / operator / burner).
2. Send a small test transaction and trace it on an explorer: status, fee, nonce.
3. Approve a small, limited allowance on a reputable protocol, then revoke it.
4. Write your stablecoin list with type, backing, who can redeem, and your depeg exit rule.

Module 2 — Trading On-Chain



Outcome: execute a swap or LP position deliberately, understanding price impact, fees, impermanent loss and MEV. Source: ATLAS "DeFi & On-Chain" ch. 6–9, 35. Educational content only. Not financial advice. Figures illustrative.

Lesson 2.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

A swap on a DEX trades against a pool of tokens priced by a formula. Providing liquidity to that pool earns fees but changes what you hold. This module teaches you to trade and provide liquidity knowing the real cost.

Words you'll need

Term	Meaning
DEX	a decentralised exchange you use from your wallet
Liquidity pool	a pot of two tokens traders swap against
Price impact	how much your trade moves the price
Slippage	the worst price you'll accept
Impermanent loss	how far an LP lags simply holding

Before you start

- ☐ Modules 0–1
- ☐ A little ETH for gas and some USDC on a low-fee network

Your first safe step

Quote a \$20 swap on one DEX and on an aggregator; compare what you'd actually receive after gas. Don't execute yet.

The mastery ladder

Level	You can...
Beginner	Makes small swaps with deliberate slippage and protected routing
Practitioner	Calculates fee APR and IL, and tracks LP vs holding
Master	Uses limit/TWAP/intent orders and runs LPs with written rules

You've mastered this module when...

...you can predict a swap's cost and an LP position's result against holding, before you enter.

Lesson 2.1 — DEXs, aggregators & routing (ch. 6)

Objective

Choose where to swap by comparing the **net amount received**, not the quoted price.

Explanation

- **AMM DEX**: you trade against a liquidity pool priced by a formula (Lesson 2.2).
- **On-chain order book**: orders matched by price and time, fully or partly on-chain.
- **Aggregator**: searches many pools and venues and may split your order across routes.
- **Price impact**: how much *your* trade moves the price. It grows with size relative to pool depth.
- **Slippage tolerance**: the worst price you'll accept before the transaction reverts.
- **Routing**: multi-hop routes (A → B → C) can give a better price but touch more contracts.

What you actually get = quoted output – price impact – pool fees – gas.

Worked example

Selling **20 ETH**:

Route	Output	Gas	Net
Single pool direct	59,100 USDC	\$8	59,092
Aggregator (split across 3 pools)	59,420 USDC	\$20	59,400

The aggregator wins by ~\$308. Now selling **0.5 ETH**: the aggregator quotes \$3 better but costs \$12 more in gas, so the direct pool wins. **Big trades benefit from routing. Small trades mostly lose to gas.**

Checklist

- ☐ Compare net output after gas, not headline price
- ☐ Price impact shown and acceptable for the size
- ☐ Route uses reputable pools and tokens you've verified
- ☐ Slippage set on purpose (see 2.5)

Quiz

Q: 1. Why can an aggregator give a better price?

Answer: It can split an order across pools, so each pool's price moves less.

Q: 2. When does the best quote lose?

Answer: When the extra gas costs more than the price improvement, which is common for small trades.

Q: 3. What drives price impact?

Answer: Trade size relative to pool liquidity.

Lesson 2.2 — AMM Mathematics ($x \cdot y = k$)

Module 2 · Trading On-Chain · Source: ATLAS ch. 7 Template every lesson follows: Objective → Explanation → Worked example → Checklist → Quiz.

Educational content only. Not financial advice. Examples are illustrative.

Objective

By the end of this lesson you can calculate what a swap will actually pay out in a constant-product pool, and explain why bigger trades get worse prices.

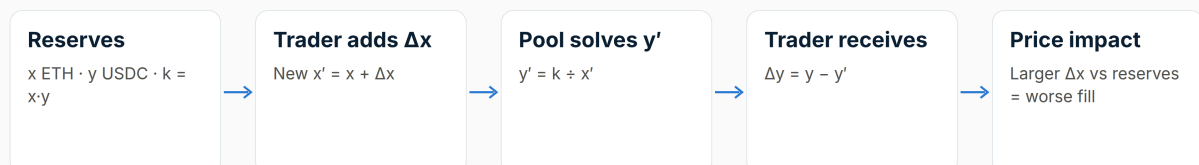
Explanation

A constant-product AMM holds two tokens in a pool. It doesn't use an order book. It keeps one rule: **reserves of token A × reserves of token B = k**, and k must stay constant (ignoring fees) after every trade.

- **Pool price** is the ratio of reserves. 100 ETH and 300,000 USDC means 1 ETH ≈ 3,000 USDC.
- When you add ETH to the pool, the pool must *remove* enough USDC to keep $x \cdot y = k$.
- The more of the pool you move, the further the price moves against you. That gap is **price impact**.
- **Arbitrageurs** then trade the pool back in line with other markets, and that changes what LPs hold. You'll see why that matters in Lesson 2.4.

How a constant-product swap is priced

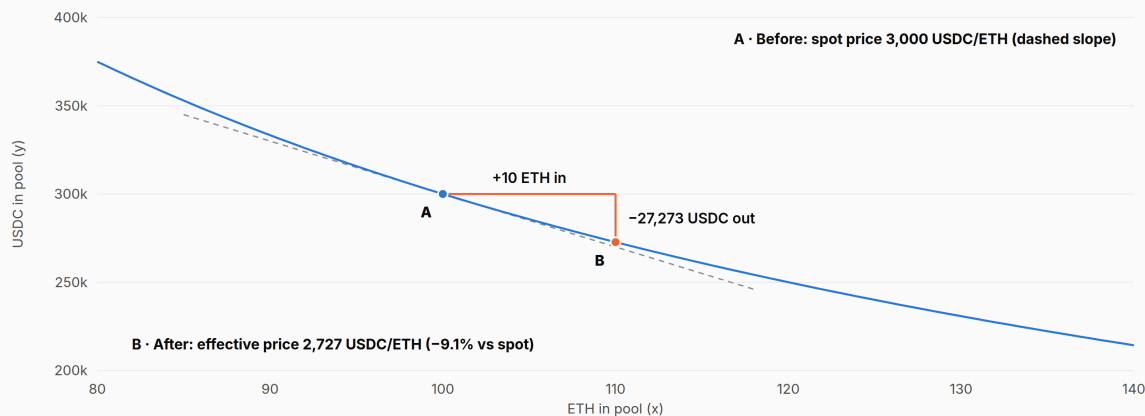
$x \cdot y = k$ stays constant; the bigger the trade relative to reserves, the worse the price



Worked example

The $x \cdot y = k$ curve: why size costs you

Pool of 100 ETH / 300,000 USDC. Swapping in 10 ETH returns $\approx 27,273$ USDC, not 30,000.



On-Chain Operator Program

Educational content only · Not financial advice · No results are guaranteed

Pool: **100 ETH** and **300,000 USDC**, so $k = 30,000,000$. Spot price = 3,000 USDC/ETH.

You swap in **10 ETH** (fees ignored):

1. $x' = 100 + 10 = 110$
2. $y' = 30,000,000 \div 110 \approx 272,727$
3. You receive $300,000 - 272,727 \approx$ **27,273 USDC**
4. Effective price $\approx 27,273 \div 10 =$ **2,727 USDC/ETH**, which is **~9.1% worse** than spot.

Now try **1 ETH**: $x' = 101 \rightarrow y' \approx 297,030 \rightarrow$ you receive $\approx 2,970$ USDC, which is $\sim 1\%$ worse. Ten times the size cost about nine times more in price impact. **Size relative to pool depth is what matters, not the size of the trade in dollars.**

Checklist before any swap

- ☐ Check the price impact the interface shows; if it isn't shown, work it out
- ☐ Compare with an aggregator quote
- ☐ Set slippage tolerance on purpose (too tight: the swap fails; too loose: you get a bad fill or a sandwich attack, see 2.5)
- ☐ Split large trades or use deeper pools
- ☐ Run the before-signing checklist (chain, contract, token, amount, spender)

Quiz

Q: 1. A pool holds 50 ETH / 150,000 USDC. What is k and the spot price?

Answer: $k = 7,500,000$; spot = 3,000 USDC/ETH.

Q: 2. Why does a 10 ETH swap get a worse average price than a 1 ETH swap in the same pool?

Answer: Each unit you add moves the reserve ratio further, so later units in the same trade are priced worse. Price impact grows with trade size relative to reserves.

Q: 3. Who moves the pool price back in line after your trade, and why does that matter to LPs?

Answer: Arbitrageurs do. Their trades rebalance what the LP holds, which is where impermanent loss comes from (Lesson 2.4).

Lesson 2.3 — Providing liquidity (ch. 8)

Objective

Estimate an LP position's fee income from real pool data.

Explanation

When you LP you deposit both tokens and receive a share of the pool (a token, or an NFT for concentrated positions). You earn your share of swap fees. In return, **your token mix changes as traders move the price**: you become the counterparty to every trade.

Fee APR estimate: $\text{fee APR} \approx (\text{daily volume} \times \text{fee tier} \times \text{your pool share} \times 365) \div \text{your deposit}$ For a full-range pool your share = deposit $\div$ pool TVL, so this simplifies to $\text{daily volume} \times \text{fee tier} \times 365 \div \text{TVL}$.

Use a **30-day average volume**, not today's. Volume spikes on volatile days, which are also the days you take the most impermanent loss.

Worked example

Pool TVL **\$10M**, 30-day average daily volume **\$2M**, fee tier **0.3%**:

- Daily pool fees = $2,000,000 \times 0.003 = \mathbf{\$6,000}$
- Your $\$10,000 = 0.1\%$ share $\rightarrow \mathbf{\$6/day} \rightarrow \mathbf{\sim 21.9\% \text{ fee APR}}$ (before impermanent loss and gas)

If volume halves, your fee APR halves. If a lot more TVL joins the pool, your share shrinks.

Checklist

- ☐ Fee APR calculated from 30-day volume
- ☐ I'm OK holding 100% of either token
- ☐ Entry prices recorded for the "vs hold" benchmark
- ☐ Exit rule written (volume drops, trend starts, fees stop covering IL)

Quiz

Q: 1. Pool TVL \$5M, daily volume \$1M, 0.05% fee. Full-range fee APR?

Answer: $1,000,000 \times 0.0005 \times 365 \div 5,000,000 = 3.65\%$.

Q: 2. Why use 30-day volume?

Answer: Single-day spikes overstate fee income, and they happen on the days you take the most impermanent loss.

Q: 3. What happens to your share if TVL doubles?

Answer: It halves, and so do your fees, at the same volume.

Lesson 2.4 — Impermanent loss & true LP P&L (ch. 9)

Objective

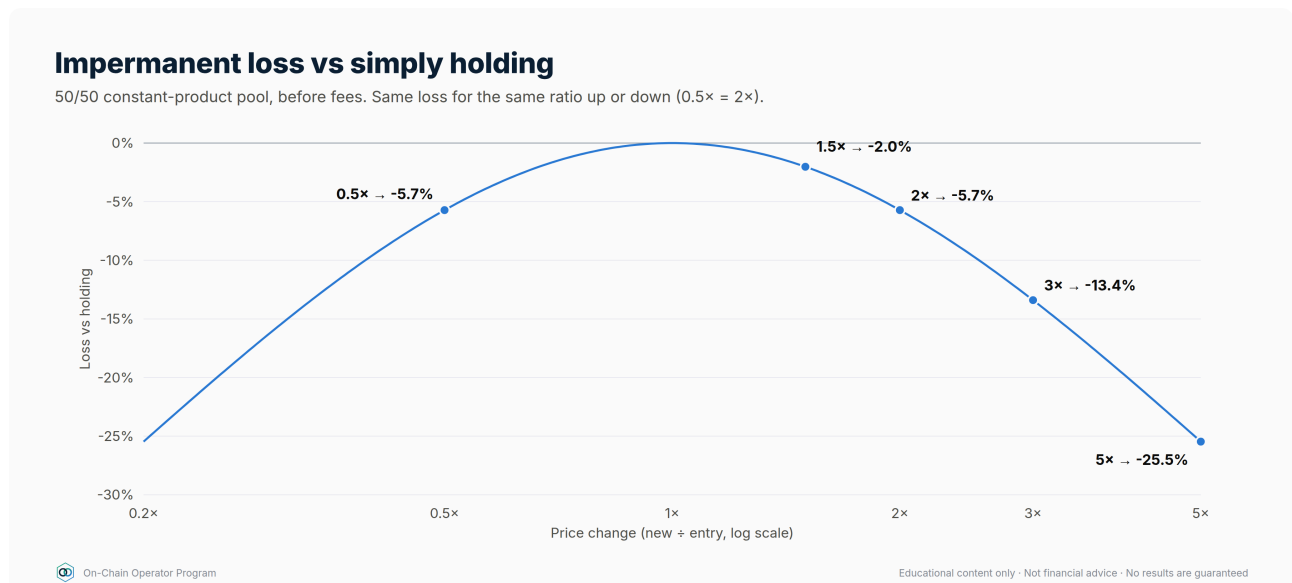
Calculate an LP position's full P&L and judge it against simply holding.

Explanation

Impermanent loss (IL) is how much the LP position lags *simply holding the same starting tokens* when their relative price changes: $IL = 2\sqrt{r} \div (1 + r) - 1$ ($r = \text{new price} \div \text{entry price}$). It's "impermanent" only if price returns. Withdraw after the move and it's permanent.

Full LP P&L = LP value + fees + incentives – gas, compared to **hold value**.

Worked example



Deposit **1 ETH + 3,000 USDC** at ETH = \$3,000 (total \$6,000). ETH rises to **\$4,000**.

- Pool rebalances you to $\sqrt{(3,000 \div 4,000)} = \mathbf{0.866 \text{ ETH}}$ and $\sqrt{(3,000 \times 4,000)} = \mathbf{3,464 \text{ USDC}}$
- LP value = $0.866 \times 4,000 + 3,464 = \mathbf{\$6,928}$
- Hold value = $4,000 + 3,000 = \mathbf{\$7,000} \rightarrow \text{IL} = -\$72 \text{ } (-1.03\%)$
- Fees earned \$150, gas \$20 $\rightarrow$ LP total **\$7,058**

Compared with	Result
Starting \$6,000	+\$1,058 (mostly from ETH going up, not from LPing)
Holding	+\$58, the actual value added by LPing

A master judges the LP on the **+\$58**, not the +\$1,058.

Calculator: `defi_calc.py il --ratio 1.3333` · `defi_calc.py lp-breakeven --ratio 1.5 --days 90 --fee-apr 12`

Checklist

- ☐ P&L tracked against hold, weekly
- ☐ IL, fees, incentives and gas recorded separately
- ☐ Exit if fees consistently fail to cover IL

Quiz

Q: 1. ETH doubles. IL on a 50/50 pool?

Answer: About -5.7% vs holding.

Q: 2. Your LP is up 20% since deposit. Did LPing work?

Answer: Unknown until you compare with holding. The 20% could be all price.

Q: 3. When does IL become permanent?

Answer: When you withdraw while the price ratio is different from entry.

Lesson 2.5 — MEV and protecting your trades (ch. 35)

Objective

Set slippage and routing so your trades aren't easy targets.

Explanation

MEV (maximal extractable value) is profit taken by whoever orders transactions in a block. Some is harmless (arbitrage realigning prices). The kind that hurts you:

- **Sandwich attack:** a bot sees your pending swap, buys just before you (pushing the price up), lets your trade execute at the worse price, then sells just after. Your slippage tolerance is its profit ceiling.

Defences:

- Tight but realistic slippage (e.g. 0.1–0.5% for liquid majors, more only when you understand why)
- MEV-protected / private transaction routing (many wallets and aggregators offer this)
- Split large trades. Avoid thin pools
- Don't broadcast huge swaps in illiquid pairs

Worked example

Swapping **\$20,000**:

Slippage tolerance	Max a sandwich can take (before its own costs)
3%	up to ~\$600

Slippage tolerance	Max a sandwich can take (before its own costs)
0.5%	up to ~\$100
Private / protected route	Transaction isn't visible in the public mempool, so it can't be sandwiched from there

Too tight a tolerance (e.g. 0.05% on a volatile pair) just makes the transaction fail, and failed transactions still cost gas (Lesson 1.2).

Checklist

- ☐ Slippage set for this pair and size, never left at a high default
- ☐ Protected routing on for larger swaps
- ☐ Large orders split or routed through deep liquidity

Quiz

Q: 1. What limits how much a sandwich bot can take?

Answer: Your slippage tolerance (and the pool's depth).

Q: 2. Why not set slippage near zero?

Answer: Normal price movement makes the transaction revert, and you still pay gas.

Q: 3. Name two MEV defences.

Answer: Any two: tight slippage, private/protected routing, splitting trades, using deep pools.

Lesson 2.6 — Advanced execution: limit, TWAP and intent-based orders (new)

Objective

Use order types that give you better prices and protection than a plain swap.

Explanation

- **On-chain limit orders:** you sign an order off-chain ("sell 1 ETH at 3,300 or better"); it fills only if the price is reached. Usually gasless until filled.
- **TWAP orders:** split a large trade into equal slices over time to reduce price impact.
- **Intent-based / RFQ trading:** you state what you want ("100,000 USDC → ETH, at least X"); competing **solvers** or market makers fill it, often with MEV protection and no failed-transaction gas.
- **Trade-offs:** limit orders may never fill; TWAP takes time and the price can move away; intents rely on the protocol's solver design.

Worked example

Buying \$50,000 of ETH in a pool where a single \$50,000 trade has ~~1.5% price impact (\$750)~~. A TWAP of 10 × \$5,000 over an hour has far less impact per slice, as long as the price doesn't trend against you during the hour. An intent-based order may beat both by sourcing liquidity from several venues at once. Compare the quotes.

Checklist

- ☐ I compare a plain swap, a TWAP and an intent quote for large trades
- ☐ Limit orders have an expiry I've chosen
- ☐ I know which order types include MEV protection

Quiz

Q: 1. Why split a large trade into a TWAP?

Answer: Smaller slices cause less price impact each.

Q: 2. What's the risk of a limit order?

Answer: It may never fill.

Q: 3. Who fills an intent-based order?

Answer: Competing solvers or market makers.

Lesson 2.7 — Advanced AMM design and LVR (expert)

Objective

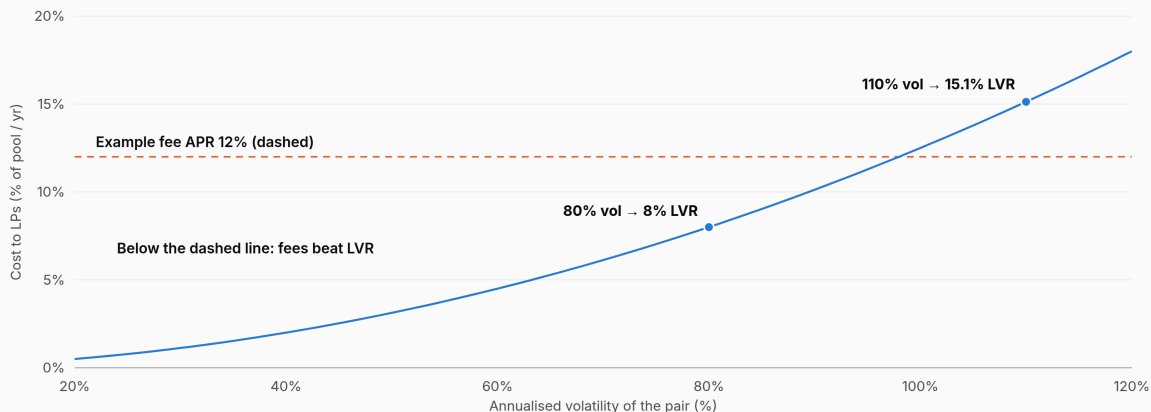
Know the major AMM designs and measure an LP's real cost with loss-versus-rebalancing (LVR).

Explanation

- **Constant product ($x \cdot y = k$):** works for any pair; spreads liquidity across all prices.
- **StableSwap:** for assets that should trade near 1:1 (stablecoins, LST/ETH). Blends constant-sum (flat, low slippage near the peg) with constant-product (safety away from it), tuned by an **amplification** parameter.
- **Weighted pools:** more than two tokens or uneven weights (e.g. 80/20), with invariant $\prod x_i^{w_i} = k$. An 80/20 pool has less impermanent loss on the 80% token.
- **Concentrated liquidity:** positions in price ranges ("ticks"); the pool tracks $\sqrt{\text{price}}$ internally.
- **Hooks and dynamic fees:** newer designs (e.g. Uniswap v4, 2025) let pools run custom code at swap time: dynamic fees, limit orders, oracles. Each hook is extra contract risk.
- **LVR (loss-versus-rebalancing):** pool prices only update when arbitrageurs trade against LPs after prices move elsewhere. LVR measures what LPs lose to that arbitrage, versus a portfolio that rebalances at market prices. For a full-range constant-product pool, **$\text{LVR} \approx \sigma^2/8$ of pool value per year** (σ = annual volatility). **Fees must beat LVR**, not just impermanent loss.

Loss-versus-rebalancing: what arbitrage costs LPs

Full-range $x=y=k$ pool: $LVR \approx \sigma^2/8$ of pool value per year. Fees must beat this line.



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Worked example

An ETH/USDC full-range pool with ETH volatility **80%/yr**: `defi_calc.py lvr --vol 80 --fee-apr 12` → LVR ≈ **8.0%/yr**. With 12% fee APR, LPs keep ≈ **+4%/yr** before gas. If volatility rises to 110%, LVR ≈ 15%/yr, and the same fees now lose money. Higher-volatility pairs need much higher fee tiers.

Checklist

- ☐ I choose pool type by pair: StableSwap for pegged pairs, weighted/CL otherwise
- ☐ I compare fee APR with LVR, not just IL
- ☐ Hook contracts reviewed like any other contract

Quiz

Q: 1. Why do stable pairs use StableSwap curves?

Answer: They give very low slippage near the peg, where these pairs trade.

Q: 2. What does LVR measure?

Answer: What LPs lose to arbitrageurs because pool prices lag the market.

Q: 3. Volatility 60%/yr: approximate LVR for a full-range pool?

Answer: $0.6^2/8 = 4.5\%$ of pool value per year.

Lesson 2.8 — The MEV supply chain (expert)

Objective

Understand who sees, orders and profits from your transactions, and how to get some of that value back.

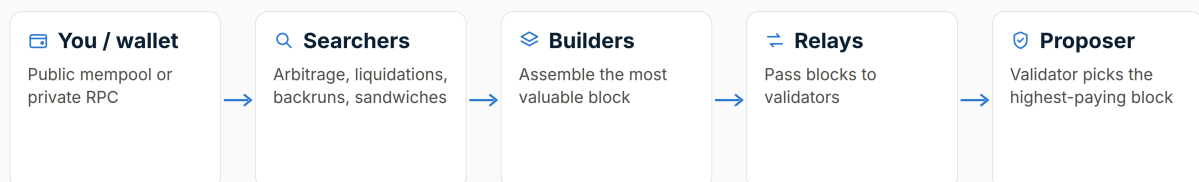
Explanation

The path of an Ethereum transaction today:

1. **You / your wallet** send it, either to the **public mempool** or to a **private RPC** (e.g. an MEV-protection endpoint).
 2. **Searchers** scan for opportunities (arbitrage, liquidations, backruns, sandwiches) and submit **bundles**.
 3. **Builders** assemble the most profitable blocks from transactions and bundles.
 4. **Relays** pass blocks to **proposers** (validators), who pick the highest-paying block (**proposer-builder separation**, via MEV-Boost).
- **Order-flow auctions / MEV rebates:** some private RPCs and wallets let searchers bid to backrun your transaction and **refund you part** of the value.
 - **Intent systems** (Lesson 2.6) move competition to solvers, often with built-in protection.
 - L2s usually have a **single sequencer** ordering transactions, which changes MEV dynamics (e.g. first-come-first-served or priority-fee ordering).

The MEV supply chain

Where your transaction goes, and who can profit from its ordering



Protect yourself: private/protected RPCs, intent systems, tight slippage, and MEV rebates where offered.

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Worked example

A \$100,000 swap sent publicly with 1% slippage could leak up to ~\$1,000 to a sandwich. Sent through a protected RPC with rebates: no sandwich, and if the trade creates a backrun opportunity (e.g. arbitrage between pools), you may receive a share of it back. Same trade, very different outcome, decided by where you send it.

Checklist

- ☐ Large trades go through a protected RPC, an intent system or an aggregator with MEV protection
- ☐ I know whether my wallet's RPC offers rebates
- ☐ I understand my L2's sequencer ordering rules

Quiz

Q: 1. What do builders do?

Answer: Assemble blocks from transactions and searcher bundles to maximise value.

Q: 2. What is an MEV rebate?

Answer: A refund of part of the value searchers extract from backrunning your transaction.

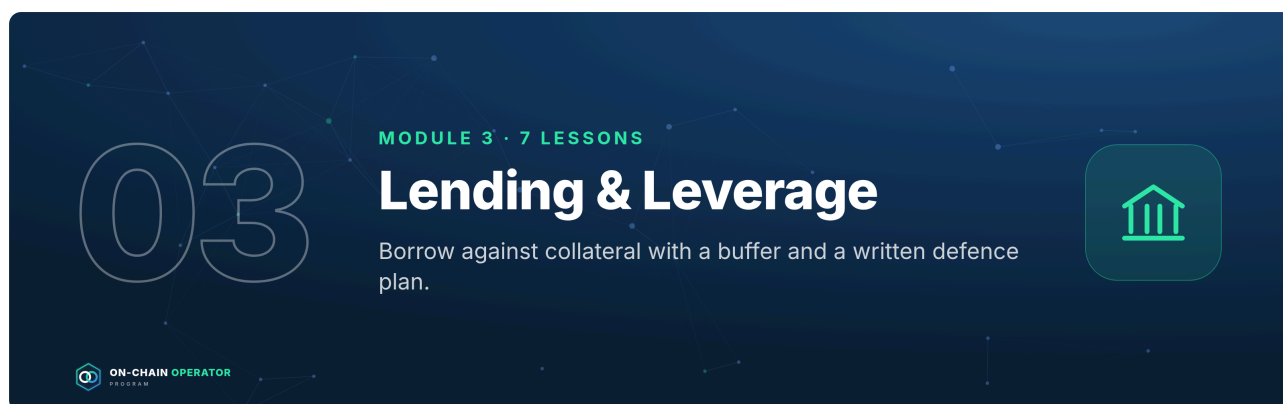
Q: 3. Who orders transactions on most L2s today?

Answer: A sequencer, often a single operator.

Module 2 practical

1. Quote the same swap on a single DEX and an aggregator. Record the net output after gas for a small and a large size.
2. Pick a real pool and calculate its full-range fee APR from 30-day volume and TVL.
3. Model an LP entry with `defi_calc.py`: IL at $\pm 25\%$ and $\pm 50\%$, and the fee APR needed to break even over 90 days.
4. Turn on protected routing in your wallet or aggregator and set a default slippage you can justify.

Module 3 — Lending & Leverage



Outcome: borrow against collateral with a buffer and a written defence plan. Stage 2 · Practitioner.
Source: ATLAS "DeFi & On-Chain" ch. 10–13. Educational content only. Not financial advice. Figures are illustrative.

Lesson 3.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Lending markets let you earn interest by lending, or borrow against what you own without selling it. Borrowing comes with one hard rule: if your collateral falls too far, it's sold automatically, at a penalty.

Words you'll need

Term	Meaning
Collateral	what you lock up to borrow
LTV	$\text{debt} \div \text{collateral value}$
Health factor	your safety margin; below 1 means liquidation
Liquidation	forced sale of collateral to repay debt
Utilisation	how much of a pool is borrowed

Before you start

- ☐ Modules 0–2
- ☐ Comfort using a DEX on a low-fee network

Your first safe step

Supply \$20 of USDC to a blue-chip lending market, read its utilisation, then withdraw it. No borrowing yet.

The mastery ladder

Level	You can...
Beginner	Supplies stablecoins and reads utilisation and rates
Practitioner	Borrows with HF ≥ 2 and a written defence ladder
Master	Runs correlated loops and perps with computed liquidation prices and break-even rates

You've mastered this module when...

...you can open, monitor and unwind a borrow without ever approaching liquidation.

Lesson 3.1 — How lending markets work (ch. 10)

Objective

Explain where lending yields come from and why rates jump when a market is nearly fully borrowed.

Explanation

A DeFi **money market** is a pool: lenders **supply** assets, borrowers **borrow** them against collateral, and a smart contract sets the rates.

- **Utilisation** = borrowed $\div$ supplied. It drives the rates.
- **Interest-rate curve**: most markets use a "kinked" curve. Rates rise gently up to an **optimal utilisation** (e.g. 80%), then steeply above it, to pull in lenders and push out borrowers.
- **Supply APY $\approx$ borrow APY $\times$ utilisation $\times$ (1 – reserve factor)**. The reserve factor is the protocol's cut.
- **Receipt tokens**: when you supply, you get a token representing your deposit plus interest.

Worked example (illustrative curve: 4% at 80% optimal, +60% slope above)

Utilisation	Borrow APY	Supply APY (10% reserve factor)
50%	2.5%	1.1%
70%	3.5%	2.2%
80%	4.0%	2.9%
90%	34.0%	27.5%
95%	49.0%	41.9%

Near 100% utilisation, rates spike **and lenders may not be able to withdraw**, because the money is lent out. A sudden high supply APY is often a warning, not a gift.

Checklist

- ☐ I check utilisation before supplying
- ☐ I know the market's optimal-utilisation kink

☐ I treat a sudden rate spike as a liquidity warning

Quiz

Q: 1. What sets the rates in a money market?

Answer: Mainly utilisation, through the protocol's interest-rate curve.

Q: 2. Why does the curve kink?

Answer: To make borrowing expensive and supplying attractive when liquidity runs low, restoring withdrawal capacity.

Q: 3. Utilisation is 99%. Risk for lenders?

Answer: Withdrawals may be blocked until borrowers repay or new supply arrives.

Lesson 3.2 — LTV, liquidation threshold & health factor (ch. 11)

Objective

Calculate your LTV, health factor and liquidation price before you borrow.

Explanation

- **LTV (loan-to-value)** = $\text{debt} \div \text{collateral value}$.
- **Max LTV:** the most you can borrow against an asset.
- **Liquidation threshold (LT):** the LTV at which your position can be liquidated. It's higher than max LTV.
- **Health factor (HF)** = $(\text{collateral value} \times \text{LT}) \div \text{debt}$. **Below 1.0 = liquidatable.**
- **Liquidation price** = $\text{debt} \div (\text{collateral quantity} \times \text{LT})$.

The liquidation cascade

Why buffers matter: one leveraged position can feed a market-wide loop



Worked example

10 ETH at \$3,000 (\$30,000), LT 0.80, borrow \$12,000:

- LTV **40%** · HF **2.0** · liquidation at **\$1,500** (−50%)
- `defi_calc.py health --qty 10 --price 3000 --lt 0.8 --debt 12000`
- Max debt for HF 2.0 = \$12,000; for HF 1.5 = \$16,000.

Checklist

- ☐ HF and liquidation price calculated before borrowing
- ☐ HF ≥ 2 for volatile collateral
- ☐ Alerts set (Module 14.1)

Quiz

Q: 1. Collateral \$50,000, LT 0.8, debt \$20,000. HF?

Answer: 2.0.

Q: 2. Why keep HF well above 1?

Answer: Collateral prices can fall fast; the buffer is your time to react.

Q: 3. Max LTV vs liquidation threshold?

Answer: Max LTV caps new borrowing; the liquidation threshold is where liquidation becomes possible.

Lesson 3.3 — Liquidations and cascades (ch. 12)

Objective

Know exactly what a liquidation costs you, and how to avoid ever reaching one.

Explanation

When HF drops below 1, a **liquidator** (usually a bot) repays part of your debt (up to the **close factor**, often 50%) and takes that value of your collateral **plus a bonus** (the liquidation penalty, often 5–10%). Liquidations sell collateral into a falling market, which can push prices lower and trigger more liquidations: a **cascade**.

Worked example

Your position from 3.2. ETH falls to **\$1,490**, HF just under 1:

- A liquidator repays **\$6,000** (50% close factor) and seizes $\$6,000 \times 1.05 = \text{\$6,300}$ of ETH $\approx$ **4.228 ETH**.
- The **\$300** bonus is a pure loss to you, on top of having sold ETH near the low.

Defence ladder: HF 2.0 → alert; HF 1.7 → repay from reserve or add collateral; HF 1.5 → sell part of the collateral yourself. Never wait for 1.1.

Checklist

- ☐ Defence ladder written with exact HF levels
- ☐ Reserve set aside to repay (Module 12.4)

□ Oracle for my collateral known (Module 5.3)

Quiz

Q: 1. What does a liquidator receive?

Answer: Collateral equal to the debt repaid plus a bonus (penalty).

Q: 2. Why is self-deleveraging better than being liquidated?

Answer: You avoid the penalty and choose the size and timing.

Q: 3. How do cascades form?

Answer: Liquidations sell collateral, pushing prices down, which pushes more positions below HF 1.

Lesson 3.4 — Borrowing strategies & looping (ch. 13)

Objective

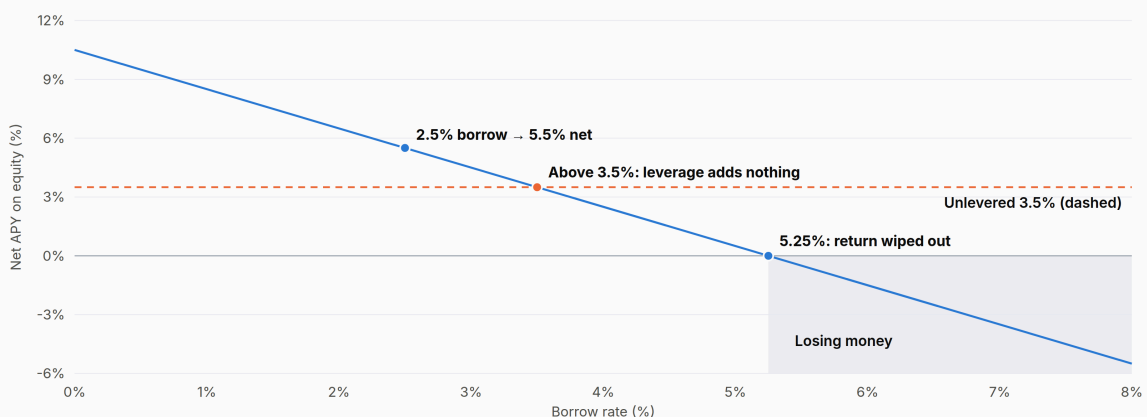
Use borrowing deliberately, and know when looping adds return and when it just adds risk.

Explanation

- **Liquidity without selling:** borrow stablecoins against assets you want to keep (Module 12.3 turns this into a credit policy).
- **Looping:** deposit → borrow → re-deposit → repeat, to multiply exposure. After n loops at LTV L :
$$\text{leverage} = \frac{(1 - L^{n+1})}{(1 - L)}$$
- **Net APY on equity** = collateral yield $\times$ leverage – borrow rate $\times$ (leverage – 1).
- **Correlated loops** (an LST against its underlying, or stablecoin against stablecoin) are far safer than volatile-against-stable loops.

Leveraged loop: the spread is everything

3× loop on 3.5% collateral yield. Net APY on equity = $3.5\% \times 3 - \text{borrow rate} \times 2$.



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Worked example

LTV 0.7, 3 loops, 3.5% collateral yield, 2.5% borrow: `defi_calc.py loop --ltv 0.7 --loops 3 --collateral-apy 3.5 --borrow-apy 2.5` → **2.53×** leverage, **5.03%** net on equity. The return is wiped

out if borrow rises to **5.78%**. Section 3.1 showed borrow rates can jump from 4% to 34% in hours. That's the risk.

Checklist

- ☐ Repayment source named before borrowing
- ☐ Loops only on correlated pairs, below max leverage
- ☐ Break-even borrow rate known and alerted

Quiz

Q: 1. Max leverage at LTV 0.75?

Answer: 4×

Q: 2. What single number decides whether a loop is worth it?

Answer: The spread between collateral yield and borrow rate.

Q: 3. Why prefer correlated loops?

Answer: Collateral and debt move together, so price moves barely change HF.

Lesson 3.5 — Perpetual futures and margin on-chain (*new*)

Objective

Understand perp positions well enough to use them for hedging (Module 11) and carry (10.3), and to see why high leverage fails.

Explanation

- A **perpetual future (perp)** tracks an asset's price with no expiry. You post **margin** and choose **leverage**.
- **Isolated margin:** only the margin in that position is at risk. **Cross margin:** your whole account backs every position (one bad trade can drain the rest).
- **Mark price** (used for liquidation) vs **index price** (spot reference). **Funding** is paid between longs and shorts, usually every hour or 8 hours.
- **Liquidation** happens when your margin falls to the **maintenance margin**. Roughly, a long is liquidated after a fall of $1 \div \text{leverage} - \text{maintenance margin}$.

Worked example

Long ETH from \$3,000 at **5×**, 0.5% maintenance margin: `defi_calc.py perp --entry 3000 --leverage 5 --mmr 0.5` → liquidation ≈ **\$2,415 (−19.5%)**. The same short at **20×** is liquidated at ≈ **\$3,135 (+4.5%)**, a move ETH can make in an hour. Leverage doesn't just magnify gains: it shrinks how wrong you're allowed to be.

Checklist

- ☐ Isolated margin by default

- ☐ Liquidation price computed before opening
- ☐ Leverage $\leq 2-3\times$ for anything but a hedge I fully understand
- ☐ Funding cost checked for how long I'll hold

Quiz

Q: 1. Isolated vs cross margin?

Answer: Isolated risks only that position's margin; cross puts the whole account behind every position.

Q: 2. Roughly how far can a $10\times$ long fall before liquidation (0.5% maintenance)?

Answer: About 9.5%.

Q: 3. Which price triggers liquidation?

Answer: The mark price.

Lesson 3.6 — Lending design deep dive: e-mode, caps, auctions, soft liquidation, bad debt (expert)

Objective

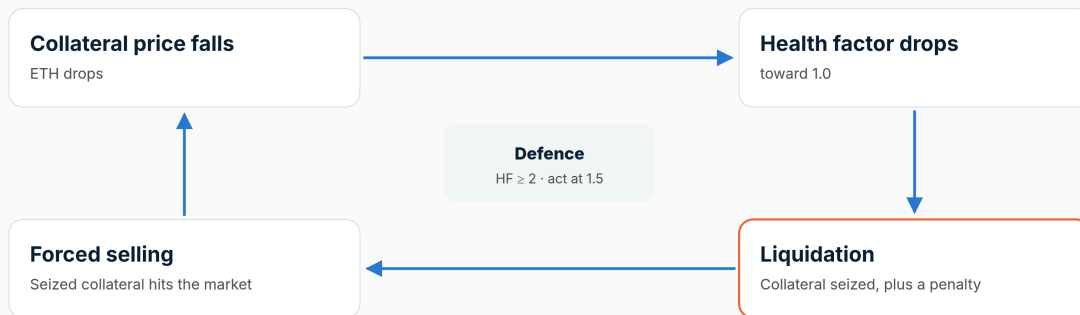
Read a lending protocol's risk parameters like a risk manager.

Explanation

- **Efficiency mode (e-mode):** higher LTVs for **correlated** assets (e.g. an LST against ETH, or stablecoin against stablecoin). Great for correlated loops; dangerous if the correlation breaks (a depeg).
- **Isolation mode / isolated markets:** newer or riskier collateral can only back limited borrowing, so its failure can't infect the whole pool.
- **Supply and borrow caps:** limits on how much of an asset can be deposited or borrowed, protecting against manipulation and illiquid collateral.
- **Liquidation mechanisms:** a **fixed bonus** (liquidators get e.g. 5% extra) vs **Dutch auctions** (the price falls until someone buys, a better price discovery in calm markets but vulnerable if keepers fail) vs **soft liquidation** (collateral is converted gradually across price bands as the price falls, and can convert back if the price recovers; losses come from the conversions rather than one penalty).
- **Bad debt:** when collateral is worth less than the debt it backs. Who pays: a reserve or insurance fund, a staking "safety module", or **lenders pro rata** (socialised).
- **Risk curators / risk managers** set these parameters; governance approves them.

The liquidation cascade

Why buffers matter: one leveraged position can feed a market-wide loop



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Worked example

An LST loop in e-mode at 90% LTV looks safe because LST and ETH move together. If the LST trades at a 6% discount on the oracle during a panic, the "correlated" position can be liquidated even though ETH's price didn't move. Check what oracle prices the LST (market price vs exchange rate) before trusting e-mode.

Checklist

- ☐ For each market: LTV, liquidation threshold, bonus/mechanism, caps, oracle
- ☐ I know who absorbs bad debt in each protocol I lend to
- ☐ E-mode positions have a depeg scenario in my stress test

Quiz

Q: 1. What is e-mode for?

Answer: Higher LTVs on correlated assets.

Q: 2. How does soft liquidation differ from a fixed-bonus liquidation?

Answer: Collateral is converted gradually across price bands (and can convert back) instead of being seized in one go with a penalty.

Q: 3. Who can end up paying for bad debt?

Answer: Protocol reserves, a safety module, or lenders pro rata.

Lesson 3.7 — CDP stablecoins: minting your own dollars (expert)

Objective

Mint stablecoins against your own collateral (as your own bank would), and manage the position safely.

Explanation

- A **CDP** (collateralised debt position) lets you lock collateral and **mint** a stablecoin against it. You owe the stablecoin back, plus a **stability fee** (interest).
- **Minimum collateral ratio** (e.g. 150%): below it, the position is liquidated.
- The stablecoin holds its peg through over-collateralisation, liquidations, interest rates and often a **peg stability module (PSM)** that swaps it 1:1 (minus a small fee) with other stablecoins.
- Compared with borrowing from a lending pool: you create new money rather than borrow someone's deposit; rates are set by governance, not utilisation.
- **Uses:** liquidity without selling (Module 12.3), funding a liquidity ladder, or a yield-covered credit line (strategy #25).

Worked example

10 ETH at \$3,000 (\$30,000) with a 150% minimum ratio: you could mint up to \$20,000. Mint **\$10,000** instead: `defi_calc.py cdp --qty 10 --price 3000 --mint 10000 --fee 6` → ratio **300%**, liquidation at **\$1,500 (-50%)**, stability fee **\$600/yr**. Operators treat the maximum as a cliff, not a target.

Checklist

- ☐ Collateral ratio target (e.g. $\geq 250\%$) and action levels written
- ☐ Stability fee vs alternatives (lending-pool borrow rates) compared
- ☐ Peg mechanism (PSM, rates) understood, and my exit if the stablecoin depegs

Quiz

Q: 1. What do you owe on a CDP?

Answer: The minted stablecoin plus the accrued stability fee.

Q: 2. Collateral \$40,000, 150% minimum ratio. Maximum mint?

Answer: About \$26,667.

Q: 3. What does a PSM do?

Answer: Swaps the stablecoin 1:1 (minus a fee) with other stablecoins, supporting the peg.

Module 3 practical

1. Pick a real lending market: record utilisation, supply and borrow APY, and the kink.
2. Model a borrow with `defi_calc.py health`; write your defence ladder.
3. Model one loop with `defi_calc.py loop` and write its break-even borrow rate.

Module 4 — Yield

04

MODULE 4 · 8 LESSONS

Yield

Split any APY into organic vs subsidised, and name the risk being paid for.

Outcome: split any APY into organic vs subsidised, and name the risk being paid for. Stage 2 · Practitioner. Source: ATLAS "DeFi & On-Chain" ch. 14–18, 37. Educational content only. Not financial advice. Figures are illustrative.

Where DeFi returns come from

Every strategy is a mix of these five. Price going up is not a strategy.

1

Service fees

Swap fees to LPs

2

Interest

Lending to borrowers

3

Security rewards

Staking

4

Structural carry

Funding, basis

5

Incentives

Emissions, points (only real once sold)

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Lesson 4.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Every yield is payment for a risk. Staking pays you to secure a network, lending pays for borrower demand, farms often pay in newly printed tokens. This module teaches you to see what each yield really pays for.

Words you'll need

Term	Meaning
APY	yearly return including compounding
Emissions	new reward tokens printed by a protocol
Staking	locking tokens to help secure a network

Term	Meaning
LST	a token representing staked assets
RWA	a token representing a real-world asset

Before you start

- ☐ Modules 0–3

Your first safe step

Take any advertised APY and split it into base yield vs incentives using the protocol's own dashboard.

The mastery ladder

Level	You can...
Beginner	Tells base yield from emissions
Practitioner	Compares vaults vs DIY and prices points/airdrops as bets
Master	Builds yield from durable sources: lending, staking, savings rates, tokenized treasuries

You've mastered this module when...

...you can name the source and the risk behind any yield in one sentence.

Lesson 4.1 — Yield farming: base yield vs emissions (*ch. 14*)

Objective

Split a farm's APY into what's earned from real activity and what's paid in newly minted tokens.

Explanation

- **Base yield:** trading fees or borrower interest. Paid by real users.
- **Emissions / incentives:** new tokens the protocol prints to attract deposits. The reward token often falls while emissions continue.
- **Mercenary liquidity:** capital that leaves when incentives end, often taking the base yield with it.

Worked example

A farm shows **60% APY**: 5% fees + 55% in reward tokens. If the reward token falls 50% on average while you harvest, the emissions are worth ~27.5%, so the real return is ~**32.5%**, before impermanent loss and gas. Remove emissions entirely and you're left with 5%. **Would you take the risk for 5%?** That's the real question.

Checklist

- ☐ APY split into base vs incentives

- ☐ Reward token valued at a realistic, falling price
- ☐ Harvest-and-sell schedule set

Quiz

Q: 1. What's the test for a farm?

Answer: Does the base yield alone justify the risk?

Q: 2. Why do emissions APYs fall?

Answer: More capital shares the rewards, and the reward token's price often declines.

Q: 3. What is mercenary liquidity?

Answer: Capital that moves on as soon as incentives end.

Lesson 4.2 — Native staking (ch. 15)

Objective

Understand what staking pays for, and its lock-up and validator risks.

Explanation

Proof-of-stake networks pay **stakers** to secure the chain. You either run a validator or **delegate** to one. Rewards come from issuance and fees. Risks: **slashing** (penalties for validator misbehaviour on some networks), **unbonding periods** (waiting time to withdraw) and validator downtime.

Worked example

Staking yields a few percent a year in the staked asset (e.g. ~3%, illustrative). Your dollar return is still dominated by the asset's price: +3% staking on an asset that falls 30% is still a large loss. **Stake what you'd hold anyway.**

Checklist

- ☐ Validator/provider chosen on uptime, fee, concentration and record
- ☐ Unbonding period known
- ☐ Only "hold-anyway" capital staked

Quiz

Q: 1. What do staking rewards pay for?

Answer: Securing the network.

Q: 2. What is unbonding?

Answer: A waiting period before unstaked assets become transferable.

Q: 3. Does staking protect against price falls?

Answer: No. It adds a small yield; price moves dominate.

Lesson 4.3 — Liquid staking (LSTs) (ch. 16)

Objective

Use liquid staking tokens knowing how they accrue value and how they can depeg.

Explanation

A **liquid staking token (LST)** represents staked assets plus rewards, and can be traded or used as collateral. Value accrues by **rebasing** (your balance grows) or an **exchange rate** that rises (each token is worth more of the underlying). In stress, an LST can trade at a **discount** to what it redeems for.

Worked example

An exchange-rate LST goes from 1.000 to 1.030 of the underlying over a year: **3%**. In a panic it trades at 0.98 on the market. Holders who can wait for redemption (queue) recover 1.03; holders forced to sell take the 2% discount. **Never borrow so close to the limit that an LST discount liquidates you.**

Checklist

- ☐ Accrual method known (rebase vs exchange rate)
- ☐ Redemption path and queue length known
- ☐ Discount alert set if used as collateral

Quiz

Q: 1. Two ways LSTs pass on rewards?

Answer: Rebasing balances, or a rising exchange rate.

Q: 2. What is an LST discount?

Answer: Its market price falling below its redemption value.

Q: 3. Who is hurt most by a discount?

Answer: Forced sellers and leveraged holders who get liquidated.

Lesson 4.4 — Restaking & shared security (ch. 17)

Objective

Map the extra risk layers restaking adds before accepting its extra yield.

Explanation

Restaking reuses staked assets to secure additional services, for extra rewards and often points. Each service adds its own **slashing conditions**, and **operators** (who run the services) add operational risk. One collateral base securing many services creates **correlated failure**. Full strategy treatment: Lesson 10.6.

Worked example

Stack for one position: ETH → staked → LST → liquid restaking token → restaking protocol → 4 services. That's **6 layers**, each able to fail. The extra yield must pay for all of them. Plan with points valued at zero.

Checklist

- ☐ Every layer of the stack written down
- ☐ Slashing conditions of each service read
- ☐ Sized in the speculative bucket

Quiz

Q: 1. What does restaking add?

Answer: Extra rewards in exchange for extra slashing and operator risk.

Q: 2. Why is correlated failure a concern?

Answer: One collateral base secures many services; a failure can hit them all.

Q: 3. How should points be valued when planning?

Answer: At zero.

Lesson 4.5 — Vaults & yield optimisers (ch. 18)

Objective

Decide when a vault's fees are worth paying compared with doing it yourself.

Explanation

A **vault** pools deposits and runs a strategy (often harvesting and reinvesting rewards). You get **shares**. Costs: **performance fee** (a share of profits), **management fee** (a share of assets per year), sometimes **withdrawal fees**. Risks: the vault's code plus **every protocol it uses**.

Worked example

Gross 10%, 20% performance fee, 0.5% management → **7.5%** net. DIY instead, compounding weekly at \$2 gas:

- \$5,000 position: gas = \$104/yr = **2.08%** drag → the vault probably wins.
- \$500,000 position: gas = **0.02%** drag, while the vault's fees cost 2.5 points → DIY probably wins.

Checklist

- ☐ Net APY after all vault fees computed
- ☐ Compared with DIY gas drag at my size
- ☐ Every underlying protocol reviewed

Quiz

Q: 1. Gross 12%, 10% performance fee, 1% management. Net?

Answer: $12 \times 0.9 - 1 = 9.8\%$.

Q: 2. Why do vaults suit small positions?

Answer: They spread gas costs across all depositors.

Q: 3. What risk does a vault add?

Answer: Its own code and strategy, on top of every protocol it deposits into.

Lesson 4.6 — Airdrops & points: opportunity cost (ch. 37)

Objective

Price airdrop and points farming as a bet, including its costs.

Explanation

Protocols reward early users with tokens (**airdrops**), often tracked by **points** first. Eligibility rules, snapshots and **sybil filters** (removing people who split into many wallets) decide who gets what. Nothing is promised. Fake claim sites are one of the most common scams (Lesson 1.6).

Worked example

`defi_calc.py airdrop --probability 0.3 --value 1500 --costs 200` → EV **\$250**. If the same \$10,000 could earn 5% (\$500/yr) risk-light elsewhere, the farm must beat that plus its extra risk, or it's the worse choice.

Checklist

- ☐ Only actions I'd take anyway, or a capped speculative budget
- ☐ Opportunity cost compared
- ☐ Claims only from verified official links

Quiz

Q: 1. Are points a promise of tokens?

Answer: No.

Q: 2. What's a sybil filter?

Answer: Rules that exclude wallets believed to belong to the same person farming many times.

Q: 3. What's the airdrop EV formula?

Answer: $\text{Probability} \times \text{value} - \text{costs}$.

Lesson 4.7 — Stablecoin savings rates and yield-bearing stablecoins (new)

Objective

Tell apart the different ways a stablecoin can pay yield, and the risk behind each.

Explanation

Yield on "dollars" on-chain comes from one of four places:

1. **Lending:** you lend stablecoins to borrowers (Module 3). Risk: the protocol, collateral and oracles.
2. **Savings rates:** some stablecoin protocols pay a set rate to holders who deposit into a savings contract, funded by their own revenue. Risk: the stablecoin's design and governance setting the rate.
3. **Treasury-backed:** the issuer holds short-term government debt and passes on part of the interest (Lesson 4.8). Risk: the issuer, custodian and redemption terms.
4. **Synthetic / basis-backed:** yield comes from a hedged derivatives position (spot long + perp short) earning funding. Risk: funding turns negative, exchange failure, and the design itself (Lesson 10.3).

Same "4–10% on dollars", four completely different risks.

Worked example

\$10,000 in a yield-bearing stablecoin paying 4.5% earns ~**\$450/yr** if the rate holds. Before buying, answer: where does the 4.5% come from (1–4 above)? What happens to it in a bear market when funding goes negative? Can you redeem for \$1, and who can?

Checklist

- ☐ Yield source identified (lending, savings rate, treasuries or basis)
- ☐ Redemption path and who can redeem known
- ☐ Split across different yield sources, not just different tickers

Quiz

Q: 1. Name the four sources of stablecoin yield.

Answer: Lending, protocol savings rates, treasury backing, and synthetic basis/funding positions.

Q: 2. Why can a basis-backed stablecoin's yield fall to zero?

Answer: Funding can turn negative in bear markets.

Q: 3. Two yield-bearing stablecoins both backed by basis trades: diversified?

Answer: Not really. They share the same yield source and failure mode.

Lesson 4.8 — Tokenized treasuries and real-world assets (RWAs) (new)

Objective

Use tokenized real-world assets knowing what you actually own and who you're trusting.

Explanation

- **RWAs** are tokens that represent off-chain assets: short-term government bills (the most common), money-market funds, credit, commodities, property.
- Tokenized treasury products typically pay close to the short-term government rate **minus a management fee**.
- **What you're trusting:** the **issuer** (legal structure), the **custodian** holding the real assets, the **redemption** process, and the token's **transfer rules**.
- **Access limits:** many RWA tokens are only for verified (KYC'd) or accredited/qualified investors, and not available in every country. Check eligibility where you live; don't try to get around restrictions.
- **Composability:** some RWA tokens can be used as collateral in DeFi; others are restricted to allowlisted wallets.

Worked example (illustrative)

If short-term government bills yield 4.0% and the product charges 0.15%, you'd earn about **3.85%** before any on-chain costs. In the own-bank ladder (Module 12.4), this can suit the T1/T2 tiers, but only if redemption times match when you'll need the money.

Checklist

- ☐ I'm eligible in my country (and it's legal to hold)
- ☐ Issuer, custodian and redemption terms read
- ☐ Transfer restrictions and minimums known
- ☐ Sized within caps, like any other issuer risk

Quiz

Q: 1. What does a tokenized treasury product's yield track?

Answer: Short-term government rates, minus the product's fees.

Q: 2. Name two parties you trust with an RWA token.

Answer: Any two: the issuer, the custodian, the redemption agent, the token's administrators.

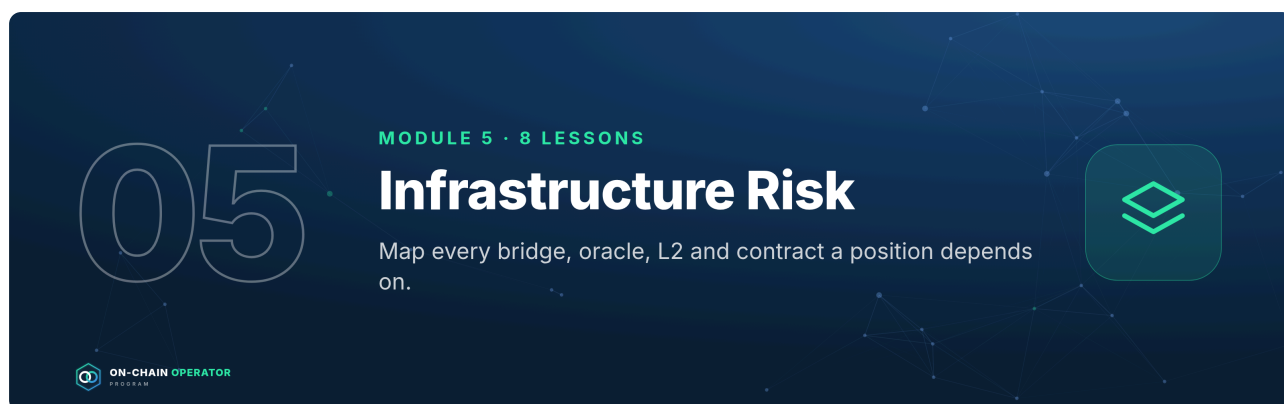
Q: 3. Why check transfer restrictions?

Answer: Some RWA tokens only move between allowlisted wallets, which limits exit and DeFi use.

Module 4 practical

1. Decompose three real APYs into base vs incentive.
2. Compare one vault's net APY with DIY at your position size.
3. Write the risk stack for one staking or restaking position.

Module 5 — Infrastructure Risk



Outcome: map every bridge, oracle, L2 and contract a position depends on. Stage 2 · Practitioner.
Source: ATLAS "DeFi & On-Chain" ch. 19–23. Educational content only. Not financial advice.

Why this module matters: most of DeFi's largest losses came from the plumbing (bridges, oracles, admin keys, contract bugs), not from price moves.

Lesson 5.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Your position is only as safe as the weakest thing it depends on: the bridge that moved your tokens, the oracle that prices your collateral, the admin key that can upgrade the contract. Most big DeFi losses came from this plumbing.

Words you'll need

Term	Meaning
Bridge	moves tokens between chains
Layer 2	a cheaper network built on Ethereum
Oracle	feeds prices into contracts
Proxy	a contract whose logic can be upgraded
Timelock	a delay before changes take effect

Before you start

☐ Modules 0–4

Your first safe step

On a block explorer, open a protocol you use: is it a proxy? Who can upgrade it? Is there a timelock?

The mastery ladder

Level	You can...
Beginner	Knows which chain, bridge and oracle each position uses
Practitioner	Draws a full dependency map and reads audits properly
Master	Operates across EVM and non-EVM chains with capped bridge and wrapper exposure

You've mastered this module when...

...you can draw the complete dependency map of any position and rate each link.

Lesson 5.1 — Bridges and trust assumptions (*ch. 19*)

Objective

Choose bridge routes by their trust model, and size bridge exposure accordingly.

Explanation

Bridges move value between chains:

- **Canonical bridges:** the chain's official route (e.g. a rollup's native bridge). Usually the strongest security, sometimes slower.
- **Lock-and-mint:** assets locked on chain A, a wrapped copy minted on chain B. The wrapped copy is only as good as the lock.
- **Liquidity networks:** market makers pay you out on the other side. Fast; you rely on their contracts and liquidity.
- **Trust model:** who can approve a transfer: a multisig of a few keys, a validator set, a light client, or fraud/validity proofs? Fewer, weaker signers = more risk.

Bridges hold large pools of value, which makes them prime targets.

Worked example

Moving \$50,000 to a Layer 2: use the canonical route for the bulk (even if slower), a reputable fast bridge for a small, time-sensitive amount, and send a test first. Never leave large balances as a **wrapped** asset from a weak bridge.

Checklist

- ☐ Trust model of every bridge I use written down
- ☐ Canonical route preferred for size
- ☐ Wrapped-asset exposure capped

Quiz

Q: 1. What backs a lock-and-mint wrapped token?

Answer: The locked assets on the source chain, and the bridge's security.

Q: 2. Why are bridges frequent hack targets?

Answer: They concentrate large amounts of value behind complex, cross-chain logic.

Q: 3. Fast bridge or canonical for \$50k?

Answer: Canonical for the bulk; fast only for small, urgent amounts.

Lesson 5.2 — Layer 2s, sequencers & withdrawal paths (ch. 20)

Objective

Understand how your L2 settles, who orders your transactions, and how you'd exit.

Explanation

- **Rollups** execute transactions off Ethereum and post data or proofs back to it.
- **Optimistic rollups** assume transactions are valid unless challenged. Native withdrawals to Ethereum wait out a **challenge window** (about 7 days on major optimistic rollups).
- **ZK rollups** post validity proofs, so native withdrawals can be faster.
- **Sequencer:** the operator that orders transactions. Often a single operator today; if it goes down, the chain may pause.
- **Escape hatches:** most rollups let you force a withdrawal via Ethereum if the sequencer censors you. Know if yours does.

Worked example

You hold \$20,000 on an optimistic rollup and need it on Ethereum. Native bridge: safest, ~7 days. Fast bridge: minutes, for a fee, relying on its liquidity. **Plan your liquidity ladder (Module 12.4) around the slow path.**

Checklist

- ☐ Rollup type and withdrawal time known for each L2 I use
- ☐ Sequencer setup and escape hatch known
- ☐ Liquidity plan assumes the slow path

Quiz

Q: 1. Why do optimistic rollup withdrawals take ~7 days?

Answer: The challenge window lets anyone dispute an invalid state before it's final.

Q: 2. What happens if a single sequencer goes down?

Answer: The L2 may pause transactions until it recovers.

Q: 3. What's an escape hatch?

Answer: A way to force a transaction or withdrawal via the base chain if the sequencer censors you.

Lesson 5.3 — Oracles, TWAPs & manipulation (ch. 21)

Objective

Know which price feed secures each position, and how it can fail.

Explanation

Smart contracts can't see market prices; **oracles** bring them in. Lending markets liquidate based on the oracle price, not the price you see on screen.

- **Push oracles:** a network updates prices on a schedule or when price moves past a **deviation threshold**.
- **Pull oracles:** prices are fetched and verified when a transaction needs them.
- **TWAP** (time-weighted average price) smooths manipulation, but **lags** fast moves.
- **Failures:** stale prices in outages, manipulation of thin markets, misconfigured feeds.

Worked example

Your collateral is priced by a 30-minute TWAP. The market drops 10% in 5 minutes. Your screen shows -10%; the oracle shows maybe -2%. You feel liquidated and aren't yet, **but you also can't rely on the lag**: it catches up. Conversely, a thinly traded collateral token can be pushed up and borrowed against by an attacker.

Checklist

- ☐ Oracle type and source known for each collateral
- ☐ Thin-market collateral avoided or capped
- ☐ Buffers sized for oracle catch-up

Quiz

Q: 1. Which price triggers a DeFi liquidation?

Answer: The oracle price used by the protocol.

Q: 2. What's the trade-off of a TWAP?

Answer: Harder to manipulate, but it lags fast moves.

Q: 3. Why is thin-market collateral dangerous for lenders?

Answer: Its price can be manipulated to borrow more than it's really worth.

Lesson 5.4 — Smart contracts: state, proxies, admin keys (ch. 22)

Objective

Read who controls a contract and whether its code can change after you deposit.

Explanation

- **State:** what a contract stores (balances, parameters). **Functions** change it; **events** log what happened.
- **Verified source:** the code published on the explorer matches what's deployed. That's good, not proof of safety.
- **Proxy / upgradeable contracts:** the address you use stays the same, but the logic behind it can be swapped. Whoever controls upgrades controls your funds' rules.
- **Admin keys and roles:** owners, guardians, multisigs; can they pause, upgrade or move funds?
- **Timelocks:** a delay between an approved change and it taking effect. Your window to exit.

Worked example

On a block explorer, the vault you use shows "Proxy", with the upgrade admin a **3-of-5 multisig** behind a **48-hour timelock**. That means no instant rug by one key, and 48 hours to exit if a change you don't like is queued. A 1-of-1 admin with no timelock is a red flag.

Checklist

- ☐ Proxy or not, and who can upgrade
- ☐ Admin roles and multisig threshold known
- ☐ Timelock length known; alert on queued changes for large positions

Quiz

Q: 1. Does verified source code mean safe?

Answer: No. It means the published code matches the deployed code.

Q: 2. Why does a timelock matter to users?

Answer: It gives time to see and exit before a change takes effect.

Q: 3. Red flag in admin setup?

Answer: A single key that can upgrade or move funds instantly.

Lesson 5.5 — Smart-contract risk & what audits don't prove (ch. 23)

Objective

Weigh contract risk honestly, and read an audit for what it actually covers.

Explanation

- **Common failure types:** logic bugs, **reentrancy** (a contract is called back before it updates its state), broken access control, and **economic exploits** (correct code, exploitable incentives or oracles).
- **Audits** review specific code at a specific time. Check the **scope** (which contracts), **date** (before later changes?), **severity of findings** and whether they were **fixed**.
- **Bug bounties** pay hackers to report instead of exploit. A large, active bounty is a good sign.
- **Time and value at risk:** code that has safely held large value for years has survived more real attacks.

Worked example

Protocol X: two audits, but the latest upgrade (last month) isn't in either scope; bounty \$50k; TVL \$400M. The newest code is unaudited and the bounty is small relative to the value. Size down, or wait.

Checklist

- ☐ Audit scope matches deployed code; date after last upgrade
- ☐ Critical/high findings resolved
- ☐ Bug bounty size relative to TVL checked

Quiz

Q: 1. What does an audit not prove?

Answer: That the code (or later changes, or economic design) is safe.

Q: 2. What is reentrancy?

Answer: A contract being called back before it finishes updating its state, letting an attacker repeat actions.

Q: 3. Why check audit dates?

Answer: Code changed after the audit wasn't reviewed.

Lesson 5.6 — Beyond Ethereum: Solana, Bitcoin and other ecosystems (new)

Objective

Operate safely on non-EVM chains and understand how Bitcoin is used in DeFi.

Explanation

- **Solana:** a separate, high-throughput chain with its own wallets (e.g. Phantom, Solflare), its own address format, very low fees (paid in SOL), and priority fees when busy. Tokens follow its own standard (SPL). Your Ethereum address **doesn't** work there. Sending between ecosystems needs a bridge or an exchange.

- **Bitcoin in DeFi:** BTC itself doesn't run DeFi apps, so it's usually used as a **wrapped** token on other chains (backed by a custodian, a group of signers, or a protocol), or on Bitcoin layer 2s and sidechains. **Each wrapper has its own trust model.** Your "BTC" is only as good as whoever holds the real BTC.
- **Other ecosystems** (e.g. Cosmos chains connected by IBC, Move-based chains) each have their own wallets, fees and bridges. Apply Module 5's dependency map to each.
- **Same rules everywhere:** official wallet from the official site, test transaction first, correct network, gas token on hand.

Worked example

You hold 1 BTC and want to lend it on Ethereum. Options: a custodial wrapper (trust one company), a decentralised wrapper (trust a signer set or protocol), or not wrapping at all. Write each option's trust model and cap wrapped BTC as bridge risk (5.1).

Checklist

- ☐ Separate, official wallet for each non-EVM chain I use
- ☐ Gas token held on each chain
- ☐ Wrapped-BTC trust model known and capped

Quiz

Q: 1. Does your Ethereum address work on Solana?

Answer: No. It's a different ecosystem with its own address format and wallets.

Q: 2. What determines a wrapped BTC token's safety?

Answer: Who holds the real BTC and how it can be redeemed: its trust model.

Q: 3. What's Solana's gas token?

Answer: SOL.

Lesson 5.7 — Operating across chains: gas, routes and chain abstraction (*new*)

Objective

Move value between chains cheaply and safely, without getting stranded without gas.

Explanation

- **Gas stranding:** tokens on a chain where you have no gas token can't move. Keep a small gas float on every chain you use.
- **Route choice:** canonical bridge (safest, sometimes slow), fast bridge/liquidity network, exchange deposit-withdraw (often simplest for large amounts, but custodial for a moment), or intent-based cross-chain swaps (solvers deliver on the other chain).
- **Chain abstraction:** wallets and apps increasingly hide chains ("pay gas in USDC", "one balance across chains") using smart accounts, paymasters and solvers. It's convenient, and every layer that

hides complexity is also a dependency.

- **Always:** check the destination network, send a test first, and record the route in your journal.

Worked example

Moving \$30,000 USDC from Arbitrum to Base: compare a canonical route via Ethereum (two transactions, slower, higher gas), a reputable fast bridge (\$5–\$15 fee, minutes), and an exchange (withdraw on Base; check the exchange supports USDC on both networks). Send \$50 first by the chosen route, keep ~\$5 of ETH on Base for gas, then send the rest.

Checklist

- ☐ Gas float on every chain I use
- ☐ Route chosen by trust model and size, not just speed
- ☐ Test amount first; route recorded

Quiz

Q: 1. What is gas stranding?

Answer: Having tokens on a chain with no gas token to move them.

Q: 2. Risk of chain abstraction?

Answer: Each hidden layer (smart accounts, paymasters, solvers) is an extra dependency.

Q: 3. First step on any new route?

Answer: Send a small test amount.

Lesson 5.8 — Reading smart-contract code: enough to verify claims (expert)

Objective

Read verified contract code well enough to check the claims a protocol makes about itself.

Explanation

You don't need to be a developer to spot the things that matter. In verified Solidity code on an explorer, look for:

- **Access control:** `onlyOwner`, `onlyRole(...)`, `onlyGovernance`. Who can call what? Search for functions guarded by them.
- **Dangerous powers:** `mint`, `pause`, `upgradeTo`, `setOracle`, `setFee`, `withdraw` / `sweep` / `rescue` functions that move user funds.
- **Upgradeability:** proxy patterns (`delegatecall`, `implementation`, `upgradeTo`), and who the admin is (5.4).
- **External calls and state order:** calling another contract before updating balances is the classic reentrancy pattern (5.5).

- **Parameters with no limits:** a fee that can be set to 100%, or an oracle address that can be swapped instantly.
- **Events:** what's logged tells you what you can monitor.

Worked example

A token claims "fixed supply". In its code you find:

```
function mint(address to, uint256 amount) external onlyOwner { _mint(to, amount); }
```

The owner can mint unlimited tokens: the claim is false unless ownership is renounced or held by a timelocked governance contract. Check the `owner()` value on the Read tab.

Checklist

- ☐ Searched the code for owner/role-guarded functions
- ☐ Listed every function that can mint, pause, upgrade, change fees/oracles or move funds
- ☐ Checked who holds those roles (Read tab) and behind what timelock

Quiz

Q: 1. What does `onlyOwner` on a mint function mean?

Answer: The owner can mint new tokens whenever they like.

Q: 2. Why check parameter limits?

Answer: An unlimited setter (fee, oracle) can be changed to harm users instantly.

Q: 3. What code pattern suggests reentrancy risk?

Answer: An external call made before the contract updates its own state.

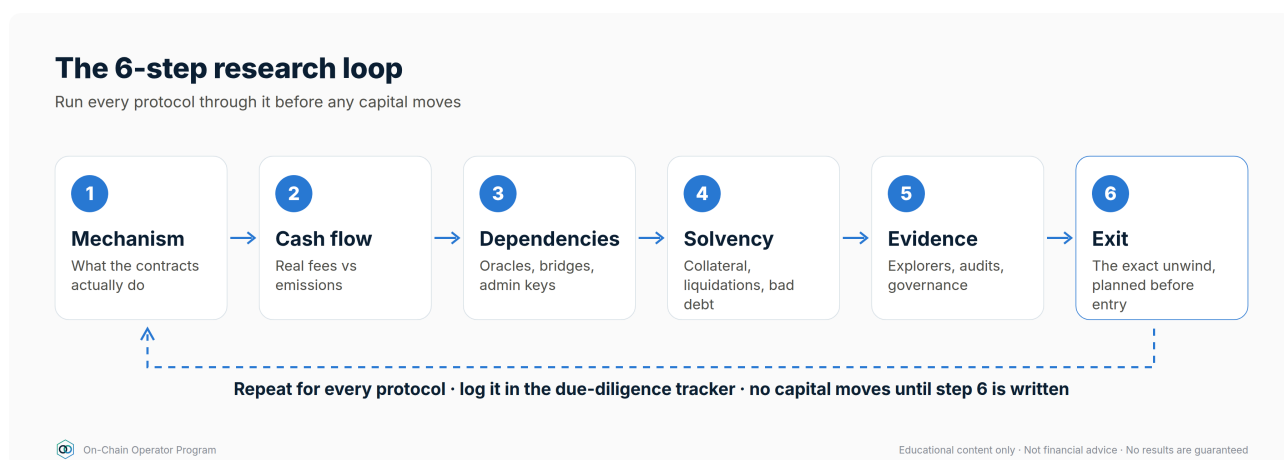
Module 5 practical

Pick one position you hold or plan. Draw its dependency map: chain, bridge, oracle, contracts, admin/timelock, audits. Mark each link Low/Medium/High risk.

Module 6 — Protocol Research



Outcome: complete a full due-diligence file on a real protocol. Stage 3 · Analyst. Source: ATLAS "DeFi & On-Chain" ch. 24–26, 41. Educational content only. Not financial advice.



Lesson 6.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Before trusting a protocol with money, a professional researches it the same way every time: what it does, where its money comes from, what it depends on, whether it's solvent, what the evidence says, and how to get out.

Words you'll need

Term	Meaning
Due diligence	structured research before investing
TVL	total value deposited in a protocol
FDV	token price × maximum supply
Governance	how holders vote on changes

Term	Meaning
Thesis	what must be true for the position to work

Before you start

☐ Modules 0–5

Your first safe step

Pick a protocol you've heard of and write one sentence for each of the six research steps, even if it's "don't know yet".

The mastery ladder

Level	You can...
Beginner	Completes the six-step loop with sources
Practitioner	Reads tokenomics and governance and writes a thesis with invalidation
Master	Spots failure patterns early and writes verdicts others can act on

You've mastered this module when...

...your due-diligence file ends in a verdict with size, conditions and an exit.

Lesson 6.1 — Protocol due diligence (*ch. 24*)

Objective

Run the 6-step research loop on a protocol and reach a written verdict.

Explanation

1. **Mechanism:** what the contracts do; users, assets, incentives.
2. **Cash flow:** real fees and interest vs emissions.
3. **Dependencies:** oracles, bridges, admin keys, stablecoins, other protocols.
4. **Solvency:** collateral quality, liquidation design, bad-debt paths.
5. **Evidence:** explorers, audits, governance, independent data.
6. **Exit:** exact unwind, gas, slippage, approvals to revoke.

Log it in the **DeFi Protocol Due Diligence Tracker** (Notion) and tick a box only when the step is **evidenced**, not assumed.

Worked example (template verdict)

"Protocol Y (lending, Arbitrum). Risk: Medium. Fees are real (borrow demand \$2M/yr), but one collateral uses a thin-market oracle and upgrades have a 24h timelock. Verdict: supply stablecoins only, cap at 10% of the DeFi book, alert on queued upgrades."

Checklist

- ☐ All six steps written with sources
- ☐ Risk rating = worst material risk, not an average
- ☐ Verdict states size, conditions and exit

Quiz

Q: 1. Why is the rating the worst risk, not the average?

Answer: One failure point is enough to lose the position.

Q: 2. Which step prevents getting stuck?

Answer: Step 6: Exit.

Q: 3. When do you tick a tracker box?

Answer: Only when there's evidence for it.

Lesson 6.2 — Tokenomics: supply, unlocks, FDV, value capture (ch. 25)

Objective

Read a token's supply schedule and judge whether protocol success reaches the token.

Explanation

- **Circulating supply:** tokens tradable now. **Max/total supply:** all that will exist.
- **Market cap** = price × circulating. **FDV** (fully diluted valuation) = price × max supply.
- **Unlocks/vesting:** team and investor tokens released on a schedule; new supply that may be sold.
- **Emissions:** ongoing new tokens (rewards).
- **Value capture:** does protocol revenue actually reach token holders (fee share, buybacks), and how is it enforced?

Worked example

Price \$2, circulating 100M, max 1B: market cap **\$200M**, FDV **\$2B**. Next month 50M unlock: circulating supply rises **50%**. Unless demand grows to match, that's heavy selling pressure. A low market cap with a huge FDV is a warning.

Checklist

- ☐ Market cap vs FDV compared
- ☐ Next 12 months of unlocks listed
- ☐ Value-capture mechanism identified (or "none")

Quiz

Q: 1. Price \$5, circulating 20M, max 100M. FDV?

Answer: \$500M.

Q: 2. Why do unlocks matter?

Answer: They add supply that recipients may sell.

Q: 3. What is value capture?

Answer: An enforceable way protocol success benefits the token.

Lesson 6.3 — Governance & DAOs (ch. 26)

Objective

Read how a protocol is controlled and spot governance risk before it hits your position.

Explanation

- **Governance tokens** vote on parameters, upgrades and treasury.
- **Proposal lifecycle:** forum discussion → vote → timelock → execution.
- **Delegation** lets holders assign votes to representatives.
- **Risks:** low turnout, concentrated voting power, borrowed votes, rushed proposals, weak timelocks.

Worked example

A proposal to raise a collateral's LTV passes with 8% turnout, 70% of it from three wallets. Lenders now carry more bad-debt risk, decided by a few voters. **Track governance for markets you're in.** It changes your risk without you doing anything.

Checklist

- ☐ Governance forum/alerts followed for protocols I use
- ☐ Voting concentration checked
- ☐ Timelock length known

Quiz

Q: 1. What's a governance attack?

Answer: Using concentrated or borrowed voting power to pass harmful changes.

Q: 2. Why follow governance as a user?

Answer: Parameter changes alter your risk.

Q: 3. Order of the proposal lifecycle?

Answer: Discussion → vote → timelock → execution.

Lesson 6.4 — The on-chain research workflow (ch. 41)

Objective

Turn research into a written thesis with invalidation and monitoring.

Explanation

Question first → primary sources (docs, contracts, governance) → economic reality (fees vs incentives) → cross-check with independent data → **thesis**: what must be true, what would prove it wrong, what to monitor.

Worked example (thesis template)

"Thesis: stablecoin lending on Protocol Z earns ~5% from real borrow demand. Must be true: utilisation 60–85%, oracle unchanged, no new risky collateral. Invalidation: utilisation > 95% for 48h, a governance change adding thin collateral, or an incident. Monitor: weekly utilisation, governance feed."

Checklist

- ☐ Research question written first
- ☐ Thesis, invalidation and monitoring written
- ☐ Evidence from at least two independent sources

Quiz

Q: 1. Why start with a question?

Answer: To avoid hunting for data that confirms what you already believe.

Q: 2. What makes a thesis useful?

Answer: A clear invalidation you'll act on.

Q: 3. What should monitoring track?

Answer: Only the metrics that would change the thesis.

Lesson 6.5 — Case studies: how DeFi failures happened (new)

Objective

Learn the patterns behind major failures so you recognise them early.

Explanation — seven failures, seven patterns

Case	What broke	Pattern	Warning signs
Black Thursday, March 2020 (Maker)	ETH crashed; congested network and failing keeper bots let some collateral auctions clear at 0 DAI bids (~\$8M lost)	Liquidation mechanism failure under stress	Auctions relying on keepers during congestion; no minimum bid
TerraUSD (UST), May 2022	Algorithmic stablecoin lost its peg and collapsed	Reflexive design: confidence was the collateral	Yield (~20%) far above any real cash flow; peg defended by a sister token
Ronin bridge, March 2022 (~\$600M)	Attackers got 5 of 9 validator keys	Weak bridge trust model	Small signer set; keys concentrated

Case	What broke	Pattern	Warning signs
Mango Markets, Oct 2022 (~\$110M)	Collateral price pumped on thin markets, then borrowed against	Oracle manipulation	Thin-market collateral valued at spot
FTX, Nov 2022	Exchange used customer funds	Custodial (CeFi) counterparty failure	Opaque reserves; "not your keys"
Euler Finance, March 2023 (~\$197M, later largely returned)	Logic flaw in a lending function	Contract bug missed by audits	Complex new function added after earlier audits
Curve pools, July 2023	Reentrancy guard broken by a compiler bug	Toolchain risk	Rare. Shows audited code can fail from below

Figures are approximate, from public reporting.

Worked example

Using the table: a new stablecoin offers 18% "sustainable" yield, backed by its own governance token. That's the UST pattern: reflexive collateral and yield with no matching cash flow. Rating: High. Verdict: don't hold.

Checklist

- ☐ For every position, I've asked which failure pattern it resembles
- ☐ My register (8.2) includes these seven patterns

Quiz

Q: 1. What pattern did Mango Markets show?

Answer: Oracle manipulation of thin-market collateral.

Q: 2. What did FTX show DeFi users?

Answer: Custodial counterparty risk: "not your keys, not your coins".

Q: 3. UST's main warning sign?

Answer: High yield with no matching real cash flow, backed by a reflexive sister token.

Lesson 6.6 — Vote-escrow tokenomics, bribes and governance markets (expert)

Objective

Understand vote-escrow ("ve") systems and bribe markets, a major source of DeFi yield and power.

Explanation

- **Vote-escrow:** you lock a governance token for a period (often up to 4 years) and get **ve-tokens** with voting power that decays as the unlock approaches. Longer lock = more power.

- **Gauges:** ve-holders vote on which pools receive the protocol's token **emissions**. Emissions attract liquidity, so votes are valuable.
- **Bribes / voting incentives:** protocols that want liquidity pay ve-holders to vote for their pool. Voters earn this as yield (plus a share of fees in some designs).
- **Liquid lockers / meta-governance:** protocols that lock on your behalf and give you a tradable token. It's liquid, but it can trade at a discount, and it adds a layer of risk.
- **Risks:** long locks (you can't sell), the token's price over the lock, emissions diluting holders, and governance capture.

Worked example

\$10,000 of a ve-token earning ~15% APR in bribes ≈ **\$1,500/yr**, paid in various tokens. But the underlying is locked for 4 years: if the token falls 60%, the position is worth \$4,000 plus the bribes collected. A liquid-locker version lets you exit, but it traded at a 20% discount during the last sell-off. Price the lock, not just the APR.

Checklist

- ☐ Lock length and decay understood
- ☐ Bribe income valued at a realistic sale price
- ☐ Liquid-locker discount history checked before using one

Quiz

Q: 1. What do ve-holders vote on?

Answer: Which pools receive emissions (gauge weights) and other governance decisions.

Q: 2. Why do protocols pay bribes?

Answer: Votes direct emissions to their pools, attracting liquidity.

Q: 3. Main cost of a 4-year lock?

Answer: You can't sell during the lock, whatever the token's price does.

Lesson 6.7 — Valuing DeFi protocols: fees, revenue, earnings and multiples (expert)

Objective

Compare protocols on their economics, not their narratives.

Explanation

- **Fees:** everything users pay. **Revenue:** the part that goes to the protocol or token holders (the rest goes to LPs/lenders). **Earnings:** revenue minus token incentives paid out.
- **Multiples:** price-to-fees (P/F), price-to-sales (P/S, on revenue), price-to-earnings (P/E). Use **FDV** as well as market cap, since unlocks will arrive (6.2).
- **Quality checks:** is revenue growing without rising incentives? Is it concentrated in one pool or chain? Does it survive a bear market? Does the token actually receive it (value capture)?

- Multiples compare protocols; they don't set a "right" price.

Worked example

FDV **\$2B**, annual revenue **\$100M**, token incentives **\$60M** → earnings **\$40M**. P/S (FDV) = **20×**; P/E (FDV) = **50×**. A rival with the same revenue but no incentives has earnings of \$100M: at the same FDV its P/E is 20×. Same revenue, very different economics.

Checklist

- ☐ Fees, revenue and earnings separated, with sources
- ☐ Multiples on FDV and market cap
- ☐ Revenue quality and value capture assessed

Quiz

Q: 1. Revenue vs earnings?

Answer: Earnings are revenue minus token incentives (and other costs).

Q: 2. Why use FDV in multiples?

Answer: Future unlocks will add supply; FDV shows the fully diluted price.

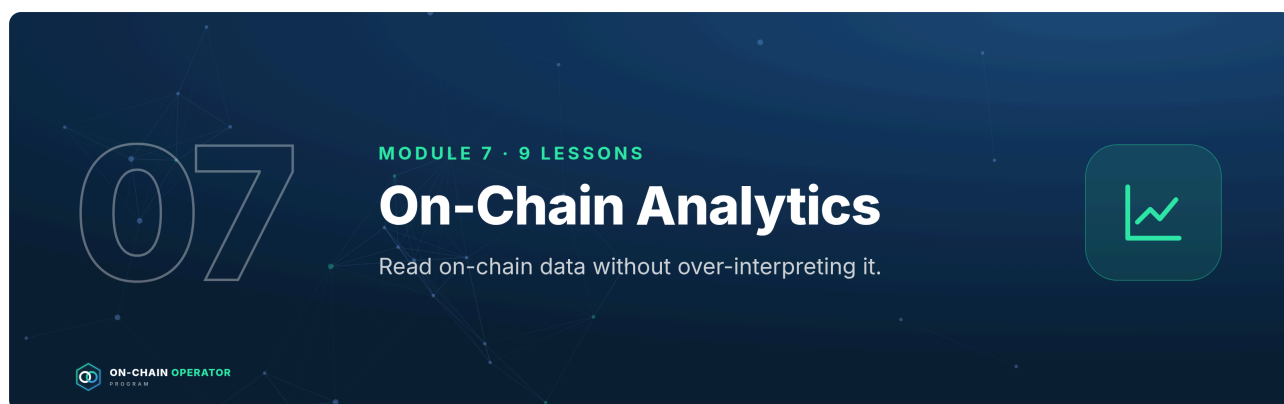
Q: 3. FDV \$600M, earnings \$20M. P/E?

Answer: 30×

Module 6 practical: analyst capstone part 1

Complete a full due-diligence file on one real protocol using lessons 6.1–6.4 and the Notion tracker. (Submitted with Module 7 as the **analyst capstone**.)

Module 7 — On-Chain Analytics



Outcome: read on-chain data without over-interpreting it. Stage 3 · Analyst. Source: ATLAS "DeFi & On-Chain" ch. 27–34. Educational content only. Not financial advice.

The analyst's rule: a transfer is observable; the owner's intention usually isn't. Every metric needs a definition, a methodology, and context.

Lesson 7.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Everything on a public blockchain can be seen, but not everything can be understood. On-chain analytics means reading that data carefully: who moved what, where liquidity sits, how positioned the market is, without jumping to conclusions.

Words you'll need

Term	Meaning
Explorer	a site showing every transaction
Netflow	exchange inflows minus outflows
MVRV	market value ÷ realised value
Open interest	total open futures contracts
Depth	how much a trade moves price

Before you start

☐ Modules 0–6

Your first safe step

Find your own last transaction on an explorer and identify: status, fee, method and token transfers.

The mastery ladder

Level	You can...
Beginner	Reads transactions and addresses on explorers
Practitioner	Uses flows, holder, activity and liquidity metrics with caveats
Master	Combines on-chain, derivatives and liquidity data into a research view without over-interpreting

You've mastered this module when...

...every metric in your research has a definition, a source and a caveat.

Lesson 7.1 — On-chain data foundations (ch. 27)

Objective

Avoid the classic misreads of on-chain data.

Explanation

- **Address ≠ person:** one person can have many addresses; one exchange address serves millions.
- **Transfer ≠ trade:** moving tokens isn't buying or selling.
- **Labels are probabilistic:** providers guess who owns what; they can be wrong.
- **Active addresses** can be inflated by bots or airdrop farming.
- **Data provenance:** know chain coverage, indexing method and delay.

Worked example

"Whale moves \$100M of BTC!" It's an exchange shuffling between its own cold and hot wallets. No trade happened. Check labels, destination and history before reacting.

Checklist

- ☐ I check what an address is before interpreting a move
- ☐ I note each metric's source and definition

Quiz

Q: 1. Does an address equal a person?

Answer: No.

Q: 2. Is a large transfer a sale?

Answer: Not necessarily. It may be internal or custodial.

Q: 3. Why can active-address counts mislead?

Answer: Bots and farming inflate them.

Lesson 7.2 — Block explorer mastery (ch. 28)

Objective

Read any transaction on an explorer without using the app's interface.

Explanation

For a transaction: **status** (success/fail), **block**, **from/to**, **value**, **fee**, **method** called, **token transfers** (events), **internal calls**, and **logs**. For a contract: **read** (state, parameters) and **write** (call functions directly, carefully) tabs; **proxy** info; **verified source**.

Worked example

Your swap on an L2: status Success · method `swap` · token transfers: 100 USDC out, 0.0331 ETH in · fee 0.00002 ETH. The event log confirms the router and pool. You've verified the trade without trusting the website.

Checklist

- ☐ I can find status, fee, method and token transfers
- ☐ I can read a contract's parameters on the Read tab
- ☐ I can check and revoke approvals

Quiz

Q: 1. Where do you see tokens that actually moved?

Answer: Token transfers (event logs).

Q: 2. What does the Read tab show?

Answer: A contract's public state and parameters.

Q: 3. Why verify on the explorer?

Answer: It's the ground truth, independent of any website.

Lesson 7.3 — Exchange flows (ch. 29)

Objective

Interpret exchange inflows and outflows with the right caveats.

Explanation

Inflows to exchanges *may* precede selling (or collateral posting, internal moves). **Outflows** *may* mean self-custody or DeFi use. **Netflow** = inflows – outflows. Exchange reserves on-chain aren't proof of solvency (liabilities aren't visible).

Worked example

Netflow +20,000 ETH in a day. Before concluding "selling pressure": check if it's one exchange reshuffling, whether derivatives open interest rose (collateral), and whether price and volume confirm. One metric is a hypothesis, not a signal.

Checklist

- ☐ Flows cross-checked with price, volume and derivatives
- ☐ Internal exchange moves ruled out

Quiz

Q: 1. What is netflow?

Answer: Inflows minus outflows over a period.

Q: 2. Do exchange reserves prove solvency?

Answer: No. Liabilities aren't visible on-chain.

Q: 3. Is a large inflow always bearish?

Answer: No. It can be collateral or an internal move.

Lesson 7.4 — Whale & entity analysis (ch. 30)

Objective

Use large-holder data without copying wallets blindly.

Explanation

Separate exchanges and contracts from individuals before measuring concentration.

Accumulation/distribution is inferred, not known. **"Smart money" labels** carry survivorship bias.

Copying wallets ignores their hedges, size and private information.

Worked example

A "smart money" wallet buys a token. You don't see its perp short elsewhere; it may be hedged and farming. Copying the visible leg makes you the unhedged one.

Checklist

- ☐ Contracts/exchanges excluded from concentration figures
- ☐ Wallet-following treated as a lead to research, not a signal

Quiz

Q: 1. Why is wallet-copying risky?

Answer: You can't see the wallet's hedges, size or intent.

Q: 2. What is survivorship bias in labels?

Answer: Only wallets that did well get labelled "smart".

Q: 3. First step in measuring holder concentration?

Answer: Remove exchange and contract addresses.

Lesson 7.5 — Holder & supply metrics (ch. 31)

Objective

Read MVRV, SOPR and HODL waves as context, not signals.

Explanation

- **Realised cap:** values each coin at the price it last moved (a cost-basis estimate).
- **MVRV** = market cap ÷ realised cap. High = holders in large aggregate profit.
- **SOPR** (spent output profit ratio): >1 means coins moved today were, on average, sold at a profit.
- **HODL waves:** supply by coin age.
- Definitions differ by provider; strongest on UTXO chains like Bitcoin.

Worked example

Market cap \$1.2T, realised cap \$0.6T → **MVRV 2.0**: the average holder is at roughly 2× their cost basis. That's context for risk management (e.g. how much to de-risk), not a precise top signal.

Checklist

- ☐ Provider methodology read before using a metric
- ☐ Metrics used for context and sizing, not as triggers alone

Quiz

Q: 1. MVRV formula?

Answer: Market cap ÷ realised cap.

Q: 2. SOPR above 1 means?

Answer: Coins moved were, on average, in profit.

Q: 3. Why read the provider's methodology?

Answer: Definitions vary, and interpretations depend on them.

Lesson 7.6 — Network activity (ch. 32)

Objective

Judge a chain's real usage beyond transaction counts.

Explanation

Transaction counts depend on architecture and bots. Better: **fees paid** (real demand for blockspace), **stablecoin activity**, **DEX volume**, **app revenue**, and retention of users over time.

Worked example

Chain A: 5M transactions/day, \$20k fees. Chain B: 500k transactions/day, \$2M fees. B's users pay 100× more for blockspace, a stronger sign of valuable activity.

Checklist

- ☐ Fees and app revenue compared, not just transaction counts
- ☐ Bot activity considered

Quiz

Q: 1. Why are fees a better usage signal than transaction counts?

Answer: Paying fees shows real demand; counts are easy to inflate.

Q: 2. What distorts transaction counts?

Answer: Bots, batching and chain design.

Q: 3. Name one strong activity metric.

Answer: Fees, app revenue, stablecoin activity or user retention.

Lesson 7.7 — DEX & liquidity analytics (ch. 33)

Objective

Measure whether a pool can actually absorb your trade and pay you as an LP.

Explanation

- **TVL** can double count and moves with token prices.
- **Depth**: the price impact for a given trade size. The number that matters for execution.
- **Volume/TVL**: rough capital turnover; higher means more fees per dollar (Lesson 2.3).
- **Liquidity migration**: incentives move liquidity quickly between pools and chains.

Worked example

Pool A: \$50M TVL, \$5M daily volume (0.1 turnover). Pool B: \$10M TVL, \$8M volume (0.8 turnover). B pays LPs far more per dollar, but a \$1M trade will move B's price much more than A's.

Checklist

- ☐ Depth checked at my trade size
- ☐ Volume/TVL used for LP fee estimates

Quiz

Q: 1. Why is depth more useful than TVL for traders?

Answer: It measures price impact at a given size.

Q: 2. High volume/TVL means?

Answer: More fees per dollar of liquidity.

Q: 3. What causes liquidity migration?

Answer: Incentives moving between pools and chains.

Lesson 7.8 — Derivatives on-chain (ch. 34)

Objective

Read funding, open interest and liquidation data as positioning context.

Explanation

- **Perpetual futures (perps):** no expiry; kept near spot by **funding** payments.
- **Positive funding:** longs pay shorts (crowded longs). **Negative:** shorts pay longs.
- **Open interest (OI):** total open contracts. Rising OI with rising price means leverage building.
- **Liquidation data:** bursts of forced closes.

Worked example

Price +8% in a week, OI +40%, funding at its highest in months: leverage is crowding into longs. That's a fragile setup (a drop could cascade) and, for Lesson 10.3, a funding-carry opportunity with a squeeze risk.

Checklist

- ☐ Funding and OI checked before leveraged or carry trades
- ☐ Liquidation clusters noted

Quiz

Q: 1. What does positive funding mean?

Answer: Longs are paying shorts.

Q: 2. Rising price + rising OI + high funding suggests?

Answer: Leveraged longs are crowding in; the setup is fragile.

Q: 3. What keeps a perp near spot?

Answer: Funding payments.

Lesson 7.9 — Querying chain data yourself (expert)

Objective

Answer your own research questions with on-chain queries, instead of relying on someone else's dashboard.

Explanation

- **Query platforms** (e.g. Dune, Flipside, Allium) index blockchain data into SQL tables: raw transactions and logs, **decoded** tables per protocol, and curated "spellbook" tables (e.g. all DEX trades).
- **Indexers / subgraphs** (e.g. The Graph) serve protocol-specific data via APIs.
- **Skills:** SELECT, WHERE, GROUP BY, date truncation, and joining a token price table to convert amounts to USD.
- **Discipline:** check your numbers against the protocol's own dashboard; know table definitions (what counts as "volume"); watch out for wash trading and double counting.

Worked example (SQL; table and column names vary by platform)

```
-- Daily DEX volume on one chain over the last 30 days
SELECT date_trunc('day', block_time) AS day,
       SUM(amount_usd)              AS volume_usd
FROM dex.trades
WHERE blockchain = 'arbitrum'
      AND block_time > now() - interval '30' day
GROUP BY 1
ORDER BY 1;
```

Then cross-check one day against another data source. If they differ a lot, find out why (different pools, wash-trade filters, price sources) before using either number.

Checklist

- ☐ I can run and adapt a basic query
- ☐ I cross-check results against a second source
- ☐ I note table definitions and filters in my research

Quiz

Q: 1. What is a decoded table?

Answer: Contract events and calls translated into readable columns for a specific protocol.

Q: 2. Why cross-check query results?

Answer: Definitions, filters and price sources differ; errors are common.

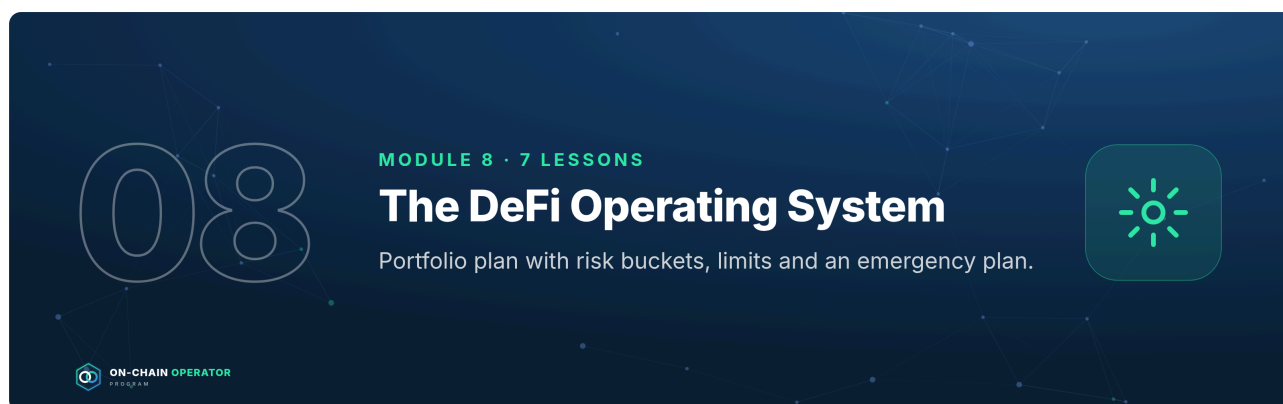
Q: 3. What SQL clause groups results by day?

Answer: GROUP BY on a date-truncated timestamp.

Module 7 practical: analyst capstone part 2

Add an on-chain section to your Module 6 file: holder concentration, flows, activity (fees/revenue), liquidity depth and derivatives positioning for the protocol's token or markets. Submit both parts as the **analyst capstone** (brief and rubric in `07-program-operations.md`).

Module 8 — The DeFi Operating System



Outcome: a written portfolio plan with risk buckets, limits and an emergency plan. Stage 4 · Strategist. Source: ATLAS "DeFi & On-Chain" ch. 38–40, 42. Lesson 8.3 (the 25-strategy library) is in `03-defi-strategy-mastery.md`. Educational content only. Not financial advice. Figures are illustrative.

Lesson 8.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Individual positions don't make a portfolio. This module turns strategies into a system: buckets and caps, a risk register, a strategy library, a routine, honest performance measurement, and rules for your own behaviour.

Words you'll need

Term	Meaning
Bucket	a portion of capital with a job and a cap
Risk register	a scored list of risks and responses
Kill rule	a pre-written exit trigger
Benchmark	the simple alternative you must beat
TWR	return that ignores deposit timing

Before you start

☐ Modules 0–7

Your first safe step

Write your current holdings into four buckets (reserve, core, productive, speculative) with a percentage for each.

The mastery ladder

Level	You can...
Beginner	Has buckets, caps and a journal
Practitioner	Keeps a scored risk register, uses the strategy library, measures TWR vs a benchmark
Master	Runs a disciplined routine with written rules for behaviour, reviewed monthly

You've mastered this module when...

...you have a written portfolio plan and a quarter of honest, benchmarked results.

Lesson 8.1 — DeFi portfolio construction (*ch. 38*)

Objective

Split capital into risk buckets with caps, so no single failure can sink the portfolio.

Explanation

- **Buckets:** liquidity reserve · core (simple, durable) · productive (LP, vaults, carry) · speculative (new, points, experiments).
- **Concentration caps:** per protocol, per chain, per stablecoin issuer, per bridge.
- **Genuine diversification:** two front ends on the same protocol, or two stablecoins with the same backing, are one risk, not two.
- **Rebalancing rules:** thresholds that trigger moving money between buckets.

Worked example — a \$100,000 DeFi book (illustrative)

Bucket	Share	Example holdings	Cap per protocol
Reserve	15%	Stablecoins split across 2 issuers, instant access	n/a
Core	45%	Blue-chip stable lending (2 protocols), ETH staking	20%
Productive	30%	Wide concentrated LP, a vault, a PT	10%
Speculative	≤10%	Points, new protocols	5%

Rebalance when any bucket drifts more than 5 points from target, or after any incident.

Checklist

- ☐ Buckets and targets written
- ☐ Caps per protocol, chain, issuer, bridge written
- ☐ Rebalancing rule written

Quiz

Q: 1. Two stablecoins backed by the same collateral: diversified?

Answer: No. They share one failure point.

Q: 2. Why keep a reserve bucket?

Answer: To defend debt, meet needs and redeploy without forced selling.

Q: 3. When should you rebalance?

Answer: When buckets drift past the threshold, or after an incident.

Lesson 8.2 — The DeFi risk framework (ch. 39)

Objective

Keep a risk register that scores every position's risks and names the response.

Explanation

The eight risk surfaces: **smart-contract** · **economic** · **oracle** · **liquidity** · **governance** · **bridge/chain** · **counterparty** · **user-operation**. Score each for **likelihood (1–5) × impact (1–5)**. Anything ≥ 15 needs a mitigation or an exit; anything ≥ 20 shouldn't be held.

Worked example — one register row

Position	Risk	L	I	Score	Response
USDC lending, Protocol Y	Oracle on one collateral is thin	3	4	12	Cap 10%, alert on governance changes
LST loop	Rate spike above break-even	3	3	9	Alert at 4.5% borrow; unwind rule
Bridged stablecoin on new L2	Bridge exploit	2	5	10	Canonical bridge; cap 5%
New farm	Contract bug (unaudited upgrade)	4	5	20	Don't hold

Checklist

- ☐ Register covers every position and all eight surfaces
- ☐ Scores ≥ 15 have a mitigation or exit
- ☐ Reviewed monthly and after incidents

Quiz

Q: 1. How is a risk scored?

Answer: Likelihood × impact, each 1–5.

Q: 2. What's a user-operation risk?

Answer: Your own mistakes: wrong network, bad signature, lost keys.

Q: 3. Score 20. Action?

Answer: Don't hold it.

Lesson 8.3 — Strategy library

The full strategy library (30 strategies in 7 levels, each with its profit engine, maths, execution, kill rules and failure modes) is **Lesson 8.3: DeFi Strategy Mastery** (`03-defi-strategy-mastery.md`).

Lesson 8.4 — The operating playbook: deploy, monitor, respond, review (ch. 42)

Objective

Run every position through the same four-phase routine.

Explanation

1. **Deploy:** due diligence done · before-signing checklist · test amount · journal row (protocol, chain, contracts, amount, entry, thesis, kill rules, approvals).
2. **Monitor:** daily HF/pegs/ranges/incidents; weekly P&L vs benchmark, harvest, revoke; monthly register review.
3. **Respond:** pre-written actions for each trigger (see Module 11.5 for incidents).
4. **Review:** net return after all costs vs benchmark; simplify anything you can't explain in one sentence.

Worked example — a journal row

Field	Entry
Position	USDC supply, Protocol Y, Arbitrum
Amount / date	\$10,000 · 2026-09-01
Thesis	~5% from real borrow demand
Kill rules	Utilisation > 95% for 48h; oracle or collateral change; any incident
Approvals	USDC → Protocol Y pool, limited to \$10,000
Weekly	Interest earned, utilisation, governance notes

Checklist

- ☐ Journal row for every position
- ☐ Daily / weekly / monthly routine in the calendar
- ☐ Response actions written per trigger

Quiz

Q: 1. The four phases?

Answer: Deploy, monitor, respond, review.

Q: 2. What goes in a journal row?

Answer: Position, amount, date, thesis, kill rules, approvals and ongoing results.

Q: 3. What to do with a strategy you can't explain simply?

Answer: Simplify it or exit.

Lesson 8.5 — Measuring performance honestly (*new*)

Objective

Measure your results so deposits, withdrawals and lucky markets don't fool you.

Explanation

- **Time-weighted return (TWR):** chains the return of each period, ignoring when you added or removed money. It measures your *decisions*.
- **Money-weighted / simple gain:** what your total capital actually did, including the timing of deposits.
- **Benchmark:** compare against what you'd have got doing something simple: holding the same assets, or plain stablecoin lending. Beating zero isn't the test.
- **Attribution:** split the result into sources: price moves, fees, interest, incentives, costs, losses. Only the parts you control show skill.

Worked example

Start \$100,000 → +10% in Q1 (\$110,000) → deposit \$50,000 (\$160,000) → -5% in Q2 → \$152,000.

```
defi_calc.py twr --period 10 --period -5 --start 100000 --end 152000 --net-deposits 50000
```

- TWR: $1.10 \times 0.95 - 1 = 4.50\%$ (your decisions)
- Simple gain on \$150,000 put in: **1.33%** (hurt by the deposit landing before a down quarter) Then compare 4.50% with holding ETH over the same period, and with stablecoin lending.

Checklist

- ☐ TWR calculated quarterly
- ☐ Benchmark chosen and compared
- ☐ Results attributed by source

Quiz

Q: 1. What does TWR remove?

Answer: The effect of when deposits and withdrawals happened.

Q: 2. Why use a benchmark?

Answer: To see whether the strategy beat doing something simple, not just beat zero.

Q: 3. Two quarters, +8% then +2%. TWR?

Answer: $1.08 \times 1.02 - 1 = 10.16\%$.

Lesson 8.6 — Psychology and discipline (*new*)

Objective

Recognise the behaviours that turn good strategies into losses, and build rules that stop them.

Explanation

Bias	What it looks like	Rule that stops it
FOMO	Jumping into a farm because everyone's posting about it	24-hour change-control wait (14.3)
Loss aversion	Holding a broken position hoping to "get back to even"	Kill rules written at entry, then followed
Sunk cost	"I've spent so much gas, I can't exit now"	Judge only the future: would you enter today?
Overconfidence	Raising size after a winning streak	Caps are percentages; size changes only at review
Revenge trading	Adding leverage to win back a loss	No new leverage within 7 days of a loss event
Anchoring	Refusing to sell below a past price	Decisions based on current data and the thesis

The operator's defence is structure: written theses, kill rules, caps, a journal, cooling-off periods and scheduled reviews. Willpower in the moment is unreliable.

Worked example

After a 20% loss on a farm, Jordan wants to put 3× leverage on the next idea "to make it back". The rulebook says: no new leverage within 7 days of a loss event, and size only changes at the monthly review. Jordan journals the urge, waits, and at review decides the idea doesn't meet the thesis standard at all.

Checklist

- ☐ Personal rulebook written (cooling-off, loss rules, sizing rules)
- ☐ Emotions noted in the journal next to decisions
- ☐ Monthly review includes "which rule did I break?"

Quiz

Q: 1. What's the sunk-cost question?

Answer: Would I enter this position today, knowing what I know now?

Q: 2. Why rely on structure rather than willpower?

Answer: Decisions under stress are unreliable; pre-written rules decide for you.

Q: 3. Rule against revenge trading?

Answer: For example, no new leverage within 7 days of a loss event.

Lesson 8.7 — Quantitative risk: volatility, VaR, drawdown, correlation and sizing (*expert*)

Objective

Put numbers on portfolio risk and size positions from them.

Explanation

- **Volatility (σ)**: annualised standard deviation of returns. **Daily $\sigma \approx \text{annual } \sigma \div \sqrt{365}$** (crypto trades every day).
- **Value at risk (VaR)**: a loss level you'd expect to exceed only rarely (e.g. 1 day in 20 at 95%). Parametric 1-day VaR $\approx 1.65 \times \text{daily } \sigma \times \text{position}$. **It assumes normal returns; crypto has fat tails**, so real losses exceed VaR more often. Use it as a floor, plus stress tests (11.4).
- **Max drawdown**: the largest peak-to-trough fall. Ask: could I live through it, and would my policies survive it?
- **Correlation**: in crashes, correlations between crypto assets rise towards 1. Diversifying within crypto helps less when you need it most.
- **Sizing: volatility targeting** (size so each position contributes similar risk); the **Kelly criterion** gives a theoretical maximum bet size from edge and odds, but estimates are noisy, so professionals use a small fraction of it or skip it.

Worked example

\$100,000 of ETH at 70% annual volatility: `defi_calc.py var --position 100000 --vol 70` → daily $\sigma \approx 3.66\%$, 1-day 95% VaR $\approx$ **\$6,046**. Real single-day falls have been far larger (in March 2020, ETH fell by roughly 40% in a day). So: VaR for day-to-day sizing, stress tests for survival.

Checklist

- ☐ Volatility and VaR computed for my largest positions
- ☐ Portfolio survives the worst historical drawdown of what it holds
- ☐ Correlation assumed to rise in stress
- ☐ Sizing rule written (e.g. volatility targeting)

Quiz

Q: 1. Annual volatility 50%. Daily σ ?

Answer: About 2.6% ($50 \div \sqrt{365}$).

Q: 2. Why is VaR a floor, not a ceiling?

Answer: It assumes normal returns; crypto's fat tails make big losses more common.

Q: 3. What happens to correlations in a crash?

Answer: They tend to rise towards 1.

Module 8 practical

Write your portfolio plan: buckets and caps (8.1), risk register (8.2), your chosen strategies from 8.3 with kill rules, and your operating calendar (8.4).

DeFi Strategy Mastery — How Strategies Make (and Lose) Money

On-Chain Operator Program · Strategy module · Built on ATLAS "DeFi & On-Chain" ch. 7–18, 38–42 and the 16-framework Strategy Library, expanded to 30 strategies.

Read this first. This is educational material, not financial advice. No DeFi strategy guarantees profit. Every return in DeFi is payment for taking a risk. This module shows **where each strategy's return comes from, the maths to test it before you enter, how to run it, and exactly how it loses money**, so you only keep positions whose expected return still makes sense after costs and risk. All numbers below are illustrative. Use live rates from protocol dashboards and independent data (e.g. DefiLlama) when you model a real position.

Calculator for every formula here: `python3 .claude/skills/defi-strategies/scripts/defi_calc.py <command> --help`

Part 1 — Where profit actually comes from

There are only five sources of return in DeFi. Every strategy is a mix of them.

Where DeFi returns come from

Every strategy is a mix of these five. Price going up is not a strategy.

1



Service fees

Swap fees to LPs

2



Interest

Lending to borrowers

3



Security rewards

Staking

4



Structural carry

Funding, basis

5



Incentives

Emissions, points (only real once sold)

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#	Source	You are paid for...	Example	Lasts?
1	Service fees	providing liquidity traders need	swap fees to LPs	As long as volume lasts
2	Interest	lending capital borrowers need	stablecoin supply APY	As long as borrow demand lasts
3	Security rewards	staking to secure a network	ETH staking yield	Structural, but changes over time
4	Structural carry	taking the other side of crowded positioning	perp funding, basis	Comes and goes with the market
5	Incentives	bringing capital to a protocol	token emissions, points, airdrops	Temporary, and often dilutive

Price moves (beta) are not a strategy. If a position only made money because ETH went up, it was a directional bet with extra steps. Strategies 1–4 can pay in flat markets. Source 5 is only real once the tokens are sold.

The one equation that matters

```
Net return = base yield (fees + interest + staking)
            + incentives (valued at the price you can actually sell at)
            - impermanent loss / inventory loss
            - borrow cost
            - gas + swap + bridge costs
            - protocol / vault fees
            - losses from risk events
            ± price change on assets held
```

A position is only worth running if **net return beats the simplest alternative** (holding the assets, or plain stablecoin lending) **by enough to pay for the extra risk**. Masters compare against a benchmark. Beginners compare against zero.

APR vs APY

APY assumes compounding. $APY = (1 + APR/n)^n - 1$. 10% APR compounded daily $\approx$ 10.52% APY, **but only if compounding is free**. On-chain, each compound costs gas. Compound only when pending rewards $\geq \sim 10\text{--}20\times$ the gas cost, or use a vault that batches it (and charges a fee for doing so).

Part 2 — Strategy selector by market regime

This uses the same thinking as the Grid Bot module: **pick the market regime first, then the strategy**.

Regime	Strategies that tend to work	Avoid / reduce
Sideways, low–moderate volatility	Concentrated LP around the range · stable-stable LP · lending · grid bots (CEX)	Leverage loops with thin buffers
Uptrend	Hold + liquid staking · delta-neutral funding carry (longs usually pay) · conservative borrow against collateral	Narrow LP ranges (they sell your winners early)
Downtrend	Stablecoin lending · stable-stable LP · cash management	Leveraged loops · volatile/volatile LP · incentive farms paid in falling tokens
High volatility / event risk	Wider LP ranges or none · larger health-factor buffers · smaller size	New positions right before major news (FOMC, CPI), new or unaudited protocols

Part 3 — The strategy playbook

The strategy library

30 strategies in 7 levels

1	Core	Stable lending · Staking & LSTs · Collateral borrowing
2	Liquidity	50/50 LP · Concentrated LP · Stable-stable LP
3	Yield	Vaults · Incentive farming · Airdrops & points
4	Advanced	Leveraged loops · LST loops · Funding carry · LP hedge
5	Treasury & research	Cash management · Accumulation & growth screens
6	Professional	PT/YT fixed rate · Basis · Options · Credit lines · Lending vaults · Hedged restaking
7	Expert	CDP minting · Bribe markets · Perp vaults · Peg arbitrage · Rates positioning

Graduate a level only when the one below is routine. Leverage and carry come last.

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Each strategy card has the same sections: **Profit engine** · **Key maths** · **Execute** · **Monitor** · **Exit/kill rules** · **How it loses** · **Size cap**. Size caps are suggested maximums as a share of the DeFi portfolio. Tighten them to fit your own risk budget.

LEVEL 1 — CORE (start here)

1. Stablecoin lending

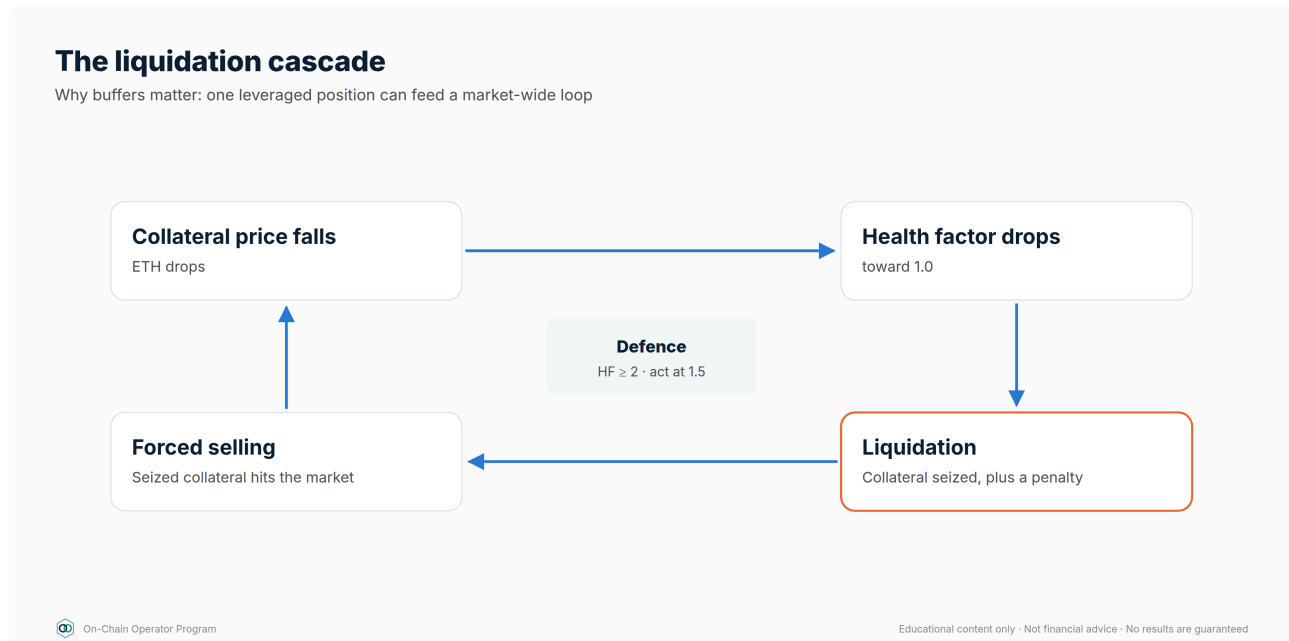
- **Profit engine:** interest paid by borrowers (source 2).
- **Key maths:** $\text{supply APY} \approx \text{borrow APY} \times \text{utilisation} \times (1 - \text{reserve factor})$. Example: $8\% \text{ borrow} \times 80\% \text{ utilisation} \times 0.9 = 5.76\%$.
- **Execute:** pick a stablecoin with a clear peg mechanism (Module 1.5) → pick a lending market that passed due diligence (*defi-due-diligence*) → check current utilisation and withdrawal liquidity → supply → record the receipt token.
- **Monitor:** utilisation (above ~90% = withdrawals may be stuck), stablecoin peg, governance changes, incidents.
- **Exit/kill:** peg below ~0.99 for a sustained period, utilisation pinned at 100%, exploit news, or the rate falls below your benchmark.
- **How it loses:** stablecoin depeg, protocol exploit, bad debt, frozen withdrawals.
- **Size cap:** can be the largest bucket, but **split across ≥2 independent protocols and ≥2 stablecoin issuers**.

2. Native / liquid staking (LSTs)

- **Profit engine:** network security rewards (source 3), minus the provider's fee.
- **Key maths:** $\text{net yield} = \text{staking APR} \times (1 - \text{provider fee})$. Your real return is still dominated by the asset's price, so this only improves on holding if you'd hold the asset anyway.

- **Execute:** choose the provider (validator spread, fee, track record, redemption path) → stake → confirm how the LST earns (rebasing, or an exchange rate that rises).
- **Monitor:** LST/underlying ratio on markets (discount), validator incidents/slashing, withdrawal queue length.
- **Exit/kill:** discount widens beyond your threshold, a slashing event, or the provider's governance or custody changes.
- **How it loses:** asset price falls (main risk), LST depeg, slashing, smart-contract risk.
- **Size cap:** "hold-anyway" capital only.

3. Borrowing against collateral (liquidity without selling)



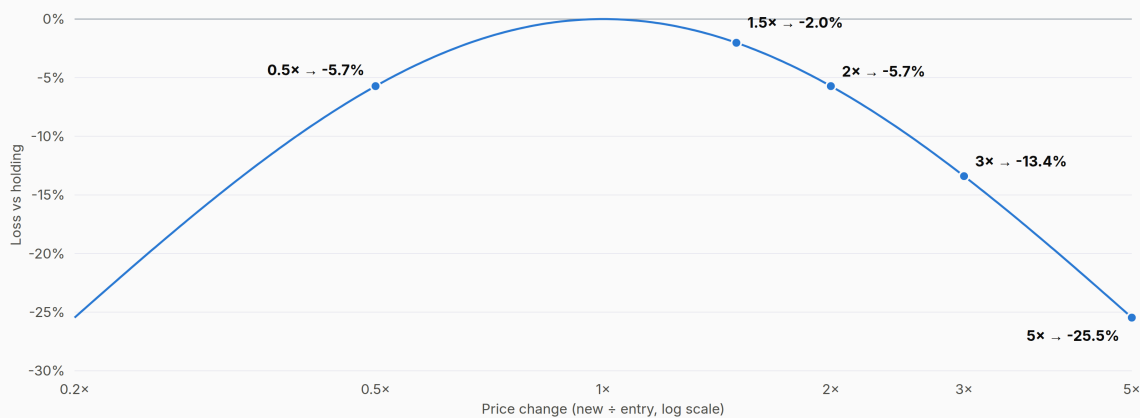
- **Profit engine:** none by itself. It's a tool. The gain comes from what the borrowed funds do, or from not triggering a sale.
- **Key maths:**
 - Health factor (HF) = $\Sigma(\text{collateral value} \times \text{liquidation threshold}) \div \text{debt}$
 - Liquidation price = $\text{debt} \div (\text{collateral qty} \times \text{liquidation threshold})$
 - Example: 10 ETH at \$3,000, LT 0.80, borrow \$12,000 → HF = 2.0 → liquidation at **\$1,500** (-50%).
- **Execute:** borrow only enough that liquidation is beyond a move you'd realistically expect to survive (HF ≥ 2 for **volatile collateral** is a sane starting rule) → write the defence plan before borrowing.
- **Monitor:** HF daily, borrow rate, oracle price.
- **Exit/kill:** HF < 1.5 → repay or add collateral. Don't wait for 1.1.
- **How it loses:** liquidation penalty (commonly 5–10%+ of the collateral seized), borrow rate spikes, oracle issues.
- **Size cap:** debt ≤ what you could repay from outside the position.

LEVEL 2 — LIQUIDITY

4. 50/50 AMM liquidity (full range)

Impermanent loss vs simply holding

50/50 constant-product pool, before fees. Same loss for the same ratio up or down ($0.5\times = 2\times$).



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- **Profit engine:** swap fees (source 1), sometimes plus incentives.
- **Key maths — impermanent loss (IL) vs just holding**, where r = price ratio change: $IL = \frac{2\sqrt{r}}{(1+r)} - 1$

Price change	1.25x	1.5x	2x	3x	4x	5x
IL vs holding	-0.6%	-2.0%	-5.7%	-13.4%	-20.0%	-25.5%

(Same IL whether price goes up or down by the same ratio. $0.5\times = 2\times$.)

- **Break-even rule:** fees earned over the holding period $\geq$ IL. Example: expect up to a 1.5x move within 90 days $\rightarrow$ IL $\approx$ 2.0% $\rightarrow$ you need fee APR $\geq$ **~8.2%** just to match holding.
- **Execute:** pick a pair you're happy to own either side of $\rightarrow$ check real fee APR over 30 days (not the peak day) $\rightarrow$ deposit $\rightarrow$ record entry prices.
- **Monitor:** fees accrued vs IL vs "just hold" benchmark, weekly.
- **Exit/kill:** fees stop covering IL, volume dies, or the pair enters a strong trend.
- **How it loses:** strong trends (you end up holding more of the losing asset), low volume, token risk.

5. Concentrated liquidity (the on-chain grid bot)

- **Profit engine:** swap fees on a chosen range. It's economically very close to a grid bot: it sells as price rises through the range and buys as it falls (Module 9).
- **Key maths:** capital efficiency vs full range (price at the geometric middle of the range) $\approx \frac{1}{(1 - (P_{low} / P_{high})^{1/4})}$. A $\pm 10\%$ range ($P_{high}/P_{low} \approx 1.22$) is **~20x** as capital-efficient, so it earns **~20x** the fees per dollar *while in range*. IL is also amplified by about the same factor, and **out of range earns 0% and leaves you holding 100% of one asset**.
- **Execute:** set the range from structure and volatility, exactly like a grid range (grid-bot-design: regime $\rightarrow$ range $\rightarrow$ invalidation) $\rightarrow$ wider in high volatility $\rightarrow$ decide the rebalance rule *before* entry.
- **Monitor:** % of time in range, fees vs IL, gas cost of each rebalance.

- **Exit/kill:** range invalidated (price breaks structure) → don't chase every move. Rebalancing too often locks in losses and burns gas.
- **How it loses:** trending markets, over-rebalancing, gas on small positions.
- **Size cap:** active capital only. Needs weekly (or better) attention.

6. Stable-stable LP

- **Profit engine:** swap fees between similarly priced assets. Very low IL *while the pegs hold*.
- **Execute:** only pair stables you'd accept holding 100% of (in a depeg the pool fills up with the weaker coin).
- **Kill:** either stable trades below ~0.99, or the pool balance goes heavily one-sided.
- **How it loses:** depeg (you end up holding the broken coin), contract risk.

LEVEL 3 — YIELD

7. Auto-compounding vaults

- **Profit engine:** the underlying strategy, plus compounding without you paying the gas.
- **Key maths:** $\text{net APY} = \text{gross APY} - \text{performance fee share} - \text{management fee}$. Compare against doing it yourself: vaults win on small positions (gas) and lose on large ones (fees).
- **Due diligence:** vault contract + **every** protocol it deposits into (the risks stack). Read withdrawal limits and fees.
- **How it loses:** any protocol in the chain fails, strategy change by the vault operator, withdrawal limits.

8. Incentive farming

- **Profit engine:** token emissions (source 5) on top of a base yield.
- **The rule:** value rewards at the price you can **actually sell at after slippage**, and assume the token price falls while emissions continue. Harvest and sell on a schedule (e.g. weekly) unless you'd buy the token with cash.
- **Quick test:** remove the incentive APR. If the base yield alone doesn't justify the risk, you're being paid in a token that's likely to keep falling.
- **How it loses:** reward token price collapse, mercenary liquidity leaving, IL on the underlying LP.

9. Airdrops and points

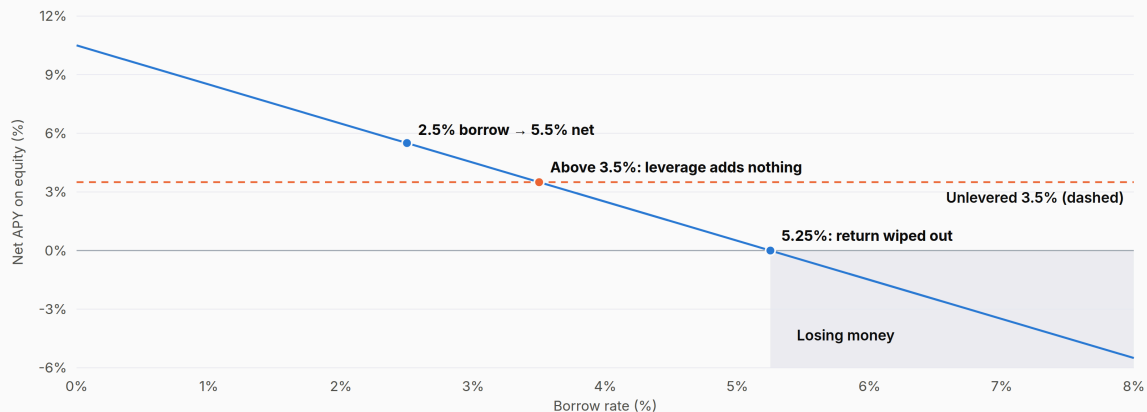
- **Profit engine:** possible future token distribution.
- **Key maths:** $\text{EV} = P(\text{airdrop}) \times \text{expected value to you} - (\text{gas} + \text{bridge} + \text{opportunity cost} + \text{risk})$.
- **Rule:** only farm with actions you'd do anyway, or with capital you've capped as speculative. Claim only from verified links (scam defence, Module 1.6).

LEVEL 4 — ADVANCED (only after Levels 1–3 are routine)

10. Leveraged lending loop

Leveraged loop: the spread is everything

3× loop on 3.5% collateral yield. Net APY on equity = 3.5% × 3 – borrow rate × 2.



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- **Profit engine:** amplifies a *positive spread* between collateral yield and borrow cost.
- **Key maths:**
 - Leverage after n loops at LTV L: $(1 - L^{(n+1)}) \div (1 - L)$, max $1 \div (1 - L)$ ($L = 0.7 \rightarrow 3.33\times$)
 - Net APY on equity = collateral yield × Lev – borrow rate × (Lev – 1)
 - Example: 3.5% collateral yield, 2.5% borrow, 3× → **5.5%**. If borrow rises to 4.5% → **1.5%**. If it rises to 5.25% → **0%**. The spread is everything.
- **Execute:** only loop assets that move closely together (e.g. an LST against its underlying, or stable against stable) → stay well below max leverage → set HF alerts.
- **Kill:** spread turns negative, HF < your floor, depeg of the collateral.
- **How it loses:** borrow rate spikes (they can happen quickly), liquidation cascades, depeg of correlated collateral, unwind costs.

11. LST collateral loop

Strategy 10 using an LST as collateral against its own underlying asset. Liquidation risk depends mainly on the **LST/underlying ratio** and on **how the protocol's oracle prices it**, less on the asset's price. Check which oracle the market uses before assuming "it can't be liquidated".

12. Delta-neutral funding carry

- **Profit engine:** perpetual funding (source 4). In bullish markets longs usually pay shorts. Hold spot long + perp short at equal size, so price moves roughly cancel and you collect funding.
- **Key maths:** funding APR $\approx$ rate per 8h × 3 × 365. 0.01%/8h $\approx$ **10.95% APR on the hedged size**. Return on *total capital* is lower because capital sits in both legs: \$10k spot + \$10k margin (1× short) → ~5.5% on \$20k; 2× short (\$5k margin) → ~7.3% on \$15k, but the short gets liquidated if price rises ~50%.
- **Execute:** open both legs at the same time, at the same size → keep a margin buffer → track funding every period.
- **Kill:** funding negative for a sustained period (you now pay), margin buffer thin, venue risk.

- **How it loses:** funding flips negative, short-leg liquidation in a squeeze, execution slippage/basis, exchange/venue failure (CEX or perp DEX).

13. LP hedge

Concentrated or 50/50 LP plus a partial perp short to offset its directional exposure. The hedge ratio changes as price moves (LP inventory changes), so it needs rebalancing. It narrows the P&L range but adds funding + execution cost. Only worth doing if fees exceed IL + hedge cost.

LEVEL 5 — TREASURY & RESEARCH

14. DeFi cash management

Build a ladder for idle stablecoins: instant-access tier (lending), core tier (split across protocols and issuers), and a cap per protocol. Goal: earn a reasonable return on cash **without any single failure costing more than you've decided you can afford to lose**.

15–16. On-chain accumulation screen / protocol growth screen

Research, not yield. Track users, fees/revenue, TVL quality, holder concentration, exchange flows and unlocks to decide *what* to hold or provide liquidity for. Never act on one metric (ATLAS ch. 27–34).

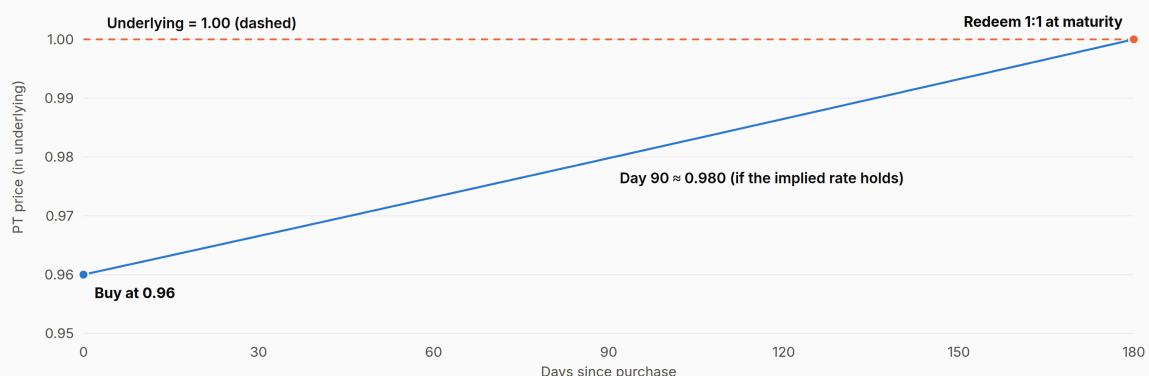
LEVEL 6 — PROFESSIONAL (structured, fixed and hedged yield)

These are the strategies professional desks and treasuries use. Each one swaps a risk you don't want for one you've chosen, so **name what you're short before you enter**. Full lessons: Module 10 (Advanced Yield Engineering), Module 11 (Hedging) and Module 12 (Own Bank).

17. Fixed-rate yield with principal tokens (PT)

Principal tokens pull to par

PT bought at 0.96 with 180 days left redeems at 1.00: an 8.63% fixed APY if held to maturity.



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- **Profit engine:** yield tokenization splits a yield-bearing asset into a **principal token (PT)**, which redeems 1:1 for the underlying at maturity, and a **yield token (YT)**, which collects all the variable yield until then. Buying PT at a discount locks in a fixed return (source 2/3, bought forward).

- **Key maths:** $\text{fixed APY} = (1 \div \text{PT price})^{(365 \div \text{days})} - 1$. PT at 0.96 with 180 days left → **8.63%** fixed if held to maturity. `defi_calc.py pt --price 0.96 --days 180`
- **Execute:** pick a PT on an underlying you'd hold anyway → check pool depth (for early exit) → buy → hold to maturity → redeem.
- **Kill:** underlying depeg or exploit. You're still exposed to the underlying asset; the rate is fixed, the asset risk isn't.
- **How it loses:** underlying fails; selling before maturity at a worse price; illiquid PT pools.

18. Yield tokens (YT): buying variable yield

- **Profit engine:** YT collects all yield (and often points) on the underlying until maturity, then goes to zero.
- **Key maths:** a YT costs $\sim 1 - \text{PT price}$ (0.04 above). It profits only if **realised variable yield plus points beats the implied rate (~8.6%)**.
- **Use:** a speculative, time-decaying position. Size it as speculation.
- **How it loses:** yields fall, points turn out worthless; the YT is worth zero at maturity by design.

19. Cash-and-carry basis

- **Profit engine:** dated futures often trade above spot in bull markets. Buy spot, short the future at the same size, and the premium (basis) converges to zero at expiry (source 4).
- **Key maths:** $\text{annualised basis} = (\text{future} \div \text{spot} - 1) \times 365 \div \text{days}$. Spot 3,000, 90-day future 3,060 → 2% → **8.11%** annualised. `defi_calc.py basis --spot 3000 --future 3060 --days 90`
- **Execute:** open both legs together → keep enough margin that the short survives a large rally → hold to expiry.
- **How it loses:** short-leg margin call in a squeeze, venue/counterparty failure, closing early at a wider basis.

20. Covered calls and options vaults

- **Profit engine:** sell upside you're willing to give away. The option premium is paid to you (source 4: selling volatility).
- **Key maths:** ETH at 3,000, sell a 7-day 3,300 call for 0.4% → **20.9%** annualised *if repeated* at that premium (it won't be exactly). Max gain per week = 0.4% + 10% = 10.4%. Break-even price 2,988. `defi_calc.py covered-call --spot 3000 --strike 3300 --premium 0.4`
- **Execute:** only on assets you'd hold anyway → strikes above levels you'd happily sell at → roll weekly or monthly.
- **How it loses:** the asset falls (the premium is a small cushion); you miss big rallies; options-vault contract risk.

21. Cash-secured puts: getting paid to wait for a lower entry

- **Profit engine:** sell a put at a price you'd buy at anyway, holding the cash to pay for it.
- **Key maths:** sell a 2,700 put for \$12 with \$2,700 of stablecoins reserved → **0.44%** for the week on the cash. If assigned, your effective entry is **\$2,688**.

- **How it loses:** price falls far below the strike (you buy at 2,688 while the market is lower); you miss the move if price rips upward.

22. Fixed-rate borrowing

- **Profit engine:** none by itself. It's insurance on your cost of credit (Module 12.3).
- **Worked comparison:** borrow \$50,000 for a year. Fixed at 6% costs **\$3,000**. Variable at 4% for 9 months, then spiking to 12% for 3 months, costs $50,000 \times (4\% \times 0.75 + 12\% \times 0.25) = \mathbf{\$3,000}$. Same cost, but fixed let you plan. If the spike lasted longer, variable costs more; if rates stayed at 4%, fixed cost \$1,000 more.
- **Use:** when the borrowing funds something with a fixed payoff, or when a rate spike would force a bad sale.

23. Being the lender: curated lending vaults

- **Profit engine:** supply stablecoins to a vault whose curator spreads them across isolated lending markets (source 2). You're the bank's depositor *and* its credit committee.
- **Key maths:** judge risk-adjusted, not headline: 7% headline with an assumed 2%/yr chance of a loss event losing half → **6.0%** risk-adjusted. `defi_calc.py expected --yield-apy 7 --loss-prob 0.02 --lgd 0.5`
- **Due diligence:** which markets, which collateral, which oracles, what liquidation LTVs, the curator's record, and how fast you can withdraw.
- **How it loses:** bad debt in any one market, oracle failure, curator error, withdrawal queues in stress.

24. Hedged restaking / points farming

- **Profit engine:** hold a liquid restaking token (staking yield + points) and short the underlying perp to remove price exposure. When funding is positive, the short *receives* funding too.
- **Key maths:** `net ≈ staking yield + funding received + points value - costs`. **Value the points at zero for planning.** Anything they pay is upside.
- **How it loses:** the restaking token depegs from the asset you hedged with (the hedge doesn't cover that), slashing, funding turns negative, short-leg liquidation.

25. The yield-covered credit line

- **Profit engine:** borrow against yield-bearing collateral where the collateral's yield pays the interest.
- **Key maths:** \$200,000 of an LST at 3.2% earns **\$6,400/yr**. A \$40,000 stablecoin loan at 5% costs **\$2,000/yr**, so yield covers interest 3.2×. LTV 20%, health factor 4.0 at an 0.8 liquidation threshold. `defi_calc.py bank --collateral 200000 --debt 40000 --borrow-apy 5`
- **Use:** access to liquidity without selling, as a bank client would (Module 12.3).
- **How it loses:** the collateral price falls (the LTV rises even if the interest is covered), the borrow rate spikes above the yield, the LST depegs.

LEVEL 7 — EXPERT (market structure)

Strategies that come from understanding how DeFi's machinery works: minting, governance markets, being the counterparty, arbitrage and rates. Full lessons in Modules 3, 6 and 10 (3.7, 6.6, 10.7, 10.8,

10.9).

26. Minting against collateral (CDP stablecoins)

- **Profit engine:** none by itself. It's how you create liquidity from your assets (Module 12's credit line), paying a stability fee.
- **Key maths:** 10 ETH at \$3,000, 150% minimum ratio: max mint \$20,000. Mint \$10,000 → ratio 300%, liquidation at \$1,500, fee \$600/yr at 6%. `defi_calc.py cdp --qty 10 --price 3000 --mint 10000 --fee 6`
- **How it loses:** collateral falls through the ratio (liquidation penalty), stability fee rises, the minted stablecoin depegs.

27. Vote-escrow and bribe income

- **Profit engine:** voting incentives (bribes) and fee shares paid to locked governance tokens (source 5, sometimes 1).
- **Key maths:** \$10,000 locked earning 15% in bribes = \$1,500/yr, *valued at the price you can sell the bribe tokens*; subtract the locked token's price risk over the whole lock.
- **How it loses:** the locked token falls while you can't sell; bribe markets dry up; liquid-locker discounts.

28. Being the house: perp liquidity vaults

- **Profit engine:** trading fees, funding and traders' losses on a perp exchange.
- **Key maths:** split history into fees vs trader P&L; the trader-P&L part can swing from +7% to -20% in a trend.
- **How it loses:** profitable traders, one-sided open interest, oracle problems, withdrawal cooldowns.

29. Peg and redemption arbitrage (patient version)

- **Profit engine:** buying an asset below its redemption value and redeeming (source 4: providing liquidity to forced sellers).
- **Key maths:** 2% LST discount with a 20-day redemption queue ≈ 36.5% annualised simple, *once*, if redemption works.
- **How it loses:** the discount was pricing a real problem; the queue lengthens; redemption is restricted.

30. Rates positioning: fixed vs floating

- **Profit engine:** choosing when to lock fixed rates (PTs, fixed borrowing) or stay floating, based on the yield curve (source 2/3).
- **Key maths:** 3-month 7% vs 12-month 9%: lock 12 months at 9% if you need the money in a year and expect rates to fall.
- **How it loses:** rates move the other way (opportunity cost); PT liquidity if you exit early; the underlying's risk remains.

Part 4 — Execution system (what turns strategies into results)

Position sizing rules (starting point, tighten as needed)

- Max **20–25%** of DeFi capital in any one protocol. Max **10%** in anything unaudited, new (< 6 months) or reliant on a single bridge.
- Max **~30–40%** on any one non-Ethereum chain / L2.
- Keep **10–20%** instantly liquid (not in any position) to defend debt and take opportunities.
- Leverage: total debt small enough that a –50% market day doesn't liquidate anything.

Before every entry (5 gates)

1. **Due diligence** file complete (`defi-due-diligence` : 6-step loop).
2. **Profit engine named.** Which of the 5 sources pays you?
3. **Maths done.** Net return after costs beats the benchmark (use `defi_calc.py`).
4. **Kill rules written down.** Exact numbers: HF floor, peg level, spread, range break.
5. **Before-signing checklist** (chain, contract, token, amount, spender, slippage, gas).

Journal (one row per position)

Protocol · chain · strategy # · entry date · amount · entry prices · profit engine · expected net APY · benchmark · kill rules · approvals granted · weekly: fees/interest earned, IL, costs, net vs benchmark.

Review rhythm

- **Daily:** health factors, pegs, LP ranges, incident feeds.
- **Weekly:** net P&L vs benchmark per position, harvest and sell incentive tokens, revoke unused approvals.
- **Monthly:** cut the bottom performer when judged against its risk, rebalance buckets, simplify anything you can't explain in one sentence.

Take profits and keep them

- Sweep realised yield to stablecoins or cold storage on a schedule. Rewards sitting in a farm are still exposed to the farm's risks.
- Compound only what still passes the 5 gates. Compounding into a deteriorating position is how gains are given back.
- Track **net realised gains** after all costs. That's the only number that counts.

Part 5 — Progression path

Stage	Strategies	Graduate when...
1. Foundation	#1 stable lending, #2 staking	You can do the full before-signing checklist without notes
2. Liquidity	#4 50/50 LP, #6 stable LP	You've tracked LP vs hold for 30+ days

Stage	Strategies	Graduate when...
3. Active	#5 concentrated LP, #7 vaults, #8 farming	You can say how much of a position's return is fees, IL and incentives
4. Leverage	#3 borrowing, #10–11 loops	You've written and rehearsed a defence plan
5. Carry & hedging	#12 funding carry, #13 LP hedge	You understand perp margin, funding and venue risk
6. Operator	#14 treasury + #15–16 research	Your portfolio has caps, a journal, and a review you actually run
7. Professional	#17–21 fixed, basis and options yield · #22–25 credit and hedged yield	You can state what every position is short, and your stress test survives a –50% day
8. Expert	#26 CDP minting · #27 ve/bribe income · #28 perp vaults · #29 peg arbitrage · #30 rates positioning	You understand the machinery (AMM design, MEV, lending internals, rates) well enough to explain why each opportunity exists

Part 6 — Mastery quiz

Q: 1. A farm shows 60% APY: 5% fees, 55% token emissions. What's your first question?

Answer: What does the base 5% pay me for the risk, and at what price can I actually sell the reward token? Judge the farm on base yield plus rewards valued at a realistic, falling price.

Q: 2. ETH doubles while you're in a 50/50 ETH/USDC pool. How did you do vs holding?

Answer: About 5.7% behind holding, before fees. Fees need to exceed that for LPing to have been worth it.

Q: 3. A loop earns 3.5% on collateral and pays 2.5% to borrow at 3×. What borrow rate wipes out the return?

Answer: $0 = 3.5 \times 3 - b \times 2 \rightarrow b = 5.25\%$.

Q: 4. Funding is +0.01% per 8 hours. What is the APR on the hedged size, and why is your return on capital lower?

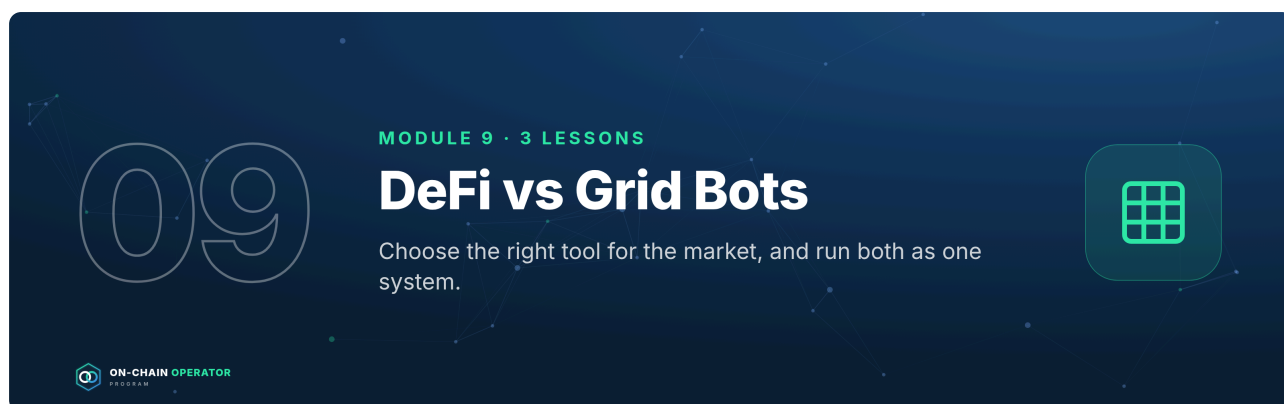
Answer: ~10.95%. Capital sits in both the spot leg and the perp margin, so the return on total capital is roughly half to three-quarters of that, depending on the short's leverage.

Q: 5. Your concentrated LP just went out of range after a breakout. What's the operator move?

Answer: Treat it as a range invalidation, just as with a grid bot: follow the pre-written rule (re-centre with a wider range, or exit). Don't chase every move with narrow rebalances.

Operating rule: if you can't name the profit engine and explain the unwind, don't enter the position.

Module 9 — DeFi vs Grid Bots



Outcome: choose the right tool for the market, and run both as one system. New content linking the On-Chain Operator Program to Grid Bot Builder. Educational only. Not financial advice. All figures illustrative.

Lesson 9.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

A grid bot on an exchange and a liquidity position on a DEX both buy as price falls and sell as it rises inside a range. This module shows when each is the better tool, and how to run both without doubling your risk.

Words you'll need

Term	Meaning
Grid bot	automated buy/sell orders at set price levels
Range	the price band a strategy works in
Concentrated liquidity	LP capital placed in a chosen range
Correlation	positions that move together
Invalidation	the event that ends a range thesis

Before you start

- ☐ Modules 0–8 (Grid Bot Builder helpful but not required)

Your first safe step

Take one asset and write the range you'd use for a grid bot and for an LP. Would they be the same?

The mastery ladder

Level	You can...
Beginner	Explains the grid–LP similarity
Practitioner	Chooses the right tool by size, custody and pair
Master	Runs grid bots and DeFi as one system with exposure counted by asset and range

You've mastered this module when...

...your exposure is counted by underlying asset and range, not by number of positions.

Lesson 9.1 — Same idea, different machine

Objective

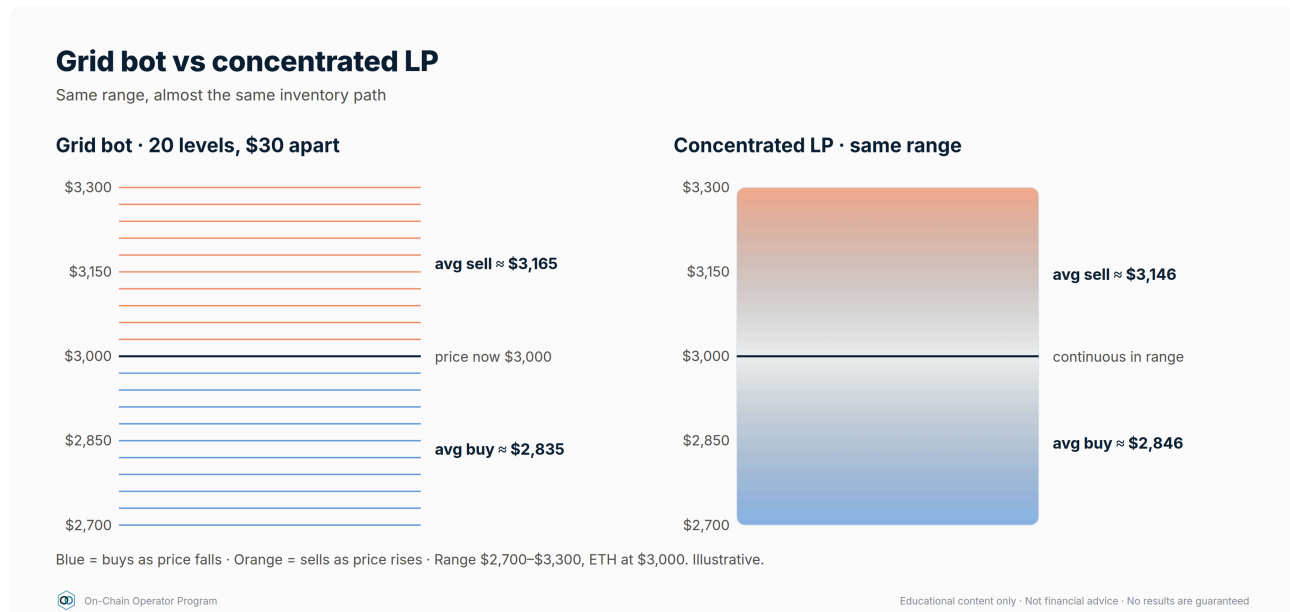
Explain why a concentrated liquidity position and a grid bot are economically close cousins, and where they differ.

Explanation

Both strategies do the same basic thing inside a price range: **sell as price rises, buy as price falls, and profit from price moving back and forth.**

	Grid bot (e.g. Grid Bot Builder on an exchange)	Concentrated LP (on-chain AMM)
How it trades	Discrete orders at set levels	Continuous. Every trade in range moves your inventory
What it earns	Spread between grid levels, minus exchange fees	Share of the pool's swap fee on every trade in range
Who pays whom	You pay trading fees	Traders pay <i>you</i> fees
At the range low	Holds mostly/all the base asset	Holds 100% of the base asset
At the range high	Holds mostly/all quote (USDT/USDC)	Holds 100% of the quote asset
Outside the range	Stops trading, holding one side	Stops earning, holding one side
Custody	Exchange holds funds (you keep API control)	You hold keys, and the contract holds the position
Main extra risk	Exchange / counterparty	Smart contract, oracle, MEV, gas
Automation	Built into the tool	Manual, or a third-party manager (extra contract risk)

Worked example — same range, same inventory behaviour



Range **\$2,700–\$3,300**, ETH at **\$3,000**.

Price falls to **\$2,700**:

- *Grid* (20 arithmetic levels, \$30 apart): buys at 2,970, 2,940 ... 2,700 → average buy ≈ **\$2,835**.
- *Concentrated LP*: converts USDC to ETH continuously on the way down at the geometric average $\sqrt{(2,700 \times 3,000)} \approx$ **\$2,846**.

Price rises to **\$3,300**:

- *Grid*: sells at 3,030 ... 3,300 → average ≈ **\$3,165**.
- *LP*: sells ETH at $\sqrt{(3,000 \times 3,300)} \approx$ **\$3,146**.

Almost the same inventory path. The real difference is **how money is made on the way**: grid profit per completed level (after paying fees), versus LP fee income on all trading volume in range (after impermanent loss).

Checklist

- ☐ I can explain why both end up 100% in one asset at a range edge
- ☐ I know which one pays fees and which one earns them
- ☐ I treat a range break as an invalidation event in both

Quiz

Q: 1. Price breaks below the range. What do the grid bot and the LP both hold?

Answer: All, or nearly all, the base asset, bought on the way down. Both stop earning until price returns or you act.

Q: 2. Who pays trading fees in each?

Answer: The grid bot pays exchange fees. The LP earns fees from traders.

Q: 3. Name one risk the LP has that the grid bot doesn't, and vice versa.

Answer: LP: smart-contract/MEV/gas risk. Grid bot: exchange/counterparty risk.

Lesson 9.2 — When each one wins

Objective

Pick grid bot, LP, both, or neither, for a given market and account.

Explanation

Factor	Favours grid bot	Favours on-chain LP
Position size	Small–medium (no gas per action)	Larger (gas becomes a smaller share of returns)
Pair	Major pairs listed on your exchange	Tokens only traded on-chain; pairs with deep pool volume
Fee economics	Low maker fees on your exchange	High-volume pools with a good fee tier (e.g. 0.05% / 0.3% / 1%)
Management	Want automation (Grid Bot Builder)	Happy to set ranges and rebalance by hand
Custody preference	Comfortable with exchange custody	Want self-custody
Counterparty view	Trust the exchange more than contracts	Trust audited contracts more than an exchange
Leverage wanted	Futures grid (advanced)	Not applicable (LP is unlevered unless you borrow)

Neither, same as in the grid module: strong breakouts, parabolic moves, major news, thin liquidity. Waiting is a position.

Worked example — three traders

1. **\$3k, trades BTC/ETH, wants automation** → grid bot. Gas would eat a small on-chain LP.
2. **\$80k, holds ETH long-term, self-custody, checks weekly** → a wide-range ETH/USDC concentrated LP could earn fees on part of the stack. Size it as ETH they're willing to sell higher and USDC they're willing to spend lower.
3. **A token that's only liquid on a DEX** → only an LP is possible. Apply full due diligence (Module 6) and small size.

Checklist

- ☐ I've matched the tool to position size, pair, custody and management time
- ☐ I've compared expected fee income to costs (gas or exchange fees)
- ☐ I've checked the regime before choosing either

Quiz

Q: 1. Why does position size matter more on-chain?

Answer: Gas is a roughly fixed cost per action, so it's a bigger percentage of a small position.

Q: 2. When is an LP the only option?

Answer: When the pair trades only on-chain.

Q: 3. Market just broke out on huge news. Grid or LP?

Answer: Neither, until a new range forms.

Lesson 9.3 — Building a combined system

Objective

Design a portfolio that uses exchange grid bots and DeFi together without doubling up on the same risk.

Explanation

Each tool is good at a different job:

- **Grid bots** → capture oscillation in range-bound major pairs, fully automated.
- **DeFi lending / stable LP** → earn on idle stablecoins and reserves.
- **Staking / LSTs** → earn on assets you'd hold anyway.
- **Concentrated LP** → earn fees on-chain in ranges, for larger, self-custodied sizes.

The trap: correlation. An ETH grid bot + an ETH/USDC LP + an LST loop are *three* bets on ETH staying in a range. If ETH breaks down, they all lose together. Add up your exposure by **underlying asset and by range**, not by number of positions.

Worked example — an illustrative allocation framework

Percentages are an example structure, not a recommendation. Set your own from your risk budget.

Bucket	Share	Tool	Job
Reserve	15%	Stablecoin lending (split across 2 protocols)	Liquidity to defend positions and redeploy after range breaks
Core yield	25%	Stable-stable LP / stable lending	Base return on cash
Hold-anyway assets	25%	Staking / LSTs	Earn on long-term holdings
Range income	30%	Grid bots on liquid majors (+ optional wide on-chain LP)	Capture oscillation
Speculative	≤5%	Burner-wallet farms, airdrops, new protocols	Capped experiments

Rules for this system:

1. **One range thesis per asset.** If a grid bot and an LP both run on ETH, their combined size must fit your ETH range-risk cap.
2. **Reserve funds the defence.** Range breaks and health-factor defence draw from reserve, never from other positions.
3. **Sweep profits up.** Realised grid profits and harvested yield go to reserve or vault on a schedule.
4. **Review together.** One weekly review across CEX and on-chain positions, using the same journal.

Checklist

- ☐ Exposure added up by underlying asset, not by bot/position count
- ☐ Reserve sized and kept out of positions
- ☐ Profit sweep schedule set
- ☐ One journal covering bots and DeFi positions

Quiz

Q: 1. ETH grid bot + ETH/USDC LP + ETH LST loop. How many independent bets?

Answer: Essentially one: ETH staying in range or rising. They're correlated.

Q: 2. What is the reserve for?

Answer: Defending debt positions and redeploying after invalidations, without forced selling.

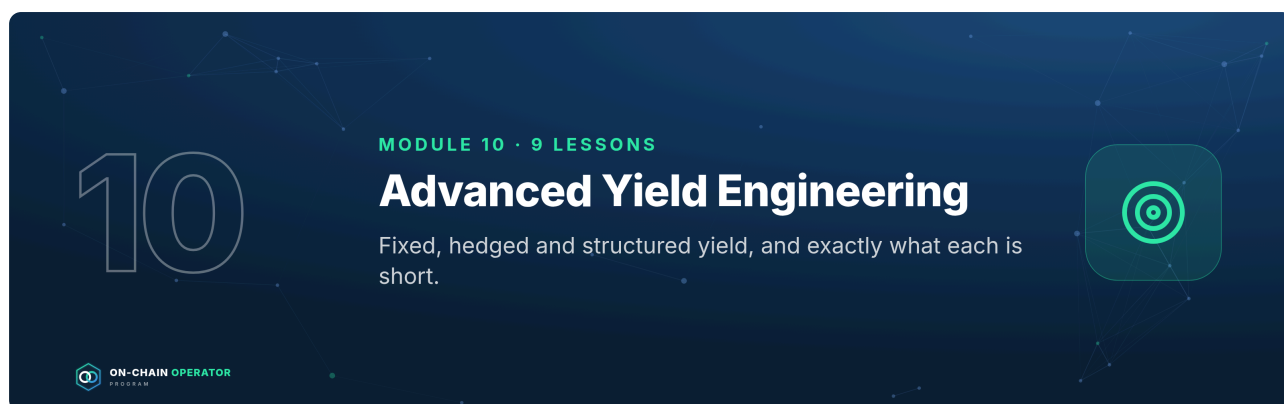
Q: 3. Why sweep profits out of positions?

Answer: Profits left in a position stay exposed to its risks. Sweeping locks them in.

Module 9 next step

- **Not a Grid Bot Builder customer yet?** Grid Bot Builder automates the "range income" bucket on your exchange: self-serve, delivered instantly. → Checkout link on the program page.
- **Want daily range and market intel?** Elite Intel Community: 3-day free trial → elite-intel-community.whop.site

Module 10 — Advanced Yield Engineering



Outcome: build fixed, hedged and structured yield, and know exactly what each position is short. Stage 4 · Strategist. Prerequisites: Modules 1–9. Educational content only. Not financial advice. Figures are illustrative; use live rates.

The professional's rule for this module: every structured position swaps one risk for another. Before entering, write one sentence: "This position is short ____." If you can't fill the blank, you're not ready to enter.

Lesson 10.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Professionals don't just take the yield on offer: they engineer it. Fixed rates, hedged carry, option income and managed ranges each swap one risk for another. This module teaches you to build them and say exactly what each one is short.

Words you'll need

Term	Meaning
PT / YT	principal and yield tokens splitting a yield-bearing asset
Basis	a future's premium over spot
Funding	payments between perp longs and shorts
Covered call	selling upside for a premium
Delta-neutral	price moves roughly cancel out

Before you start

- ☐ Modules 0–9, including perps (3.5) and LP maths (Module 2)

Your first safe step

Price one live PT with `defi_calc.py pt` and write: "This position is short ____."

The mastery ladder

Level	You can...
Beginner	Understands each structure and its payoff
Practitioner	Sizes and runs one structured position with written exits
Master	Combines fixed, carry, options and LP yield inside caps, naming every short exposure

You've mastered this module when...

...you can complete "this position is short ____" for every structured position you hold.

Lesson 10.1 — Fixed-rate yield: principal and yield tokens (PT/YT)

Objective

Lock a fixed return using principal tokens, and price a yield token against its implied rate.

Explanation

Yield tokenization takes a yield-bearing asset (a staked asset, a lending receipt, a stablecoin savings token) and splits it into two tokens that mature on a set date:

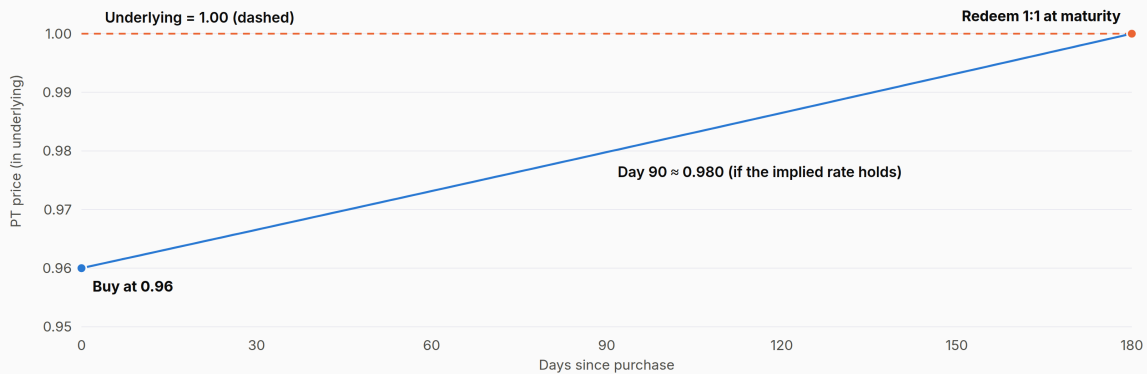
- **PT (principal token):** redeems 1:1 for the underlying at maturity. It trades at a **discount** before then. Buy the discount, hold to maturity, and your return is fixed.
- **YT (yield token):** receives *all* the variable yield (and often points) the underlying earns until maturity, then is worth zero.

PT + YT = one unit of the underlying. The PT price implies a fixed rate; the YT is a bet that realised yield beats it.

```
fixed APY = (1 ÷ PT price)^(365 ÷ days) - 1
```

Principal tokens pull to par

PT bought at 0.96 with 180 days left redeems at 1.00: an 8.63% fixed APY if held to maturity.



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Worked example

A stablecoin savings token has a PT maturing in **180 days**, priced at **0.96**.

- `defi_calc.py pt --price 0.96 --days 180` → **8.63%** fixed APY if held to maturity.
- \$100,000 buys ~104,167 PT → redeems for ~104,167 of the underlying at maturity.
- The YT costs ~0.04 per unit. It only profits if the underlying's realised yield (plus any points) averages above ~8.6% for the 180 days.

Near maturity, small discounts are big rates: PT at 0.985 with 60 days left → **9.63%** APY. Check liquidity before chasing these; exiting early means selling into the pool.

What this position is short

PT: the underlying's credit/depeg risk, and liquidity if you exit early. YT: falling yields, and time (it decays to zero).

Checklist

- ☐ I'd hold the underlying asset anyway
- ☐ Maturity date matches when I'll need the money (Module 12.4 ladder)
- ☐ Pool depth checked for an early exit
- ☐ YT sized as speculation, not income

Quiz

Q: 1. PT at 0.97, 120 days to maturity. Fixed APY?

Answer: $(1/0.97)^{(365/120)} - 1 \approx 9.7\%$.

Q: 2. Does buying PT remove the underlying's risk?

Answer: No. It fixes the rate. If the underlying depegs or is exploited, the PT redeems into a damaged asset.

Q: 3. What must happen for a YT to profit?

Answer: Realised variable yield plus points must beat the implied rate priced into the PT.

Lesson 10.2 — Cash-and-carry basis trades

Objective

Capture the premium of a dated future over spot without taking a view on price.

Explanation

In rising markets, dated futures usually trade **above** spot. The gap (basis) shrinks to zero at expiry because the future settles at spot. **Buy spot + short the future at the same size** and you lock that gap, whatever the price does, as long as both legs survive to expiry.

```
annualised basis = (future ÷ spot - 1) × 365 ÷ days
```

Worked example

ETH spot **3,000**, 90-day future **3,060**.

- Basis = 2.0% → **8.11%** annualised (`defi_calc.py basis --spot 3000 --future 3060 --days 90`).
- At expiry, if ETH is 2,000: spot leg -1,000, short future +1,060 → +60 per ETH.
- At expiry, if ETH is 4,500: spot leg +1,500, short future -1,440 → +60 per ETH.
- **The catch:** before expiry, a rally to 4,500 means the short leg shows -1,440 per ETH. Without enough margin on the futures venue, it's closed out and you're left long spot at the top.

What this position is short

Margin and venue risk: a squeeze on the short leg, or the venue failing.

Checklist

- ☐ Both legs opened together, same size
- ☐ Futures margin survives at least a +100% move, or a rule to top up
- ☐ Venue risk sized (cap per venue)
- ☐ Plan to hold to expiry; early exit may be at a worse basis

Quiz

Q: 1. Spot 2,000, 180-day future 2,080. Annualised basis?

Answer: $4\% \times 365/180 \approx 8.1\%$.

Q: 2. Why is the return "locked" at expiry?

Answer: The future settles at spot, so the gain on one leg offsets the loss on the other, leaving the entry basis.

Q: 3. What breaks the trade before expiry?

Answer: A margin call or liquidation on the short future during a rally, or venue failure.

Lesson 10.3 — Delta-neutral funding carry, done properly

Objective

Run a funding-carry position with proper sizing, venue limits and exit rules.

Explanation

Perpetual futures pay **funding** between longs and shorts to keep the perp near spot. In bullish markets longs usually pay shorts. Long spot + short perp at the same size = no net price exposure, while collecting funding.

Professional rules:

1. **Measure funding over time, not today.** Use a 7- and 30-day average.
2. **Exit rule:** close when the 7-day average turns negative (you'd start paying).
3. **Margin buffer:** the short leg's leverage decides how big a rally it survives.
4. **Venue limits:** split across venues; cap each one.

Worked example

Funding **+0.01% per 8h** $\approx$ **10.95% APR** on the hedged size.

Short-leg leverage	Capital per \$1 hedged	Return on capital	Short liquidates on a rally of roughly
1×	\$2.00	~5.5%	+100%
2×	\$1.50	~7.3%	+50%
3×	\$1.33	~8.2%	+33%

```
defi_calc.py carry --rate-8h 0.01 --short-lev 3
```

Higher leverage improves return on capital a little and makes the position much more fragile. Most operators stay at 1–2× and top up margin from reserve.

What this position is short

Funding turning negative, short squeezes, and venue failure.

Checklist

- ☐ 7- and 30-day funding averages checked
- ☐ Exit rule written (e.g. 7-day average < 0)
- ☐ Short-leg leverage $\leq$ 2×, top-up plan from reserve
- ☐ Per-venue cap set

Quiz

Q: 1. Funding is +0.005%/8h. APR on hedged size?

Answer: $\approx 5.5\%$.

Q: 2. Why not run the short at 5×?

Answer: A ~20% rally would liquidate the short and leave you unhedged long.

Q: 3. What's the exit signal?

Answer: Average funding turning negative, meaning shorts start paying.

Lesson 10.4 — Options income: covered calls, cash-secured puts, options vaults

Objective

Earn option premium on assets you'd hold or buy anyway, knowing exactly what you give up.

Explanation

- **Covered call:** you hold the asset and sell someone the right to buy it from you at a higher **strike**. You keep the premium, and your upside is capped at the strike.
- **Cash-secured put:** you hold stablecoins and sell someone the right to sell you the asset at a lower strike. You keep the premium. If price falls below the strike, you buy at the strike.
- **The wheel:** sell puts until assigned → hold the asset → sell calls until called away → repeat.
- **Options vaults** automate this on-chain. They add contract risk and you don't choose the strikes.

Worked example

ETH at 3,000.

- **Covered call:** sell a 7-day 3,300 call for 0.4% (\$12). $\approx 20.9\%$ annualised *if* repeated at that premium, which it won't be exactly. Max gain for the week 10.4%. Break-even 2,988. `defi_calc.py`
`covered-call --spot 3000 --strike 3300 --premium 0.4`
- **Cash-secured put:** sell a 7-day 2,700 put for \$12 with \$2,700 reserved → 0.44% on the cash for the week. If assigned, effective entry \$2,688.

What this position is short

Covered call: upside above the strike, plus the full downside of the asset. Cash-secured put: a crash through the strike.

Checklist

- ☐ Only on assets I'd hold (calls) or buy at the strike (puts)
- ☐ Strike chosen from levels, not from the premium
- ☐ Annualised premium treated as "if repeated", never promised
- ☐ Vault due diligence done if automating

Quiz

Q: 1. What do you give up by selling a covered call?

Answer: Gains above the strike for that period.

Q: 2. Put strike 2,700, premium \$15. Effective entry if assigned?

Answer: \$2,685.

Q: 3. Does premium protect you in a crash?

Answer: Only by the size of the premium. The asset's downside remains.

Lesson 10.5 — Active concentrated-liquidity management

Objective

Choose a range width and a rebalance rule that fee income can actually pay for.

Explanation

Narrower ranges earn more fees per dollar *while in range* and go out of range sooner.

Range around \$3,000	Capital efficiency vs full range
\$2,400–\$3,600 ($\pm 20\%$)	$\sim 10.4\times$
\$2,700–\$3,300 ($\pm 10\%$)	$\sim 20.4\times$
\$2,850–\$3,150 ($\pm 5\%$)	$\sim 40.5\times$

```
defi_calc.py cl --low 2850 --high 3150
```

Every rebalance costs gas and swap fees and **locks in** the impermanent loss so far. Professionals set a rule before entering:

- **Width from volatility:** at least $\sim 1.5\times$ the typical weekly move.
- **Rebalance trigger:** out of range for 24h+, *and* the range thesis still holds.
- **Invalidation:** a structural break (as with a grid bot) means exit, not re-centre.
- **Budget:** rebalancing costs capped at a share of expected fees (e.g. $\leq 25\%$).

Worked example

ETH typically moves $\pm 7\%$ a week. A $\pm 5\%$ range would be out of range most weeks. The rule points to **$\pm 10\text{--}12\%$** : $\sim 20\times$ efficiency, and time in range to earn.

What this position is short

Trending markets and volatility above what the range was sized for.

Checklist

- ☐ Width $\geq 1.5\times$ typical weekly move

- ☐ Rebalance trigger and invalidation written
- ☐ Rebalance cost budget set
- ☐ P&L tracked against holding (Lesson 2.4)

Quiz

Q: 1. Why does a $\pm 5\%$ range earn $\sim 2\times$ the fees of $\pm 10\%$ while in range?

Answer: The same capital is concentrated in half the price range, so it provides about twice the liquidity there.

Q: 2. What does a rebalance lock in?

Answer: The impermanent loss accrued so far, plus gas and swap costs.

Q: 3. Price breaks structure and exits the range. Re-centre?

Answer: No. That's an invalidation. Exit per plan.

Lesson 10.6 — Restaking and points: pricing speculative yield

Objective

Value restaking and points programmes without letting unknown rewards drive position size.

Explanation

Restaking reuses staked assets to secure more services, paying extra rewards and often **points**: an off-chain score that may or may not become a token. The risk stack grows at each layer: staking → liquid restaking token → restaking protocol → each service it secures → slashing conditions.

Professional rules:

- **Plan with points at zero.** Base the decision on yield you can measure.
- **Treat points as an airdrop bet:** $EV = P(\text{payout}) \times \text{value} - \text{costs}$.
- **Hedge price if you only want the yield:** short the underlying perp (strategy #24), knowing the hedge doesn't cover a depeg of the restaking token.
- **Cap it:** speculative bucket only ($\leq 5\text{--}10\%$ of DeFi capital).

Worked example

\$20,000 in a liquid restaking token: base staking yield $\sim 3\%$ (\$600/yr). Points: you estimate a 30% chance of a payout worth \$1,500, costing \$200 in gas and bridging: `defi_calc.py airdrop --probability 0.3 --value 1500 --costs 200` → EV **\$250**. Worth doing only if the position still makes sense at \$600/yr and the risk stack is acceptable.

What this position is short

Slashing, depeg of the restaking token, and the chance that points never become anything.

Checklist

- ☐ Decision still works with points valued at zero
- ☐ Full risk stack written, layer by layer
- ☐ Sized inside the speculative bucket
- ☐ Hedge (if any) and its gap understood

Quiz

Q: 1. Why value points at zero?

Answer: They're not guaranteed to become anything, so a decision that needs them is a speculation.

Q: 2. What does a perp hedge on the underlying miss?

Answer: The restaking token depegging from the underlying.

Q: 3. Name three layers of the restaking risk stack.

Answer: Any three: staking/validators, the liquid restaking token, the restaking protocol, each service secured, slashing conditions.

Lesson 10.7 — Being the house: perp-exchange liquidity vaults (expert)

Objective

Understand what you're really taking on when you provide liquidity to a perpetuals exchange.

Explanation

Many on-chain perp exchanges let you deposit into a **liquidity vault** that acts as the **counterparty** to traders (the "house"). The vault earns trading fees, borrowing/funding fees and part of liquidations, and **loses when traders win**.

- **Returns:** fees + traders' losses – traders' profits.
- **Risks:** a run of profitable traders (or a few very large ones), open interest concentrated on one side, oracle manipulation or latency, the market-making strategy of the vault (some vaults actively trade), and exchange contract risk.
- **Check:** the vault's historical P&L split (fees vs trader P&L), its open-interest caps, how it prices assets (oracle), and withdrawal rules (cooldowns, fees).

Worked example

A vault shows 25% APR: 18% from fees, 7% from traders' net losses over the last 6 months. In a strong one-directional trend, traders who are long win, so the vault's "trader P&L" component can swing to –20% or worse, wiping out the fees. Size it as an **active trading exposure**, not as fixed income, and check whether withdrawals are delayed in stress.

What this position is short

Trader skill and trending markets; oracle quality; the exchange's own risk controls.

Checklist

- ☐ Vault returns split into fees vs trader P&L, with history
- ☐ Open-interest caps and oracle design reviewed
- ☐ Withdrawal cooldowns and stress behaviour known

Quiz

Q: 1. When does a perp liquidity vault lose money?

Answer: When traders are net profitable, e.g. in strong trends.

Q: 2. Why isn't it fixed income?

Answer: Part of the return is traders' losses, which can reverse sharply.

Q: 3. Name two things to check before depositing.

Answer: Any two: fee vs trader-P&L history, OI caps, oracle design, withdrawal rules, contract risk.

Lesson 10.8 — DeFi rates: term structure, fixed vs floating (expert)

Objective

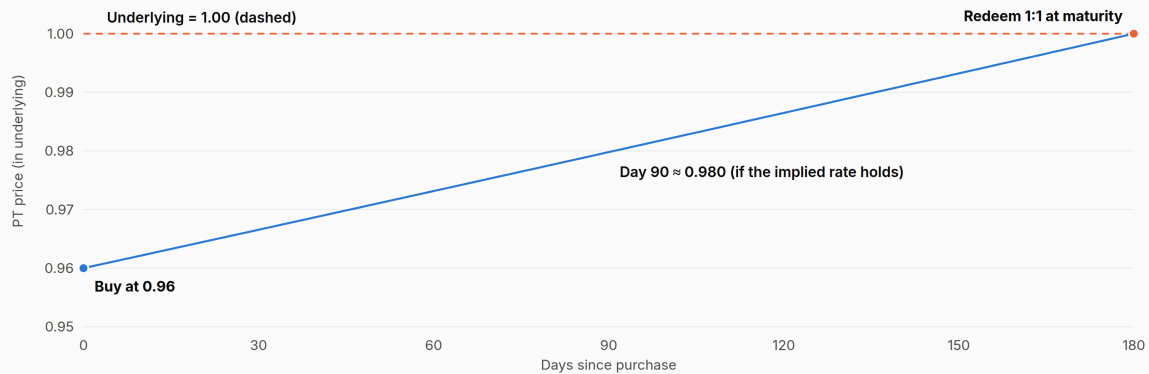
Read DeFi's interest-rate curve and position for fixed or floating rates deliberately.

Explanation

- **Floating rates:** lending and borrowing APYs that change with utilisation (Module 3).
- **Fixed rates:** PTs (10.1), fixed-rate lending markets and fixed-rate borrowing (#22).
- **Term structure:** PT implied yields across maturities form a **yield curve** (e.g. 3-month vs 12-month). Upward sloping: the market expects rates to stay high or rise. Inverted: it expects them to fall.
- **Positioning:** buying PT = receiving a fixed rate (you win if floating rates fall). Buying YT = receiving the floating rate (you win if floating rates rise). Fixed-rate borrowing = paying a fixed rate (you win if floating borrow rates rise).
- **Basis between venues:** the same asset's rates can differ across protocols and chains; the gap reflects risk, liquidity and friction as much as opportunity.

Principal tokens pull to par

PT bought at 0.96 with 180 days left redeems at 1.00: an 8.63% fixed APY if held to maturity.



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Worked example

Stablecoin PT implied yields: 3-month **7%**, 12-month **9%**. You need the money in 12 months and think rates will fall: buy the 12-month PT and lock 9%. If you think rates will rise instead, stay floating (supply to lending) or buy YT with a small, capped amount.

Checklist

- ☐ I can state whether each position is fixed or floating
- ☐ I compare implied rates across maturities before locking
- ☐ Maturities match my liquidity ladder (12.4)

Quiz

Q: 1. You buy a PT. Are you receiving fixed or floating?

Answer: Fixed.

Q: 2. An inverted curve suggests?

Answer: The market expects rates to fall.

Q: 3. You think borrow rates will spike. How do you protect a loan?

Answer: Switch to fixed-rate borrowing.

Lesson 10.9 — Peg and redemption arbitrage (expert)

Objective

Understand the arbitrages that keep stablecoins and LSTs near their value, and when a retail operator can take part.

Explanation

- **Stablecoin peg arbitrage:** if a stablecoin trades at 0.995 on a DEX and a **PSM** or issuer redeems it at 1.00 (minus a fee), buy-and-redeem closes the gap. It's competitive, needs fast execution, and is often limited to whoever can redeem.
- **LST discount arbitrage:** if an LST trades below its redemption value, buy it and **queue a redemption**. Your return is the discount over the waiting time, with the risk that the discount reflects a real problem (slashing, a bug) or that the queue lengthens.
- **Who wins:** professional searchers take most instant arbitrage (14.6). What's left to patient operators is the **slow** kind: buying discounts and waiting, sized small.

Worked example

An LST trades at a **2% discount**; the redemption queue is about 20 days. Return $\approx 2\%$ over 20 days $\approx$ **36.5% annualised (simple)**, but only this once, only if redemption works as expected, and only if the discount wasn't pricing a real problem. Research why the discount exists first (Module 6); cap it in the speculative bucket.

What this position is short

The reason the discount exists: an actual problem with the asset, or redemption delays.

Checklist

- ☐ Redemption path and eligibility confirmed
- ☐ Reason for the discount researched
- ☐ Queue length and worst case written

Quiz

Q: 1. What closes a stablecoin's discount?

Answer: Arbitrageurs buying below peg and redeeming at 1.00 (e.g. via a PSM or issuer).

Q: 2. 1% discount, 10-day redemption. Simple annualised return?

Answer: About 36.5%.

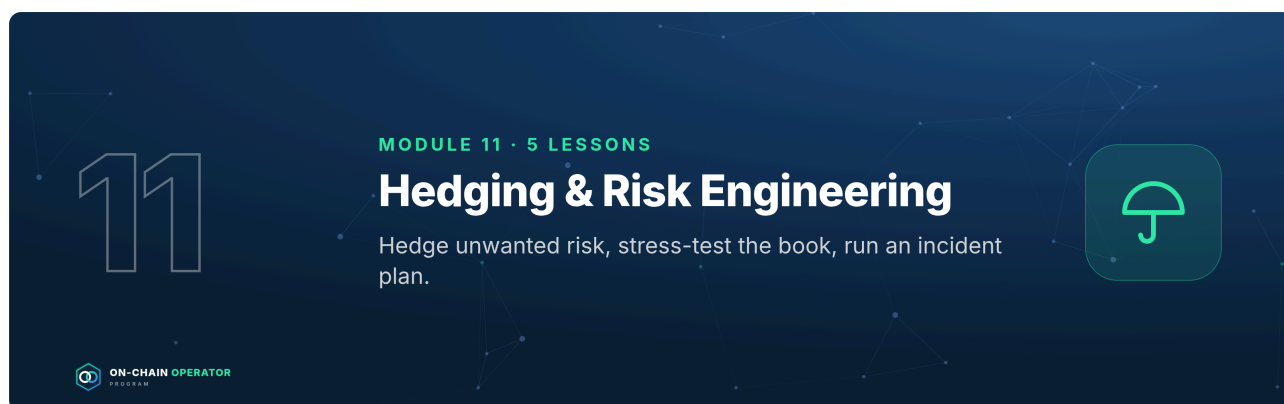
Q: 3. Why might a discount be a warning, not an opportunity?

Answer: It may reflect a real problem (slashing, exploit, insolvency).

Module 10 practical

1. Price three live PTs with `defi_calc.py pt` and choose the one whose maturity fits your liquidity ladder.
2. Paper-trade a basis position: record entry basis, margin, and the rally it would survive.
3. Write the "this position is short ____" sentence for every strategy in this module.
4. Size a concentrated LP range from a real asset's weekly volatility and write its rebalance rule.

Module 11 — Hedging & Risk Engineering



Outcome: hedge the risks you don't want, stress-test the portfolio, and have an incident plan ready. Stage 4 · Strategist. Prerequisites: Modules 1–10. Educational content only. Not financial advice. Figures are illustrative.

Lesson 11.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Risk you don't want can often be hedged, insured or engineered away, at a cost. This module covers hedging, cover, liquidation protection, stress testing and what to do in the first hour of a crisis.

Words you'll need

Term	Meaning
Hedge	a position that offsets another's risk
Protective put	an option capping downside below a price
Cover	on-chain insurance for defined events
Stress test	a what-if shock on your book
Incident	an exploit, depeg or compromise

Before you start

☐ Modules 0–10

Your first safe step

Write down the single event that would hurt your portfolio most, and what you'd do in the first 10 minutes.

The mastery ladder

Level	You can...
Beginner	Knows the hedge tools and what they cost
Practitioner	Hedges one real exposure and runs the five stress tests
Master	Runs a stress-tested book with a rehearsed incident playbook

You've mastered this module when...

...your book survives all five stress scenarios and your incident playbook is saved offline.

Lesson 11.1 — Hedging price exposure with perps and options

Objective

Reduce price exposure deliberately, and know what each hedge costs.

Explanation

- **Perp hedge:** short a perpetual future against spot you hold. Cheap to open; pays or earns funding; needs margin; the hedge ratio is your choice.
- **Protective put:** buy the right to sell at a strike. Costs a premium, and your loss is capped below the strike.
- **Collar:** buy a put and sell a call to pay for it. Cheaper protection in exchange for capped upside.
- A hedge **costs** something (premium, funding, capped upside). Hedge the risk you can't afford, not every risk.

Worked example

You hold **10 ETH at \$3,000** (\$30,000).

- **50% perp hedge** (short 5 ETH): ETH falls 30% → spot -\$9,000, short +\$4,500 → **-\$4,500** (-15%) instead of -30%.
- **Protective put:** buy a 30-day \$2,700 put for 2% (\$60/ETH). Worst case per ETH = \$300 drop to the strike + \$60 premium = **\$360 (12%)**, however far ETH falls in that month.

Checklist

- ☐ Hedge target stated (which risk, how much)
- ☐ Cost computed (premium, funding, capped upside)
- ☐ Margin buffer for perp hedges set

Quiz

Q: 1. What does a protective put cap?

Answer: Your loss below the strike, for the option's term, at the cost of the premium.

Q: 2. What does a collar trade away?

Answer: Upside above the call strike, to pay for the put.

Q: 3. 20 ETH, 25% perp hedge, ETH -40%. Net loss if ETH was \$3,000?

Answer: Spot -\$24,000, short +\$6,000 → -\$18,000.

Lesson 11.2 — Depeg, protocol and smart-contract cover

Objective

Evaluate on-chain cover products as insurance: what they pay, when, and what they don't.

Explanation

Cover protocols sell protection against defined events (a protocol hack, a depeg beyond a threshold). Read: **what's covered** (exact wording), **exclusions**, **claim process** (who decides, how long), **payout asset**, and the cover provider's **own** solvency and contract risk.

Worked example

Cover \$50,000 in a lending protocol at 2.5%/yr = **\$1,250/yr**. If your risk-adjusted yield there is 4.5%, cover takes more than half of it. Worth it for a large, concentrated position; often not for a small, diversified one.

Checklist

- ☐ Cover wording and exclusions read
- ☐ Claim process and payout asset understood
- ☐ Cost compared with the yield it protects

Quiz

Q: 1. Name two things to check in cover terms.

Answer: Any two: covered events, exclusions, claim process, payout asset, provider solvency.

Q: 2. Cover costs 3%/yr on a 5% yield. Net?

Answer: About 2%.

Q: 3. Does cover remove all risk?

Answer: No. The cover provider has its own claim, solvency and contract risks.

Lesson 11.3 — Liquidation protection: buffers, alerts and automated deleveraging

Objective

Make liquidation practically impossible with buffers, alerts and pre-computed repayment.

Explanation

Three layers: **buffer** (HF $\geq 2-2.5$ at entry) → **alerts** (HF 2.0 and 1.7, Module 14.1) → **action** (repay, add collateral, or automated deleveraging tools that repay debt when HF crosses a trigger; these add their own contract and permission risk). **Pre-compute** the repayment needed to restore your target HF:

$$\text{debt to keep} = \text{collateral value} \times \text{LT} \div \text{target HF}.$$

Worked example

10 ETH falls to \$2,400 → collateral \$24,000, LT 0.8, debt \$12,000 → **HF 1.6**. To restore HF 2.5: debt must be $24,000 \times 0.8 \div 2.5 = \$7,680$ → **repay \$4,320** from the reserve.

Checklist

- ☐ Alerts at HF 2.0 and 1.7
- ☐ Repayment to restore target HF pre-computed
- ☐ Reserve covers that repayment

Quiz

Q: 1. Collateral \$40,000, LT 0.8, target HF 2. Max debt?

Answer: \$16,000.

Q: 2. What do automated deleveraging tools add?

Answer: Their own contract and permission risk.

Q: 3. Why pre-compute the repayment?

Answer: So you act immediately, not while calculating under stress.

Lesson 11.4 — Stress-testing a portfolio

Objective

Run standard shock scenarios on your book and fix what fails before markets test it.

Explanation

Scenarios every operator runs: **crypto -50% in a day** · **a stablecoin depegs 10%** · **borrow rates spike to 20%** · **your largest protocol is hacked** · **your main L2 halts for 48h**. For each: loss, liquidations, liquidity, and whether the payout policy survives.

Worked example — \$200,000 book

Holdings: ETH spot \$60k · LST collateral \$60k with \$12k debt · stable lending \$50k (\$25k each in two protocols) · ETH/USDC LP \$20k · PT stable \$10k.

Scenario	Impact	Passes?
ETH -50%	Spot -\$30k; collateral -\$30k (HF 4.0 → 2.0, no liquidation); LP \$20k → \$14.1k (-\$5.9k) → -\$65.9k (-32.9%)	Survives, but the payout must fall
One stablecoin -10%	-\$2.5k on \$25k	Yes
Borrow rate 20%	Interest on \$12k rises from ~\$600 to \$2,400/yr	Yes: covered by reserve
One lending protocol hacked (total loss)	-\$25k (-12.5%)	Survives; cap was 12.5%
L2 halted 48h	Positions frozen; the reserve on another chain covers needs	Yes, <i>if</i> the reserve isn't on the same L2

Checklist

- ☐ Five scenarios run on the current book
- ☐ Every "fail" has a fix (smaller cap, bigger buffer, move the reserve)
- ☐ Re-run quarterly and after big changes

Quiz

Q: 1. Why did the collateral position survive -50%?

Answer: It started at HF 4.0; halving the price took it to 2.0.

Q: 2. What does the L2-halt scenario test?

Answer: Whether your liquidity depends on a single chain.

Q: 3. What's the output of a stress test?

Answer: Fixes: smaller caps, bigger buffers, better-placed reserves.

Lesson 11.5 — Incident response: the first 60 minutes

Objective

Know exactly what to do when a protocol you use is exploited, a stablecoin depegs, or your wallet is compromised.

Explanation — the 60-minute playbook

0–5 min: Verify. Official channels only (bookmarked X/Discord/status page, security firms). Don't click links in DMs, and don't sign anything "to protect funds". **5–20 min: Contain.**

- *Protocol exploit:* withdraw if still possible and safe; revoke approvals to affected contracts.
- *Depeg:* follow your written peg rule (e.g. exit below 0.99 for X hours); don't panic-sell far below it on a thin book.
- *Wallet compromise:* move remaining assets to a **fresh** wallet (new seed, clean device); revoke approvals; assume the old seed is burned.

20–60 min: Stabilise. Check HFs across all positions (prices may be moving), top up from the reserve, and journal every action with timestamps. **After: Review.** What was the signal, what worked, and what changes in the caps and the register?

Worked example

A lending protocol you use announces a paused market after an exploit. You verify on its bookmarked status page, see withdrawals still work on unaffected markets, withdraw your USDC, revoke the pool approval, check your other HFs, and journal it. Total time: 25 minutes. Loss: none. Because the steps were written in advance.

Checklist

- ☐ Incident playbook printed/saved offline
- ☐ Official channels bookmarked for every protocol I use
- ☐ A clean "fresh wallet" procedure written

Quiz

Q: 1. First thing to do in an incident?

Answer: Verify through official, bookmarked channels.

Q: 2. Someone offers a link to "rescue your funds". What do you do?

Answer: Ignore it. It's a common scam during incidents.

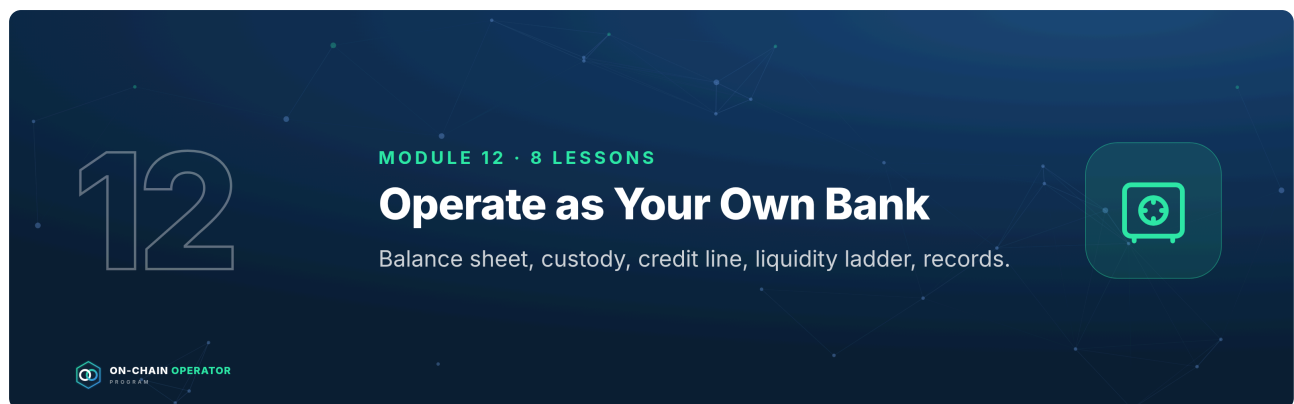
Q: 3. Wallet compromised: can you keep using the seed?

Answer: No. Move to a fresh wallet with a new seed; the old one is burned.

Module 11 practical

1. Price one hedge (perp and put) for your largest price exposure.
2. Run the five stress scenarios on your book and write the fixes.
3. Write and save your incident playbook, with bookmarks.

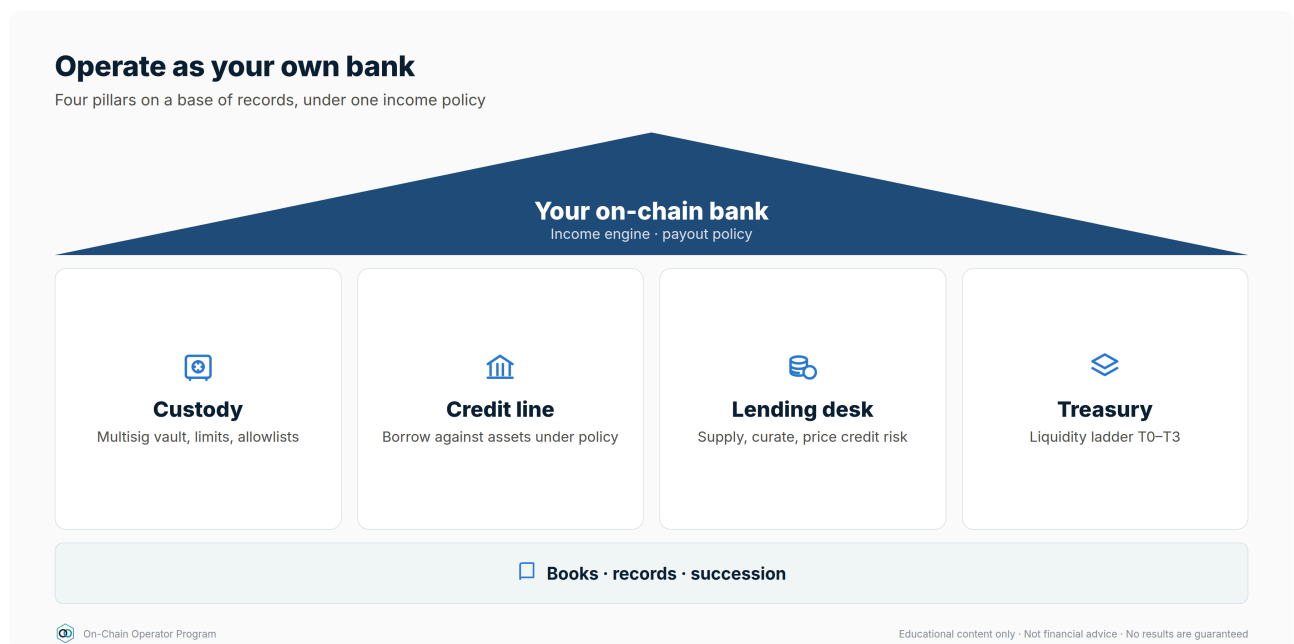
Module 12 — Operate as Your Own Bank



Outcome: run your crypto the way a bank runs its book: a balance sheet, a custody policy, a credit line, a liquidity ladder, a lending desk and records someone else could follow. Stage 5 · Operator.

Prerequisites: Modules 1–11. Educational content only. Not financial, tax or legal advice. Figures are illustrative.

A bank does four things: it **keeps assets safe**, **lends**, **borrow**s, and **manages liquidity** so it can always meet what it owes. DeFi lets you do all four yourself, with no one to call when something goes wrong. This module turns that into written policy.



Lesson 12.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

A bank keeps assets safe, lends, borrows and always has cash for what it owes. With DeFi you can run those four jobs yourself, with written policies, because there's nobody to call when something goes

wrong.

Words you'll need

Term	Meaning
Balance sheet	assets, liabilities and equity
Liquidity ladder	reserves tiered by how fast you can access them
Credit line	borrowing against assets under a policy
Runway	months your reserve can cover
Succession	how others recover your assets if you can't

Before you start

☐ Modules 0–11

Your first safe step

Fill in the balance sheet worksheet (W6) with rough numbers: equity, LTV and runway.

The mastery ladder

Level	You can...
Beginner	Has a balance sheet and a basic custody split
Practitioner	Runs written custody, credit, ladder and lending policies
Master	Operates a complete bank with books, tax-ready records, succession and quarterly reviews

You've mastered this module when...

...a stranger could run your bank from your written policies, with no keys in any document.

Lesson 12.1 — Your balance sheet

Objective

Build a personal on-chain balance sheet and read the three numbers that matter: equity, LTV and liquidity runway.

Explanation

- **Assets:** everything you own, at market value: collateral, yield positions, reserves.
- **Liabilities:** everything you owe: loans, margin, any payables.
- **Equity** = assets – liabilities. This is what's actually yours.
- **LTV** = debt ÷ collateral. Your bank's leverage.
- **Liquidity runway** = liquid reserve ÷ monthly obligations (spending + interest).

Banks fail from **liquidity** (can't pay today) more often than from **solvency** (owe more than they own). Track both.

Worked example

\$300,000 collateral, \$60,000 debt at 5%, \$40,000 stablecoin reserve, \$5,000/month spending:

```
defi_calc.py bank --collateral 300000 --debt 60000 --borrow-apy 5 --reserve 40000 --monthly-spend 5000
```

- Equity **\$280,000** · LTV **20%** · health factor **4.0**
- Obligations \$5,250/month (\$250 of it interest) → reserve covers **7.6 months**
- Policy checks: $LTV \leq 30\%$ ✓ · $HF \geq 2$ ✓ · $reserve \geq 6 \text{ months}$ ✓

Checklist

- ☐ Balance sheet updated weekly
- ☐ Equity, LTV and runway tracked over time
- ☐ Policy limits written (max LTV, min HF, min runway)

Quiz

Q: 1. Assets \$500k, debt \$100k. Equity?

Answer: \$400k.

Q: 2. Reserve \$30k, obligations \$6k/month. Runway?

Answer: 5 months, below a 6-month policy.

Q: 3. Why track liquidity separately from equity?

Answer: You can be solvent and still be forced to sell at the worst time if you can't meet obligations today.

Lesson 12.2 — Custody architecture: vault, multisig and spending policies

Objective

Design custody so that no single lost device, stolen key or mistaken signature can drain the bank.

Explanation

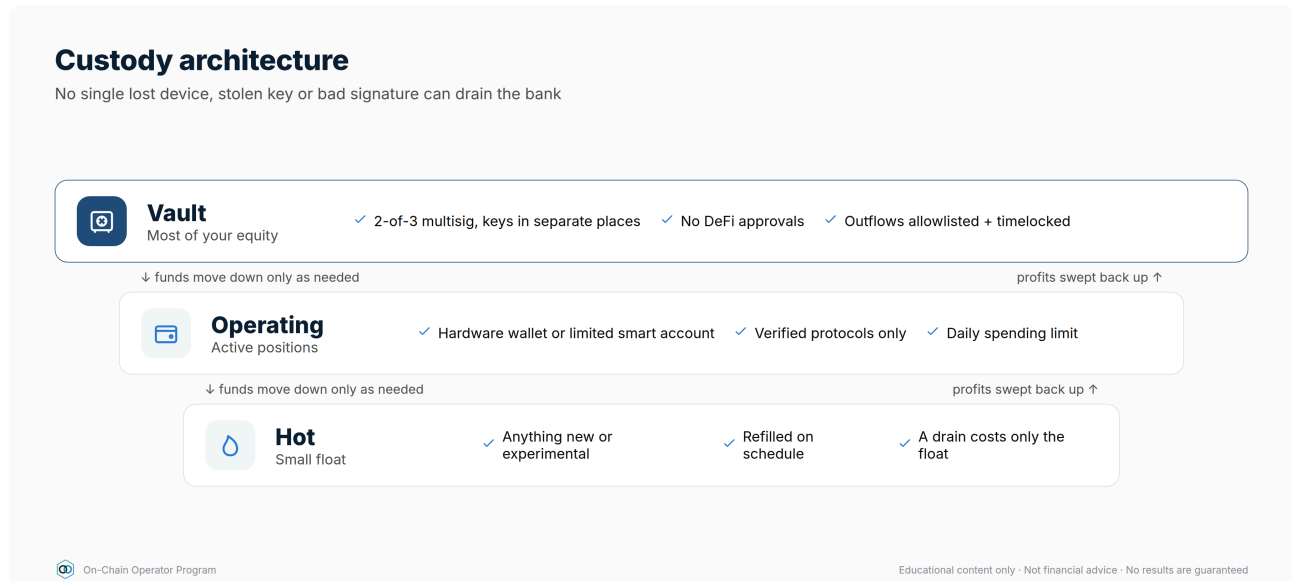
Scale Module 1's three wallets into a bank-grade setup:

Tier	Holds	Control	Rules
Vault	Most of your equity	Multisig , e.g. 2-of-3 hardware keys in separate locations	No DeFi approvals. Moves only to the Operating tier, to allowlisted addresses
Operating	Active positions	Hardware wallet or a smart account with limits	Verified protocols only. Spending limit per day

Tier	Holds	Control	Rules
Hot	Small float	Hot wallet	Anything new or experimental. Refilled on schedule

Smart-account features to use where available:

- **Spending limits** per day/week
- **Allowlists** (it can only send to addresses you pre-approved)
- **Timelocks** on large moves (a delay you can cancel if it wasn't you)
- **Recovery paths** agreed and tested in advance



Worked example

Vault: 2-of-3 multisig, with key A at home, key B in a safe-deposit box and key C with a trusted person or a professional co-signer. Losing any one key is survivable; stealing any one key is useless. **Test it:** move \$10 through every path once a quarter.

Checklist

- ☐ Vault is multisig; keys in separate physical locations
- ☐ Operating tier has a daily limit
- ☐ Allowlist on vault outflows
- ☐ Quarterly recovery test done and logged

Quiz

Q: 1. Why 2-of-3 rather than 1-of-1?

Answer: One key can be lost or stolen without losing the funds or giving an attacker control.

Q: 2. What does a timelock buy you?

Answer: Time to notice and cancel an unauthorised large move.

Q: 3. Why test recovery with a small amount?

Answer: A recovery plan you've never run is an assumption. Testing proves every key and path works.

Lesson 12.3 — The credit line: borrowing like a bank client

Objective

Use your assets as collateral for liquidity without selling, under a written credit policy.

Explanation

Wealthy clients rarely sell appreciating assets to raise cash; they borrow against them. DeFi money markets offer the same: deposit collateral, borrow stablecoins, repay when it suits you, with no credit check. **The collateral is liquidated if you breach the threshold**, and nobody calls first.

A credit policy (write yours):

- Max LTV **25–30%** on volatile collateral (liquidation is often near 80%+)
- Health factor floor **2.5**; act at **2.0**
- Prefer collateral whose **yield covers the interest** (strategy #25)
- Consider **fixed-rate** borrowing when a rate spike would force a sale (strategy #22)
- **Repayment source** named before borrowing (income, reserve, maturing PT)
- Tax treatment of borrowing varies by country: **check with a tax professional**

Worked example

\$200,000 of liquid-staked ETH earning 3.2% (\$6,400/yr). Borrow **\$40,000** stablecoins at 5% (\$2,000/yr).

- Yield covers interest **3.2×**. LTV 20%. HF **4.0** at a 0.8 threshold.
- ETH would have to fall **~75%** before liquidation (HF falls in proportion to price).
- Defence plan: at HF 2.0, repay from reserve; at HF 1.7, sell part of the collateral deliberately rather than let a liquidator do it at a penalty.

Checklist

- ☐ Credit policy written: max LTV, HF floor, action levels
- ☐ Repayment source named
- ☐ Alerts set at the action levels (Module 14.1)
- ☐ Fixed vs variable decision made on purpose

Quiz

Q: 1. Why borrow against assets instead of selling them?

Answer: You keep the asset and its upside (and in some jurisdictions avoid a taxable sale), at the cost of interest and liquidation risk.

Q: 2. HF is 4.0. Roughly how far can collateral fall before liquidation?

Answer: About 75%, since HF scales with collateral price (HF 1.0 at a quarter of the current price).

Q: 3. What should happen at your action level?

Answer: Repay or add collateral per the written plan. Sell deliberately before a liquidator does it at a penalty.

Lesson 12.4 — Treasury and the liquidity ladder

Objective

Structure reserves so money is available when it's needed, earning a return at every tier.

Explanation

Banks match the timing of assets to the timing of what they owe. Your ladder:

Tier	Access	Holds	Size (example: \$5,000/month spend)
T0 Instant	Seconds	Stablecoins in the operating wallet	1 month: \$5,000
T1 Same day	Hours	Blue-chip stablecoin lending, split across 2 protocols	5 months: \$25,000
T2 Term	Scheduled	PTs maturing on dates you'll need the money (Lesson 10.1)	Next 6–12 months of planned spending
T3 Growth	Days–weeks	Strategy positions, LPs, staking	The rest, within caps

The liquidity ladder

Match the timing of your assets to the timing of what you owe

T0	Instant Seconds	Stablecoins in the operating wallet	1 month · \$5,000
T1	Same day Hours	Blue-chip lending, split across 2 protocols	5 months · \$25,000
T2	Term Scheduled	PTs maturing when money is needed	Next 6–12 months of plans
T3	Growth Days–weeks	Strategy positions, LPs, staking	The rest, within caps

Refill downward on a schedule. Never fund a T0 need by selling T3 in a bad market. Example: \$5,000/month spending.

Refill downward on a schedule: T1 tops up T0 monthly; maturing PTs top up T1. Never fund a T0 need by selling a T3 position in a bad market. That's what the ladder exists to prevent.

Worked example

T0 \$5,000 + T1 \$25,000 = **6 months** instantly available. T2 holds three PTs maturing in 3, 6 and 9 months, each sized for 3 months of spending. A -50% market day touches none of it.

Checklist

- ☐ T0 + T1 $\geq$ 6 months of obligations
- ☐ T2 maturities matched to planned spending
- ☐ Monthly refill schedule set
- ☐ T3 never used for day-to-day needs

Quiz

Q: 1. Why hold T0 in plain stablecoins earning little?

Answer: Instant, certain access. It's the tier that pays today's bills without selling anything.

Q: 2. What's T2 for?

Answer: Known future spending, with maturities matched to when the money is needed and a fixed return locked in.

Q: 3. What does the ladder prevent?

Answer: Forced selling of growth positions at bad prices to meet short-term needs.

Lesson 12.5 — Being the lender: supplying, curating and pricing credit risk

Objective

Act as the lending side of the bank: choose markets, price the risk, and know when to pull liquidity.

Explanation

When you supply to a money market, you're the bank's depositor, and effectively its credit desk. Your return: $\text{supply APY} \approx \text{borrow APY} \times \text{utilisation} \times (1 - \text{reserve factor})$.

Modern designs split lending into **isolated markets** (one collateral, one loan asset, one oracle, one liquidation LTV). **Curated vaults** spread deposits across several of these; the curator decides the allocation.

What a lender must check:

- **Collateral quality and liquidation LTV:** the higher the LTV, the thinner the buffer before bad debt
- **Oracle:** what prices the collateral, and can it be manipulated? (Module 5.3)
- **Utilisation:** near 100% means you may not be able to withdraw
- **Curator:** track record, how allocations are changed, timelocks
- **Risk-adjusted return**, not headline

Worked example

Vault A: 7% headline, exposure to newer collateral; you assume a 2%/yr chance of a loss event losing half the deposit → **6.0%** risk-adjusted. Vault B: 4.5% headline, blue-chip collateral; 0.5%/yr, total loss assumed → **4.0%**. `defi_calc.py expected --yield-apy 7 --loss-prob 0.02 --lgd 0.5`

A pays more after the haircut, *if* your loss estimates are honest. Split between them, and cap A lower because its tail is fatter and less known.

Checklist

- ☐ Every market's collateral, oracle and LTV listed
- ☐ Loss probability and loss-given-default written down (your assumptions)
- ☐ Utilisation alert (e.g. > 90%)
- ☐ Per-vault and per-curator caps

Quiz

Q: 1. Borrow APY 10%, utilisation 70%, reserve factor 10%. Supply APY?

Answer: 6.3%.

Q: 2. Why is utilisation at 100% a lender's problem?

Answer: All liquidity is lent out, so withdrawals wait until borrowers repay or new deposits arrive.

Q: 3. What's a curated vault's extra risk?

Answer: The curator's allocation decisions and permissions, on top of each market's risk.

Lesson 12.6 — Books, records and succession

Objective

Keep books a stranger could follow, and make sure your family could recover everything if you couldn't.

Explanation

Books (update weekly from your journal, Module 8):

- Every position: protocol, chain, contract, amount, entry date and price, cost basis
- Every loan: collateral, debt, rate, HF, action levels
- Every approval granted, and when it was revoked
- Income received, by source and date
- Keep exportable records for tax. Rules differ by country, so **use a crypto-aware tax professional**.

Succession: self-custody means that if you're gone and nobody can sign, the money is gone too.

- A **letter of instruction** that explains the setup (wallets, multisig, where the documents are, who to contact), **stored separately from any seed or key**
- Multisig with a trusted co-signer or professional service, so recovery doesn't depend on one person
- Legal documents (will, powers of attorney) that reference the digital assets: **use a lawyer**

- **Annual drill:** walk a trusted person through the letter without revealing secrets

Worked example

A 2-of-3 vault: you hold two keys (home + safe-deposit box); a professional co-signer holds the third under an agreed recovery process. Your letter of instruction, stored with your will, explains how your executor works with the co-signer. No single document contains enough to steal the funds.

Checklist

- ☐ Books current within a week
- ☐ Tax records exportable; professional engaged
- ☐ Letter of instruction written and stored apart from keys
- ☐ Annual succession drill done

Quiz

Q: 1. Why keep the letter of instruction apart from seeds and keys?

Answer: So finding the letter isn't enough to steal the funds.

Q: 2. What's the risk of single-key self-custody for your family?

Answer: If you can't sign and no one else can, the assets are unrecoverable.

Q: 3. What goes in the books for each loan?

Answer: Collateral, debt, rate, health factor and your action levels.

Lesson 12.7 — Tax, regulation and compliance awareness (new)

Objective

Know which on-chain actions may have tax or legal consequences, keep the records to handle them, and know when to get professional help.

Explanation

This lesson is awareness, not advice. Rules differ by country and change. Use a crypto-aware tax professional and, for legal questions, a lawyer.

- **Events that may be taxable** in many countries: selling crypto for cash, swapping one token for another, spending crypto, and receiving income (interest, staking rewards, LP fees, airdrops). Some treat borrowing, wrapping, LP deposits or bridging differently. Ask.
- **Cost basis:** what you paid, including fees. Methods (e.g. FIFO, specific identification) depend on local rules.
- **Records:** every buy, sell, swap, transfer, fee and income event, with dates, amounts and prices (W1 records sheet). On-chain data can be exported, but self-transfers must be labelled or they may look like disposals.
- **Regulation:** use services legally available where you live; don't use tools to get around geo-restrictions; never interact with sanctioned addresses or services; exchanges must run KYC/AML

checks, and some DeFi front ends block certain regions.

- **Reporting:** some countries require reporting of foreign accounts or digital assets. Ask your professional.

Worked example

In one year: bought ETH, swapped half to USDC, lent USDC for interest, received an airdrop, bridged to an L2 and moved funds between your own wallets. For your tax professional, you'd prepare: the records sheet, labelled self-transfers (not disposals), interest and airdrop income with dates and values, and exchange statements. **They** tell you what's taxable where you live.

Checklist

- ☐ Records sheet complete, with self-transfers labelled
- ☐ Crypto-aware tax professional engaged
- ☐ Only legally available services used; no geo-restriction workarounds
- ☐ Sanctions awareness: never interact with sanctioned addresses or services

Quiz

Q: 1. Name three events that are often taxable.

Answer: Any three: selling for cash, swapping tokens, spending crypto, receiving interest, staking rewards, LP fees or airdrops.

Q: 2. Why label self-transfers?

Answer: So moves between your own wallets aren't mistaken for disposals.

Q: 3. Who decides what's taxable for you?

Answer: Your local rules, applied by a qualified tax professional.

Lesson 12.8 — Institutional-grade custody: MPC, custodians and policy engines (expert)

Objective

Know the custody options institutions use, and when they make sense for a large personal or family book.

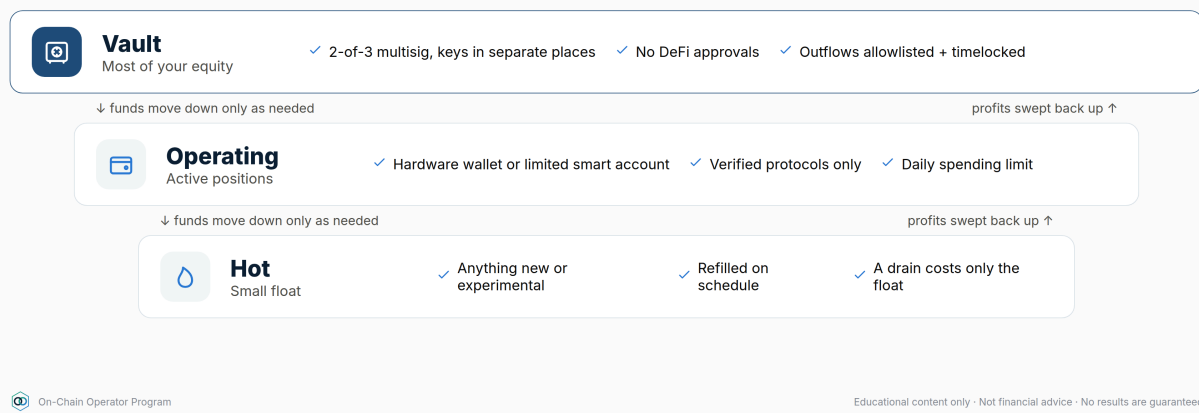
Explanation

- **Multisig** (12.2): several keys, rules enforced **on-chain**, transparent and verifiable.
- **MPC (multi-party computation)**: one key is split into shares held by different devices or parties; signatures are created together without ever assembling the key. The rules are enforced **off-chain** by the provider's software. It works on any chain, but you depend on the vendor and can't verify the policy on-chain.
- **Qualified custodians**: regulated firms that hold assets for clients. Strong legal protections in some jurisdictions; you give up self-custody and on-chain flexibility.

- **Policy engines:** approval workflows (who can approve what, above which amount), allowlists, time windows and velocity limits, available in both institutional MPC platforms and smart-account setups.
- **Hybrid:** many large holders use a custodian or MPC for long-term reserves and multisig/smart accounts for active DeFi.

Custody architecture

No single lost device, stolen key or bad signature can drain the bank



Worked example

A \$5M family book: cold reserves (\$3M) with a qualified custodian (legal title, insurance terms read); active DeFi (\$1.5M) in a 2-of-3 multisig with a policy of two approvers above \$50,000 and an allowlist; hot float (\$20,000) in a limited smart account. The succession plan (12.6) covers all three.

Checklist

- ☐ Custody option chosen per tier, with reasons
- ☐ Vendor/custodian terms, insurance and recovery read
- ☐ Policy engine rules written: approvers, thresholds, allowlists

Quiz

Q: 1. MPC vs multisig: where are the rules enforced?

Answer: MPC: off-chain by the provider's software. Multisig: on-chain by the contract.

Q: 2. What do you give up with a qualified custodian?

Answer: Self-custody and on-chain flexibility.

Q: 3. What is a policy engine?

Answer: Rules for approvals, thresholds, allowlists and limits on transactions.

Module 12 practical: your bank's founding documents

1. Balance sheet with equity, LTV and runway (`defi_calc.py bank`).
2. Custody policy: tiers, multisig setup, limits, allowlists, recovery test log.
3. Credit policy: max LTV, HF floor, action levels, repayment source.

4. Liquidity ladder: T0–T3 sizes and refill schedule.
5. Lending policy: markets, caps, risk assumptions.
6. Books template and letter of instruction (stored separately).

Module 13 — The Income Engine



Outcome: design an income portfolio, measure expected income after expected losses, and set a payout you can sustain. Stage 5 · Operator. Prerequisites: Modules 1–12. Educational content only. Not financial or tax advice. No income is guaranteed. Loss probabilities are your own assumptions; figures are illustrative.

The operator's income rule: plan on **expected** income (yield minus expected losses), pay out **less** than that, and never pay out of principal.

Lesson 13.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

Income from DeFi should be planned like a pension, not chased like a jackpot: build it from durable sources, subtract the losses you should expect, and pay yourself less than that.

Words you'll need

Term	Meaning
Durability	how long and reliably a source tends to pay
Risk-adjusted yield	yield minus expected losses
LGD	how much you lose if a loss event happens
Payout ratio	share of expected income you take out
Principal	the capital itself, never spent as income

Before you start

☐ Modules 0–12

Your first safe step

Take one position and compute its risk-adjusted yield with `defi_calc.py expected`.

The mastery ladder

Level	You can...
Beginner	Ranks income sources by durability
Practitioner	Builds an income portfolio with risk-adjusted expected income
Master	Runs a payout policy that survives bad years, with annual reviews and scaling rules

You've mastered this module when...

...your payout is set below risk-adjusted expected income and never draws on principal.

Lesson 13.1 — Income sources ranked by durability

Objective

Rank income sources by how long and how reliably they're likely to pay, not by their headline rate.

Explanation

Source	Typical driver	Durability	Variability
Blue-chip stablecoin lending	Borrower demand	High	Medium: moves with demand
Staking / LSTs	Network security rewards	High	Low–medium
Fixed-rate PTs to maturity	Locked at purchase	Fixed for the term	None until maturity
Curated lending vaults	Borrower demand in isolated markets	Medium	Medium
LP fees (major pairs)	Trading volume	Medium	High
Funding / basis carry	Market positioning	Low–medium	High: can turn negative
Options premium	Volatility	Medium	High
Incentives / points	Protocol budgets	Low	Very high

A durable income engine is built mostly from the top of this table. The bottom rows are **boosters** you size small and expect to switch off.

Checklist

- ☐ Every income position tagged with its source and durability
- ☐ Core income from high-durability sources

Quiz

Q: 1. Why is funding carry low durability?

Answer: It depends on market positioning and can turn negative for long periods.

Q: 2. Which source gives a known rate for a set term?

Answer: Fixed-rate PTs held to maturity.

Q: 3. How should incentive income be treated?

Answer: As a temporary booster, sold on a schedule and not relied on.

Lesson 13.2 — Risk-adjusted yield: subtracting expected losses

Objective

Convert every headline yield into an expected yield after the losses it statistically carries.

Explanation

$\text{risk-adjusted yield} = \text{headline} - (\text{annual loss probability} \times \text{loss given default}) - \text{costs}$

You don't know the true loss probability, so **write down an assumption and be conservative**. Newer protocols, more dependencies and higher leverage deserve higher numbers. The point isn't precision; it's making you compare positions on the same footing.

Worked example

Position	Headline	Assumed loss prob.	LGD	Risk-adjusted
New protocol vault	9.0%	3%/yr	100%	6.0%
Blue-chip lending	4.5%	0.5%/yr	100%	4.0%

```
defi_calc.py expected --yield-apy 9 --loss-prob 0.03
```

The 9% vault still wins on expectation, but by 2 points, not 4.5. Its bad outcome is also far worse, so size it smaller (Lesson 13.3).

Checklist

- ☐ Loss probability and LGD written for every position
- ☐ Decisions made on risk-adjusted, not headline

Quiz

Q: 1. 12% headline, 5%/yr loss probability, 60% LGD. Risk-adjusted?

Answer: $12 - 3 = 9\%$.

Q: 2. Why assume conservatively?

Answer: Losses cluster in stress, when many positions fail together; optimistic assumptions overstate income.

Q: 3. Two positions, same risk-adjusted yield. Which gets more capital?

Answer: The one with the smaller and better-understood worst case.

Lesson 13.3 — Building the income portfolio

Objective

Assemble an income portfolio and calculate its blended risk-adjusted yield and expected income.

Explanation

Build in layers: **core** (high-durability, 60–80%), **term** (fixed PTs matched to the liquidity ladder), **boosters** (carry, options, LP; small). Apply Module 8's caps: per protocol, per chain, per stablecoin issuer.

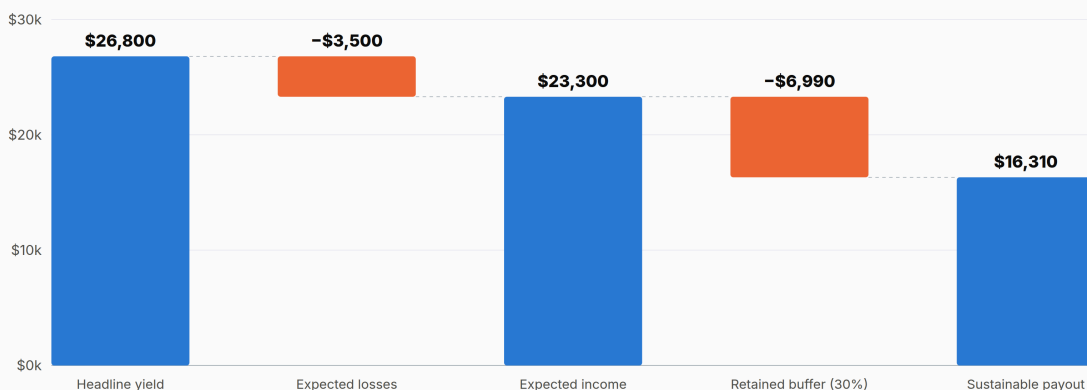
Worked example — \$500,000 income portfolio

```
defi_calc.py income --pos "Stable lending A:100000:5:0.005:1" --pos "Stable lending  
B:100000:4.5:0.005:1" --pos "LST staking:150000:3.2:0.01:0.5" --pos "PT fixed  
stable:100000:8:0.02:0.5" --pos "Funding carry:50000:9:0.05:0.3"
```

Position	Amount	Yield	Expected loss	Net	Income/yr
Stable lending A	100,000	5.0%	0.5%	4.5%	4,500
Stable lending B	100,000	4.5%	0.5%	4.0%	4,000
LST staking	150,000	3.2%	0.5%	2.7%	4,050
PT fixed stable	100,000	8.0%	1.0%	7.0%	7,000
Funding carry	50,000	9.0%	1.5%	7.5%	3,750
Total	500,000			4.66%	23,300

From headline yield to a sustainable payout

Illustrative \$500,000 income portfolio (Module 13). Loss rates are assumptions; income is not guaranteed.



Note: the LST line is priced in ETH, so its dollar value, and therefore the dollar income, moves with ETH. The rest is stablecoin-denominated.

Checklist

- ☐ Core $\geq$ 60% from high-durability sources
- ☐ Caps respected per protocol, chain and issuer
- ☐ Blended risk-adjusted yield calculated

Quiz

Q: 1. Which line contributes the most expected income, and why?

Answer: PT fixed stable (\$7,000): a high locked rate on a sizeable allocation, even after the haircut.

Q: 2. Why is the carry line only \$50,000?

Answer: It's a low-durability booster with a higher assumed loss rate.

Q: 3. What makes the LST income variable in dollars?

Answer: It's denominated in ETH, so the dollar value moves with the ETH price.

Lesson 13.4 — The payout policy: how much you can take out

Objective

Set a payout you can sustain through bad years, and the rules that change it.

Explanation

- **Pay out a share of *expected* income, not headline** (e.g. 70%).
- **Retain the rest** as a loss buffer; it absorbs the losses you assumed in 13.2.
- **Pay from the T0/T1 ladder** (Module 12.4), refilled by income, so payouts never depend on selling anything that day.
- **Never pay from principal.** If income falls short, the payout falls.
- **Review quarterly:** if realised income is below expected for two quarters, cut the payout to 70% of the new realised level.

Worked example

Expected income **\$23,300/yr** $\times$ 70% = **\$16,310/yr** paid out (**\$1,359/month**). **\$6,990/yr** retained. If a position fails as assumed, the buffer absorbs it and the payout continues.

Checklist

- ☐ Payout ratio written ($\leq$ 70–80% of expected income)
- ☐ Paid from the ladder, not from positions
- ☐ Quarterly review rule written
- ☐ "Never from principal" rule written

Quiz

Q: 1. Expected income \$40,000. Payout at 70%?

Answer: \$28,000/yr (~\$2,333/month).

Q: 2. What's the retained 30% for?

Answer: Absorbing expected losses and bad years without cutting into principal.

Q: 3. Realised income lags expected for two quarters. What happens?

Answer: Cut the payout to the policy share of the new realised level.

Lesson 13.5 — Scaling, compounding and the annual review

Objective

Grow the engine safely and review it like a bank's annual report.

Explanation

Compounding: reinvesting the retained share grows principal, and income with it. At a 4.66% risk-adjusted yield with 30% retained, principal grows ~**1.4%/yr** from retention alone. The payout grows at about the same rate: \$16,310 → ~\$17,480 after five years (before any new capital).

Scaling rules as the book grows:

- Caps are **percentages**, so position sizes rise, and so does market impact. Check pool depth and exit liquidity at the new size.
- Add independent protocols and issuers before adding size to existing ones.
- More size means more value at stake in custody: revisit Module 12.2.

Annual review (one page):

1. Balance sheet: start vs end equity, LTV, runway
2. Income: expected vs realised, by source
3. Losses and near-misses: what happened, what changed
4. Payout: paid vs policy
5. Risk assumptions: update loss probabilities from the year's incidents
6. Custody and succession drill: done?

Checklist

- ☐ Retained income reinvested per policy
- ☐ Exit liquidity checked at current size
- ☐ Annual review completed and filed with the books

Quiz

Q: 1. Yield 5%, 40% retained. Growth from retention alone?

Answer: ~2% a year.

Q: 2. Why check exit liquidity as the book grows?

Answer: A position that was easy to exit at \$50k may move the market at \$500k.

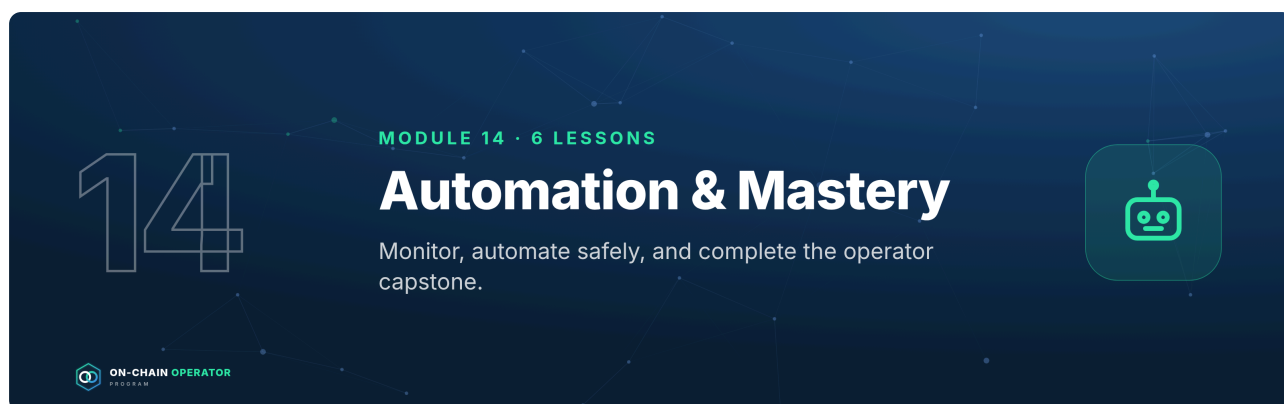
Q: 3. What should the annual review update?

Answer: Loss-probability assumptions, from the year's incidents and near-misses.

Module 13 practical

1. Tag every position by source and durability (13.1).
2. Write loss assumptions and compute risk-adjusted yields (13.2).
3. Build your income portfolio with `defi_calc.py income` (13.3).
4. Write your payout policy (13.4) and first annual-review template (13.5).

Module 14 — Automation & Mastery



Outcome: monitor and automate safely, run multisig operations, and complete the operator capstone. Stage 5 · Operator. Prerequisites: Modules 0–13. Educational content only. Not financial advice.

The automation rule: automate *watching* freely; automate *acting* only with limited permissions you can revoke, and never by handing over your keys.

Lesson 14.0 — Mastery Starter

New to this topic? Start here. It takes about 10 minutes and gets you from zero to ready for this module.

The 60-second version

At mastery, you watch everything automatically and act through rules and limited permissions, never by handing over keys. Then you prove it with the operator capstone.

Words you'll need

Term	Meaning
Alert	an automatic notification tied to an action
Keeper	an automated service that executes conditions
Session key	a limited, revocable permission
Change control	a written, delayed process for new risks
RPC	a connection that lets software read the chain

Before you start

☐ Modules 0–13

Your first safe step

Set one alert today: a health factor, a peg or any outflow from your vault wallet.

The mastery ladder

Level	You can...
Beginner	Has alerts mapped to actions
Practitioner	Automates one task with scoped permissions and writes signing procedures
Master	Builds read-only tools and passes the operator capstone

You've mastered this module when...

...you've passed the operator capstone.

Lesson 14.1 — Monitoring: dashboards, alerts and on-chain watchers

Objective

Set up monitoring so you hear about problems before they cost money.

Explanation

Watch four things:

1. **Positions:** health factors, LP ranges, pegs, utilisation of markets you lend to.
2. **Protocols:** governance proposals, queued upgrades (timelocks), incidents.
3. **Wallets:** any outgoing transaction or new approval from your vault/operating wallets.
4. **Market:** funding, large depegs, liquidations.

Tools: portfolio dashboards, protocol-native alerts, wallet-activity watchers, block-explorer address alerts, and official status/X accounts. Alerts should reach your **phone**.

Worked example — alert sheet

Alert	Threshold	Action (from your policies)
HF, credit line	< 2.0 / < 1.7	Repay per 11.3
Stablecoin peg	< 0.99 for 2h	Peg rule
Market utilisation	> 95% for 24h	Withdraw per kill rule
Vault wallet outflow	Any	Verify immediately (11.5)
Queued upgrade	Any, on protocols > 10% of book	Review within timelock

Checklist

- ☐ Alerts for all four categories, delivered to phone
- ☐ Every alert maps to a written action

Quiz

Q: 1. Why alert on any vault-wallet outflow?

Answer: The vault should almost never move; any movement could be a compromise.

Q: 2. Why watch queued upgrades?

Answer: The timelock is your window to exit before the change takes effect.

Q: 3. An alert with no action attached is...?

Answer: Noise. Every alert needs a written response.

Lesson 14.2 — Automation: keepers, bots and agents without handing over the keys

Objective

Automate routine actions with the least permission possible.

Explanation

- **Keeper / automation networks** execute pre-defined actions when conditions are met (e.g. repay when $HF < 1.8$, rebalance an LP range).
- **Smart-account permissions / session keys:** grant a bot the right to do *one thing* (e.g. repay debt on one protocol, up to \$X per day), revocable at any time.
- **AI agents and bots:** treat them like any other contract permission. They get scoped rights, never your seed phrase or unlimited approvals.
- **Test on a testnet or with tiny amounts first; log every automated action.**

Worked example

Automated repay: a session key allowed only to call `repay` on your lending position, from your operating wallet, capped at \$5,000/day, expiring in 30 days. Worst case if the bot is compromised: it repays your debt, which is annoying, not catastrophic.

Checklist

- ☐ Each automation has a scoped, capped, expiring permission
- ☐ No automation holds seed phrases or unlimited approvals
- ☐ Tested small; actions logged; revoke procedure known

Quiz

Q: 1. What's a session key?

Answer: A limited, revocable permission for a specific action.

Q: 2. Should an AI agent ever get your seed phrase?

Answer: No. Give it scoped, revocable permissions only.

Q: 3. Worst case of a well-scoped repay bot?

Answer: It repays debt within its cap. No funds leave your control.

Lesson 14.3 — Operating procedures: multisig signing, change control, reviews

Objective

Run your bank with procedures that prevent single-point mistakes.

Explanation

- **Multisig signing procedure:** every signer independently verifies the transaction (destination, amount, calldata) on their **own device screen**, not a shared screenshot.
- **Change control:** new protocol, new chain, new strategy, or a cap change is written up (thesis, risk register row, size) and waits 24h before execution.
- **Separation:** the person proposing a vault transaction isn't the only one approving it.
- **Review calendar:** weekly books · monthly register · quarterly stress test and recovery drill · annual review (Module 13.5).

Worked example

A 2-of-3 vault moves \$40,000 to the operating wallet. Signer 1 proposes; Signer 2 checks the destination against the allowlist and the amount against the ladder refill schedule on their hardware wallet screen, then signs. Logged in the books with both names.

Checklist

- ☐ Signing procedure written for every signer
- ☐ Change-control template and 24h rule
- ☐ Review calendar set

Quiz

Q: 1. Why verify on your own device screen?

Answer: Screens and screenshots can be faked; the hardware display shows what you're actually signing.

Q: 2. What is change control for?

Answer: Forcing new risks through thesis, register and a cooling-off period.

Q: 3. How often is the stress test re-run?

Answer: Quarterly, and after big changes.

Lesson 14.4 — Operator capstone and certification

Objective

Assemble your complete on-chain bank and have it assessed.

Explanation — the operator capstone

Submit one document (template in `07-program-operations.md`) containing:

1. **Balance sheet** with equity, LTV and runway (12.1)
2. **Custody policy** and recovery-test log (12.2)
3. **Credit policy** with action levels (12.3)
4. **Liquidity ladder** T0–T3 (12.4)
5. **Lending policy** with risk assumptions (12.5)
6. **Books template** and letter of instruction location (12.6; never the seed)
7. **Income portfolio** with risk-adjusted expected income (13.3)
8. **Payout policy** (13.4)
9. **Stress test**, five scenarios, with fixes (11.4)
10. **Incident playbook** (11.5), **alert sheet** (14.1) and **automation permissions** (14.2)

Certification levels (assessed against the rubric in `07-program-operations.md`):

- **Analyst:** passed the analyst capstone (Modules 6–7)
- **Operator:** passed the operator capstone
- **Operator with distinction:** operator capstone scored $\geq 90\%$, plus one quarter of books kept to standard

Checklist

- ☐ All ten sections complete
- ☐ Calculators used for every number
- ☐ Submitted (Live tier: booked for review)

Quiz

Q: 1. What must never appear in the capstone?

Answer: Seed phrases, private keys or anything that grants access.

Q: 2. Which lesson's output shows income is sustainable?

Answer: 13.4: the payout policy, based on risk-adjusted expected income.

Q: 3. What's needed for "with distinction"?

Answer: $\geq 90\%$ on the operator capstone plus a quarter of books kept to standard.

Lesson 14.5 — Building your own tools: reading contracts, data and simple scripts *(new)*

Objective

Read on-chain data directly and build simple read-only tools, so you're not dependent on any one dashboard.

Explanation

- **Reading without code:** block explorers' Read tabs show a contract's public state (e.g. a lending market's parameters, your position).
- **Public data APIs** (e.g. protocol analytics sites, explorers) return TVL, fees, yields and prices as JSON.
- **RPC:** a node endpoint that lets a script read chain state directly (balances, contract calls). Use a reputable provider; reading is free or cheap.
- **Scripts that only read** (never hold keys) can check health factors, pegs and utilisation, and send you alerts.
- **Rule:** tools that *read* are low-risk; tools that *sign* follow Lesson 14.2 (scoped, capped, revocable permissions). Never put a seed phrase or private key in a script, a file or an environment variable on a shared machine.

Worked example (read-only, pseudo-code)

```
every 15 minutes:
    position = lending_protocol.getUserAccountData(my_address)  # public read call
    if position.health_factor < 1.8: send_phone_alert("HF " + position.health_factor)
    peg = price_api.get("USDC")
    if peg < 0.99: send_phone_alert("USDC peg " + peg)
```

Uses only public reads, holds no keys, and can't move funds. Worst case if it breaks: you miss an alert, so keep your other alerts (14.1) as a backup.

Checklist

- ☐ I can read a contract's state on an explorer
- ☐ My monitoring scripts are read-only and hold no keys
- ☐ Backup alerts exist if my script fails

Quiz

Q: 1. Why prefer read-only tools?

Answer: They can't move funds, so a bug or breach costs little.

Q: 2. Where should a private key go in a script?

Answer: Nowhere. Use scoped permissions (14.2) if a tool must act.

Q: 3. What's an RPC endpoint?

Answer: A node connection that lets software read (and submit to) the blockchain.

Lesson 14.6 — Searchers, keepers and arbitrage bots: how they work (expert)

Objective

Understand the professional bot economy that shapes DeFi prices, and why competing in it is hard.

Explanation

- **Searchers** run bots that find and capture opportunities: **DEX arbitrage** (price gaps between pools/venues), **liquidations** (repay debt, take the bonus), **backruns** (trading right after a large swap), and, harmfully, sandwiches.
- **Keepers** perform maintenance jobs for protocols (liquidations, auction bids, rebalancing) for a fee.
- **The economics:** bots bid for inclusion by paying builders/validators (2.8). Competition drives profits towards zero; the winners have the lowest latency, best infrastructure, private order flow and capital.
- **Why it matters to you:** these bots are why pool prices track the market (and cause LVR, 2.7), why liquidations happen within seconds (3.3), and why tight slippage and protected routing matter (2.5).
- **For a learner:** build read-only monitors (14.5) and understand the mechanics. Running competitive bots is a professional engineering business with real losses from failed transactions and bugs.

Worked example

A liquidation worth a \$300 bonus (3.3) appears. Dozens of bots see it in the same block; the winner pays most of the \$300 to the block builder to be included first. The "profit" left after fees and gas might be a few dollars, and losing bots may still pay gas. That's why liquidations are near-instant, and why your buffer (not a liquidator's delay) is your protection.

Checklist

- ☐ I can name the main searcher strategies and their effect on me
- ☐ My protections (slippage, private routing, HF buffers) assume bots act within seconds

Quiz

Q: 1. Where does most of a competitive searcher's profit go?

Answer: To builders/validators, through bids for inclusion.

Q: 2. What do keepers do?

Answer: Paid maintenance for protocols: liquidations, auctions, rebalancing.

Q: 3. Why do liquidations happen so fast?

Answer: Competing bots race to capture the bonus.

Module 14 practical

Build your alert sheet, scope one automation, write your signing procedure, then complete the operator capstone.

11. Skills archive

Claude skills for building and running the Grid Bot Builder business and the new DeFi program on Whop. Everything lives in `.claude/skills/`, so Claude Code loads these skills automatically in any session opened on this repo.

Start here

Ask Claude: "Use whop-defi-program to plan the DeFi program." That skill walks through offer → curriculum → Whop setup → funnel → guardrails → launch, and calls the others as needed.

On-Chain Operator Program

Everything for the DeFi program is in one file: `ON-CHAIN-OPERATOR-PROGRAM.md`, also as [Word](#) and [PDF](#). All 107 lessons and 15 Mastery Starters are written (topic map: `programs/defi-program/09-mastery-map.md`); VSLs and module intro videos are in `programs/defi-program/video/`. To finish and launch it, paste `programs/defi-program/BUILD-PROMPT.md` into Whop's AI (Grok 4.6) with the files it lists, or run it in Claude Code.

Skills

Your business skills (custom)

Skill	What it does	Source
<code>whop-defi-program</code>	Master playbook: turn the 42-chapter DeFi module into a Whop product and wire it into the GBB funnel	New, built from Notion + Zapier
<code>defi-strategies</code>	30 DeFi strategies in 7 levels (up to fixed-rate, basis, options, credit lines, hedged yield, CDP minting, bribe markets, perp vaults, arbitrage and rates), plus <code>defi_calc.py</code> with 20 calculators including income portfolios and a personal balance sheet	Notion strategy library, expanded
<code>defi-due-diligence</code>	6-step protocol risk loop, before-signing checklist, logs to Notion tracker	Notion: DeFi & On-Chain module
<code>grid-bot-design</code>	Regime/range/spacing/fees/risk method + <code>grid_calc.py</code> calculator, logs to Notion checklist	Notion: Grid Bot Builder module
<code>gbb-new-lead</code>	Score, tag and email new leads	Zapier (archive copy)
<code>gbb-customer-onboarding</code>	Post-purchase onboarding + Elite Intel upsell	Zapier (archive copy)
<code>gbb-ad-performance</code>	Weekly CAC / ROAS / budget report	Zapier (archive copy)

Skill	What it does	Source
gbb-pipeline-check	Funnel stages, rates, stale leads	Zapier (archive copy)

Building blocks (from [anthropics/skills](#), Apache 2.0)

Skill	Use it for
mcp-builder	Building an MCP server, e.g. to expose Grid Bot / DeFi tools to members' AI agents
web-artifacts-builder	Interactive tools, e.g. a DeFi Command Centre or grid simulator for the Whop app
frontend-design	Sales pages and course UI that don't look generic
webapp-testing	Playwright testing for anything you build
skill-creator	Writing, testing and improving new skills in this archive

Notes

- **Zapier is the source of truth** for the four `gbb-*` skills. This repo is public, so internal IDs (Mailchimp list, Whop business ID, plan IDs, owner email) are `<PLACEHOLDERS>`. To keep real values locally, put them in `config.local.md`, which git ignores.
- **Pricing conflict to resolve:** `gbb-ad-performance` still describes a "Grid Bot Starter" (A\$47–97) and GBB at A\$497. The newer skills say \$997 and that there's no Starter tier.
- Provenance and licences: see `SOURCES.md`.

12. Build Prompt: On-Chain Operator Program (for Whop AI · Grok 4.6)

How to run it

1. Open **Whop's AI** (it runs on Grok 4.6).
2. **Attach the core files** from the GitHub repo `kyleashton88-cyber/claude`, branch `claude/grid-bot-builder-skills-bmrez2`, folder `programs/defi-program/`:
 - `01-offer-and-curriculum.md` (decisions + full curriculum)
 - `09-mastery-map.md` (every topic, where it's taught, at what level)
 - `10-video-production-plan.md` (every course video, with length and script file)
 - `06-whop-store-listing.md`, `07-program-operations.md` (includes the refund policy), `08-worksheets.md`
 - `assets/brand/brand-guide.png`, `assets/store/banner-1920x1080.png` (so the AI can see the brand)
 - `video/SCRIPTS.md` (VSL and intro scripts) and one example lesson script: `video-scripts/lessons/lesson-00-1.json`
3. Paste everything below the line as your first message.
4. **The lessons come in six section files** in `sections/`, one per stage. Attach one section per pass when the AI asks: `section-0-zero.md` → `section-1-foundations.md` → `section-2-practitioner.md` → `section-3-analyst.md` → `section-4-strategist.md` → `section-5-operator.md`. The matching lesson video scripts are in `video-scripts/lessons/` (attach the ones for that section).
5. Put what the AI returns back into the repo: lesson files in `lessons/`, video scripts in `video-scripts/lessons/`, new files next to the others. Then rebuild and render: `python3 programs/defi-program/build_master.py && cd programs/defi-program/export && npm install && npm run build && node build_video.js lessons`

You're finishing and launching my **On-Chain Operator Program** on Whop: a DeFi education program that takes someone who knows nothing about crypto to complete mastery: a safe setup, 30 professional strategies, a risk-adjusted income engine, and operating as their own on-chain bank. It sits in my Whop store alongside Grid Bot Builder (\$997) and Elite Intel Community (\$67/mo).

Everything is built: 6 stages, 15 modules, **107 lessons (15 of them expert-level) plus a Mastery Starter (N.0) opening every module**, 30 strategy playbooks in 7 levels, 2 capstones, 17 worksheets, 20 calculators, a store listing, an operations kit with the refund policy, 55 images, 18 finished videos (VSLs, welcome, module intros) and **121 lesson video scripts**. The lessons come in six section files, which I'll attach one at a time when you ask. Read the core files fully before doing anything.

Decided (don't ask about these, don't change them)

- Name: **On-Chain Operator Program**. Tagline: **From zero to your own on-chain bank**.

- Course tier **\$15,000 one-time**. **Live tier: price on application**, set on the sales call. Lifetime access.
- **Refund policy: 14-day conditional refund** (full text in `07-program-operations.md §9`): a full refund within 14 days if $\leq 10\%$ of lessons are completed (10 of 107), no capstone has been submitted and no live session or review has been used. Duplicate purchases and access problems not fixed within 7 days are always refunded. No refunds based on investment results. Chargebacks end access. Show it on the listing, at checkout and in the application acknowledgement.
- Launch to everyone; cold traffic goes **application** → **call** → **checkout**.
- 6 stages · 15 modules (0–14) · 107 lessons + 15 Mastery Starters · 30 strategy playbooks · 2 capstones.
- **Every module opens with a Mastery Starter (N.0)**: 60-second version, words you'll need, before you start, first safe step, beginner → practitioner → master ladder, "you've mastered this module when...". Keep this in every module.
- Every lesson: Objective → Explanation → Worked example → Checklist → 3-question quiz, with at least one image.

Brand (lead with the logo)

- Logo: two interlocked chain rings inside a hexagonal block with corner nodes. Colours: ink `#0B1F33`, navy `#12355B`, brand `#1F4E79`, teal `#2ee6a6` (on dark only), blue `#2a78d6`, orange `#eb6834`. Type: Inter; formulas in JetBrains Mono.
- Dark navy gradient with a faint node-network motif for marketing and video; `#FCFCFB` light surface for documents and diagrams.
- Never recolour or stretch the logo. Every marketing asset and video carries the logo and the footer "Educational content only · Not financial advice · No results are guaranteed".

Rules

1. **No promised, projected or guaranteed returns or income** anywhere: lessons, videos, ads, emails, sales page, call script. Show how every strategy loses money next to how it earns. No fake scarcity or countdowns.
2. "Operate as your own bank" is a **method**, never presented as a licence or financial service.
3. Never ask for or accept seed phrases, private keys, exchange API keys or account access.
4. Beginner standard: define every term on first use; do-it-now steps with a small test amount first; name products only as examples and tell learners to check availability where they live.
5. Every number is computed, not typed. Show the calculation.
6. In lessons, tax and legal points tell learners to consult a professional.
7. No Discord, Skool or Slack. Use only the Calendly links named in the files.
8. Never put internal IDs, plan IDs, list IDs or private emails in files. Use `<PLACEHOLDERS>`.

Work to do, in order

1. Section-by-section mastery review (six passes, 0 → 5)

Ask me for one section at a time. For each section:

- **Coverage:** check it against `09-mastery-map.md` ; write any missing or thin topic a complete DeFi mastery course needs (new lesson or addition, keeping the numbering).
- **Mastery Starter:** make sure each N.0 takes a complete beginner to ready.
- **Accuracy:** check facts are current (products, networks, mechanisms, dates); recompute every worked example; flag anything you can't verify.
- **Depth:** every lesson reaches its level (B/P/M), follows the template, and has at least one image.
- **Output:** change list, then each changed module file in full (`lessons/module-NN-name.md`), and an updated `09-mastery-map.md` if coverage changed.

2. Course video production: every lesson, maximum 25 minutes each

Do this in the same six passes, right after each section's review.

- **One video per lesson and per Mastery Starter** (121 videos: all lessons except 8.3), plus the **strategy library (8.3) as a 7-video series**, one per level.
- **Length:** each video at most **25 minutes**. Targets: Mastery Starters 5–10 min; lessons 8–20 min; expert lessons and the 8.3 level videos up to 25 min. If a lesson needs more, split it into Part 1 and Part 2, each ≤ 25 min.
- **Teach, don't read:** the baseline scripts in `video-scripts/lessons/` read the lesson text (3–5 min each). Rewrite each into a full teaching script: explain the idea in plain words, walk through the worked example step by step with the numbers on screen, show every diagram and say what to look at, add a real-world scenario, read the checklist, and do the quiz as question → pause → answer. Use the lesson file as the source of truth; don't add claims that aren't in it.
- **Structure (every video):** title card (module banner + lesson number/title) → objective → teaching segments (one idea per scene) → diagrams/charts → worked example → common mistakes → checklist → quiz → outro ("Do the checklist before moving on"). Add chapter timestamps.
- **Visuals:** use the existing images in `assets/` (module banners, diagrams, charts). Where a new visual would teach better, describe it precisely (title, what it shows, data) and name the file `assets/diagrams/<name>.png` or `assets/charts/<name>.png` so it can be added to the renderer.
- **Captions and brand:** burned-in captions, logo, footer disclaimer, the brand colours above.
- **Thumbnail:** for each video, a thumbnail title (≤ 5 words) and subtitle in the brand style.
- **Format:** deliver each video as a JSON file in exactly this schema (the renderer turns it into an MP4 with narration and captions), named `video-scripts/lessons/lesson-NN-M.json` :

```
{
  "id": "lesson-03-2", "title": "Lesson 3.2: LTV, liquidation threshold & health factor", "size": [1920, 1080],
  "group": "lessons", "maxMinutes": 25, "use": "Lesson 3.2 page in the Whop course.",
  "chapters": [
    [
      {
        "time": "0:00", "text": "Objective"
      },
      {
        "time": "1:10", "text": "Health factor explained"
      }
    ]
  ],
  "thumbnail": {
    "title": "Never get liquidated",
    "subtitle": "Lesson 3.2"
  },
  "scenes": [
    {
      "type": "image",
      "src": "assets/modules/module-03.png",
      "eyebrow": "Lesson 3.2",
      "vo": "Spoken narration...",
      "cap": "Written caption..."
    },
    {
      "type": "statement",
      "lines": ["Objective"],
      "sub": "On-screen text",
      "vo": "...",
      "cap": "..."
    },
    {
      "type": "bullets",
      "title": "Key ideas",
      "items": ["Short point", "Short point"],
      "vo": "...",
      "cap": "..."
    },
    {
      "type": "statement",
      "lines": ["Quiz", "Question?"],
      "sub": "Answer: ...",
      "subAt": 0.62,
      "vo": "Question? ... Pause and think. ... The answer: ...",
      "cap": "..."
    },
    {
      "type": "cta",
      "button": "Do the checklist",
      "sub": "Educational content only · Not financial advice",
      "vo": "...",
      "cap": "..."
    }
  ]
}
```

Scene types: `image`, `statement`, `bullets`, `stats`, `strike`, `logo`, `cta`. `vo` is what the narrator says (spell out abbreviations and numbers the way they should be spoken, e.g. "A.P.Y.", "health factor"); `cap` is the on-screen caption in normal written form. About 150 spoken words per minute, so a 15-minute video is about 2,200 words of `vo`.

- **If you can generate video directly** (voice and visuals) in Whop, also produce the finished videos in this brand and structure, each ≤ 25 minutes, and tell me which tool you used.

3. Build the course in Whop

Create (or give me exact values and steps for) the product, both plans (Course \$15,000 one-time; Live on application), the refund policy text at checkout, the course with 15 chapters (Start here + Modules 0–14), every lesson page (lesson text, images, its video), the Day-1 Setup Kit PDF, capstone pages, and the drip rule (each stage unlocks when the previous stage's quizzes are passed). Upload the logo, banner and 7 gallery images in the order in `06-whop-store-listing.md`, and the main VSL as the listing video.

4. Sales page

A complete, responsive HTML page in the brand: hero with logo and the main VSL (`video/vsl-main.mp4`), the path (6 stages), curriculum, expert lessons, own-bank section, strategy library, what's included (Course vs Live), who it's for/not for, FAQ (including the refund policy), application CTA, disclaimer footer. One file, Google Fonts (Inter) only.

5. Application form

Build-ready spec from operations kit §4 (questions, field types, required flags, scoring, and the acknowledgement that includes the refund policy).

6. Marketing pack

20 ad hooks, 5 ad scripts (15–30s) in the VSL style, a 5-email launch sequence, 10 organic post ideas, and 60s + 15s VSL cuts in the JSON scene schema above. All with no return claims.

7. Final checklist

Everything left to launch, in order.

Output format

- Deliver complete files, each under a heading with its file name, ready to drop into `programs/defi-program/`.
- Keep existing file names; new files get the next number (`11-sales-page.html` , `12-application-form.md` , `13-marketing-pack.md` , ...).
- After each item, list what changed and anything you couldn't verify.

Done when

All six sections reviewed · 121 lesson videos + the 7-part strategy series scripted (or produced), each ≤ 25 minutes · course built in Whop with the refund policy · sales page, application form and marketing pack delivered · final checklist written.